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DEPARTMENT OF DEFENSE HANDBOOK

CABLE COMPARISON HANDBOOK; DATA PERTAINING TO ELECTRIC SHIPBOARD CABLE

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AMSC N/A FSC 6145

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FOREWORD

1. This handbook is approved for use by all Departments and Agencies of the Department of Defense.

2. This document supplements departmental manuals, directives, and military standards and provides basic information on shipboard cables. This handbook contains listings of shipboard cables, data tables, and supersession information and should provide valuable facts and guidance to personnel responsible for the design, handling, installation, and maintenance of shipboard cable.

3. The Cable Comparison Handbook was first published in 1946 to facilitate utilization of electrical shipboard cable and to assist in the planning of cable installations. A reprint was issued in 1948 and revised editions were printed in 1953, 1956, 1960, 1964, 1975, 1977, and 1989. This edition contains information on current cables. The information is based on current data, section 304 of applicable ship specifications, and applicable military specifications, and detailed specifications for shipboard cable, design, construction, installation, and utilization.

4. For many years, most shipboard power and lighting cables for fixed installation used silicone-glass insulation, polyvinyl chloride jackets, aluminum armor, and watertight construction. It was determined that cables with all these features were not necessary for many applications, especially for applications within watertight compartments and non-critical areas above the watertightness level. Therefore, for applications within watertight compartments and non-critical areas, a new family of lower-cost cables was designed. This new family of cables is electrically and dimensionally interchangeable with silicone-glass insulated cables of equivalent sizes and is covered in MIL-DTL-915.

Events in ships have determined that cables jacketed with polyvinyl chloride presented the dangers of toxic fumes and dense, impenetrable smoke when undergoing combustion. These hazards became increasingly evident when an electrical fire smoldered through the cableways aboard a U. S. Navy surface ship. Because of the overwhelming amount of smoke and fumes, firefighters were unable to effectively control the fire, and a large amount of damage resulted. A new family of low smoke, low toxicity cable, constructed with a polyolefin jacket vice polyvinyl chloride jacket, conforms to rigid toxicity and smoke indexes and effectively reduces the hazards associated with the polyvinyl chloride jacketed cables. Since the polyolefin jacket provides protection equivalent to cable armor, the requirement for cable armor for most cable types has been deleted. The new low smoke, unarmored cable is covered by MIL-DTL-24643.

A family of lightweight cables has also been introduced to aid in the elimination of excessive weight from ships. Considering the substantial amount of cable present in a ship or submarine, a reduction in cable weight will significantly reduce the overall load, thus improving performance and increasing efficiency. This new family of lightweight cables is constructed from cross-linked polyalkene and mica polyimide insulation with a cross-linked polyolefin jacket. The lightweight cable is also low smoke and is covered by MIL-DTL-24640.

5. Cables are listed in this handbook by general classification according to application and design. Electrical shipboard cables are under control of the Naval Supply Systems Command and are covered in Federal Supply Class (FSC) 6145.

6. Comments, suggestions, or questions on this document should be addressed to Commander, Naval Sea Systems Command, ATTN: SEA 05T, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard, DC 20376-5160 or emailed to CommandStandards@navy.mil, with the subject line “Document Comment”. Since contact information can change, you may want to verify the currency of this address information using ASSIST at https://assist.dla.mil.

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1. SCOPE

1.1 Scope. This handbook is for guidance only and cannot be cited as a requirement. This handbook provides technical guidance that may help the Government, design agents, and the fleet with cable applications, performance, and selection. This handbook is intended to aid supply, installation, and maintenance activities in utilization of electric shipboard cable, particularly in the selection of alternate or substitute cables for use in lieu of specified types and sizes that might not be immediately available. This handbook is also intended to aid in selecting currently available cables for replacement of obsolete cables. This handbook does not cover shore-use cable, magnet wire, fiberoptic cable, coaxial cables, or radio frequency special cables used in connection with minesweeping and harbor defense.

2. APPLICABLE PUBLICATIONS

2.1 General. The publications listed below are not necessarily all of the publications referenced herein, but are those needed to understand the information provided by this handbook.

2.2 Government publications.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein.

COMMERCIAL ITEM DESCRIPTIONS

A-A-59608	-	Cable, Power and Welding, Special Purpose
A-A-59609	-	Cable, Welding, Special Purpose
A-A-59610	-	Cable, Communications, Special Purpose
A-A-59612	-	Cable, Instrumentation and Communication, Electronic
A-A-59615	-	Cable, Specialty Communications

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-DTL-915	-	Cable, Electrical, for Shipboard Use, General Specification for
MIL-DTL-915/6	-	Cable, Power Electrical, 600 Volts, for Outboard Use Only (not for Inboard Use), Type THOF
MIL-DTL-915/7	-	Cable, Power Electrical, for Outboard Use Only (not for Inboard Use), Type DSWS
MIL-DTL-915/8	-	Cable, Power Electrical, 600 Volts, for Outboard Use Only (not for Inboard Use), Types DSS, TSS, FSS, and 7SS
MIL-DTL-915/9	-	Cable, Power Electrical, 600 Volts, for Outboard Use Only (not for Inboard Use), Type JAS
MIL-DTL-915/10	-	Cable, Power Electrical, 600 Volts, for Outboard Use Only (not for Inboard Use), Type MCSF-4
MIL-DTL-915/22	-	Cable, Special Purpose Electrical, 300 Volts, for Outboard Use Only (not for Inboard Use), Type TSP (Including Variation Type TPSA)
MIL-DTL-915/47	-	Cable, Special Purpose, Electrical, for Outboard Use Only (not for Inboard Use), Type 1SWF
MIL-DTL-915/48	-	Cable, Special Purpose, Electrical, for Outboard Use Only (not for Inboard Use), Type 2SWF
MIL-DTL-915/58	-	Cable, Power Electrical, 600 Volts, for Outboard Use Only (not for Inboard Use), Type MWF
MIL-DTL-915/66	-	Cable, Special Purpose, Electrical, 59-Conductor, for Outboard Use Only (not for Inboard Use), Type MSPW

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MIL-DTL-915/67	-	Cable, Special Purpose Electrical, 59-Conductor, for Outboard Use Only (not for Inboard Use), Type MSP
MIL-C-915/79	-	Cable, Power Electrical, for Outboard use Only (not for Inboard use), Type TPUM-6
MIL-DTL-915/81	-	Cable, Electrical, One Twisted Pair AWG-20, for Submarine Outboard Use Only (not for Inboard Use), Type 1PR-A20E
MIL-C-915/82	-	Cable, Power Electrical, One Twisted Pair No. 16 Gauge Fluorpolymer Insulation, for Submarine Outboard use Only (Not for Inboard use), Type 1PR-16
MIL-C-915/83	-	Cable, Power Electrical, Seven Twisted Pairs No. 16 Gauge Fluoropolymer Insulation, for Submarine Outboard use Only (not for Inboard use), Type 7PR-16
MIL-C-915/84	-	Cable, Power Electrical, Two Shielded Twisted Pairs No. 16 Gauge, Flouropolymer Insulation, for Submarine Outboard use Only (not for Inboard use), Type 2SPR-16
MIL-C-915/85	-	Cable, Power Electrical, Three Twisted Pairs No. 16 Gauge, Fluoropolymer Insulation, for Submarine Outboard use Only (not for Inboard use), Type 3PR-16
MIL-C-915/86	-	Cable, Power Electrical, Twisted Quad No. 16 Gauge, Fluoropolymer Insulation for Submarine Outboard use Only (not for Inboard use), Type 1Q-16
MIL-C-915/87	-	Cable, Power Electrical, Twisted Triad, No. 16 Gauge, Fluoropolymer Insulation, for Submarine Outboard use Only (not for Inboard use), Type 1TR-16
MIL-C-915/88	-	Cable, Power Electrical, Seven Shielded Twisted Pairs with Overall Shield, No. 16 Gauge, Fluoropolymer Insulation, for Submarine Outboard use Only (not for Inboard use), Type 7SPR-16S
MIL-C-915/89	-	Cable, Power Electrical, One Shielded Twisted Pair No. 16 Gauge, Fluoropolymer Insulation, for Submarine Outboard use Only (not for Inboard use), Type 1SPR-16
MIL-DTL-24640	-	Cables, Lightweight, Low Smoke, Electric, for Shipboard Use, General Specification for
MIL-DTL-24640/1	-	Cable, Electric, -20 to +105 °C, 600 Volts, Lightweight, Two-Conductor, Low Power, Types DX and DXWB
MIL-DTL-24640/2	-	Cable, Electric, -20 to +105 °C, 600 Volts, Lightweight, Three-Conductor, Low Power, Types TX and TXWB
MIL-DTL-24640/3	-	Cable, Electric, -20 to +105 °C, 600 Volts, Lightweight, Four-Conductor, Low Power, Types FX and FXWB
MIL-DTL-24640/4	-	Cable, Electric, -20 to +105 °C, 600 Volts, Lightweight, Multi-Pair, 20 AWG, Types TTX and TTXWB
MIL-DTL-24640/5	-	Cable, Electric, -20 to +105 °C, 600 Volts, Lightweight, Multi-Pair, 20 AWG, Types TTXS or TTXSW (Including Variation Type TTXSO or TTXSOW)
MIL-DTL-24640/6	-	Cable, Electric, -20 to +105 °C, 600 Volts, Lightweight, Multi-Pair, 22 AWG, Types 2XAO and 2XAOW
MIL-DTL-24640/7	-	Cable, Electric, -20 to +105 °C, 600 Volts, Lightweight, Multi-Conductor, 22 AWG, Types 1XMSO and 1XMSOW

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MIL-DTL-24640/8	-	Cable, Electric, -20 to +105 °C, 600 Volts, Lightweight, Multi-Conductor, 20 AWG, Types MXO and MXOW
MIL-DTL-24640/9	-	Cable, Electric, -20 to +105 °C, 600 Volts, Lightweight, Multi-Pair, 22 AWG, Type 2XS and 2XSW (Including Variation Types 2XSO and 2XSOW)
MIL-DTL-24640/10	-	Cable, Electric, -20 to +105 °C, 600 Volts, Lightweight, Multi-Conductor, 16 AWG, Types MXSO and MXSOW
MIL-DTL-24640/11	-	Cable, Electric, -20 to +105 °C, 600 Volts, Lightweight, Multi-Triad, 18 AWG, Types 3XS and 3XSW
MIL-DTL-24640/12	-	Cable, Electric, -20 to +105 °C, 600 Volts, Lightweight, Multi-Pair, 26 AWG, Types 2XO and 2XOW
MIL-DTL-24640/13	-	Cable, Electric, -20 to +105 °C, 600 Volts, Lightweight, Multi-Pair, 26 AWG, Types 2XSXO and 2XSXOW
MIL-DTL-24640/14	-	Cable, Electric, -20 to +105 °C, 600 Volts, Lightweight, Multi-Conductor, 22 AWG, Type 1XSOW
MIL-DTL-24640/15	-	Cable, Electric, -20 to +105 °C, 600 Volts, Lightweight, Multi-Pair, 22 AWG, Type 2XSAW (Including Variation Types 2XSAOW and 2XSAWA)
MIL-DTL-24640/16	-	Cable, Electric, -20 to +105 °C, 600 Volts, Lightweight, Multi-Pair, 18 AWG, Type 2XSW (Including Variation Type 2XSOW)
MIL-DTL-24640/17	-	Cable, Electric, -20 to +105 °C, 600 Volts, Lightweight, Multi-Pair, 26 AWG, Type 2XOW
MIL-DTL-24640/18	-	Cable, Electric, -20 to +105 °C, 600 Volts, Lightweight, Multi-Triad, 18 AWG, Type 3XSW (Including Variation Type 3XSOW)
MIL-DTL-24640/19	-	Cable, Electric, -20 to +105 °C, 600 Volts, Lightweight, Two-Conductor, Power, Type DXW (Including Variation Type DXOW)
MIL-DTL-24640/20	-	Cable, Electric, -20 to +105 °C, 600 Volts, Lightweight, Three-Conductor, Low Power, Type TXW (Including Variation Type TXOW)
MIL-DTL-24640/21	-	Cable, Electric, -20 to +105 °C, 600 Volts, Lightweight, Four-Conductor, Low Power, Type FXW (Including Variation Type FXOW)
MIL-DTL-24640/22	-	Cable, Electric, -20 to +105 °C, 600 Volts, Lightweight, Seven-Conductor, Low Power, Type 7XW
MIL-DTL-24640/23	-	Cable, Electric, -20 to +105 °C, 600 Volts, Lightweight, Multi-Conductor, 18 AWG, Type MXCW (Including Variation Type MXCOW)
MIL-DTL-24640/24	-	Cable, Electric, -20 to +105 °C, 600 Volts, Lightweight, Multi-Pair, 22 AWG, Type TTXW (Including Variation Type TTXOW)
MIL-DTL-24640/25	-	Cable, Electrical, -20 to +105 °C, 600 Volts, Type 9XS
MIL-DTL-24640/26	-	Cable, Electrical, -20 to +105 °C, 600 Volts, Type 5XO
MIL-DTL-24643	-	Cables, Electric, Low Smoke, Halogen-Free, for Shipboard Use, General Specification for
MIL-DTL-24643/2	-	Cable, Electrical, -4 Degrees Fahrenheit (-20 Degrees Celsius) to +194 Degrees Fahrenheit (+90 Degrees Celsius), 300 Volts, Types LSDCOP and LSTCOP
MIL-DTL-24643/3	-	Cable, Electrical, -4 Degrees Fahrenheit (-20 Degrees Celsius) to +194 Degrees Fahrenheit (+90 Degrees Celsius), 600 Volts, Types LSSHOF, LSDHOF, LSTHOF, and LSFHOF

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MIL-DTL-24643/4	-	Cable, Electrical, -4 Degrees Fahrenheit (-20 Degrees Celsius) to +194 Degrees Fahrenheit (+90 Degrees Celsius), 600 Volts, Type LSMCOS
MIL-DTL-24643/5	-	Cable, Electrical, -4 Degrees Fahrenheit (-20 Degrees Celsius) to +194 Degrees Fahrenheit (+90 Degrees Celsius), 600 Volts, Type LSMDU
MIL-DTL-24643/7	-	Cable, Electrical, -4 Degrees Fahrenheit (-20 Degrees Celsius) to +194 Degrees Fahrenheit (+90 Degrees Celsius), 600 Volts, Type LSMHOF
MIL-DTL-24643/8	-	Cable, Electrical, -4 Degrees Fahrenheit (-20 Degrees Celsius) to +194 Degrees Fahrenheit (+90 Degrees Celsius), 300 Volts, Type LSMMOP
MIL-DTL-24643/9	-	Cable, Electrical, -4 Degrees Fahrenheit (-20 Degrees Celsius) to +194 Degrees Fahrenheit (+90 Degrees Celsius), 600 Volts, Type LSMRI
MIL-DTL-24643/10	-	Cable, Electrical, -4 Degrees Fahrenheit (-20 Degrees Celsius) to +194 Degrees Fahrenheit (+90 Degrees Celsius), 600 Volts, Type LSPBTMU
MIL-DTL-24643/11	-	Cable, Electrical, -4 Degrees Fahrenheit (-20 Degrees Celsius) to +194 Degrees Fahrenheit (+90 Degrees Celsius), 600 Volts, Type LSSSF
MIL-DTL-24643/12	-	Cable, Electrical, -4 Degrees Fahrenheit (-20 Degrees Celsius) to +194 Degrees Fahrenheit (+90 Degrees Celsius), 300 Volts, Type LSTTOP
MIL-DTL-24643/13	-	Cable, Electrical, -4 Degrees Fahrenheit (-20 Degrees Celsius) to +194 Degrees Fahrenheit (+90 Degrees Celsius), 300 Volts, Type LSTTRS
MIL-DTL-24643/14	-	Cable, Electrical, -4 Degrees Fahrenheit (-20 Degrees Celsius) to +221 Degrees Fahrenheit (+105 Degrees Celsius), 1,000 Volts, Type LSSSGU
MIL-DTL-24643/15	-	Cable, Electrical, -4 Degrees Fahrenheit (-20 Degrees Celsius) to +221 Degrees Fahrenheit (+105 Degrees Celsius), 1,000 Volts, Type LSDSGU
MIL-DTL-24643/16	-	Cable, Electrical, -4 Degrees Fahrenheit (-20 Degrees Celsius) to +221 Degrees Fahrenheit (+105 Degrees Celsius), 1,000 Volts, Type LSTSGU
MIL-DTL-24643/17	-	Cable, Electrical, -4 Degrees Fahrenheit (-20 Degrees Celsius) to +221 Degrees Fahrenheit (+105 Degrees Celsius), 1,000 Volts, Type LSFSGU
MIL-DTL-24643/18	-	Cable, Electrical, -4 Degrees Fahrenheit (-20 Degrees Celsius) to +221 Degrees Fahrenheit (+105 Degrees Celsius), 1,000 Volts, Type LSMSCU (Including Variation LSMSCS)
MIL-DTL-24643/19	-	Cable, Electrical, -4 Degrees Fahrenheit (-20 Degrees Celsius) to +221 Degrees Fahrenheit (+105 Degrees Celsius), 1,000 Volts, Type LS6SGU
MIL-DTL-24643/20	-	Cable, Electrical, -4 Degrees Fahrenheit (-20 Degrees Celsius) to +221 Degrees Fahrenheit (+105 Degrees Celsius), 1,000 Volts, Type LS7SGU
MIL-DTL-24643/21	-	Cable, Electrical, -4 Degrees Fahrenheit (-20 Degrees Celsius) to +221 Degrees Fahrenheit (+105 Degrees Celsius), 300 Volts, Types LSTCJU and LSTCTU
MIL-DTL-24643/22	-	Cable, Electrical, -4 Degrees Fahrenheit (-20 Degrees Celsius) to +221 Degrees Fahrenheit (+105 Degrees Celsius), 5,000 Volts, Type LS5KVTSGU (Including Variation LS5KVTSGUS)
MIL-DTL-24643/23	-	Cable, Electrical, -4 Degrees Fahrenheit (-20 Degrees Celsius) to +221 Degrees Fahrenheit (+105 Degrees Celsius), 300 Volts, Type LSTTSU
MIL-DTL-24643/24	-	Cable, Electrical, -4 Degrees Fahrenheit (-20 Degrees Celsius) to +302 Degrees Fahrenheit (+150 Degrees Celsius), 300 Volts, Types LSTCJX, LSTCKX, LSTCTX, LSTCJXN, LSTCKXN, and LSTCTXN

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MIL-DTL-24643/25	-	Cable, Electrical, -4 Degrees Fahrenheit (-20 Degrees Celsius) to +302 Degrees Fahrenheit (+150 Degrees Celsius), 300 Volts, Type LSPI
MIL-DTL-24643/26	-	Cable, Electrical, -4 Degrees Fahrenheit (-20 Degrees Celsius) to +302 Degrees Fahrenheit (+150 Degrees Celsius), 600 Volts, Types LSDPS, LSTPS, LSFPS, LS7PS, LSDPSN, LSTPSN, LSFPSN, and LS7PSN
MIL-DTL-24643/27	-	Cable, Electrical, -4 Degrees Fahrenheit (-20 Degrees Celsius) to +194 Degrees Fahrenheit (+90 Degrees Celsius), 600 Volts, Type LS2AU (Including Variation LS2AUS)
MIL-DTL-24643/28	-	Cable, Electrical, -4 Degrees Fahrenheit (-20 Degrees Celsius) to +194 Degrees Fahrenheit (+90 Degrees Celsius), 300 Volts, Type LS1S50MU (Including Variation LS1S50MUS)
MIL-DTL-24643/29	-	Cable, Electrical, -4 Degrees Fahrenheit (-20 Degrees Celsius) to +194 Degrees Fahrenheit (+90 Degrees Celsius), 300 Volts, Type LSMU (Including Variation LSMUS)
MIL-DTL-24643/30	-	Cable, Electrical, -4 Degrees Fahrenheit (-20 Degrees Celsius) to +194 Degrees Fahrenheit (+90 Degrees Celsius), 600 Volts, Type LS1SWU
MIL-DTL-24643/31	-	Cable, Electrical, -4 Degrees Fahrenheit (-20 Degrees Celsius) to +194 Degrees Fahrenheit (+90 Degrees Celsius), 600 Volts, Type LS2SU (Including Variation LS2SUS)
MIL-DTL-24643/32	-	Cable, Electrical, -4 Degrees Fahrenheit (-20 Degrees Celsius) to +194 Degrees Fahrenheit (+90 Degrees Celsius), 600 Volts, Type LS2SWAU
MIL-DTL-24643/33	-	Cable, Electrical, -4 Degrees Fahrenheit (-20 Degrees Celsius) to +194 Degrees Fahrenheit (+90 Degrees Celsius), 600 Volts, Type LS2SWU
MIL-DTL-24643/34	-	Cable, Electrical, -4 Degrees Fahrenheit (-20 Degrees Celsius) to +194 Degrees Fahrenheit (+90 Degrees Celsius), 300 Volts, Type LSMS
MIL-DTL-24643/35	-	Cable, Electrical, -4 Degrees Fahrenheit (-20 Degrees Celsius) to +194 Degrees Fahrenheit (+90 Degrees Celsius), 600 Volts, Type LS3SU (Including Variation LS3SUS)
MIL-DTL-24643/36	-	Cable, Electrical, -4 Degrees Fahrenheit (-20 Degrees Celsius) to +194 Degrees Fahrenheit (+90 Degrees Celsius), 600 Volts, Type LS3SWU (Including Variation LS3SWUS)
MIL-DTL-24643/37	-	Cable, Electrical, -4 Degrees Fahrenheit (-20 Degrees Celsius) to +194 Degrees Fahrenheit (+90 Degrees Celsius), 300 Volts, Type LS3U
MIL-DTL-24643/38	-	Cable, Electrical, -4 Degrees Fahrenheit (-20 Degrees Celsius) to +194 Degrees Fahrenheit (+90 Degrees Celsius), 600 Volts, Type LSECM
MIL-DTL-24643/39	-	Cable, Electrical, -4 Degrees Fahrenheit (-20 Degrees Celsius) to +194 Degrees Fahrenheit (+90 Degrees Celsius), 300 Volts, Type LS1S75MU
MIL-DTL-24643/40	-	Cable, Electrical, -4 Degrees Fahrenheit (-20 Degrees Celsius) to +194 Degrees Fahrenheit (+90 Degrees Celsius), 600 Volts, Type LS1SMU
MIL-DTL-24643/41	-	Cable, Electrical, -4 Degrees Fahrenheit (-20 Degrees Celsius) to +194 Degrees Fahrenheit (+90 Degrees Celsius), 600 Volts, Type LS1SAU
MIL-DTL-24643/42	-	Cable, Electrical, -4 Degrees Fahrenheit (-20 Degrees Celsius) to +194 Degrees Fahrenheit (+90 Degrees Celsius), 600 Volts, Type LS1SU
MIL-DTL-24643/43	-	Cable, Electrical, -4 Degrees Fahrenheit (-20 Degrees Celsius) to +194 Degrees Fahrenheit (+90 Degrees Celsius), 600 Volts, Types LS2SJ, LS3SJ, and LS4SJ

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MIL-DTL-24643/44	-	Cable, Electrical, -4 Degrees Fahrenheit (-20 Degrees Celsius) to +194 Degrees Fahrenheit (+90 Degrees Celsius), 600 Volts, Type LS3SF
MIL-DTL-24643/45	-	Cable, Electrical, -4 Degrees Fahrenheit (-20 Degrees Celsius) to +194 Degrees Fahrenheit (+90 Degrees Celsius), 300 Volts, Type LS2U (Including Variation LS2WU)
MIL-DTL-24643/46	-	Cable, Electrical, -4 Degrees Fahrenheit (-20 Degrees Celsius) to +194 Degrees Fahrenheit (+90 Degrees Celsius), 600 Volts, Type LS2WAU
MIL-DTL-24643/47	-	Cable, Electrical, -4 Degrees Fahrenheit (-20 Degrees Celsius) to +194 Degrees Fahrenheit (+90 Degrees Celsius), 600 Volts, Type LS1SMWU
MIL-DTL-24643/48	-	Cable, Electrical, -4 Degrees Fahrenheit (-20 Degrees Celsius) to +194 Degrees Fahrenheit (+90 Degrees Celsius), 1,000 Volts, Type LSDNW
MIL-DTL-24643/49	-	Cable, Electrical, -4 Degrees Fahrenheit (-20 Degrees Celsius) to +194 Degrees Fahrenheit (+90 Degrees Celsius), 1,000 Volts, Type LSTNW
MIL-DTL-24643/50	-	Cable, Electrical, -4 Degrees Fahrenheit (-20 Degrees Celsius) to +194 Degrees Fahrenheit (+90 Degrees Celsius), 1,000 Volts, Type LSFNW
MIL-DTL-24643/51	-	Cable, Electrical, -4 Degrees Fahrenheit (-20 Degrees Celsius) to +194 Degrees Fahrenheit (+90 Degrees Celsius), 1,000 Volts, Type LSMNW
MIL-DTL-24643/52	-	Cable, Electrical, -4 Degrees Fahrenheit (-20 Degrees Celsius) to +194 Degrees Fahrenheit (+90 Degrees Celsius), 300 Volts, Type LSTPNW
MIL-DTL-24643/53	-	Cable, Electrical, -4 Degrees Fahrenheit (-20 Degrees Celsius) to +194 Degrees Fahrenheit (+90 Degrees Celsius), 3,000 Volts, Types LSSRW, LSDRW, and LSTRW
MIL-DTL-24643/54	-	Cable, Electrical, -4 Degrees Fahrenheit (-20 Degrees Celsius) to +194 Degrees Fahrenheit (+90 Degrees Celsius), 1,000 Volts, Type LS8NW
MIL-DTL-24643/55	-	Cable, Electrical, -4 Degrees Fahrenheit (-20 Degrees Celsius) to +194 Degrees Fahrenheit (+90 Degrees Celsius), 1,000 Volts, Type LS4NW
MIL-DTL-24643/56	-	Cable, Electrical, -4 Degrees Fahrenheit (-20 Degrees Celsius) to +194 Degrees Fahrenheit (+90 Degrees Celsius), 600 Volts, Type LS2SWL
MIL-DTL-24643/57	-	Cable, Electrical, -4 Degrees Fahrenheit (-20 Degrees Celsius) to +194 Degrees Fahrenheit (+90 Degrees Celsius), 300 Volts, Type LS2UW (Including Variation LS2UWS)
MIL-DTL-24643/58	-	Cable, Electrical, -4 Degrees Fahrenheit (-20 Degrees Celsius) to +194 Degrees Fahrenheit (+90 Degrees Celsius), 300 Volts, Type LS2CS
MIL-DTL-24643/59	-	Cable, Electrical, -4 Degrees Fahrenheit (-20 Degrees Celsius) to +176 Degrees Fahrenheit (+80 Degrees Celsius), 150 Volts, Type LSC5FS (Including Variations LSC5FSW, LSC5OS, and LSC5OSW)
MIL-DTL-24643/61	-	Cable, Electrical, -4 Degrees Fahrenheit (-20 Degrees Celsius) to +176 Degrees Fahrenheit (+80 Degrees Celsius), 150 Volts, Types LSC5P, LSC5POS, and LSC5POSR
MIL-DTL-24643/62	-	Cable, Electrical, -4 Degrees Fahrenheit (-20 Degrees Celsius) to +176 Degrees Fahrenheit (+80 Degrees Celsius), 150 Volts, Type LSPB2SD (Including Variations LSPB2SDW, LSPB2SDOS, LSPB2SDOSW)
MIL-DTL-24643/63	-	Cable, Electrical, -4 Degrees Fahrenheit (-20 Degrees Celsius) to +221 Degrees Fahrenheit (+105 Degrees Celsius), 1,000 Volts, Type LSYSGU

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MIL-DTL-24643/64 - Cable, Electrical, Twisted Shielded Pair(s), -4 Degrees Fahrenheit (-20 Degrees Celsius) to +194 Degrees Fahrenheit (+90 Degrees Celsius), 600 Volts, Type LSTPSJ

MIL-DTL-24643/65 - Cable, Electrical, -4 Degrees Fahrenheit (-20 Degrees Celsius) to +194 Degrees Fahrenheit (+90 Degrees Celsius), 150 Volts, Type LS3C179DT (Including Variation LS6C179DT)

MIL-DTL-24643/66 - Cable, Electrical, -4 Degrees Fahrenheit (-20 Degrees Celsius) to +221 Degrees Fahrenheit (+105 Degrees Celsius), 1,000 Volts, Type LS2OW

MIL-DTL-24643/67 - Cable, Electrical, -4 Degrees Fahrenheit (-20 Degrees Celsius) to +221 Degrees Fahrenheit (+105 Degrees Celsius), 1,000 Volts, Type LS3OW

MIL-DTL-24643/68 - Cable, Electrical, -4 Degrees Fahrenheit (-20 Degrees Celsius) to +221 Degrees Fahrenheit (+105 Degrees Celsius), 1,000 Volts, Type LS4OW

MIL-DTL-24643/69 - Cable, Electrical, -4 Degrees Fahrenheit (-20 Degrees Celsius) to +194 Degrees Fahrenheit (+90 Degrees Celsius), 300 Volts, Type LSC264

MIL-DTL-24643/71 - Cable, Electrical, -4 Degrees Fahrenheit (-20 Degrees Celsius) to +194 Degrees Fahrenheit (+90 Degrees Celsius), 1,000 Volts, Flexible, Type LSSCF

MIL-DTL-24643/72 - Cable, Electrical, -4 Degrees Fahrenheit (-20 Degrees Celsius) to +194 Degrees Fahrenheit (+90 Degrees Celsius), 1,000 Volts, Flexible, for Inboard Use Only (Not for Outboard Use), Type LSFCF

MIL-DTL-24643/73 - Cable, Electrical, -4 Degrees Fahrenheit (-20 Degrees Celsius) to +194 Degrees Fahrenheit (+90 Degrees Celsius), 1,000 Volts, Flexible, Type LSTCF

MIL-DTL-24643/75 - Cable, Electrical, -4 Degrees Fahrenheit (-20 Degrees Celsius) to +194 Degrees Fahrenheit (+90 Degrees Celsius), 300 Volts, Type LSLCFS

MIL-DTL-24643/76 - Cable, Electrical, -20 °C to +105 °C, 5,000 Volts, 133 Percent Rated, Type LS5KVTEPSG

MIL-DTL-24643/77 - Cable, Electrical, -4 Degrees Fahrenheit (-20 Degrees Celsius) to +176 Degrees Fahrenheit (+80 Degrees Celsius), 150 Volts, Type LSC6FS, LSC6OS (Including Watertight Variation LSC6OSW)

MIL-DTL-24643/78 - Cable, Electrical, -4 Degrees Fahrenheit (-20 Degrees Celsius) to +176 Degrees Fahrenheit (+80 Degrees Celsius), 300 Volts, Type LSCBPSJ (Including Variant LSCBPSJW)

MIL-DTL-24643/79 - Cable, Electrical, -4 Degrees Fahrenheit (-20 Degrees Celsius) to +221 Degrees Fahrenheit (+105 Degrees Celsius), 2,000 Volts, 133 Percent Rated, Type LS2KVTSGSG

MIL-DTL-24643/80 - Cable, Electrical, -4 Degrees Fahrenheit (-20 Degrees Celsius) to +194 Degrees Fahrenheit (+90 Degrees Celsius), 1,000 Volts, Single Conductor, Type LST and Type LSTW

MIL-DTL-24643/81 - Cable, Electrical, -4 Degrees Fahrenheit (-20 Degrees Celsius) to +194 Degrees Fahrenheit (+90 Degrees Celsius), 1,000 Volts, Topside, Multi Conductor, Types LSE1E, LSE2E, LSE3E, LSE4E, LSE7E, LSE10E, LSE14E, LSE19E, LSE24E, LSE30E, LSE33E, LSE37E, LSE44E, and LSE61E (Including Variants LSE1EW, LSE2EW, LSE3EW, LSE4EW, LSE7EW, LSE10EW, LSE14EW, LSE19EW, LSE24EW, LSE30EW, LSE33EW, LSE37EW, LSE44EW, and LSE61EW)

MIL-DTL-24643/82 - Cable, Electrical, -4 Degrees Fahrenheit (-20 Degrees Celsius) to +194 Degrees Fahrenheit (+90 Degrees Celsius), 300 Volts, Topside, Multi-Pair,

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Types LSE6U, LSE10U, LSE15U, LSE19U, LSE30U, LSE45U, LSE60U, LSE7AU, LSE19AU, LSE40AU, LSE3TTOP, LSE5TTOP, LSE10TTOP, and LSE15TTOP (Including Variants LSE7UW, LSE10UW, LSE19UW, LSE37UW, LSE42UW, LSE7AUW, LSE10AUW, LSE19AUW, LSE37AUW, and LSE40AUW)

MIL-DTL-24643/83 - Cable, Electrical, -4 Degrees Fahrenheit (-20 Degrees Celsius) to +194 Degrees Fahrenheit (+90 Degrees Celsius), 300 Volts, Topside, 75 Ohms, Multi Shielded Pairs, Types LSE1SPE, LSE2SPE, LSE3SPE, LSE4SPE, LSE6SPE, LSE7SPE, LSE8SPE, LSE10SPE, LSE12SPE, LSE14SPE, LSE16SPE, LSE19SPE, LSE24SPE, LSE30SPE, LSE37SPE, LSE44SPE, and LSE61SPE (Including Variants LSE1SPEW, LSE2SPEW, LSE3SPEW, LSE4SPEW, LSE6SPEW, LSE7SPEW, LSE8SPEW, LSE10SPEW, LSE12SPEW, LSE14SPEW, LSE16SPEW, LSE19SPEW, LSE24SPEW, LSE30SPEW, LSE37SPEW, LSE44SPEW, and LSE61SPEW)

MIL-DTL-24643/84 - Cable, Electrical, -4 Degrees Fahrenheit (-20 Degrees Celsius) to +176 Degrees Fahrenheit (+80 Degrees Celsius), 150 Volts, Topside, Type LSEC6FS (Including Variants LSEC6OS and LSEC6OSW)

MIL-DTL-24643/85 - Cable, Electrical, -22 Degrees Fahrenheit (-30 Degrees Celsius) to +221 Degrees Fahrenheit (+105 Degrees Celsius), 1,000 Volts, Topside, Multi Conductor Circuit Integrity Types LSE1CI, LSE2CI, LSE3CI, LSE4CI, LSE6CI, LSE7CI, LSE10CI, LSE14CI, LSE19CI, LSE30CI, LSE37CI, LSE44CI, LSE61CI, and LSE91CI (Including Variants LSE1CIW, LSE2CIW, LSE3CIW, LSE4CIW, LSE6CIW, LSE7CIW, LSE10CIW, LSE14CIW, LSE19CIW, LSE30CIW, LSE37CIW, LSE44CIW, LSE61CIW, and LSE91CIW)

MIL-DTL-24643/86 - Cable, Electrical, -4 Degrees Fahrenheit (-20 Degrees Celsius) to +221 Degrees Fahrenheit (+105 Degrees Celsius), 5,000 Volts, Type LS5KVTSSGUSHR

MIL-DTL-32180 - Cable Assembly, Aircraft Electrical Service

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-2003-1 - Electric Plant Installation Standard Methods for Surface Ships and Submarines (Cable)

MIL-STD-2003-4 - Electric Plant Installation Standard Methods for Surface Ships and Submarines (Cableways)

DEPARTMENT OF DEFENSE HANDBOOKS

MIL-HDBK-454 - General Guidelines for Electronic Equipment

(Copies of these publications are available online at https://quicksearch.dla.mil/.)

2.2.2 Other Government publications. The following other Government publications form a part of this document to the extent specified herein.

NAVAL SEA SYSTEMS COMMAND (NAVSEA) PUBLICATIONS

S9407-AB-HBK-010 - Handbook of Shipboard Electromagnetic Shielding Practices

(Copies of this publication are available online via Model Based Product Support (MBPS) at https://mbps.navsea.cloud.navy.mil/Windchill. To request an MBPS account, submit a ticket to the ServiceNow Assistance Portal (SNAP), https://snap.navair.navy.mil/mbps (DoD PKI Certificate required).)

2.3 Non-Government publications. The following publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these publications are those cited in the solicitation or contract.

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IEEE

IEEE 45.8	-	IEEE Recommended Practice for Electrical Installations on Shipboard--Cable Systems
IEEE 400	-	IEEE Guide for Field Testing and Evaluation of the Insulation of Shielded Power Cable Systems Rated 5 kV and Above
IEEE 1580	-	IEEE Recommended Practice for Marine Cable for Use on Shipboard and Fixed or Floating Facilities
IEEE 1637	-	IEEE Guide for Selection and Application of Terminations for Shielded AC Power Cable Rated 5 kV to 46 kV

(Copies of these publications are available online at www.ieee.org.)

TELECOMMUNICATIONS INDUSTRY ASSOCIATION (TIA)

TIA ANSI/TIA 568.2	-	Balanced Twisted-Pair Telecommunications Cabling and Components Standard

(Copies of this publication are available online at www.tiaonline.org.)

3. DEFINITIONS

3.1 Ampacity. Ampacity is an electrical property denoting current carrying capacity. The ampacity of the cable is normally defined as the current, in amperes, a conductor can carry continuously under the conditions of use (conditions of the surrounding medium in which the cables are installed) without exceeding the cable temperature rating limit.

3.2 Circuit integrity. Circuit integrity indicates cable construction and provides added protection that will allow that cable to function for a longer period under fire conditions.

3.3 Ambient temperature. The ordinary temperature (°C) of the air in the compartment where the cable is located.

4. GENERAL GUIDANCE

4.1 General description of data and cable types. General information concerning cable types is specified in 4.1.1 through 4.1.8, below.

4.1.1 Cable types and construction. A list of cable types and construction characteristics is provided in 5.1, 5.2, and 5.3. Cables listed are in accordance with MIL-DTL-24643 (for low smoke cable), MIL-DTL-24640 (for lightweight and low smoke cable), and MIL-DTL-915 (for specific cables). Cables are listed alphabetically within application and design characteristics. This listing provides a brief description of the number of conductors, the type of insulation and jacketing employed in construction, and watertightness. In addition, the applicable specification sheet for the cable type is provided.

4.1.2 Identification information. Conductor identification information employed for various types of cables is provided in 5.4.

4.1.3 Cable available through military specifications (tables I, II, and III). Tables I, II, and III provide a listing of cables, available through military specifications, according to general application. This list is intended to aid in the selection of cables for different applications.

4.1.4 Commercially available cables (table IV). Table IV provides a list of commercially available cables, according to general application.

4.1.5 Supersession data (table V). Table V lists cable types alphabetically using MIL-DTL-915 cable designations. Cable types that were covered by previous specifications are listed with the corresponding present type.

4.1.6 Physical characteristics and electrical ratings of cables (tables VI, VII, and VIII). Tables VI, VII, and VIII contain information concerning physical characteristics and electrical ratings at specific operating temperatures. Cables should not be loaded in excess of these ratings. Additional explanations applicable to tables VI, VII, and VIII are as follows:

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a. Cables are listed according to military specification sheet, functional category within that military specification sheet (watertight and non-watertight, flexing and non-flexing, power and lighting, control, or communications and electronics), and size and type designation within each table.

b. Rated voltages are not listed for some of the cable types designed for voice communication, analog or digital data transmission, or sending circuits, such as sonar and pyrometer. The applicability of these cable types should be determined from additional circuit parameters (signal waveform, frequency and peak amplitude, signal fidelity, pulse duration and recurrence frequency, attenuation, and frequency bandwidth).

c. Individual (Ind) and average (Avg): The notation Ind/Avg denotes that individual conductors can carry the current listed under Ind, provided the average of all currents in the individual conductors does not exceed the value listed under Avg.

d. The measurement point for minimum radius of bend should be the surface of the cable jacket that is on the innermost portion of the cable bend. Dimensions listed are approximately eight times the overall diameter of the cable or cord. However, during the installation or operation, a dimension of approximately 12 times the cable overall diameter for conduit bends, sheaves, and other curved surfaces around which the cable or cord may be pulled under tension should be used.

e. Unless otherwise indicated herein, all conductors are of the same size and soft, annealed copper. For additional data covering conductor stranding and conductor dimensions, see MIL-DTL-915, MIL-DTL-24640, and MIL-DTL-24643.

4.1.7 Ampacity derating factors (table IX). Table IX lists ampacity derating factors for ambient temperatures above 50 degrees Celsius (°C) (see 5.9).

4.1.8 Ampacity ratings for degaussing cable. Table X lists the ampacity ratings (maximum amperes per conductor) for degaussing cables.

4.1.9 Guidance for installation and maintenance of medium voltage cable. Due to physical and performance differences between higher voltage rated cables (5 kilovolts and greater) and lower voltage rated cables (less than 5 kilovolts), installation of higher voltage cables should include precautions in addition to the methods normally used for cable installation. In addition, due to physical and performance differences between shielded and unshielded medium voltage cables, installation of shielded cables should include precautions in addition to the methods normally used for cable installation. The following guidance applies for higher voltage rated cables:

a. Installation instructions in the applicable ship drawings and in applicable ship specifications should be followed.

b. The partial discharge (PD) (corona) inception voltage (CIV), and PD (corona) extinction voltage (CEV), and basic insulation level as defined in MIL-HDBK-454, should not be reduced or degraded by cable installation.

c. IEEE 1637 should be consulted for methods for selecting terminations for shielded alternating current (AC) power cables to limit partial discharge.

d. Proper cable type selection should be considered to limit the effect on the cable of partial discharge and the effect of complex waveforms from the power system. For example, the semiconductor tape layer required in MIL-DTL-24643/76 for type LS5KVTEPSG cable serves to control the partial discharge.

e. The electric field stress at the point of cable and conductor shield termination should be managed. Excess stress in the dielectric at the shield termination may be above the design stress and may lead to premature insulation failure. Termination methods that control electric field distribution may be necessary to limit PD at the point of shield termination for some medium voltage cables.

f. PD can occur due to defects such as gas-filled gaps in the cable or in insulation applied during high-voltage cable termination. PD occurs due to capacitive discharge when source voltage reaches a threshold value during the AC power cycle. PD can be the result of internal weak spots in power cables, such as voids, cracks, or particles. PD is also caused by damage to the outer semi-conductive layer of cables or to joints and terminations during installation. PD may gradually decompose organic material in insulation at the edge of the gap, thereby prematurely aging the insulation. Thermal degradation processes inside cable joints with improperly performed cable jointing workmanship can lead to PD inception. The following steps apply:

(1) The installed cable should have undergone PD testing, if applicable, by the cable supplier in accordance with the applicable specification sheet of MIL-DTL-24643, MIL-DTL-24640, or MIL-DTL-915.

(2) Detailed manufacturers' instructions for termination insulation material should be followed to prevent defects and faults in the insulation that could cause PD.

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(3) During all steps of the termination process, care should be taken to avoid gaps and discontinuities in the insulation assembly.

(4) After cable termination, the system should certainly be checked by performing dielectric testing in accordance with IEEE 400.

g. When the cable termination is connected to a circuit subject to high-voltage switching, the termination should be rated for the system basic switching impulse level. Components that cause switching impulses are circuit breakers and high-voltage power converters.

h. Cable grouping requirements of MIL-STD-2003-4 should be met to ensure cables operate within temperature and current limits.

i. Cables between variable frequency drives (VFD) and motors are subjected to higher voltage stress than power cables. Additionally, unless proper measures are taken, VFD output cables can have unacceptable radiated and conducted emissions. VFD cables should incorporate an overall electrostatic shield to reduce electric field emissions. VFD cables should incorporate additional conductors (e.g., using a four-conductor cable on a three-phase VFD output or incorporate dedicated ground return wires) to reduce common mode emissions.

j. MIL-STD-2003-1 includes methods for insulation repair and splicing of higher voltage cables.

k. See figures in MIL-STD-2003-4 for guidance for installation of higher voltage cables in cableways.

4.1.10 Cable bundling, ampacity, and cable derating considerations to avoid cable heating. The guidance for determining cable ampacity and cable derating for power cables bundled in raceways, trays, cableways, or otherwise bundled together, is as follows. MIL-STD-2003-4 includes technical information for determining cable derating for specific applications. See 5.9 for derating guidance for specific cable types.

a. The ampacities for most cable conductors are covered in tables VI, VII, and VIII for various source voltage and ambient temperature conditions. These values should be used as the cable ampacity rating when the cable is physically or thermally isolated from other cables and as the starting point for other applications. Prior to detail design, if the ambient temperature of the space is not known, the ampacity for an ambient temperature of 50 °C should be assumed.

b. Unless cables rated for temperatures above 105 °C are physically or thermally isolated from other cables of a lower temperature rating, the 105 °C rated cables should be limited to 90 °C rated service regardless of whether the cables are in raceways, trays, cableways, or otherwise bundled together.

c. To the maximum extent practical, power cables in cableways should be separated from control and instrumentation cables and from other power cables to prevent overheating of the cables.

d. Where cables enter electrical equipment such as switchboards or machinery, care should be exercised to avoid bunching or closely grouping the cables in any manner that may restrict dissipation of heat from the cables.

e. Cables should be derated if installed in locations where the ambient temperature may exceed the temperature rating of the cable. Installation of cables in locations subject to excessive heat should be avoided. Where needed, heat insulating barriers should be installed. Cables should not be installed adjacent to machinery, piping, or other surfaces having an exposed surface temperature greater than 70 °C. Cables should not be run over boilers in the upper portions of firerooms or in other locations where they may be surrounded by hot air. Cables should not be run in locations where they will be exposed to hot stack gases. Unless necessary to support ship arrangement requirements, fire detection and aqueous film forming foam (AFFF) system cabling should not be routed in close proximity to engine and gas turbine exhaust piping. If the fire detection and AFFF cabling needs to be in proximity of engine and gas turbine exhaust piping, heat protection for the cabling should be provided.

f. For situations where power cables are grouped in close proximity, the cable ampacity derating may be determined by a test method or an analytical calculation method.

4.1.11 Calculating cable ampacity. Cable conductor ampacity for Navy cables is listed in tables VI, VII, and VIII. These conductor ampacities are used to further calculate overall cable ampacities needed to ensure cable and conductor temperature limits are not exceeded. The ability of a cable conductor to conduct current depends on several factors. Important factors that are normally considered by the designers of electrical distribution systems are, but not limited to, the following:

a. Thermal details of the surrounding medium (e.g., conduit or bare cable)

b. Ambient compartment temperature

c. Heat generated by adjacent conductors and cable grouping in a cableway

d. Heat generated by the conductor due to conductor losses

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e. Dielectric losses due to cable construction

Cable sizing is one of the most important concerns when designing distribution and transmission systems. Once the load has been sized and confirmed, the cable system is designed in a way to transfer required power from the generation to the end user. Calculation of the cable ampacity normally considers only power cables, since control and instrumentation cables transmit very little current that has negligible effect to the overall temperature rise. Other important factors that need to be considered when selecting power cables are voltage drop, short circuit capability, and future load growth. Overall installation information about power cables needs to be collected, collated, and preserved in case of future system changes, including added cables.

4.1.12 Watertightness of cables. Cable types are designated as watertight or non-watertight in MIL-DTL-24643, MIL-DTL-24640, and MIL-DTL-915. Requirements for installing watertight versus non-watertight cable are provided in MIL-STD-2003-1 and are summarized as follows:

a. Surface ship cables terminating in specific equipment defined in MIL-STD-2003-1, located below FWL-II, should be watertight.

b. Watertight cables should meet the watertight requirements of MIL-DTL-24640, MIL-DTL-24643, or MIL-DTL-915.

c. Cables that penetrate FWL-II but do not pass through a watertight deck or bulkhead below FWL-II are not required to be watertight.

d. Cables that do not pass through a watertight deck or bulkhead are not required to be watertight.

e. See other watertight cable exceptions in MIL-STD-2003-1 for more information.

4.1.13 Cable spacing and shielding requirements. See S9407-AB-HBK-010 for information on cable spacing and shielding requirements for cables described in certain military specifications, information on flexible and rigid shielding conduit, and conduit fittings.

4.1.14 Unarmored cable. See MIL-STD-2003-4 for information on installation of unarmored versus armored cable in ships. Also, see table V for supersession data for detailed specification sheets regarding the transition from armored to unarmored cable.

4.1.15 Time domain reflectometry (TDR). TDR should be considered for the determination of the total length of a cable, the location of low resistive cable faults, the location of cable interruptions, and the location of joints along the cable. This method is useful for preventive maintenance, corrective maintenance, and cable termination checks after installation. A TDR sends a low voltage pulse into the cable under test and at any impedance change within the cable a reflection will be seen. The TDR measures the time between release and return of the low voltage pulse from any reflections. By measuring the time and knowing the propagation velocity of the pulse, the distance to the reflection can be calculated. This helps to give a user information on the type of impedance change or failure that could be present in the cable. TDR work on any type of metallic conductors (e.g., single conductors, triads, and pairs). TDR requires two conductors to measure impedance changes. The pair of conductors may include a shield, if necessary.

4.1.16 Recommended practices. Users should consult IEEE 1580 and IEEE 45.8 for recommended practices for shipboard cable; Navy standards take precedence.

4.1.17 Cable voltage drop. See 128 A for calculations of cable voltage drop.

5. DETAILED GUIDANCE

5.1 Cable types and construction characteristics (as specified in MIL-DTL-24643) list. Cable types and construction characteristics (as specified in MIL-DTL-24643) are as follows:

LSCBPSJ	-	One or two pairs, metallic foil/polyester tape, braided shield, cross-linked polyolefin jacket, unarmored, non-watertight (in accordance with MIL-DTL-24643/78).
LSCBPSJW	-	One or two pairs, metallic foil/polyester tape, braided shield, cross-linked polyolefin jacket, unarmored, watertight (in accordance with MIL-DTL-24643/78).
LSCVSF	-	400-Hertz (Hz) aircraft servicing: three ethylene propylene rubber insulated conductors and one uninsulated conductor, overall cross-linked polyolefin jacket (in accordance with MIL-DTL-32180; previously MIL-DTL-24643/1, then superseded by MIL-DTL-915/1).

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LSC264	-	Four coaxial grouped conductors, braided shield on each, thermoset insulation, braided shield, overall cross-linked polyolefin jacket, unarmored, watertight (in accordance with MIL-DTL-24643/69).
LSC5FS	-	Four pairs, polyolefin insulation, metallic foil shield facing in, copper drain wire, cross-linked polyolefin jacket, unarmored, non-watertight (in accordance with MIL-DTL-24643/59).
LSC5FSW	-	Four pairs, polyolefin insulation, metallic foil shield facing in, copper drain wire, cross-linked polyolefin jacket, unarmored, watertight (in accordance with MIL-DTL-24643/59).
LSC5OS	-	Four pairs, polyolefin insulation, metallic foil shield facing out, copper drain wire, cross-linked polyolefin jacket unarmored, non-watertight (in accordance with MIL-DTL-24643/59).
LSC5OSW	-	Four pairs, polyolefin insulation, metallic foil shield facing out, copper drain wire, cross-linked polyolefin jacket, unarmored, watertight (in accordance with MIL-DTL-24643/59).
LSC5P	-	Four pairs, polyolefin insulation, cross-linked polyolefin jacket, unarmored, non-watertight (in accordance with MIL-DTL-24643/61).
LSC5POS	-	Four pairs, polyolefin insulation, metallic foil side facing out, cross-linked polyolefin jacket, unarmored, non-watertight (in accordance with MIL-DTL-24643/61).
LSC5POSR	-	Four pairs, polyolefin insulation, metallic foil side facing out, cross-linked polyolefin jacket, unarmored, non-watertight (in accordance with MIL-DTL-24643/61).
LSC6FS	-	Four pairs, copper drain wire, solid or foamed polyolefin insulation, metallic foil/polyester tape (metallic foil side facing in and in contact with the drain wire), cross-linked polyolefin jacket, unarmored, non-watertight (in accordance with MIL-DTL-24643/77).
LSC6OS	-	Four pairs, copper drain wire, solid or foamed polyolefin insulation, metallic foil/polyester tape (metallic foil side facing out and in contact with the drain wire), cross-linked polyolefin jacket, unarmored, non-watertight (in accordance with MIL-DTL-24643/77).
LSC6OSW	-	Four pairs, copper drain wire, solid or foamed polyolefin insulation, metallic foil/polyester tape (metallic foil side facing out and in contact with the drain wire), cross-linked polyolefin jacket, unarmored, watertight (in accordance with MIL-DTL-24643/77).
LSDCOP	-	Double conductor, oil-resistant, portable cord: thermoset insulation, and cross-linked polyolefin jacket, unarmored, non-watertight (in accordance with MIL-DTL-24643/2).
LSDHOF	-	Double conductor, heat and oil-resistant, flexible: thermoset insulation, cross-linked polyolefin jacket, unarmored, non-watertight (in accordance with MIL-DTL-24643/3).
LSDNW	-	Double conductor: thermoset insulation, cross-linked polyolefin jacket, unarmored, non-watertight (in accordance with MIL-DTL-24643/48).
LSDNWA	-	Double conductor: ethylene propylene rubber or cross-linked polyethylene insulation, cross-linked polyolefin jacket, armored (in accordance with MIL-DTL-24643/48; note that, at the time of this document's publication, armored version is not in most recent revision of specification sheet).
LSDPS	-	Double conductor, power: silicone rubber insulation, glass braid, silicone rubber jacket, armored, watertight with circuit integrity (in accordance with MIL-DTL-24643/26).

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LSDPSN	-	Double conductor, power: silicone rubber insulation, glass braid, silicone rubber jacket, armored, non-watertight with circuit integrity (in accordance with MIL-DTL-24643/26).
LSDRW	-	Double conductor: thermoset insulation, cross-linked polyolefin jacket, unarmored, watertight (in accordance with MIL-DTL-24643/53).
LSDRWA	-	Double conductor: cross-linked polyethylene insulation, cross-linked polyolefin jacket, armored (in accordance with MIL-DTL-24643/53; note that, at the time of this document's publication, armored version is not in most recent revision of specification sheet).
LSDSGA	-	Double conductor: silicone rubber and glass braid insulated, cross-linked polyolefin jacket, armored (in accordance with MIL-DTL-24643/15; note that, at the time of this document's publication, armored version is not in most recent revision of specification sheet).
LSDSGU	-	Double conductor: thermoset and glass braid (optional), cross-linked polyolefin jacket, unarmored, watertight with circuit integrity (in accordance with MIL-DTL-24643/15).
LSECM	-	Eight pairs shielded, and eight groups of seven conductors for each group: thermoset insulation for the conductors of each pair, braided shield over each pair; thermoset insulation for the conductors of the groups of seven; crosslinked polyolefin jacket, unarmored (in accordance with MIL-DTL-24643/38).
LSECMA	-	Eight pairs shielded, and eight groups of seven conductors for each group: cross-linked polyethylene insulation for the conductors of each pair, braided shield over each pair; ethylene propylene rubber or cross-linked polyethylene insulation for the conductors of the groups of seven; cross-linked polyolefin jacket, armored (in accordance with MIL-DTL-24643/38; note that, at the time of this document's publication, armored version is not in most recent revision of specification sheet).
LSEC6FS	-	Four pairs, topside, copper drain wire, polyolefin insulation, metallic foil side facing in, cross-linked polyolefin jacket, copper foil/polyester tape, braided shield, copper foil/polyester tape, braided shield, cross-linked polyolefin jacket, unarmored, non-watertight (in accordance with MIL-DTL-24643/84).
LSEC6OS	-	Four pairs, topside, metallic foil side facing out, braided shield, cross-linked polyolefin jacket, copper foil/polyester tape, braided shield, copper foil/polyester tape, braided shield, cross-linked polyolefin jacket, unarmored, non-watertight (in accordance with MIL-DTL-24643/84).
LSEC6OSW	-	Four pairs, topside, metallic foil side facing out, braided shield, cross-linked polyolefin jacket, copper foil/polyester tape, braided shield, copper foil/polyester tape, braided shield, solid or semi-solid water-block materials, cross-linked polyolefin jacket, unarmored, watertight (in accordance with MIL-DTL-24643/84).
LSE**CI	-	Copper conductors (see table VI), thermoset insulation, copper foil/polyester tape, braided shield, copper foil/polyester tape/copper foil, braided shield, cross-linked polyolefin jacket, unarmored, non-watertight with circuit integrity (in accordance with MIL-DTL-24643/85).
LSE**CIW	-	Copper conductors (see table VI), thermoset insulation, copper foil/polyester tape, braided shield, copper foil/polyester tape/copper foil, braided shield, solid or semi-solid water-block materials, cross-linked polyolefin jacket, unarmored, watertight with circuit integrity (in accordance with MIL-DTL-24643/85).
LSE**E	-	Multiple component wires (see table VI), topside, copper foil/polyester tape, braided shield, copper foil/polyester tape/copper foil, braided shield, cross-linked polyolefin jacket, unarmored, non-watertight (in accordance with MIL-DTL-24643/81).

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LSE**EW - Multiple component wires (see table VI), topside, copper foil/polyester tape, braided shield, copper foil/polyester tape/copper foil, braided shield, cross-linked polyolefin jacket, unarmored, watertight (in accordance with MIL-DTL-24643/81).

LSE**SPE - Multiple shielded pairs (see table VI), topside, thermoset insulation, braided shield, tape for shield insulation, copper foil/polyester tape, braided shield of tin-coated copper, copper foil/polyester tape/copper foil, braided shield of tin-coated copper, cross-linked polyolefin jacket, unarmored, non-watertight (in accordance with MIL-DTL-24643/83).

LSE**SPEW - Multiple shielded pairs (see table VI), topside, thermoset insulation, braided shield, tape for shield insulation, copper foil/polyester tape, braided shield of tin-coated copper, copper foil/polyester tape/copper foil, braided shield of tin-coated copper, solid or semi-solid water-block materials, cross-linked polyolefin jacket, unarmored, watertight (in accordance with MIL-DTL-24643/83).

LSE**U - Conductor pairs (see table VI), topside, thermoset insulation, copper foil/polyester tape, braided shield copper foil/polyester tape/copper foil, braided shield, cross-linked polyolefin jacket, unarmored, non-watertight (in accordance with MIL-DTL-24643/82).

LSE**UW - Conductor pairs (see table VI), topside, thermoset insulation, copper foil/polyester tape, braided shield copper foil/polyester tape/copper foil, braided shield, cross-linked polyolefin jacket, unarmored, watertight (in accordance with MIL-DTL-24643/82).

LSE**AU - Conductor pairs (see table VI), topside, thermoset insulation, copper foil/polyester tape, braided shield copper foil/polyester tape/copper foil, braided shield, cross-linked polyolefin jacket, unarmored, non-watertight (in accordance with MIL-DTL-24643/82).

LSE**AUW - Conductor pairs (see table VI), topside, thermoset insulation, copper foil/polyester tape, braided shield copper foil/polyester tape/copper foil, braided shield, cross-linked polyolefin jacket, unarmored, watertight (in accordance with MIL-DTL-24643/82).

LSFCF - Eight conductors four in outer interstices of the other four, two layers of thermoset insulation, cross-linked polyolefin jacket, unarmored, flexible, non-watertight (in accordance with MIL-DTL-24643/72).

LSFHOF - Four conductors, heat and oil-resistant, flexible: thermoset insulation, cross-linked polyolefin jacket, unarmored, non-watertight (in accordance with MIL-DTL-24643/3).

LSFNW - Four conductors: thermoset insulation, cross-linked polyolefin jacket, unarmored, non-watertight (in accordance with MIL-DTL-24643/50).

LSFNWA - Four conductors: ethylene propylene rubber or cross-linked polyethylene insulation, cross-linked polyolefin jacket, armored (in accordance with MIL-DTL-24643/50; note that, at the time of this document's publication, armored version is not in most recent revision of specification sheet).

LSFPS - Four conductors, power: silicone rubber insulation, glass braid, silicone rubber jacket, armored, watertight with circuit integrity (in accordance with MIL-DTL-24643/26).

LSFPSN - Four conductors, power: silicone rubber insulation, glass braid, silicone rubber jacket, armored, non-watertight with circuit integrity (in accordance with MIL-DTL-24643/26).

LSFSGA - Four conductors: silicone rubber and glass insulated, cross-linked polyolefin jacket, armored (in accordance with MIL-DTL-24643/17; note that, at the time of this

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LSFSGU - Four conductors: thermoset and glass braid (optional), cross-linked polyolefin jacket, unarmored, watertight with circuit integrity (in accordance with MIL-DTL-24643/17).

LSLCFS - Conductor pairs in layers, thermoset insulation, cross-linked polyolefin jacket, unarmored, non-watertight (in accordance with MIL-DTL-24643/75).

LSMA - Multiple conductor: ethylene propylene rubber or cross-linked polyethylene insulation, cross-linked polyolefin jacket, armored (in accordance with MIL-DTL-24643/29; note that, at the time of this document's publication, armored version is not in most recent revision of specification sheet).

LSMCOS - Multiple conductor, oil-resistant, shielded: thermoset insulation over pairs, or over assembly, cross-linked polyolefin jacket, unarmored, non-watertight (in accordance with MIL-DTL-24643/4).

LSMDU - Multiple conductor: thermoset insulation, cross-linked polyolefin jacket, unarmored, watertight (in accordance with MIL-DTL-24643/5).

LSMDY - Multiple conductor, degaussing: ethylene propylene or cross-linked polyethylene insulation, cross-linked polyolefin jacket, armored between double-layer jacket (in accordance with MIL-DTL-24643/5; previously MIL-DTL-24643/6).

LSMHOF - Multiple conductor, heat and oil-resistant, flexible: thermoset insulation, cross-linked polyolefin jacket, unarmored, non-watertight (in accordance with MIL-DTL-24643/7).

LSMMOP - Multiple conductor, microphone, oil-resistant, portable: thermoset insulation, cross-linked polyolefin jacket, unarmored, non-watertight (in accordance with MIL-DTL-24643/8).

LSMNW - Multiple conductor: thermoset insulation, cross-linked polyolefin jacket, unarmored, non-watertight (in accordance with MIL-DTL-24643/51).

LSMNWA - Multiple conductor: ethylene propylene rubber or cross-linked polyethylene insulation, cross-linked polyolefin jacket, armored (in accordance with MIL-DTL-24643/51; note that, at the time of this document's publication, armored version is not in most recent revision of specification sheet).

LSMRI - Multiple conductor: thermoset insulated, without fillers, no overall jacket, non-watertight (in accordance with MIL-DTL-24643/9).

LSMS - Multiple conductor, shielded: thermoset insulation, overall braided shield, cross-linked polyolefin jacket, unarmored (in accordance with MIL-DTL-24643/34).

LSMSA - Multiple conductor, shielded: ethylene propylene rubber or cross-linked polyethylene insulation, overall braided shield, cross-linked polyolefin jacket, armored (in accordance with MIL-DTL-24643/34; note that, at the time of this document's publication, armored version is not in most recent revision of specification sheet).

LSMSCA - Multiple conductor: silicone rubber insulated-glass braided conductors, cross-linked polyolefin jacket, armored (in accordance with MIL-DTL-24643/18; note that, at the time of this document's publication, armored version is not in most recent revision of specification sheet).

LSMSCS - Multiple conductor: thermoset insulation, cross-linked polyolefin jacket, double overall shielded, watertight with circuit integrity (in accordance with MIL-DTL-24643/18).

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LSMSCU	-	Multiple conductor: thermoset insulation, cross-linked polyolefin jacket, unarmored, watertight with circuit integrity (in accordance with MIL-DTL-24643/18).
LSMU	-	Multiple conductor: thermoset insulation, cross-linked polyolefin jacket, unarmored, non-watertight (in accordance with MIL-DTL-24643/29).
LSMUS	-	Multiple conductor: thermoset insulation, cross-linked polyolefin jacket, double overall shielded, unarmored, non-watertight (in accordance with MIL-DTL-24643/29).
LSPBTM	-	Pyrometer base multiple pairs: ethylene propylene rubber or cross-linked polyethylene insulation on one copper and one constantan conductor, cross-linked polyolefin jacket, armored (in accordance with MIL-DTL-24643/10; note that, at the time of this document's publication, armored version is not in most recent revision of specification sheet).
LSPBTMU	-	Pyrometer base multiple pairs: thermoset insulation on one copper and one constantan conductor, cross-linked polyolefin jacket, unarmored, watertight (in accordance with MIL-DTL-24643/10).
LSPB2SD	-	Two conductor, polyolefin insulation, copper drain wire, braided shield, cross-linked polyolefin jacket, unarmored, non-watertight (in accordance with MIL-DTL-24643/62).
LSPB2SDW	-	Two conductor, polyolefin insulation, copper drain wire, braided shield, cross-linked polyolefin jacket, unarmored, watertight (in accordance with MIL-DTL-24643/62).
LSPB2SDOS	-	Two conductor, polyolefin insulation, copper drain wire, braided shield, first cross-linked polyolefin jacket, second braided shield, second cross-linked polyolefin jacket unarmored, non-watertight (in accordance with MIL-DTL-24643/62).
LSPB2SDOSW	-	Two conductor, polyolefin insulation, copper drain wire, braided shield, first cross-linked polyolefin jacket, second braided shield, second cross-linked polyolefin jacket unarmored, watertight (in accordance with MIL-DTL-24643/62).
LSPI	-	Position indicator: silicone rubber insulation, glass braid, shielded pairs, silicone rubber jacket, armored, non-watertight with circuit integrity (in accordance with MIL-DTL-24643/25).
LSSCF	-	Single conductor, flexible, thermoset insulation, cross-linked polyolefin jacket, unarmored, non-watertight (in accordance with MIL-DTL-24643/71).
LSSHOF	-	Single conductor, heat and oil-resistant, flexible: thermoset insulation, cross-linked polyolefin jacket, unarmored, non-watertight (in accordance with MIL-DTL-24643/3).
LSSRW	-	Single conductor, radio, thermoset insulation, cross-linked polyolefin jacket, unarmored, watertight (in accordance with MIL-DTL-24643/53).
LSSRWA	-	Single conductor, radio: cross-linked polyethylene insulation, cross-linked polyolefin jacket, armored (per MIL-DTL-24643/53; note that, at the time of this document's publication, armored version is not in most recent revision of specification sheet).
LSSSF	-	Single conductor: thermoset insulation, cross-linked polyolefin jacket, unarmored, non-watertight (in accordance with MIL-DTL-24643/11).
LSSSGA	-	Single conductor: silicone rubber and glass tape insulated, cross-linked polyolefin jacket, armored (in accordance with MIL-DTL-24643/14; note that, at the time of this document's publication, armored version is not in most recent revision of specification sheet).
LSSSGU	-	Single conductor: thermoset insulated, cross-linked polyolefin jacket, unarmored, watertight with circuit integrity (in accordance with MIL-DTL-24643/14).

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LST	-	Single conductor, thermoset insulation, unarmored, non-watertight (in accordance with MIL-DTL-24643/80).
LSTW	-	Single conductor, thermoset insulation, unarmored, watertight (in accordance with MIL-DTL-24643/80).
LSTCF	-	Three conductors, thermoset insulation, cross-linked polyolefin jacket, unarmored, non-watertight (in accordance with MIL-DTL-24643/73).
LSTCJA	-	Thermocouple, type J, single pair: one iron and one constantan conductor, extruded silicone rubber insulated, glass braided, cross-linked polyolefin jacket, armored (in accordance with MIL-DTL-24643/21; note that, at the time of this document's publication, armored version is not in most recent revision of specification sheet).
LSTCJU	-	Thermocouple, type J, single pair: one iron and one constantan conductor, thermoset insulated, optional glass braided, cross-linked polyolefin jacket, unarmored (in accordance with MIL-DTL-24643/21).
LSTCJX	-	Thermocouple, type J, multiple pairs: extruded silicone rubber insulated, glass braid on one iron and one constantan conductor for each pair, silicone rubber jacket, armored, watertight with circuit integrity (in accordance with MIL-DTL-24643/24).
LSTCJXN	-	Thermocouple, type J, multiple pairs: extruded silicone rubber insulated, glass braid on one iron and one constantan conductor for each pair, silicone rubber jacket, armored, non-watertight with circuit integrity (in accordance with MIL-DTL-24643/24).
LSTCKX	-	Thermocouple, type K, multiple pairs: extruded silicone rubber insulated, glass braid on one chromel and one alumel conductor for each pair, silicone rubber jacket, armored, watertight with circuit integrity (in accordance with MIL-DTL-24643/24).
LSTCKXN	-	Thermocouple, type K, multiple pairs: extruded silicone rubber insulated, glass braid on one chromel and one alumel conductor for each pair, silicone rubber jacket, armored, non-watertight with circuit integrity (in accordance with MIL-DTL-24643/24).
LSTCOP	-	Three conductors, oil-resistant, portable cord: thermoset insulation, cross-linked polyolefin jacket, unarmored, non-watertight (in accordance with MIL-DTL-24643/2).
LSTCTA	-	Thermocouple, type T, single pair: one copper and one constantan conductor, extruded silicone rubber insulated, glass braided, cross-linked polyolefin jacket, armored (in accordance with MIL-DTL-24643/21; note that, at the time of this document's publication, armored version is not in most recent revision of specification sheet).
LSTCTU	-	Thermocouple, type T, single pair: one copper and one constantan conductor, thermoset insulated, optional glass braided, cross-linked polyolefin jacket, unarmored.(in accordance with MIL-DTL-24643/21).
LSTCTX	-	Thermocouple, type T, multiple pairs: extruded silicone rubber insulated, glass braid on one copper and one constantan conductor for each pair, silicone rubber jacket, armored, watertight with circuit integrity (in accordance with MIL-DTL-24643/24).
LSTCTXN	-	Thermocouple, type T, multiple pairs: extruded silicone rubber insulated, glass braid on one copper and one constantan conductor for each pair, silicone rubber jacket, armored, non-watertight with circuit integrity (in accordance with MIL-DTL-24643/24).
LSTHOF	-	Three conductors, heat and oil-resistant, flexible: thermoset insulation, cross-linked polyolefin jacket, unarmored, non-watertight (in accordance with MIL-DTL-24643/3).

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LSTNW	-	Three conductors: thermoset insulation, cross-linked polyolefin jacket, unarmored, non-watertight (in accordance with MIL-DTL-24643/49).
LSTNWA	-	Three conductors: ethylene propylene rubber or cross-linked polyethylene insulation, cross-linked polyolefin jacket, armored (in accordance with MIL-DTL-24643/49; note that, at the time of this document's publication, armored version is not in most recent revision of specification sheet).
LSTPNW	-	Twisted pairs: thermoset insulation, cross-linked polyolefin jacket, unarmored, non-watertight (in accordance with MIL-DTL-24643/52).
LSTPNWA	-	Twisted pairs: ethylene propylene or cross-linked polyethylene insulation, cross-linked polyolefin jacket, armored (in accordance with MIL-DTL-24643/52; note that, at the time of this document's publication, armored version is not in most recent revision of specification sheet).
LSTPS	-	Three conductors, power supply: silicone rubber insulation, glass braid, silicone rubber jacket, armored with circuit integrity, watertight (in accordance with MIL-DTL-24643/26).
LSTPSN	-	Three conductors, power supply: silicone rubber insulation, glass braid, silicone rubber jacket, armored with circuit integrity, non-watertight (in accordance with MIL-DTL-24643/26).
LSTPSJ	-	Twisted pairs, thermoset insulation, braided shield, cross-linked polyolefin jacket, unarmored, non-watertight (in accordance with MIL-DTL-24643/64).
LSTRW	-	Three conductors: thermoset insulation, cross-linked polyolefin jacket, unarmored, watertight (in accordance with MIL-DTL-24643/53).
LSTRWA	-	Three conductors, radio: cross-linked polyethylene insulation, cross-linked polyolefin jacket, armored (in accordance with MIL-DTL-24643/53; note that, at the time of this document's publication, armored version is not in most recent revision of specification sheet).
LSTSGA	-	Three conductors: extruded silicone rubber and glass insulation, cross-linked polyolefin jacket, armored (in accordance with MIL-DTL-24643/16; note that, at the time of this document's publication, armored version is not in most recent revision of specification sheet).
LSTSGU	-	Three conductors: thermoset and glass braid (optional), cross-linked polyolefin jacket, unarmored, watertight with circuit integrity (in accordance with MIL-DTL-24643/16).
LSTTOP	-	Twisted pairs, oil-resistant, portable: thermoset insulation, cross-linked polyolefin jacket, unarmored, non-watertight (in accordance with MIL-DTL-24643/12).
LSE**TTOP	-	Conductor pairs (see table VI), topside, thermoset insulation, copper foil/polyester tape, braided shield copper foil/polyester tape/copper foil, braided shield, cross-linked polyolefin jacket, unarmored, non-watertight (in accordance with MIL-DTL-24643/82).
LSTTRS	-	Twisted pairs, radio, shielded, flexible: thermoset insulation, braided shield for each pair, cross-linked polyolefin jacket, unarmored, non-watertight (in accordance with MIL-DTL-24643/13).
LSTTRSA	-	Twisted pairs, radio, shielded, flexible: cross-linked polyethylene insulation, braided shield for each pair, cross-linked polyolefin jacket, armored (in accordance with MIL-DTL-24643/13; note that, at the time of this document's publication, armored version is not in most recent revision of specification sheet).
LSTTSA	-	Twisted pairs: extruded silicone rubber and polyamide special purpose, cross-linked polyolefin jacket, armored (in accordance with MIL-DTL-24643/23; note that, at the

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LSTTSU - Twisted pairs: thermoset and polyamide special purpose insulation, cross-linked polyolefin jacket, unarmored, watertight with circuit integrity (in accordance with MIL-DTL-24643/23).

LSYSGU - Designed to connect to Y-connected power systems, seven copper conductors, six for three phases with two parallel conductors in each phase, one neutral copper conductor, cross-linked polyolefin jacket, unarmored, watertight with circuit integrity (in accordance with MIL-DTL-24643/63).

LS1SA - Single, shielded: cross-linked polyethylene insulation, braided shield on each conductor, cross-linked polyolefin jacket, armored (in accordance with MIL-DTL-24643/41; note that, at the time of this document's publication, armored version is not in most recent revision of specification sheet).

LS1SAU - Singles, shielded: thermoset insulation, braided shield on each conductor, cross-linked polyolefin jacket, unarmored, non-watertight (in accordance with MIL-DTL-24643/41).

LS1SMU - Singles, shielded, multiple conductor: thermoset insulation, braided shield on each conductor, cross-linked polyolefin jacket, unarmored, non-watertight (in accordance with MIL-DTL-24643/40).

LS1SMWA - Singles, shielded, multiple conductor: cross-linked polyethylene insulation, braided shield over each conductor, cross-linked polyolefin jacket, armored (in accordance with MIL-DTL-24643/47; note that, at the time of this document's publication, armored version is not in most recent revision of specification sheet).

LS1SMWU - Singles, shielded, multiple conductor: thermoset insulation, braided shield over each conductor, cross-linked polyolefin jacket, unarmored, watertight (in accordance with MIL-DTL-24643/47).

LS1SU - Singles, shielded: thermoset insulated, braided shield on each conductor, cross-linked polyolefin jacket, for LS1SU-36, thirty-two 20 American Wire Gage (AWG) conductors and four 18 AWG conductors; for LS1SU-60, sixty 20 AWG conductors, unarmored, non-watertight (in accordance with MIL-DTL-24643/42).

LS1SUA - Singles, shielded: cross-linked polyethylene insulated, braided shield on each conductor, cross-linked polyolefin jacket, armored (per MIL-DTL-24643/42; note that, at the time of this document's publication, armored version is not in most recent revision of specification sheet).

LS1SWA - Singles, shielded: cross-linked polyethylene insulation, braided shield on each conductor, cross-linked polyolefin jacket, armored (in accordance with MIL-DTL-24643/30; note that, at the time of this document's publication, armored version is not in most recent revision of specification sheet).

LS1SWU - Singles, shielded: thermoset insulation, braided shield on each conductor, cross-linked polyolefin jacket, unarmored, watertight (in accordance with MIL-DTL-24643/30).

LS1S50MA - Singles, shielded, 50-ohm, multiple conductor: cross-linked polyethylene insulation, braided shield on each conductor, cross-linked polyolefin jacket, armored (in accordance with MIL-DTL-24643/28; note that, at the time of this document's publication, armored version is not in most recent revision of specification sheet).

LS1S50MU - Singles, shielded, 50-ohm, multiple conductor: thermoset insulation, braided shield on each conductor, cross-linked polyolefin jacket, unarmored, non-watertight (in accordance with MIL-DTL-24643/28).

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LS1S50MUS	-	Singles, shielded, 50-ohm, multiple conductor: thermoset insulation, braided shield on each conductor, cross-linked polyolefin jacket, double overall shielded, unarmored, non-watertight (in accordance with MIL-DTL-24643/28).
LS1S75MA	-	Singles, shielded, 75-ohm, multiple conductor: cross-linked polyethylene insulation, braided shield over each conductor, cross-linked polyolefin jacket, armored (in accordance with MIL-DTL-24643/39; note that, at the time of this document's publication, armored version is not in most recent revision of specification sheet).
LS1S75MU	-	Singles, shielded, 75-ohm, multiple conductor: thermoset insulation, braided shield over each conductor, cross-linked polyolefin jacket, unarmored non-watertight (in accordance with MIL-DTL-24643/39).
LS2A	-	Twisted pairs, shielded: cross-linked polyethylene insulation, overall braided shield, cross-linked polyolefin jacket, armored (in accordance with MIL-DTL-24643/27; note that, at the time of this document's publication, armored version is not in most recent revision of specification sheet).
LS2AU	-	Twisted pairs, shielded: thermoset insulation, overall braided shield, cross-linked polyolefin jacket, unarmored, non-watertight (in accordance with MIL-DTL-24643/27).
LS2AUS	-	Twisted pairs, shielded: thermoset insulation, overall braided shield, cross-linked polyolefin jacket, double overall shielded, unarmored, non-watertight (in accordance with MIL-DTL-24643/27).
LS2CS	-	Pairs, shielded: thermoset insulation, double braided shield overall, cross-linked polyolefin jacket, unarmored, non-watertight (in accordance with MIL-DTL-24643/58).
LS2OW	-	Two conductors, thermoset insulation, braided shield, cross-linked polyolefin jacket, unarmored, watertight with circuit integrity (in accordance with MIL-DTL-24643/66).
LS2KVTSGSG	-	Three conductors, extruded silicone-glass tape insulation with an extruded covering of polyester elastomer, three ground wires with extruded thermoset insulation, cross-linked polyolefin jacket, unarmored, watertight with circuit integrity (in accordance with MIL-DTL-24643/79).
LS2SA	-	Pairs, shielded: cross-linked polyethylene insulation, braided shield over each pair, cross-linked polyolefin jacket, armored (in accordance with MIL-DTL-24643/31; note that, at the time of this document's publication, armored version is not in most recent revision of specification sheet).
LS2SJ	-	Two conductors, shielded: thermoset insulation, overall braided shield, cross-linked polyolefin jacket, unarmored, non-watertight (in accordance with MIL-DTL-24643/43).
LS2SJA	-	Pairs, shielded: ethylene propylene rubber or cross-linked polyethylene insulation, overall braided shield, cross-linked polyolefin jacket, armored (in accordance with MIL-DTL-24643/43; note that, at the time of this document's publication, armored version is not in most recent revision of specification sheet).
LS2SU	-	Pairs, shielded: thermoset insulation, braided shield over each pair, cross-linked polyolefin jacket, unarmored, non-watertight (in accordance with MIL-DTL-24643/31).
LS2SUS	-	Pairs, shielded: thermoset insulation, braided shield over each pair, cross-linked polyolefin jacket, double overall shielded, unarmored, non-watertight (in accordance with MIL-DTL-24643/31).
LS2SWA	-	Pairs, shielded: cross-linked polyethylene insulation, braided shield over each pair, cross-linked polyolefin jacket, armored (in accordance with MIL-DTL-24643/32;

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LS2SWAU - Pairs, shielded: thermoset insulation, braided shield over each pair, cross-linked polyolefin jacket, unarmored, watertight (in accordance with MIL-DTL-24643/32).

LS2SWL - Pairs, shielded: thermoset insulation, braided shield over each pair, cross-linked polyolefin jacket, unarmored, watertight (in accordance with MIL-DTL-24643/56).

LS2SWLA - Pairs, shielded: cross-linked polyethylene insulation, braided shield over each pair, cross-linked polyolefin jacket, armored (in accordance with MIL-DTL-24643/56; note that, at the time of this document's publication, armored version is not in most recent revision of specification sheet).

LS2SWU - Pairs, shielded: thermoset insulation, braided shield over each pair, cross-linked polyolefin jacket, unarmored, watertight (in accordance with MIL-DTL-24643/33).

LS2SWUA - Pairs, shielded: cross-linked polyethylene insulation, braided shield over each pair, cross-linked polyolefin jacket, armored (in accordance with MIL-DTL-24643/33; note that, at the time of this document's publication, armored version is not in most recent revision of specification sheet).

LS2U - Pairs: thermoset insulation, overall braided shield, cross-linked polyolefin jacket, unarmored, non-watertight (in accordance with MIL-DTL-24643/45).

LS2UA - Pairs: cross-linked polyethylene insulation, overall braided shield, cross-linked polyolefin jacket, armored (in accordance with MIL-DTL-24643/45; note that, at the time of this document's publication, armored version is not in most recent revision of specification sheet).

LS2UW - Twisted pairs: thermoset insulation, overall braided shield, cross-linked polyolefin jacket, unarmored, watertight (in accordance with MIL-DTL-24643/57).

LS2UWA - Twisted pairs: cross-linked polyethylene insulation, overall braided shield, cross-linked polyolefin jacket, armored (in accordance with MIL-DTL-24643/57; note that, at the time of this document's publication, armored version is not in most recent revision of specification sheet).

LS2UWS - Twisted pairs: thermoset insulation, overall braided shield, cross-linked polyolefin jacket, double overall shielded, unarmored, watertight (in accordance with MIL-DTL-24643/57).

LS2WA - Pairs: cross-linked polyethylene insulation, overall braided shield, cross linked polyolefin jacket, armored (per MIL-DTL-24643/46; note that, at the time of this document's publication, armored version is not in most recent revision of specification sheet).

LS2WAU - Pairs: thermoset insulation, overall braided shield, cross linked polyolefin jacket, unarmored, watertight (in accordance with MIL-DTL-24643/46).

LS2WU - Pairs: thermoset insulation, overall braided shield, cross-linked polyolefin jacket, unarmored, watertight (in accordance with MIL-DTL-24643/45).

LS3C179DT - Three coaxes, thermoset insulation, braided shield, cross-linked polyolefin jacket over each coax, overall braided shield, cross-linked polyolefin jacket, unarmored, non-watertight (in accordance with MIL-DTL-24643/65).

LS3OW - Three conductors, thermoset insulation, braided shield, cross-linked polyolefin jacket, unarmored, watertight with circuit integrity (in accordance with MIL-DTL-24643/67).

LS3SA - Triads, shielded: cross-linked polyethylene insulation, braided shield over each triad, cross-linked polyolefin jacket, armored (in accordance with

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		MIL-DTL-24643/35; note that, at the time of this document's publication, armored version is not in most recent revision of specification sheet).
LS3SF	-	Triads, shielded, flexible: thermoset insulation, braided shield over each triad, polyester tape over the assembled triads, cross-linked polyolefin jacket, unarmored, non-watertight (in accordance with MIL-DTL-24643/44).
LS3SJ	-	Triads, shielded: thermoset insulation, overall braided shield, cross-linked polyolefin jacket, unarmored, non-watertight (in accordance with MIL-DTL-24643/43).
LS3SJA	-	Three conductors, shielded: ethylene propylene rubber or cross- linked polyethylene insulation, overall braided shield, cross-linked polyolefin jacket, armored (in accordance with MIL-DTL-24643/43; note that, at the time of this document's publication, armored version is not in most recent revision of specification sheet).
LS3SU	-	Triads, shielded: thermoset insulation, braided shield over each triad, cross-linked polyolefin jacket, unarmored, non-watertight (in accordance with MIL-DTL-24643/35).
LS3SUS	-	Triads, shielded: thermoset insulation, braided shield over each triad, cross-linked polyolefin jacket, double overall shielded, unarmored, non-watertight (in accordance with MIL-DTL-24643/35).
LS3SWA	-	Triads, shielded: cross-linked polyethylene insulation, braided shield over each triad, cross-linked polyolefin jacket, armored (in accordance with MIL-DTL-24643/36; note that, at the time of this document's publication, armored version is not in most recent revision of specification sheet).
LS3SWU	-	Triads, shielded: thermoset insulation, braided shield over each triad, cross-linked polyolefin jacket, unarmored, watertight (in accordance with MIL-DTL-24643/36).
LS3SWUS	-	Triads, shielded: thermoset insulation, braided shield over each triad, cross-linked polyolefin jacket, double overall shielded, unarmored, watertight (in accordance with MIL-DTL-24643/36).
LS3U	-	Triads: thermoset insulation, marker braid on each triad, cross-linked polyolefin jacket, unarmored, non-watertight (in accordance with MIL-DTL-24643/37).
LS3UA	-	Triads: cross-linked polyethylene insulation, marker braid on each triad, cross-linked polyolefin jacket, armored (in accordance with MIL-DTL-24643/37; note that, at the time of this document's publication, armored version is not in most recent revision of specification sheet).
LS4NW	-	Four conductors: thermoset insulation, cross-linked polyolefin jacket, unarmored, non-watertight (in accordance with MIL-DTL-24643/55).
LS4NWA	-	Four conductors: cross-linked polyethylene insulation or ethylene propylene rubber, cross-linked polyolefin jacket, armored (in accordance with MIL-DTL-24643/55; note that, at the time of this document's publication, armored version is not in most recent revision of specification sheet).
LS4OW	-	Four conductors, thermoset insulation, braided shield, cross-linked polyolefin jacket, unarmored, watertight with circuit integrity (in accordance with MIL-DTL-24643/68).
LS4SJ	-	Four conductors, shielded: thermoset insulation, overall braided shield, cross-linked polyolefin jacket, unarmored, non-watertight (in accordance with MIL-DTL-24643/43).
LS4SJA	-	Four conductors, shielded: ethylene propylene rubber or cross-linked polyethylene insulation, overall braided shield, cross-linked polyolefin jacket, armored (in

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accordance with MIL-DTL-24643/43; note that, at the time of this document's publication, armored version is not in most recent revision of specification sheet).

LS5KVTSGA - 5,000-volt, three conductors: silicone rubber and glass tape insulation, cross-linked polyolefin jacket, armored (in accordance with MIL-DTL-24643/22; note that, at the time of this document's publication, armored version is not in most recent revision of specification sheet).

LS5KVTSGU - 5,000-volt, three conductors: thermoset insulation, cross-linked polyolefin jacket, unarmored, watertight with circuit integrity (in accordance with MIL-DTL-24643/22).

LS5KVTSSGUSHR- 5,000-volt, three conductors, extruded silicone rubber insulation, silicone rubber/fiberglass tape insulation, black semi-conducting layer, braided shield, extruded low smoke, zero halogen, cross-linked polyolefin jacket, unarmored, watertight with circuit integrity (in accordance with MIL-DTL-24643/86).

LS5KVTEPSG - 5,000-volt, three conductors, extruded thermoset semi-conducting conductor shield, thermoset ethylene propylene rubber insulation, extruded semi-conducting insulation shield, tin-coated copper braided wire shield, three insulated ground wires, cross-linked polyolefin jacket, unarmored, watertight with circuit integrity (in accordance with MIL-DTL-24643/76).

LS5KVTSGUS - 5,000-volt, three conductors and three ground wires: thermoset insulation, single overall shield, cross-linked polyolefin jacket, unarmored, watertight with circuit integrity (in accordance with MIL-DTL-24643/22).

LS6C179DT - Six coaxes, thermoset insulation, braided shield, cross-linked polyolefin jacket over each coax, overall braided shield, cross-linked polyolefin jacket, unarmored, non-watertight (in accordance with MIL-DTL-24643/65).

LS6SGA - Six conductors: silicone and glass insulation, cross-linked polyolefin jacket, armored (in accordance with MIL-DTL-24643/19; note that, at the time of this document's publication, armored version is not in most recent revision of specification sheet).

LS6SGU - Six conductors: thermoset insulation, cross-linked polyolefin jacket, unarmored, watertight with circuit integrity (in accordance with MIL-DTL-24643/19).

LS7PS - Seven conductors, power supply: silicone rubber insulation, glass braid, silicone rubber jacket, armored, watertight with circuit integrity (in accordance with MIL-DTL-24643/26).

LS7PSN - Seven conductors, power supply: silicone rubber insulation, glass braid, silicone rubber jacket, armored, non-watertight with circuit integrity (in accordance with MIL-DTL-24643/26).

LS7SGA - Seven conductors: silicone rubber and glass insulation, cross-linked polyolefin jacket, armored (in accordance with MIL-DTL-24643/20; note that, at the time of this document's publication, armored version is not in most recent revision of specification sheet).

LS7SGU - Seven conductors: thermoset insulation, cross-linked polyolefin jacket, unarmored, watertight with circuit integrity (in accordance with MIL-DTL-24643/20).

LS8NW - Eight conductors: thermoset insulation, cross-linked polyolefin jacket, unarmored, non-watertight (in accordance with MIL-DTL-24643/54).

LS8NWA - Eight conductors: cross-linked polyethylene or ethylene propylene rubber insulation, cross-linked polyolefin jacket, armored (in accordance with MIL-DTL-24643/54; note that, at the time of this document's publication, armored version is not in most recent revision of specification sheet).

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LS15KVSSU - Single conductor: ethylene propylene rubber insulation, cross-linked polyolefin jacket, unarmored, watertight (note that, at the time of this document's publication, the corresponding specification sheet for MIL-DTL-24643 has not been issued).

LS15KVTSU - Three conductors: ethylene propylene rubber insulation, cross-linked polyolefin jacket, unarmored, watertight (note that, at the time of this document's publication, the corresponding specification sheet for MIL-DTL-24643 has not been issued).

5.2 Cable types and construction characteristics (as specified in MIL-DTL-24640) list. Cable types and construction characteristics (as specified in MIL-DTL-24640) are as follows:

DX - Double conductor, power supply: cross-linked polyalkene insulation, cross-linked polyolefin jacket, unarmored, non-watertight (in accordance with MIL-DTL-24640/1).

DXA - Double conductor, power supply: cross-linked polyalkene insulation, cross-linked polyolefin jacket, armored (in accordance with MIL-DTL-24640/1; note that, at the time of this document's publication, armored version is not in most recent revision of specification sheet).

DXW - Two conductors: mica polyimide insulation, cross-linked polyolefin jacket, unarmored, watertight with circuit integrity (in accordance with MIL-DTL-24640/19).

DXOW - Two conductors: mica polyimide insulation, cross-linked polyolefin jacket, overall shielded, unarmored, watertight with circuit integrity (in accordance with MIL-DTL-24640/19).

DXWA - Two conductors: mica polyimide insulation, cross-linked polyolefin jacket, armored (in accordance with MIL-DTL-24640/19; note that, at the time of this document's publication, armored version is not in most recent revision of specification sheet).

DXWB - Double conductor, power supply: cross-linked polyalkene insulation, cross-linked polyolefin jacket, unarmored, watertight (in accordance with MIL-DTL-24640/1).

FX - Four conductors, power supply: cross-linked polyalkene insulation, cross-linked polyolefin jacket, unarmored, non-watertight (in accordance with MIL-DTL-24640/3).

FXA - Four conductors, power supply: cross-linked polyalkene insulation, cross-linked polyolefin jacket, armored (in accordance with MIL-DTL-24640/3; note that, at the time of this document's publication, armored version is not in most recent revision of specification sheet).

FXW - Four conductors: mica polyimide insulation, cross-linked polyolefin jacket, unarmored, watertight with circuit integrity (in accordance with MIL-DTL-24640/21).

FXOW - Four conductors: mica polyimide insulation, cross-linked polyolefin jacket, overall shielded, unarmored, watertight with circuit integrity (in accordance with MIL-DTL-24640/21).

FXWA - Four conductors: mica polyimide insulation, cross-linked polyolefin jacket, armored (in accordance with MIL-DTL-24640/21; note that, at the time of this document's publication, armored version is not in most recent revision of specification sheet).

FXWB - Four conductors, power supply: cross-linked polyalkene insulation, cross-linked polyolefin jacket, unarmored, watertight (in accordance with MIL-DTL-24640/3).

MXCW - Multi-conductor, control: mica polyimide insulation, cross-linked polyolefin jacket, unarmored, watertight (in accordance with MIL-DTL-24640/23).

MXCOW - Multi-conductor, control: mica polyimide insulation, cross-linked polyolefin jacket, overall shielded, watertight (in accordance with MIL-DTL-24640/23).

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MXCWA	-	Multi-conductor, control: mica polyimide insulation, cross-linked polyolefin jacket, armored (in accordance with MIL-DTL-24640/23; note that, at the time of this document's publication, armored version is not in most recent revision of specification sheet).
MXO	-	Multi-conductor, power supply: cross-linked polyalkene insulation, cross-linked polyolefin jacket, overall shielded, unarmored, non-watertight (in accordance with MIL-DTL-24640/8).
MXOW	-	Multi-conductor, power supply: cross-linked polyalkene insulation, cross-linked polyolefin jacket, overall shielded, unarmored, watertight (in accordance with MIL-DTL-24640/8).
MXSO	-	Multi-conductor: cross-linked polyalkene insulation, cross-linked polyolefin jacket, overall shielded, unarmored, non-watertight (in accordance with MIL-DTL-24640/10).
MXSOW	-	Multi-conductor: cross-linked polyalkene insulation, cross-linked polyolefin jacket, overall shielded, unarmored, watertight (in accordance with MIL-DTL-24640/10).
TTX	-	Multi-pair, twisted: cross-linked polyalkene insulation, cross-linked polyolefin jacket, unarmored, non-watertight (in accordance with MIL-DTL-24640/4).
TTXA	-	Multi-pair, twisted: cross-linked polyalkene insulation, cross-linked polyolefin jacket, armored (in accordance with MIL-DTL-24640/4; note that, at the time of this document's publication, armored version is not in most recent revision of specification sheet).
TTXS	-	Multi-pair, twisted, shielded: cross-linked polyalkene insulation, cross-linked polyolefin jacket, unarmored, non-watertight (in accordance with MIL-DTL-24640/5).
TTXSA	-	Multi-pair, twisted, shielded: cross-linked polyalkene insulation, cross-linked polyolefin jacket, armored (in accordance with MIL-DTL-24640/5; note that, at the time of this document's publication, armored version is not in most recent revision of specification sheet).
TTXSO	-	Multi-pair, twisted, shielded: cross-linked polyalkene insulation, cross-linked polyolefin jacket, overall shielded, unarmored, non-watertight (in accordance with MIL-DTL-24640/5).
TTXSOW	-	Multi-pair, twisted, shielded: cross-linked polyalkene insulation, cross-linked polyolefin jacket, overall shielded, unarmored, watertight (in accordance with MIL-DTL-24640/5).
TTXSW	-	Multi-pair, twisted, shielded: cross-linked polyalkene insulation, cross-linked polyolefin jacket, unarmored, watertight (in accordance with MIL-DTL-24640/5).
TTXW	-	Multi-pair: mica polyimide insulation, cross-linked polyolefin jacket, unarmored, watertight with circuit integrity (in accordance with MIL-DTL-24640/24).
TTXOW	-	Multi-pair: mica polyimide insulation, cross-linked polyolefin jacket, overall shielded, watertight with circuit integrity (in accordance with MIL-DTL-24640/24).
TTXWA	-	Multi-pair: mica polyimide insulation, cross-linked polyolefin jacket, armored (in accordance with MIL-DTL-24640/24; note that, at the time of this document's publication, armored version is not in most recent revision of specification sheet).
TTXWB	-	Multi-pair, twisted: cross-linked polyalkene insulation, cross-linked polyolefin jacket, unarmored, watertight (in accordance with MIL-DTL-24640/4).
TX	-	Three conductor, power supply: cross-linked polyalkene insulation, cross-linked polyolefin jacket, unarmored, non-watertight (in accordance with MIL-DTL-24640/2).

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TXA	-	Three conductor, power supply: cross-linked polyolefin insulation, cross-linked polyolefin jacket, armored (in accordance with MIL-DTL-24640/2; note that, at the time of this document's publication, armored version is not in most recent revision of specification sheet).
TXW	-	Three conductor, power: mica polyimide insulation; cross-linked polyolefin jacket, unarmored, watertight with circuit integrity (in accordance with MIL-DTL-24640/20).
TXOW	-	Three conductor, power: mica polyimide insulation; cross-linked polyolefin jacket, overall shielded, unarmored, watertight with circuit integrity (in accordance with MIL-DTL-24640/20).
TXWA	-	Three conductor, power: mica polyimide insulation; cross-linked polyolefin jacket, armored (in accordance with MIL-DTL-24640/20; note that, at the time of this document's publication, armored version is not in most recent revision of specification sheet).
TXWB	-	Three conductor, power supply: cross-linked polyalkene insulation, cross-linked polyolefin jacket, unarmored, watertight (in accordance with MIL-DTL-24640/2).
1XMSO	-	Multi-conductor, shielded: cross-linked polyalkene insulation, cross-linked polyolefin jacket, overall shielded, unarmored, non-watertight (in accordance with MIL-DTL-24640/7).
1XMSOW	-	Multi-conductor, shielded: cross-linked polyalkene insulation, cross-linked polyolefin jacket, overall shielded, unarmored, watertight (in accordance with MIL-DTL-24640/7).
1XSOW	-	Multi-conductor, shielded: cross-linked polyalkene insulation, cross-linked polyolefin jacket, overall shielded, unarmored, watertight (in accordance with MIL-DTL-24640/14).
2XAO	-	Multi-pair, twisted: cross-linked polyalkene insulation, cross-linked polyolefin jacket, overall shielded, unarmored, non-watertight (in accordance with MIL-DTL-24640/6).
2XAOW	-	Multi-pair, twisted: cross-linked polyalkene insulation, cross-linked polyolefin jacket, overall shielded, unarmored, watertight (in accordance with MIL-DTL-24640/6).
2XO	-	Multi-pair, twisted: cross-linked polyalkene insulation, cross-linked polyolefin jacket, overall shielded, unarmored, non-watertight (in accordance with MIL-DTL-24640/12).
2XOW	-	Multi-pair, twisted: cross-linked polyalkene insulation, cross-linked polyolefin jacket, overall shielded, unarmored, watertight (in accordance with MIL-DTL-24640/12 and MIL-DTL-24640/17).
2XS	-	Multi-pair, twisted, shielded: cross-linked polyalkene insulation, cross-linked polyolefin jacket, unarmored, non-watertight (in accordance with MIL-DTL-24640/9).
2XSA	-	Multi-pair, twisted, shielded: cross-linked polyalkene insulation, cross-linked polyolefin jacket, armored (in accordance with MIL-DTL-24640/9; note that, at the time of this document's publication, armored version is not in most recent revision of specification sheet).
2XSO	-	Multi-pair, twisted, shielded: cross-linked polyalkene insulation, cross-linked polyolefin jacket, overall shielded, unarmored, non-watertight (in accordance with MIL-DTL-24640/9).

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2XSAW	-	Multi-pair, shielded: cross-linked polyalkene insulation, cross-linked polyolefin jacket, unarmored, watertight (in accordance with MIL-DTL-24640/15).
2XSAOW	-	Multi-pair, shielded: cross-linked polyalkene insulation, cross-linked polyolefin jacket, overall shielded, unarmored, watertight (in accordance with MIL-DTL-24640/15).
2XSAWA	-	Multi-pair, shielded: cross-linked polyalkene insulation, cross-linked polyolefin jacket, armored, watertight (in accordance with MIL-DTL-24640/15).
2XSXO	-	Four-pair, twisted, shielded: cross-linked polyalkene insulation, cross-linked polyolefin jacket, overall shielded, unarmored, non-watertight (in accordance with MIL-DTL-24640/13).
2XSXOW	-	Four-pair, twisted, shielded: cross-linked polyalkene insulation, cross-linked polyolefin jacket, overall shielded, unarmored, watertight (in accordance with MIL-DTL-24640/13).
2XSW	-	Multi-pair, shielded: cross-linked polyalkene insulation, cross-linked polyolefin jacket, unarmored, watertight (in accordance with MIL-DTL-24640/9 and MIL-DTL-24640/16).
2XSOW	-	Multi-pair, shielded: cross-linked polyalkene insulation, cross-linked polyolefin jacket, overall shielded unarmored, watertight (in accordance with MIL-DTL-24640/9 and MIL-DTL-24640/16).
2XSWA	-	Multi-pair, shielded: cross-linked polyalkene insulation, cross-linked polyolefin jacket, armored (in accordance with MIL-DTL-24640/16; note that, at the time of this document's publication, armored version is not in most recent revision of specification sheet).
3XS	-	Multi-triad, twisted, shielded: cross-linked polyalkene insulation, cross-linked polyolefin jacket, unarmored, non-watertight (in accordance with MIL-DTL-24640/11).
3XSA	-	Multi-triad, twisted, shielded: cross-linked polyalkene insulation, cross-linked polyolefin jacket, armored (in accordance with MIL-DTL-24640/11; note that, at the time of this document's publication, armored version is not in most recent revision of specification sheet).
3XSW	-	Multi-triad, twisted, shielded: cross-linked polyalkene insulation, cross-linked polyolefin jacket, unarmored, watertight (in accordance with MIL-DTL-24640/11 and MIL-DTL-24640/18).
3XSOW	-	Multi-triad, twisted, shielded: cross-linked polyalkene insulation, cross-linked polyolefin jacket, overall shielded, unarmored, watertight (in accordance with MIL-DTL-24640/18).
3XSWA	-	Multi-triad, twisted, shielded: cross-linked polyalkene insulation, cross-linked polyolefin jacket, armored (in accordance with MIL-DTL-24640/18; note that, at the time of this document's publication, armored version is not in most recent revision of specification sheet).
5XO	-	Five conductors, shielded, cross-linked polyolefin jacket, unarmored, non-watertight (in accordance with MIL-DTL-24640/26).
7XW	-	Seven conductors: mica polyimide insulation, cross-linked polyolefin jacket, unarmored, watertight with circuit integrity (in accordance with MIL-DTL-24640/22).
7XWA	-	Seven conductors: mica polyimide insulation, cross-linked polyolefin jacket, armored (in accordance with MIL-DTL-24640/22; note that, at the time of this

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document's publication, armored version is not in most recent revision of specification sheet).

9XS - Four pairs, one single, shielded, cross-linked polyolefin jacket, unarmored, non-watertight (in accordance with MIL-DTL-24640/25).

5.3 Cable types and construction characteristics (as specified in MIL-DTL-915) list. Cable types and construction characteristics (as specified in MIL-DTL-915) are as follows:

CVSF - 400-Hz aircraft servicing: three synthetic rubber insulated conductors and one uninsulated conductor, overall polychloroprene jacket (in accordance with MIL-DTL-32180; previously MIL-DTL-915/1).

DLT - Divers lifeline and telephone: four rubber insulated conductors cabled around an insulated steel core, reinforced polychloroprene jacket overall (in accordance with A-A-59615; previously MIL-DTL-915/5).

DSS - Double conductor, shielded: rubber insulation, overall braided shield, polychloroprene or chlorosulfonated polyethylene jacket, not for inboard use, watertight (in accordance with MIL-DTL-915/8).

DSWS - Double conductor, shielded: rubber insulation, overall braided shield, polychloroprene jacket, not for inboard use, watertight (in accordance with MIL-DTL-915/7).

FSS - Four conductors, shielded: rubber insulated, overall braided shield, polychloroprene or chlorosulfonated polyethylene jacket, not for inboard use, watertight (in accordance with MIL-DTL-915/8).

JAS - Jet aircraft servicing: four rubber insulated conductors, two conductors Navy size 250, two conductors size AWG 12, reinforced polychloroprene jacket, not for inboard use, non-watertight (in accordance with MIL-DTL-915/9).

MCSF-4 - Multiple conductor, acoustic minesweeping, power: two AWG 6 and two AWG 1 conductors, rubber insulation, reinforced polychloroprene jacket, not for inboard use, non-watertight (in accordance with MIL-DTL-915/10).

MSP - Multiple conductor: 59 conductors, 16 AWG 22 having fluoropolymer insulation and a braided copper shield, 18 AWG 20 having polyvinyl chloride insulation and a braided copper shield (nine singles, one triad and three pairs, each shielded), 25 AWG 16 having polyvinyl insulation (eight pairs and three triads, each shielded), polychloroprene jacket, not for inboard use, non-watertight (in accordance with MIL-DTL-915/67).

MSPW - Multiple conductor: 59 conductors; 16 AWG 22 having fluorocarbon insulation and a braided copper shield, 18 AWG 20 having polyvinyl chloride insulation and a braided copper shield (nine singles, one triad and three pairs, each shielded), 25 AWG 16 having polyvinyl insulation (eight pairs and three triads, each shielded), polychloroprene jacket, not for inboard use, watertight (in accordance with MIL-DTL-915/66).

MWF - Multiple conductor: rubber or cross-linked polyethylene insulation, arctic type neoprene jacket, watertight, not for inboard use (in accordance with MIL-DTL-915/58).

S2S - Two conductors, shielded: cross-linked polyethylene insulations, braided shield, rubber insulation over shield, outer-braided shield; reinforced rubber, insulated, arctic type polychloroprene jacket (in accordance with A-A-59610; previously MIL-DTL-915/61).

THOF - Three conductors, heat and oil resistant, flexible: ethylene propylene rubber insulation, polychloroprene jacket, non-watertight, not for inboard use (in accordance with MIL-DTL-915/6).

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TRF	-	Single conductor, flexible: rubber insulation, polychloroprene jacket (in accordance with A-A-59608; previously MIL-C-915/20).
TPUM-6	-	Telephone, portable, multiple conductor: copper-clad steel conductors, polypropylene insulation, six pairs cabled, polyurethane jacket applied in two layers, watertight, not for inboard use (in accordance with MIL-C-915/79 [inactive for new design]).
TRXF	-	Single conductor: polychloroprene jacket (in accordance with A-A-59609; previously MIL-C-915/21).
TSP	-	Twisted pairs: polyvinyl chloride insulated, special thermoplastic jacket, watertight, unarmored (in accordance with MIL-DTL-915/22).
TSPA	-	Twisted pairs: polyvinyl chloride insulated, special thermoplastic jacket, watertight, armored (in accordance with MIL-DTL-915/22; note that, at the time of this document's publication, armored version is not in most recent revision of specification sheet).
TSS	-	Three conductors: special purpose, shielded: rubber insulation, overall braided shield, polychloroprene or chlorosulfonated polyethylene jacket, not for inboard use, watertight (in accordance with MIL-DTL-915/8).
1PR-A20E	-	One twisted pair, high strength copper alloy conductors, ethylene tetrafluoroethylene insulation, clear natural polyethylene insulation jacket, clear natural polyethylene jacket, for submarine outboard use only, non-watertight (in accordance with MIL-DTL-915/81).
1PR-16	-	One twisted pair, ethylene tetrafluoroethylene insulation, cable jacket of polyether polyurethane, for submarine outboard use only. Specification sheet inactive for new design (MIL-C-915/82).
7PR-16	-	Seven twisted pairs, ethylene tetrafluoroethylene insulation, cable jacket of polyether polyurethane, for submarine outboard use only. Specification sheet inactive for new design (MIL-C-915/83).
2SPR-16	-	Two twisted shielded pairs, ethylene tetrafluoroethylene insulation, braided shield, cable jacket of polyether polyurethane, for submarine outboard use only. Specification sheet inactive for new design (MIL-C-915/84).
3SPR-16	-	Three twisted shielded pairs, ethylene tetrafluoroethylene insulation, cable jacket of polyether polyurethane, for submarine outboard use only. Specification sheet inactive for new design (MIL-C-915/85).
1Q-16	-	Four conductors, ethylene tetrafluoroethylene insulation, cable jacket of polyether polyurethane, for submarine outboard use only. Specification sheet inactive for new design (MIL-C-915/86).
1TR-16	-	Three conductors, ethylene tetrafluoroethylene insulation, cable jacket of polyether polyurethane, for submarine outboard use only. Specification sheet inactive for new design (MIL-C-915/87).
7SPR-16S	-	Seven twisted shielded pairs, ethylene tetrafluoroethylene insulation, overall braided shield, cable jacket of polyether polyurethane, for submarine outboard use only. Specification sheet inactive for new design (MIL-C-915/88).
1SPR-16	-	Two conductors, ethylene tetrafluoroethylene insulation, braided shield, cable jacket of polyether polyurethane, for submarine outboard use only. Specification sheet inactive for new design (MIL-C-915/89).
1SWF	-	Two singles, shielded: polyethylene insulation, braided shield on each conductor, arctic-type polychloroprene jacket, not for inboard use, watertight (in accordance with MIL-DTL-915/47).

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2SWF	-	Pairs, shielded, watertight, flexible: polyethylene insulation, braided shield over each pair, arctic-type polychloroprene jacket, not for inboard use (in accordance with MIL-DTL-915/48).
5SS	-	Five conductors, shielded, sonar: rubber insulation, braided shield on one conductor only, and a braided shield over the assembled five conductors, polychloroprene jacket overall (in accordance with A-A-59612; previously MIL-C-915/74).
7SS	-	Seven conductors, shielded: rubber insulation, overall braided shield, polychloroprene or chlorosulfonated polyethylene jacket, not for inboard use, watertight (in accordance with MIL-DTL-915/8).

5.4 Identification information. Cable and cable conductor information and requirements are included in MIL-DTL-24643, MIL-DTL-24640, and MIL-DTL-915.

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TABLE I. MIL-DTL-24643 cable application data. [1/]

Application	Cable type [2/]	
	Non-flexing service	Repeated flexing service
General usage:		
For all portions of power, lighting, interior communication, weapons control, and electronics systems, except where circuit parameters (such as audio or radio frequency, low-level microphone, synchro, scale voltage, and other types of signals) require special types of cable. See 4.1.12 for guidance on installation of watertight and non-watertight cable types. Type SG cable should be used for connections between the ship service generators rated 1,000 volts or less and their respective switchboards and between sections of the ship service switchboards.	LSDNW LSTNW LSFNW LSMNW LSSSGU LSDSGU LSTSGU LSFSGU LS6SGU LS7SGU LSMSCU LSMU LS3U LS3SU LSSSF LSDPS LSTPS LSFPS LS7PS LSDPSN LSTPSN LSFPSN LS7PSN LSECM LS8NW LS4NW LSYSGU LSSCF LSFCF LSTCF LST LSTW	LSSHOF LSDHOF LSTHOF LSFHOF LSMHOF LSDCOP LSTCOP LSMCOS
Greater than 1,000 volts – 60 Hz:		
For 2,000 volts and higher, three-phase power applications.	LS5KVTSGU LS5KVTSGUS LS5KVTEPSG LS2KVTSGSG LS5KVTSSGUSHR LS15KVSSU LS15KVTSU	
Casualty power		LSTHOF-42

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TABLE I. MIL-DTL-24643 cable application data [1/] – Continued.

Application	Cable type [2/]	
	Non-flexing service	**Repeated flexing service**
400-Hz power:		
For 400-Hz service for static frequency changer cables, bus ties, and feeders where cable of lower impedance is required to reduce voltage drop.	LS6SGU LS2U LS3SJ LS3SU LS3SWU LS4SJ LSMS	
Audio and telephone:		
For audio, telephone, call bell, announcing, and alarm systems. May also be used for other interior communication and weapons control systems, provided ampere rating of the cable and voltage drop for the system are not exceeded. See 4.1.12 for guidance on installation of watertight and non-watertight cable types.	LSTPNW LSTTSU LSMRI (no jacket)	LSTTOP
Radio high power	LSSRW LSDRW LSTRW	
Radio frequency:		
For application up to 2 megahertz (MHz) Maximum total copper operating temperature should not exceed 75 °C.		LSTTRS
Degaussing (see table X)	LSSSGU LSDSGU LSTSGU LSFSGU LS6SGU LSMDU LSMSCU LS7SGU	
Thermocouple and pyrometer temperature range:		
Type TCTX, 125 to 260 °C Type TCJX, 150 to 540° C Type TCKX, 260 to 870 °C	LSTCTX LSTCJX LSTCKX LSTCJU LSTCTU LSPBTMU LSTCTXN LSTCJXN LSTCKXN	

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TABLE I. MIL-DTL-24643 cable application data [1/] – Continued.

Application	Cable type [2/]	
	Non-flexing service	Repeated flexing service
Shielded circuits: For combat systems, interior communications, lighting, and power circuits, where shielding is required. See 4.1.12 for guidance on installation of watertight and non-watertight cable types.	LS1SAU LS1SMU LS1SMWU LS1SWU LS1SU LS1S50MU LS1S50MUS LS1S75MU LS2AU LS2AUS LS2SJ LS2SWAU LS2SWU LS2SU LS2UW LS2UWS LS2WAU LS2U LS2WU LS3SJ LS3SUS LS3SWU LS3SWUS LS4SJ LSMS LSMSCS LSMUS LS3SF LS2SWL LS2CS LSTPSJ LS2OW LS3OW LS4OW LSC264	LSTTRS
Microphone circuits		LSMCOS LSMMOP
Position indicator	LSPI	

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TABLE I. MIL-DTL-24643 cable application data [1/] – Continued.

Application	Cable type [2/]	
	Non-flexing service	**Repeated flexing service**
Low voltage (150 V)	LSC5FS LSC5FSW LSC5OS LSC5OSW LSC5P LSC5POS LSC5POSR LS3C179DT LS6C179DT LSC6FS LSC6OS LSC6OSW	
PROFIBUS	LSPB2SD LSPB2SDW LSPB2SDOS LSPB2SDOSW	
Computer applications (CAT cables)	LSLCFS LSCBPSJ [3/] LSCBPSJW [3/]	

NOTES:

1/ The order of listing of cables for general applications data has no significant meaning for their usage. Also, the listings of cable types for specific applications are for guidance; the decision on the selection of cable types is made by the design agent, the supervisor, or NAVSEA.

2/ Many cable types are manufactured in variations of unarmored, unarmored with overall shields, and with individual conductor shields (see 5.1). Some cable types are armored. The use of armored cable on non-nuclear ships is optional and to be determined by the overhaul shipyard, except armored cables should not be installed in weather locations due to EMC considerations.

3/ Specifically for Controlled Area Network bus data communication.

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TABLE II. MIL-DTL-24640 cable application data. [1/]

Application	Cable type [2/]	
	Non-flexing service	**Repeated flexing service**
General usage:		
For all portions of power, lighting, interior communication, weapons control, and electronics systems, except where circuit parameters (such as, audio or radio frequency, low-level microphone, synchro, scale voltage, and other types of signals) require special types of cable. Types DX, TX, FX, and MXO should be used only for runs that are either totally within one compartment or totally within two contiguous compartments. However, these types of cable should not be used where a watertight deck or watertight bulkhead below FWL-II is penetrated. See 4.1.12 for guidance on installation of watertight and non-watertight cable types.	DX TX FX DXW/DXWB TXW/TXWB FXW/FXWB 7XW	
Audio and telephone:		
For audio, telephone, call bell, announcing, and alarm systems. May also be used for other interior communication and weapons control systems, provided ampere rating of the cable and voltage drop for the system are not exceeded. See 4.1.12 for guidance on installation of watertight and non-watertight cable types.	TTX/TTXWB TTXW/TTXOW MXCW/MXCOW	
Radio frequency:		
For applications up to two MHz Maximum total copper operating temperature should not exceed 75 °C.	TTXS/TTXSO TTXSW/TTXSOW	

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TABLE II. MIL-DTL-24640 cable application data [1/] – Continued.

Application	Cable type [2/]	
	Non-flexing service	Repeated flexing service
Shielded circuits: For combat systems, interior communications, lighting, and power circuits, where shielding is required. See 4.1.12 for guidance on installation of watertight and non-watertight cable types.	2XAO/2XAOW 1XMSO/1XMSOW 2XS/2XSO MXSO/MXSOW 3XS 2XO/2XOW 2XSXO/2XSXOW 1XSOW 2XSAW/2XSAOW 2XSAWA 2XSW/2XSOW 3XSW/3XSOW DXOW FXOW TXOW MXO/MXOW 9XS 5XO	

NOTES:

1/ The order of listing of cables for general applications data has no significant meaning for their usage. Also, the listings of cable types for specific applications is for guidance; the decision on the selection of cable types is made by the design agent, the supervisor, or NAVSEA.

2/ Many cable types are manufactured in variations of unarmored, unarmored with overall shields, and with individual conductor shields (see 5.1). Some cable types are armored. The use of armored cable on non-nuclear ships is optional and to be determined by the overhaul shipyard, except armored cables should not be installed in weather locations due to EMC considerations.

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TABLE III. MIL-DTL-915 cable application data.[1/]

Application	Cable type [2/]	
	Non-flexing service	Repeated flexing service
Outboard submersible:		
For hydrophones, transducers, outboard dial telephones, retractable antennae, and similar equipment. Types MWF, 1SWF, and 2SWF are intended for hydrophones, transducers, and telephone lines in the weather. Types 1PR-A20E, 1PR-16, 7PR-16, 3PR-16, 1Q-16, ITR-16, 1SPR-16, and 7SPR-16S are intended only for submarine outboard use.	MSPW TSPA 1PR-A20E 1PR-16 7PR-16 2SPR-16 3PR-16 1Q-16 ITR-16 1SPR-16 7SPR-16S	MSP TSP 5SS S2S DSS FSS TSS 7SS MWF DSWS MCSF 1SWF 2SWF TPUM
Welding electrode circuit		TRF TRXF
Shore-to-shore power		THOF-400 THOF-500
Diver's line and telephone		DLT
400-Hz aircraft servicing		CVSF-4
DC aircraft servicing		JAS-250

NOTES:

1/ The order of listing of cables for general application data has no significant meaning for their usage.

2/ Many cables are manufactured in variations of armored, unarmored, and unarmored with overall shielding (see 5.3). In some cases, the armored version is no longer included in the respective MIL-DTL-915 specification sheet. In some cases, the associated specification sheet has been inactivated or canceled; however, the cable types may be installed in ships. See table V for applicable supersession data.

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TABLE IV. Commercial cable application data.

Application	Cable type repeated flexing service
Cords for portable tools and equipment: For power supply to typewriters, office machines, electric drills, sanders, portable extension lights, and similar equipment. Safety ground conductors should be green.	Underwriters approved S, SO, ST, SJ, SJO, SJT
NOTE: The order of listing of cables for general application data has no significant meaning for their usage.	

TABLE V. Supersession data.

Previous or present type, and applicable MIL-DTL-915 (or MIL-C-915) specification sheet	Present type and applicable specification sheet number	Obsolete type and detail specification number
CVSF-4……...../1	MIL-DTL-32180	None
DCOP……/3	LSDCOP…….MIL-DTL24643/2	None
DHOF…… /6	LSDHOF…… MIL-DTL-24643/3	None
DLT……..…… /5	A-A-59615	None
DNW/A……/68	LSDNW……..MIL-DTL-24643/48	None
DPS……......./40	LSDPS/N…… MIL-DTL24643/26	PS…… MIL-C-23206/3
DRW/A……..../73	LSDRW……..MIL-DTL24643/53	DHFR…….. MIL-C-915/4
DSGU/A……/29	LSDSGU…… MIL-DTL-24643/15	DSGA……..MIL-C-2194/2
DSGU/A /29	LSDSGU…… MIL-DTL-24643/15	DHFA……..MIL-C-2194/2
DSS…… /8	DSS……..MIL-DTL-915/8	None
DSWS……/7	DSWS……MIL-DTL-915/7	None
ECM/A……../54	LSECM……...MIL-DTL-24643/38	ECM…… MIL-C-24145/14
FHOF……/6	LSFHOF……. MIL-DTL-24643/3	None
FNW/A…… /70	LSFNW…….. MIL-DTL-24643/50	None
FPS……/40	LSFPS/N…….MIL-DTL-24643/26	PS…… MIL-C-23206/3
FSGU/A……/31	LSFSGU……. MIL-DTL-24643/17	FSGA……...MIL-C-2194/4
FSGU/A……/31	LSFSGU……. MIL-DTL-24643/17	FHFA……...MIL-C-2194/4
FPS……/40	LSFPS/N…….MIL-DTL-24643/26	None
FSS……/8	FSS……. MIL-DTL-915/8	None
JAS……/9	JAS……. MIL-DTL-915/9	None
MCOS…… /11	LSMCOS…… MIL-DTL-24643/4	None
MCSF…….. /10	MCSF……. MIL-DTL-915/10	None
MDU……/12	LSMDU……..MIL-DTL-24643/5	None
MDY……/13	LSMDU……..MIL-DTL-24643/5	None
LSMDY…….. 24643/6	LSMDU……..MIL-DTL-24643/5	None
MHOF…….. /14	LSMHOF…… MIL-DTL-24643/7	MHFF…….. MIL-C-915
MMOP……../15	LSMMOP……... MIL-DTL-24643/8	None
MNW/A……/71	LSMNW……. MIL-DTL-24643/51	MNW…….. MIL-C-915/71
MRI-D…….. /16	LSMRI-D…… MIL-DTL-24643/9	None
MRI-T……../16	LSMRI-T…… MIL-DTL-24643/9	None

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TABLE V. Supersession data – Continued.

Previous or present type, and applicable MIL-DTL-915 (or MIL-C-915) specification sheet	Present type and applicable specification sheet number	Obsolete type and detail specification number
MS/A……/50	LSMS…… MIL-DTL-24643/34	MS……... MIL-C-24145/10
MS/A……/50	LSM.S…… MIL-DTL-24643/34	MA……... MIL-C-24145/10
MSCU/A/S…….. /32	LSMSCU/S…….MIL-DTL-24643/18	MSCA……. MIL-C-2194/5
MSCU/A/S…….. /32	LSMSCU/S…… MIL-DTL-24643/18	MHFA……. MIL-C-2194/5
MSP……/67	MSP……MIL-DTL-915/67	None
MSPW……/66	MSPW…… MIL-DTL-915/66	None
MU/A/S……/43	LSMU/S……. MIL-DTL-24643/29	MA……... MIL-C-24145/3
MWF……/58	MWF…….. MIL-DTL-915/58	MWF……... MIL-C-24145/18
PBTM…….. /17	LSPBTM…… MIL-DTL-24643/10	None
PBTMU……/17	LSPBTMU……. MIL-DTL-24643/10	None
PI…….. /39	LSPI……MIL-DTL-24643/25	PI…… MIL-C-23206/2
SHOF……/6	LSSHOF……. MIL-DTL-24643/3	None
SRW/A……../73	LSSRW……... MIL-DTL-24643/53	SHFR……...MIL-C-915/4
SSF…….. /18	LSSSF…… MIL-DTL-24643/11	None
SSGU/A……/28	LSSSGU……. MIL-DTL-24643/14	SHGA……..MIL-C-2194/1
SSGU/A……/28	LSSSGU……. MIL-DTL-24643/14	SSGA……...MIL-C-2194/1
S2S…….. /61	S2S…… A-A-59610	S2S……... MIL-C-24145/21
TCJU/A……/35	LSTCJU……. MIL-DTL-24643/21	TCJA……... MIL-C-2194/10
TCJX……/38	LSTCJX/N……. MIL-DTL-24643/24	TCJX……... MIL-C-23206/1
TCKX…….. /38	LSTCKX/N…… MIL-DTL-24643/24	TCKX……..MIL-C-23206/1
TCOP……/3	LSTCOP……. MIL-DTL-24643/2	None
TCTU/A…….. /35	LSTCTU…….MIL-DTL-24643/21	TCTA…….. MIL-C-2194/10
TCTU/A…….. /35	LSTCTU…….MIL-DTL-24643/21	PBTX……...MIL-C-915
TCTX…….. /38	LSTCTX/N…….MIL-DTL-24643/24	TCTX…….. MIL-C-23206/1
THOF…….. /6	LSTHOF…….MIL-DTL-24643/3	None
THOF-42…….. /6	LSTHOF-42……MIL-DTL-24643/3	None
THOF-400……/6	THOF-400…….. MIL-DTL-915/6	None
THOF-500……/6	THOF-500…….. MIL-DTL-915/6	None
TNW/A……/69	LSTNW……... MIL-DTL-24643/49	TNW……MIL-C-915/69
TPNW/A…….. /72	LSTPNW……. MIL-DTL-24643/52	None
TPS…….. /40	LSTPS/N…… MIL-DTL-24643/26	PS…… MIL-C-23206/3
TPUM-6…….. /79	TPUM-6……. MIL-C-915/79	TPU……. MIL-C-915/19
TRF…….. /20	TRF……..A-A-59608	None
TRW/A…….. /73	LSTRW……... MIL-DTL-24643/53	THFR…….. MIL-C-915/4
TRXF…….. /21	A-A-59609	None
TSGU/A…….. /30	LSTSGU…… MIL-DTL-24643/16	THFA…….. MIL-C-2194/3
TSGU/A……../30	LSTSGU…….MIL-DTL-24643/16	TSGA…….. MIL-C-2194/3
TSP/A…….. /22	TSP/A……MIL-DTL-915/22	TSP……..MIL-C-915/22

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TABLE V. Supersession data – Continued.

Previous or present type, and applicable MIL-DTL-915 (or MIL-C-915) specification sheet	Present type and applicable specification sheet number	Obsolete type and detail specification number
TSS……../8	TSS……MIL-DTL-915/8	None
TTOP……./24	LSTTOP……MIL-DTL-24643/12	None
TTRS/A……./25	LSTTRS……MIL-DTL-24643/13	TTRS……..MIL-C-915/25
TRSA……./26	LSTTRS……MIL-DTL-24643/13	
TTSU/A……./37	LSTTSU……MIL-DTL-24643/23	TTSA……...MIL-C-2194/12
TTSU/A……./37	LSTTSU……MIL-DTL-24643/23	TTHFWA…MIL-C-915/23
1SA/U……../57	LS1SAU……MIL-DTL-24643/41	1SA……..MIL-C-24145/17
1SMU/A……/56	LS1SMU……..MIL-DTL-24643/40	1SMU…….. MIL-C-24145/16
1SMWU/A…….. /65	LS1SMWU…….MIL-DTL-24643/47	1SMWA….. MIL-C-24145/27
1SU/A……../59	LS1SU…… MIL-DTL-24643/42	1SU……..MIL-C-24145/19
1SWF…….. /47	1SWF…….. MIL-DTL-915/47	1SWF……...MIL-C-24145/7
1SWU/A…….. /44	LS1SWU/A…… MIL-DTL-24643/30	1SWA……..MIL-C-24145/4
1S50MU/A/S……./42	LS1S50MU/S…. MIL-DTL-24643/28	1S50MA….. MIL-C-24145/2
1S75MU/A…….. /55	LS1S75MU…… MIL-DTL-24643/39	1S75MA….. MIL-C-24145/15
2AU/A/A…….. /41	LS2AU/S…… MIL-DTL-24643/27	2A…… MIL-C-24145/1
2CS…….. /80	LS2CS…… MIL-DTL-24643/58	None
2SJ/A……/60	LS2SJ……..MIL-DTL-24643/43	DBSP……...MIL-C-915/2
2SU/A/S……/45	LS2SU/S…….MIL-DTL-24643/31	None
2SWA/U…….. /46	LS2SWAU……. MIL-DTL-24643/32	2SWA……..MIL-C-24145/6
2SWF……/48	2SWF……..MIL-DTL-915/48	2SWF……...MIL-C-24145/8
2SWL/A…….. /77	LS2SWL…….MIL-DTL-24643/56	None
2SWU/A…….. /49	LS2SWU…… MIL-DTL-24643/33	2SWU……..MIL-C-24145/9
2U/A…… /63	LS2U…….. MIL-DTL-24643/45	2U…… MIL-C-24145/23
2UW/A/S……./78	LS2UW/S……... MIL-DTL-24643/57	None
2WAU/A…… /64	LS2SWAU……. MIL-DTL-24643/46	2WU…… MIL-C-24145/24
3SF…….. /62	LS3SF……MIL-DTL-24643/44	3SF…….. MIL-C-24145/22
3SJ/A……/60	LS3SJ…… MIL-DTL-24643/43	TBSP……... MIL-C-915/2
3SJ/A……/60	LS3SJ……..MIL-DTL-24643/43	3SJ……... MIL-C-24145/20
3SU/A/S…….. /51	LS3SU/S…….MIL-DTL-24643/35	3SA……. MIL-C-24145/11
3SWU/A/S…….. /52	LS3SWU/S……. MIL-DTL-24643/36	3SWA……..MIL-C-24145/12
3U/A…… /53	LS3U…….. MIL-DTL-24643/37	3U…… MIL-C-24145/13
4NW8/A…….. /76	LS4NW……...MIL-DTL-24643/55	None
4SJ/A……/60	LS4SJ…… MIL-DTL-24643/43	FBSP……... MIL-C-915/2
4SJ/A……/60	LS4SJ…… MIL-DTL-24643/43	4SJ……. MIL-C-24145/20
5KVTSGU/A…….. /36	LS5KVTSGU/S..MIL-DTL-24643/22	5KVTSGA...MIL-C-2194/11
5SS…….. /74	A-A-59612	None
6SGU/A…….. /33	LS6SGU……. MIL-DTL-24643/19	6SGA……...MIL-C-2194/6

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TABLE V. Supersession data – Continued.

Previous or present type, and applicable MIL-DTL-915 (or MIL-C-915) specification sheet	Present type and applicable specification sheet number	Obsolete type and detail specification number
7PS…….. /40	LS7PS/N……MIL-DTL-24643/26	PS…… MIL-C-23206/3
7SGU/A……./34	LS7SGU……. MIL-DTL-24643/20	7SGA……...MIL-C-2194/9
7SS…….. /8	7SS……. MIL-DTL-915/8	None
8NW6/A…….. /75	LS8NW…….. MIL-DTL-24643/54	None

5.5 Brief explanation of cable ratings and characteristics tables.

5.5.1 First five columns. Each cable is identified by the specification and specification sheet number in the left-hand column, followed by the type designation, conductor size (AWG or MCM), number of conductors, and conductor cross sectional area (circular mils).

5.5.2 Overall diameter. The overall diameter is the overall measurement of the finished cable and should be the determining dimension in selecting the proper deck or bulkhead stuffing tube size, or multi-cable transit inserts. This diameter is also the determining dimension for stuffing tubes for equipment.

5.5.3 Rated voltage, ampacity, and minimum radius of bend. Electrical characteristics are given under columns headed “Rated voltage” and “Ampacity”. “Minimum radius of bend”, which is approximately eight times the overall diameter, is also given. Cables should not be used in applications where these rated voltages or ampacity may be exceeded or where a bend radius less than the minimum radius of the bend would be needed..

5.5.4 Conductor identification. The letters in the conductor identification column represent the identification and the number represents the method of applying the identification. For example, STD-1 means standard identification applied by method 1, which is printing of the number and color designation on the outer surface of the insulation or jacket of each conductor. TEL-3 means telephone identification applied by method 3, which is colored insulation on each conductor. The abbreviations used are as follows:

STD	-	Standard identification code
TEL	-	Telephone identification code
SPL	-	Special identification code
LTR	-	Letter identification code

5.6 Cable classification (MIL-DTL-24643). Cables specified in MIL-DTL-24643 are listed in table VI under the following general classifications:

a. Watertight (with circuit integrity), non-flexing service:
   (1) Power and lighting
   (2) Control
   (3) Electronic, communication, and instrumentation
b. Watertight, non-flexing service:
   (1) Power and lighting
   (2) Electronic, communication, and instrumentation
   (3) Control
c. Non-watertight (with circuit integrity), non-flexing service:
   (1) Electronic, communication, and instrumentation
   (2) Power and lighting
   (3) Control

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d. Non-watertight, non-flexing service:
   (1) Power and lighting
   (2) Control
   (3) Electronic, communication, and instrumentation
e. Non-watertight, flexing service:
   (1) Power and lighting
   (2) Control
   (3) Electronic, communication, and instrumentation

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TABLE VI (a)(1). MIL-DTL-24643 cable ratings and characteristics, watertight (with circuit integrity), non-flexing service (power and lighting).

MIL-DTL-	Cable type designation	Conductor size: AWG or MCM	Number of conductors, pairs, or triads in cable	Area of each conductor (MCM)	Cable overall diameter max. (inches)	Cable weight per ft approx. (lbs)	Radius of bend min. (inches)	CDR ID	Rated voltage, max. (RMS)	Ampacity, each conductor (amps, max.)				NSN 6145-01
										DC or 60 Hz		400 Hz		
										40 °C Ambient	50 °C Ambient	40 °C Ambient	50 °C Ambient	
24643/19 [4/]	LS6SGA- 100	0	6	105.600	1.650	2.919	10.0	LTR-5	1,000	136	127	272	250	203-0370
	LS6SGA- 125	00	6	133.100	1.840	3.600	11.5	LTR-5	1,000	160	142	299	275	201-9513
	LS6SGA- 150	000	6	167.800	2.010	4.468	12.0	LTR-5	1,000	188	173	326	300	201-9514
	LS6SGA- 200	0000	6	211.600	2.250	5.613	13.5	LTR-5	1,000	219	202	639	340	201-9515
24643/19	LS6SGU- 100	0	6	105.600	1.600	2.820	10.0	LTR-5 or STD 1 or 3	1,000	136	127	272	250	201-9511
	LS6SGU- 125	00	6	133.100	1.790	3.490	11.5	LTR-5 or STD 1 or 3	1,000	160	147	299	275	201-9512
	LS6SGU- 150	000	6	167.800	1.960	4.339	12.0	LTR-5 or STD 1 or 3	1,000	188	173	326	300	203-2370
	LS6SGU- 200	0000	6	211.600	2.200	5.479	13.5	LTR-5 or STD 1 or 3	1,000	219	202	369	340	203-0369
24643/26	LS7PS- 6	12	7	6.530	0.775	0.361	4.5	STD-4	600	35/20 [1/]	---	---	---	811-8376 [3/]
	LS7PS- 14	9	7	13.090	0.986	0.361	5.7	STD-4	600	---	---	---	---	---
24643/20 [4/]	LS7SGA- 3	16	7	2.580	0.595	0.177	4.0	STD-1	1,000	15/11	14/10 [1/]	15/11 [1/]	14/10 [1/]	201-9518
	LS7SGA- 4	14	7	4.110	0.645	0.231	4.0	STD-1	1,000	26/14	24/13	26/14	24/13	201-9519
24643/20	LS7SGU- 3	16	7	2.580	0.545	0.152	4.0	STD-1	1,000	15/11	14/10	15/11	14/10	201-9516
	LS7SGU- 4	14	7	4.110	0.595	0.198	4.0	STD-1	1,000	26/14	24/13	26/14	24/13	201-9517
	LS7SGU- 6	12	7	6.530	0.630	0.335	4.2	STD-1	1,000	41/22	38/21	41/22	38/21	---
24643/26	LSDPS- 3	16	2	2.580	0.455	0.121	3.0	STD-4	600	10	---	10	---	156-9511
	LSDPS- 4	14	2	4.110	0.489	0.141	3.0	STD-4	600	20	---	20	---	977-6159 [3/]
	LSDPS- 6	12	2	6.530	0.585	0.194	3.5	STD-4	600	28	---	28	---	158-2118
	LSDPS- 9	10	2	10.380	0.628	0.232	4.0	STD-4	600	41	---	41	---	542-6699 [3/]
	LSDPS- 14	9	2	13.090	0.730	0.323	4.5	STD-4	600	54	---	54	---	154-4372
24643/15 [4/]	LSDSGA- 3	16	2	2.580	0.441	0.109	3.0	STD-1	1,000	13	12	13	12	202-9501
	LSDSGA- 4	14	2	4.110	0.477	0.133	3.0	STD-1	1,000	22	20	22	20	203-0367
	LSDSGA- 9	10	2	10.380	0.594	0.230	4.0	STD-1	1,000	44	41	44	41	202-9502
	LSDSGA- 14	9	2	13.090	0.720	0.297	4.0	STD-1	1,000	60	55	60	55	202-9503
	LSDSGA- 23	7	2	20.820	0.831	0.410	5.0	STD-1 [2/]	1,000	78	72	78	72	202-9504
	LSDSGA- 50	3	2	52.620	0.961	0.740	6.0	STD-1	1,000	126	116	126	116	204-4855

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TABLE VI (a)(1). MIL-DTL-24643 cable ratings and characteristics, watertight (with circuit integrity), non-flexing service (power and lighting) – Continued.

MIL-DTL-	Cable type designation	Conductor size: AWG or MCM	Number of conductors, pairs, or triads in cable	Area of each conductor (MCM)	Cable overall diameter max. (inches)	Cable weight per ft approx. (lbs)	Radius of bend min. (inches)	CDR ID	Rated voltage, max. (RMS)	Ampacity, each conductor (amps, max.)				NSN 6145-01
										DC or 60 Hz		400 Hz		
										40 °C Ambient	50 °C Ambient	40 °C Ambient	50 °C Ambient	
24643/15 [4]	LSDSGA- 75	1	2	83.690	1.174	1.086	7.0	LTR-5	1,000	168	155	168	155	210-2343
	LSDSGA- 100	0	2	105.600	1.217	1.289	7.5	LTR-5	1,000	199	183	199	183	204-4856
	LSDSGA- 200	0000	2	211.600	1.633	2.458	10.0	LTR-5	1,000	308	284	288	266	204-4857
	LSDSGA- 300	300	2	300.000	1.891	3.326	11.5	LTR-5	1,000	413	380	347	319	204-4858
	LSDSGA- 400	400	2	413.600	2.119	4.408	13.0	LTR-5	1,000	492	453	337	310	203-0368
24643/15	LSDSGU- 3	16	2	2.580	0.391	0.086	3.0	STD-1 or 3 or LTR-5	1,000	13	12	13	12	202-8463
	LSDSGU- 4	14	2	4.110	0.427	0.106	3.0	STD-1 or 3 or LTR-5	1,000	22	20	22	20	202-2795
	LSDSGU- 9	10	2	10.380	0.544	0.194	4.0	STD-1 or 3 or LTR-5	1,000	44	41	44	41	202-2796
	LSDSGU- 14	9	2	13.090	0.670	0.259	4.0	STD-1 or 3 or LTR-5	1,000	60	55	60	55	202-2797
	LSDSGU- 23	7	2	20.820	0.781	0.363	5.0	STD-1 or 3 or LTR-5	1,000	78	72	78	72	202-2798
	LSDSGU- 50	3	2	52.620	0.911	0.681	6.0	STD-1or 3 or LTR-5	1,000	126	116	126	116	202-3476
	LSDSGU- 75	1	2	83.690	1.074	1.018	7.0	STD-1 or 3 or LTR-5	1,000	168	155	168	155	202-3477
	LSDSGU- 100	0	2	105.600	1.167	1.217	7.5	STD-1 or 3 or LTR-5	1,000	199	183	199	183	203-5390
	LSDSGU- 200	0000	2	211.600	1.583	2.360	10.0	STD-1 or 3 or LTR-5	1,000	308	241	288	266	203-3628
	LSDSGU- 300	300	2	300.000	1.841	3.213	11.5	STD-1 or 3 or LTR-5	1,000	413	380	347	319	203-5391
	LSDSGU- 400	400 (Navy STD)	2	413.600	2.069	4.782	13.0	STD-1 or 3 or LTR-5	1,000	492	453	337	310	202-8464
24643/26	LSFPS- 14	9	4	13.090	0.815	0.452	5.0	STD-4	600	42	---	---	---	156-8790
24643/17 [4]	LSFSGA- 3	16	4	2.580	0.447	0.126	3.0	STD-1	1,000	11	10	11	10	202-3481
	LSFSGA- 4	14	4	4.110	0.513	0.172	3.5	STD-1	1,000	18	17	18	17	202-3482
	LSFSGA- 9	10	4	10.380	0.630	0.296	4.5	STD-1	1,000	39	36	39	36	202-3483

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TABLE VI (a)(1). MIL-DTL-24643 cable ratings and characteristics, watertight (with circuit integrity), non-flexing service (power and lighting) – Continued.

MIL-DTL-	Cable type designation	Conductor size: AWG or MCM	Number of conductors, pairs, or triads in cable	Area of each conductor (MCM)	Cable overall diameter max. (inches)	Cable weight per ft approx. (lbs)	Radius of bend min. (inches)	CDR ID	Rated voltage, max. (RMS)	Ampacity, each conductor (amps, max.)				NSN 6145-01
										DC or 60 Hz		400 Hz		
										40 °C Ambient	50 °C Ambient	40 °C Ambient	50 °C Ambient	
24643/17 [4/]	LSFSGA- 23	7	4	20.820	0.890	0.460	5.5	STD-1	1,000	69	64	69	64	202-3484
	LSFSGA- 50	3	4	52.620	1.050	1.015	6.5	LTR-5	1,000	110	101	110	101	203-3299
	LSFSGA- 75	1	4	83.690	1.240	1.486	8.0	LTR-5	1,000	148	136	148	136	202-3485
	LSFSGA- 100	0	4	105.600	1.358	1.820	8.5	LTR-5	1,000	174	160	170	157	202-8465
	LSFSGA- 150	000	4	167.800	1.625	3.105	10.0	LTR-5	1,000	235	216	224	206	202-8466
	LSFSGA- 200	0000	4	211.600	1.820	3.819	11.0	LTR-5	1,000	271	250	254	234	202-3511
24643/17	LSFSGU- 3	16	4	2.580	0.447	0.126	3.0	STD-1 or 3 or LTR-5	1,000	11	10	11	10	202-3509
	LSFSGU- 4	14	4	4.110	0.513	0.172	3.5	STD-1 or 3 or LTR-5	1,000	18	17	18	17	210-2344
	LSFSGU- 9	10	4	10.380	0.630	0.296	4.5	STD-1 or 3 or LTR-5	1,000	39	36	39	36	202-3510
	LSFSGU- 23	7	4	20.820	0.890	0.460	5.5	STD-1 or 3 or LTR-5	1,000	69	64	69	64	202-3479
	LSFSGU- 50	3	4	52.620	1.050	1.015	6.5	STD-1 or 3 or LTR-5	1,000	110	101	110	101	206-3629
	LSFSGU- 75	1	4	83.690	1.240	1.486	8.0	STD-1or 3 or LTR-5	1,000	148	136	148	136	202-6955
	LSFSGU- 100	0	4	105.600	1.358	1.820	8.5	STD-1 or 3 or LTR-5	1,000	174	160	170	157	202-6956
	LSFSGU- 150	000	4	167.800	1.625	3.105	10.0	STD-1 or 3 or LTR-5	1,000	235	216	224	206	202-6957
	LSFSGU- 200	0000	4	211.600	1.820	3.819	11.0	STD-1 or 3 or LTR-5	1,000	271	250	254	234	202-3480
24643/14 [4/]	LSSSGA- 50	3	1	52.620	0.570	0.316	3.5	---	1,000	149	137	149	137	202-9492
	LSSSGA- 75	1	1	83.690	0.652	0.454	4.0	---	1,000	197	181	197	181	202-9493
	LSSSGA- 100	0	1	105.600	0.719	0.550	4.5	---	1,000	232	214	232	214	202-9494
	LSSSGA- 200	0000	1	211.600	0.922	1.022	5.5	---	1,000	361	332	---	---	202-9495
	LSSSGA- 300	300	1	300.000	1.051	1.369	6.5	---	1,000	467	430	---	---	202-9496
	LSSSGA- 400	400	1	413.600	1.168	1.833	7.0	---	1,000	575	530	---	---	202-9497
	LSSSGA- 650	650	1	650.000	1.421	2.801	8.5	---	1,000	785	722	---	---	204-4854

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TABLE VI (a)(1). MIL-DTL-24643 cable ratings and characteristics, watertight (with circuit integrity), non-flexing service (power and lighting) – Continued.

MIL-DTL-	Cable type designation	Conductor size: AWG or MCM	Number of conductors, pairs, or triads in cable	Area of each conductor (MCM)	Cable overall diameter max. (inches)	Cable weight per ft approx. (lbs)	Radius of bend min. (inches)	CDR ID	Rated voltage, max. (RMS)	Ampacity, each conductor (amps, max.)				NSN 6145-01
										DC or 60 Hz		400 Hz		
										40 °C Ambient	50 °C Ambient	40 °C Ambient	50 °C Ambient	
24643/14 [4/]	LSSSGA- 800	800	1	800.000	1.535	3.429	9.5	---	1,000	940	865	---	---	202-9498
	LSSSGA- 1000	1,000	1	1,000.000	1.670	4.223	10.0	---	1,000	1,090	950	---	---	202-9499
	LSSSGA- 1600	1,600	1	1,600.000	2.060	6.489	12.5	---	1,000	1,450	1,270	---	---	202-9500
	LSSSGA- 2000	2,000	1	2,000.000	2.260	7.755	13.5	---	1,000	1,630	1,450	---	---	201-9496
24643/14	LSSSGU- 50	3	1	52.620	0.520	0.281	3.5	---	1,000	149	137	149	137	203-5396
	LSSSGU- 75	1	1	83.690	0.602	0.414	4.0	---	1,000	197	181	197	181	203-0379
	LSSSGU- 100	0	1	105.600	0.669	0.510	4.5	---	1,000	232	214	232	214	205-3647
	LSSSGU- 200	0000	1	211.600	0.872	0.967	5.5	---	1,000	361	332	---	---	202-8471
	LSSSGU- 300	300	1	300.000	1.001	1.309	6.5	---	1,000	467	430	---	---	203-6576
	LSSSGU- 400	400	1	413.600	1.118	1.763	7.0	---	1,000	575	530	---	---	206-2230
	LSSSGU- 650	650	1	650.000	1.371	2.716	8.5	---	1,000	785	722	---	---	203-6577
	LSSSGU- 800	800	1	800.000	1.485	3.338	9.5	---	1,000	940	865	---	---	204-2246
	LSSSGU- 1000	1,000	1	1,000.000	1.620	4.123	10.0	---	1,000	1090	950	---	---	203-6578
	LSSSGU- 1600	1,600	1	1,600.000	2.010	6.368	12.5	---	1,000	1450	1,270	---	---	203-3648
	LSSSGU- 2000	2,000	1	2,000.000	2.210	7.622	13.5	---	1,000	1630	1,450	---	---	204-4869
24643/26	LSTPS- 3	16	3	2.580	0.475	0.129	3.0	STD-4	600	10	---	10		088-9141 [4/]
	LSTPS- 4	14	3	4.110	0.553	0.160	3.5	STD-4	600	17	---	17	---	159-6656
	LSTPS- 6	12	3	6.530	0.620	0.229	4.0	STD-4	600	23	---	23	---	155-6691
	LSTPS- 9	10	3	10.380	0.657	0.299	4.0	STD-4	600	36	---	36	---	542-6700 [4/]
	LSTPS- 14	9	3	13.090	0.751	0.411	4.5	STD-4	600	47	---	47	---	155-6692
	LSTPS- 23	7	3	20.820	0.866	0.579	5.5	STD-4	600	64	---	64	---	154-0989
	LSTPS- 30	5	3	33.090	0.989	0.738	6.0	STD-4	600	77	---	77	---	157-0923
24643/16 [4/]	LSTSGA- 3	16	3	2.580	0.461	0.128	3.0	STD-1	1,000	11	10	11	10	202-0678
	LSTSGA- 4	14	3	4.410	0.499	0.156	3.0	STD-1	1,000	18	17	18	17	202-0679
	LSTSGA- 9	10	3	10.380	0.625	0.278	4.0	STD-1	1,000	39	36	39	36	202-0680
	LSTSGA- 14	9	3	13.090	0.768	0.356	4.5	STD-1	1,000	51	47	51	47	202-0681
	LSTSGA- 23	7	3	20.820	0.862	0.492	5.0	STD-1	1,000	69	64	69	64	202-0682
	LSTSGA- 50	3	3	52.620	1.019	0.947	6.5	STD-1	1,000	110	101	110	101	202-0683

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TABLE VI (a)(1). MIL-DTL-24643 cable ratings and characteristics, watertight (with circuit integrity), non-flexing service (power and lighting) – Continued.

MIL-DTL-	Cable type designation	Conductor size: AWG or MCM	Number of conductors, pairs, or triads in cable	Area of each conductor (MCM)	Cable overall diameter max. (inches)	Cable weight per ft approx. (lbs)	Radius of bend min. (inches)	CDR ID	Rated voltage, max. (RMS)	Ampacity, each conductor (amps, max.)				NSN 6145-01
										DC or 60 Hz		400 Hz		
										40 °C Ambient	50 °C Ambient	40 °C Ambient	50 °C Ambient	
24643/16 [4/]	LSTSGA- 75	1	3	83.690	1.184	1.388	7.5	STD-1	1,000	148	136	148	136	203-4030
	LSTSGA- 100	0	3	105.600	1.316	1.696	8.0	STD-1	1,000	174	160	174	160	202-0684
	LSTSGA- 150	000	3	167.800	1.565	2.558	9.5	STD-1	1,000	235	216	224	206	202-0685
	LSTSGA- 200	0000	3	211.600	1.719	3.197	10.5	STD-1	1,000	271	250	254	234	202-0686
	LSTSGA- 300	300	3	300.000	2.007	4.356	12.0	STD-1	1,000	348	320	292	269	202-0687
	LSTSGA- 400	400	3	413.600	2.253	5.829	13.5	STD-1	1,000	435	400	298	274	202-0688
24643/16	LSTSGU- 3	16	3	2.580	0.411	0.099	3.0	STD-1 or 3 or LTR-5	1,000	11	10	11	10	201-9497
	LSTSGU- 4	14	3	4.110	0.449	0.125	3.0	STD-1 or 3 or LTR-5	1,000	18	17	18	17	201-9498
	LSTSGU- 9	10	3	10.380	0.575	0.241	4.0	STD-1 or 3 or LTR-5	1,000	39	36	39	36	201-9499
	LSTSGU- 14	9	3	13.090	0.718	0.313	4.5	STD-1 or 3 or LTR-5	1,000	51	47	51	47	202-3478
	LSTSGU- 23	7	3	20.820	0.812	0.443	5.0	STD-1 or 3 or LTR-5	1,000	69	64	69	64	201-9500
	LSTSGU- 30	5	3	33.098	0.852	---	5.2	STD-1 or 3 or LTR-5	1,000	84	78	84	78	---
	LSTSGU- 40	4	3	41.735	0.900	---	5.5	STD-1 or 3 or LTR-5	1,000	96	89	96	89	---
	LSTSGU- 50	3	3	52.620	0.969	0.886	6.5	STD-1 or 3 or LTR-5	1,000	110	101	110	101	201-9053
	LSTSGU- 60	2	3	66.381	1.060	---	7.1	STD-1 or 3 or LTR-5	1,000	127	110	127	110	---
	LSTSGU- 75	1	3	83.69	1.134	1.313	7.5	STD-1 or 3 or LTR-5	1,000	148	136	148	136	201-9501
	LSTSGU- 100	0	3	105.600	1.266	1.618	8	STD-1 or 3 or LTR-5	1,000	160	147	160	147	---
	LSTSGU 125	00	3	133.056	1.408	---	8.9	STD-1 or 3 or LTR-5	1,000	174	160	174	160	201-9502
	LSTSGU- 150	000	3	167.800	1.515	2.465	9.5	STD-1 or 3 or LTR-5	1,000	235	216	224	206	202-9505

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TABLE VI (a)(1). MIL-DTL-24643 cable ratings and characteristics, watertight (with circuit integrity), non-flexing service (power and lighting) – Continued.

MIL-DTL-	Cable type designation	Conductor size: AWG or MCM	Number of conductors, pairs, or triads in cable	Area of each conductor (MCM)	Cable overall diameter max. (inches)	Cable weight per ft approx. (lbs)	Radius of bend min. (inches)	CDR ID	Rated voltage, max. (RMS)	Ampacity, each conductor (amps, max.)				NSN 6145-01
										DC or 60 Hz		400 Hz		
										40 °C Ambient	50 °C Ambient	40 °C Ambient	50 °C Ambient	
24643/16	LSTSGU- 200	0000	3	211.600	1.669	3.086	10.5	STD-1 or 3 or LTR-5	1,000	271	250	254	234	202-0676
	LSTTGU- 250	250	3	250.000	1.794	---	11.3	STD-1 or 3 or LTR-5	1,000	304	280	285	262	---
	LSTSGU- 300	300	3	300.000	1.957	4.237	12.0	STD-1 or 3 or LTR-5	1,000	348	320	292	259	202-0677
	LSTTGU- 350	350	3	350	2.073	---	12.7	STD-1 or 3 or LTR-5	1,000	386	355	324	287	---
	LSTSGU- 400	400 (Navy STD)	3	413.600	2.203	5.695	13.5	STD-1 or 3 or LTR-5	1,000	435	400	298	274	202-9506
24643/22 [4/]	LS5KVTSGA- 100	0	3	105.600	1.790	2.417	11.8	LTR-5	5,000	174	160	---	---	201-6514
	LS5KVTSGA- 150	000	3	167.800	2.000	3.073	13.0	LTR-5	5,000	235	216	---	---	201-6515
	LS5KVTSGA- 250	250	3	250.000	2.270	4.429	14.5	LTR-5	5,000	315	290	---	---	201-9054
	LS5KVTSGA- 350	350	3	350.000	2.500	5.562	15.9	LTR-5	5,000	391	360	---	---	201-9055
	LS5KVTSGA- 400	400	3	413.600	2.650	6.551	16.5	LTR-5	5,000	435	400	---	---	202-2050
24643/22	LS5KVTSGU- 100	0	3	105.600	1.740	2.310	11.8	STD-1 or 3 or LTR-5	5,000	174	160	---	---	201-6510
	LS5KVTSGU- 150	000	3	167.800	1.950	2.964	13.0	STD-1 or 3 or LTR-5	5,000	235	216	---	---	203-6566
	LS5KVTSGU- 250	250	3	250.000	2.220	4.294	14.5	STD-1 or 3 or LTR-5	5,000	315	290	---	---	201-6511
	LS5KVTSGU- 350	350	3	350.000	2.450	5.417	15.9	STD-1 or 3 or LTR-5	5,000	391	360	---	---	201-6512
	LS5KVTSGU- 400	400 (Navy STD)	3	413.600	2.600	6.190	16.5	STD-1 or 3 or LTR-5	5,000	435	400	---	---	201-6513
	LS5KVTSGUS- 100	0	3	105.600	1.82	---	12.5	STD-1 or 3 or LTR-5	5,000	174	160	---	---	---
	LS5KVTSGUS- 150	000	3	167.800	2.03	---	13.5	STD-1 or 3 or LTR-5	5,000	235	216	---	---	---
	LS5KVTSGUS- 250	250	3	250.000	2.3	---	15	STD-1 or 3 or LTR-5	5,000	315	290	---	---	---
	LS5KVTSGUS- 350	350	3	350.000	2.53	---	16.4	STD-1 or 3 or LTR-5	5,000	391	360	---	---	---

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TABLE VI (a)(1). MIL-DTL-24643 cable ratings and characteristics, watertight (with circuit integrity), non-flexing service (power and lighting) – Continued.

MIL-DTL-	Cable type designation	Conductor size: AWG or MCM	Number of conductors, pairs, or triads in cable	Area of each conductor (MCM)	Cable overall diameter max. (inches)	Cable weight per ft approx. (lbs)	Radius of bend min. (inches)	CDR ID	Rated voltage, max. (RMS)	Ampacity, each conductor (amps, max.)				NSN 6145-01
										DC or 60 Hz		400 Hz		
										40 °C Ambient	50 °C Ambient	40 °C Ambient	50 °C Ambient	
24643/22	LS5KVTSGUS- 400	400 (Navy STD)	3	413.600	2.68	---	17	STD-1 or 3 or LTR-5	5,000	435	400	---	---	---
24643/63	LSYSGU- 3	16	7	2.582	0.535	0.188	3.2	STD-1, 3, or 5	1,000	18	16	18	16	---
	LSYSGU- 4	14	7	4.106	0.58	0.255	3.5	STD-1, 3, or LTR-5	1,000	29	27	29	27	---
	LSYSGU- 6	12	7	6.526	0.64	---	3.8	STD-1, 3, or LTR-5	1,000	39	36	39	36	---
	LSYSGU- 9	10	7	10.382	0.73	0.314	4.4	STD-1, 3, or LTR-5	1,000	62	58	62	58	---
	LSYSGU- 16	8	7	16.52	1	---	6	STD-1, 3, or LTR-5	1,000	84	77	84	77	---
	LSYSGU- 40	4	7	41.84	1.206	---	7.2	STD-1, 3, or LTR-5	1,000	145	133	145	133	---
	LSYSGU- 60	2	7	66.31	1.426	2	8.6	STD-1, 3, or LTR-5	1,000	193	178	193	178	---
	LSYSGU- 75	1	7	83.68	1.541	2.4	9.2	STD-1, 3, or LTR-5	1,000	237	218	237	218	---
	LSYSGU- 100	0	7	105.58	1.669	3	10	STD-1, 3, or LTR-5	1,000	272	254	272	254	---
	LSYSGU- 125	00	7	133.02	1.828	3.75	11	STD-1, 3, or LTR-5	1,000	320	294	320	294	---
	LSYSGU- 150	000	7	167.75	1.99	4.5	11.9	STD-1, 3, or LTR-5	1,000	376	346	376	346	---
24643/66	LS2OW- 3	16	2	2.582	0.442	---	2.7	STD-1 or 3 or LTR-5	1,000	16	15	16	15	---
	LS2OW- 4	14	2	4.106	0.46	---	2.8	STD-1 or 3 or LTR-5	1,000	26	24	26	24	---
	LS2OW- 9	10	2	10.382	0.551	---	3.3	STD-1 or 3 or LTR-5	1,000	36	33	36	33	694-6291
	LS2OW- 14	9	2	13.08	0.686	---	4.1	STD-1 or 3 or LTR-5	1,000	44	39	44	39	---
	LS2OW- 23	7	2	20.92	0.797	---	4.8	STD-1 or 3 or LTR-5	1,000	59	53	59	53	---

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TABLE VI (a)(1). MIL-DTL-24643 cable ratings and characteristics, watertight (with circuit integrity), non-flexing service (power and lighting) – Continued.

MIL-DTL-	Cable type designation	Conductor size: AWG or MCM	Number of conductors, pairs, or triads in cable	Area of each conductor (MCM)	Cable overall diameter max. (inches)	Cable weight per ft approx. (lbs)	Radius of bend min. (inches)	CDR ID	Rated voltage, max. (RMS)	Ampacity, each conductor (amps, max.)				NSN 6145-01
										DC or 60 Hz		400 Hz		
										40 °C Ambient	50 °C Ambient	40 °C Ambient	50 °C Ambient	
24643/66	LS2OW- 50	3	2	52.69	0.959	---	5.8	STD-1 or 3 or LTR-5	1,000	105	94	105	94	---
	LS2OW- 75	1	2	83.68	1.108	---	6.6	STD-1 or 3 or LTR-5	1,000	132	119	132	119	---
	LS2OW- 100	0	2	105.58	1.232	---	7.4	STD-1 or 3 or LTR-5	1,000	155	139	155	139	---
	LS2OW- 200	0000	2	211.17	1.63	---	9.8	STD-1 or 3 or LTR-5	1,000	139	125	139	125	---
24643/67	LS3OW- 3	16	3	2.582	0.463	---	2.8	STD-1 or 3 or LTR-5	1,000	16	15	16	15	---
	LS3OW- 4	14	3	4.106	0.482	---	2.9	STD-1 or 3 or LTR-5	1,000	23	21	23	21	---
	LS3OW- 9	10	3	10.382	0.58	---	3.5	STD-1 or 3 or LTR-5	1,000	36	33	36	33	694-6308
	LS3OW- 14	9	3	13.08	0.733	---	4.4	STD-1 or 3 or LTR-5	1,000	44	39	44	39	---
	LS3OW- 23	7	3	20.92	0.853	---	5.1	STD-1 or 3 or LTR-5	1,000	59	53	59	53	---
	LS3OW- 50	3	3	52.69	1.027	---	6.2	STD-1 or 3 or LTR-5	1,000	105	94	105	94	---
	LS3OW- 75	1	3	83.68	1.187	---	7.1	STD-1 or 3 or LTR-5	1,000	132	119	132	119	---
	LS3OW- 100	0	3	105.58	1.321	---	7.9	STD-1 or 3 or LTR-5	1,000	155	139	155	139	---
	LS3OW- 200	0000	3	211.17	1.748	---	10.5	STD-1 or 3 or LTR-5	1,000	237	213	237	213	---
	LS3OW- 300	300 MCM	3	300	2.055	---	12.3	STD-1 or 3 or LTR-5	1,000	291	262	291	262	---
	LS3OW- 400	400 (Navy STD)	3	413.6	2.349	1.3	14.1	STD-1 or 3 or LTR-5	1,000	435	400	346	312	---
24643/68	LS4OW- 3	16	4	2.582	0.497	---	3	STD-1 or 3 or LTR-5	1,000	16	15	16	15	---
	LS4OW- 4	14	4	4.106	0.521	---	3.1	STD-1 or 3 or LTR-5	1,000	23	21	23	21	---

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TABLE VI (a)(1). MIL-DTL-24643 cable ratings and characteristics, watertight (with circuit integrity), non-flexing service (power and lighting) – Continued.

MIL-DTL-	Cable type designation	Conductor size: AWG or MCM	Number of conductors, pairs, or triads in cable	Area of each conductor (MCM)	Cable overall diameter max. (inches)	Cable weight per ft approx. (lbs)	Radius of bend min. (inches)	CDR ID	Rated voltage, max. (RMS)	Ampacity, each conductor (amps, max.)				NSN 6145-01
										DC or 60 Hz		400 Hz		
										40 °C Ambient	50 °C Ambient	40 °C Ambient	50 °C Ambient	
24643/68	LS4OW- 9	10	4	10.382	0.631	---	3.8	STD-1 or 3 or LTR-5	1,000	36	33	36	33	---
	LS4OW- 14	9	4	13.08	0.828	---	5	STD-1 or 3 or LTR-5	1,000	44	39	44	39	614-2500
	LS4OW- 23	7	4	20.92	0.965	---	5.8	STD-1 or 3 or LTR-5	1,000	59	53	59	53	---
	LS4OW- 50	3	4	52.69	1.163	---	7	STD-1 or 3 or LTR-5	1,000	105	94	105	94	---
	LS4OW- 75	1	4	83.68	1.345	---	8.1	STD-1 or 3 or LTR-5	1,000	132	119	132	119	---
	LS4OW- 100	0	4	105.58	1.498	---	9	STD-1 or 3 or LTR-5	1,000	155	139	155	139	---
	LS4OW- 150	00	4	133.056	1.725	---	10.4	STD-1 or 3 or LTR-5	1,000	177	160	177	160	---
	LS4OW- 200	0000	4	211.17	1.984	---	11.9	STD-1 or 3 or LTR-5	1,000	237	213	237	213	---
24643/76	LS5KVTEPSG- 250	250 (Navy STD)	3	250	2.9	---	17.4	STD-1, 3, or 5	5,000	373	405	373	405	---
	LS5KVTEPSG- 400	400 (Navy STD)	3	400	3.3	---	19.8	STD-1, 3, or 5	5,000	504	546	504	546	---
24643/79	LS2KVTSGSG-400	400 (Navy STD)	3	405.4	2.34	---	14	18.7	2,000	346	319	346	319	---
24643/85	LSE1CIW- 2	18	1	1.62	0.359	---	2.2	STD-1 or 3, or LTR-5	1,000	13	12	13	12	---
	LSE2CIW- 2	18	2	1.62	0.464	---	2.8	STD-1 or 3, or LTR-5	1,000	13	12	13	12	---
	LSE3CIW- 2	18	3	1.62	0.479	---	2.9	STD-1 or 3, or LTR-5	1,000	13	12	13	12	---
	LSE4CIW- 2	18	4	1.62	0.505	---	3	STD-1 or 3, or LTR-5	1,000	13	12	13	12	---
	LSE7CIW- 2	18	7	1.62	0.564	---	3.4	STD-1 or 3, or LTR-5	1,000	13	12	13	12	---

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TABLE VI (a)(1). MIL-DTL-24643 cable ratings and characteristics, watertight (with circuit integrity), non-flexing service (power and lighting) – Continued.

MIL-DTL-	Cable type designation	Conductor size: AWG or MCM	Number of conductors, pairs, or triads in cable	Area of each conductor (MCM)	Cable overall diameter max. (inches)	Cable weight per ft approx. (lbs)	Radius of bend min. (inches)	CDR ID	Rated voltage, max. (RMS)	Ampacity, each conductor (amps, max.)				NSN 6145-01
										DC or 60 Hz		400 Hz		
										40 °C Ambient	50 °C Ambient	40 °C Ambient	50 °C Ambient	
24643/85	LSE2CIW- 3	16	2	2.58	0.492	---	3	STD-1 or 3, or LTR-5	1,000	16	13	16	13	---
	LSE3CIW- 3	16	3	2.58	0.509	---	3.1	STD-1 or 3, or LTR-5	1,000	16	13	16	13	---
	LSE4CIW- 3	16	4	2.58	0.539	---	3.2	STD-1 or 3, or LTR-5	1,000	16	13	16	13	---
	LSE7CIW- 3	16	7	2.58	0.606	---	3.6	STD-1 or 3, or LTR-5	1,000	16	13	16	13	---
	LSE2CIW- 4	14	2	4.11	0.522	---	3.1	STD-1 or 3, or LTR-5	1,000	23	21	23	21	---
	LSE3CIW- 4	14	3	4.11	0.541	---	3.2	STD-1 or 3, or LTR-5	1,000	23	21	23	21	---
	LSE4CIW- 4	14	4	4.11	0.575	---	3.5	STD-1 or 3, or LTR-5	1,000	23	21	23	21	---
	LSE7CIW- 4	14	7	4.11	0.666	---	4	STD-1 or 3, or LTR-5	1,000	23	21	23	21	---
	LSE2CIW- 9	10	2	10.38	0.608	---	3.6	STD-1 or 3, or LTR-5	1,000	32	29	32	29	---
	LSE3CIW- 9	10	3	10.38	0.639	---	3.8	STD-1 or 3, or LTR-5	1,000	32	29	32	29	---
	LSE4CIW- 9	10	4	10.38	0.7	---	4.2	STD-1 or 3, or LTR-5	1,000	32	29	32	29	---
	LSE2CIW- 14	9	2	13.09	0.66	---	4	STD-1 or 3, or LTR-5	1,000	38	35	38	35	---
	LSE3CIW- 14	9	3	13.09	0.685	---	4.1	STD-1 or 3, or LTR-5	1,000	38	35	38	35	---
	LSE1CIW- 23	7	1	20.82	0.485	---	2.9	STD-1 or 3, or LTR-5	1,000	53	49	53	49	---
	LSE2CIW- 23	7	2	20.82	0.72	---	4.3	STD-1 or 3, or LTR-5	1,000	53	49	53	49	---
	LSE3CIW- 23	7	3	20.82	0.765	---	4.6	STD-1 or 3, or LTR-5	1,000	53	49	53	49	---
	LSE4CIW- 23	7	4	20.82	0.835	---	5	STD-1 or 3, or LTR-5	1,000	53	49	53	49	---

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TABLE VI (a)(1). MIL-DTL-24643 cable ratings and characteristics, watertight (with circuit integrity), non-flexing service (power and lighting) – Continued.

MIL-DTL-	Cable type designation	Conductor size: AWG or MCM	Number of conductors, pairs, or triads in cable	Area of each conductor (MCM)	Cable overall diameter max. (inches)	Cable weight per ft approx. (lbs)	Radius of bend min. (inches)	CDR ID	Rated voltage, max. (RMS)	Ampacity, each conductor (amps, max.)				NSN 6145-01
										DC or 60 Hz		400 Hz		
										40 °C Ambient	50 °C Ambient	40 °C Ambient	50 °C Ambient	
24643/85	LSE1CIW- 50	3	1	52.62	0.62	---	3.7	STD-1 or 3, or LTR-5	1,000	90	84	90	84	---
	LSE2CIW- 50	3	2	52.62	1.015	---	6.1	STD-1 or 3, or LTR-5	1,000	90	84	90	84	---
	LSE2CIW- 50	3	3	52.62	1.075	---	6.5	STD-1 or 3, or LTR-5	1,000	90	84	90	84	---
	LSE4CIW- 50	3	4	52.62	1.18	---	7.1	STD-1 or 3, or LTR-5	1,000	90	84	90	84	---
	LSE2CIW- 60	2	2	66.36	1.09	---	6.5	STD-1 or 3, or LTR-5	1,000	105	88	105	88	---
	LSE3CIW- 60	2	3	66.36	1.15	---	6.9	STD-1 or 3, or LTR-5	1,000	105	88	105	88	---
	LSE1CIW- 75	1	1	83.69	0.705	---	4.2	STD-1 or 3, or LTR-5	1,000	132	122	132	122	---
	LSE2CIW- 75	1	2	83.69	1.16	---	7	STD-1 or 3, or LTR-5	1,000	132	122	132	122	---
	LSE3CIW- 75	1	3	83.69	1.235	---	7.4	STD-1 or 3, or LTR-5	1,000	132	122	132	122	---
	LSE4CIW- 75	1	4	83.69	1.36	---	8.2	STD-1 or 3, or LTR-5	1,000	132	122	132	122	---
	LSE1CIW- 100	0	1	105.6	0.745	---	4.5	STD-1 or 3, or LTR-5	1,000	155	143	155	143	---
	LSE2CIW- 100	0	2	105.6	1.25	---	7.5	STD-1 or 3, or LTR-5	1,000	155	143	155	143	---
	LSE3CIW- 100	0	3	105.6	1.32	---	7.9	STD-1 or 3, or LTR-5	1,000	155	143	155	143	---
	LSE4CIW- 100	0	4	105.6	1.46	---	8.8	STD-1 or 3, or LTR-5	1,000	155	143	155	143	---
	LSE6CIW- 100	0	6	105.6	1.735	---	10.4	STD-1 or 3, or LTR-5	1,000	155	143	155	143	---
	LSE1CIW- 125	00	1	133.1	0.825	---	5	STD-1 or 3, or LTR-5	1,000	177	164	177	164	---

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MIL-DTL-	Cable type designation	Conductor size: AWG or MCM	Number of conductors, pairs, or triads in cable	Area of each conductor (MCM)	Cable overall diameter max. (inches)	Cable weight per ft approx. (lbs)	Radius of bend min. (inches)	CDR ID	Rated voltage, max. (RMS)	Ampacity, each conductor (amps, max.)				NSN 6145-01
										DC or 60 Hz		400 Hz		
										40 °C Ambient	50 °C Ambient	40 °C Ambient	50 °C Ambient	
24643/85	LSE2CIW- 125	000	2	133.1	1.38	---	8.3	STD-1 or 3, or LTR-5	1,000	177	164	177	164	---
	LSE3CIW- 125	00	3	133.1	1.46	---	8.8	STD-1 or 3, or LTR-5	1,000	177	164	177	164	---
	LSE1CIW- 150	000	1	167.8	0.87	---	5.2	STD-1 or 3, or LTR-5	1,000	205	189	205	189	---
	LSE2CIW- 150	000	2	167.8	1.47	---	8.8	STD-1 or 3, or LTR-5	1,000	205	189	205	189	---
	LSE3CIW- 150	000	3	167.8	1.555	---	9.3	STD-1 or 3, or LTR-5	1,000	205	189	205	189	---
	LSE4CIW- 150	000	4	167.8	1.75	---	10.5	STD-1 or 3, or LTR-5	1,000	205	189	205	189	---
	LSE6CIW- 150	000	6	167.8	2.065	---	12.4	STD-1 or 3, or LTR-5	1,000	205	189	205	189	---
	LSE1CIW- 200	0000	1	211.6	0.965	---	5.8	STD-1 or 3, or LTR-5	1,000	237	218	237	218	---
	LSE2CIW- 200	0000	2	211.6	1.64	---	9.8	STD-1 or 3, or LTR-5	1,000	237	218	237	218	---
	LSE3CIW- 200	0000	3	211.6	1.76	---	10.6	STD-1 or 3, or LTR-5	1,000	237	218	237	218	---
	LSE4CIW- 200	0000	4	211.6	1.955	---	11.7	STD-1 or 3, or LTR-5	1,000	237	218	237	218	---
	LSE6CIW- 200	0000	6	211.6	2.32	---	13.9	STD-1 or 3, or LTR-5	1,000	237	218	237	218	---
	LSE1CIW- 300	300	1	300	1.065	---	6.4	STD-1 or 3, or LTR-5	1,000	291	269	291	269	---
	LSE2CIW- 300	300	2	300	1.85	---	11.1	STD-1 or 3, or LTR-5	1,000	291	269	291	269	---
	LSE3CIW- 300	300	3	300	1.965	---	11.8	STD-1 or 3, or LTR-5	1,000	291	269	291	269	---
	LSE1CIW- 400	400 (Navy STD)	1	405.4	1.18	13	7.1	STD-1 or 3, or LTR-5	1,000	346	319	346	319	---

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MIL-DTL-	Cable type designation	Conductor size: AWG or MCM	Number of conductors, pairs, or triads in cable	Area of each conductor (MCM)	Cable overall diameter max. (inches)	Cable weight per ft approx. (lbs)	Radius of bend min. (inches)	CDR ID	Rated voltage, max. (RMS)	Ampacity, each conductor (amps, max.)				NSN 6145-01
										DC or 60 Hz		400 Hz		
										40 °C Ambient	50 °C Ambient	40 °C Ambient	50 °C Ambient	
24643/85	LSE2CIW- 400	400 (Navy STD)	2	405.4	2.1	13	12.6	STD-1 or 3, or LTR-5	1,000	346	319	346	319	---
	LSE3CIW- 400	400 (Navy STD)	3	405.4	2.23	13	13.4	STD-1 or 3, or LTR-5	1,000	346	319	346	319	---
24643/86	LS5KVTSSGUSHR- 100	100	3	100	1.87	---	14.5	See spec sheet	5,000	428	395	428	395	---
	LS5KVTSSGUSHR- 250	250	3	250	2.3	---	17.5	See spec sheet	5,000	450	416	450	416	---
	LS5KVTSSGUSHR- 400	400 (Navy STD)	3	405.4	2.75	---	20.5	See spec sheet	5,000	683	630	683	630	---
See Note 7/	LS15KVSSU	777 MCM	1	777	2.18	4.1	17.4	---	15,000	542	542	---	---	---
	LS15KVTSU	0	3	105.6	3	5.25	24	---	15,000	210	210	---	---	---

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TABLE VI (a)(2). MIL-DTL-24643 cable ratings and characteristics, watertight (with circuit integrity), non-flexing service (control).

MIL-DTL-	Cable type designation	Conductor size: AWG or MCM	Number of conductors, pairs, or triads in cable	Area of each conductor (MCM)	Cable overall diameter max. (inches)	Cable weight per ft approx. (lbs)	Radius of bend min. (inches)	CDR ID	Rated voltage, max. (RMS)	Ampacity, each conductor (amps, max)				NSN 6145-01
										DC or 60 Hz		400 Hz		
										40 °C Ambient	50 °C Ambient	40 °C Ambient	50 °C Ambient	
24643/18 [4/]	LSMSCA- 7	18	7	1.620	0.534	0.176	3.5	STD-1	1,000	12/8 [1/]	9/6 [1/]	12/8 [1/]	9/6 [1/]	202-2046
	LSMSCA- 10	18	10	1.620	0.672	0.267	4.0	STD-1	1,000	12/8	9/6	12/8	9/6	202-2047
	LSMSCA- 14	18	14	1.620	0.718	0.324	4.5	STD-1	1,000	12/8	9/6	12/8	9/6	202-2048
	LSMSCA- 19	18	19	1.620	0.788	0.401	5.0	STD-1	1,000	12/8	9/6	12/8	9/6	202-2049
	LSMSCA- 24	18	24	1.620	0.905	0.522	5.5	STD-1	1,000	12/6	9/5	12/6	9/5	203-5392
	LSMSCA- 30	18	30	1.620	0.951	0.596	5.5	STD-1	1,000	12/6	9/5	12/6	9/5	205-3630
	LSMSCA- 37	18	37	1.620	1.055	0.707	6.0	STD-1	1,000	12/6	9/5	12/6	9/5	205-3631
	LSMSCA- 44	18	44	1.620	1.164	0.861	7.0	STD-1	1,000	12/5	9/4	12/5	9/4	201-9506
	LSMSCA- 61	18	61	1.620	1.300	1.058	8.0	STD-1	1,000	12/4	9/3	12/4	9/3	201-9507
	LSMSCA- 91	18	91	1.620	1.530	1.600	9.0	STD-1	1,000	12/4	9/3	12/4	9/3	201-9508
24643/18	LSMSCS- 7	18	7	1.620	0.544	0.244	6.5	STD-1	1,000	12/8	9/6	12/8	9/6	201-9509
	LSMSCS- 10	18	10	1.620	0.682	0.349	8.5	STD-1	1,000	12/8	9/6	12/8	9/6	202-3486
	LSMSCS- 14	18	14	1.620	0.728	0.403	9.0	STD-1	1,000	12/8	9/6	12/8	9/6	202-6960
	LSMSCS- 19	18	19	1.620	0.798	0.509	9.5	STD-1	1,000	12/8	9/6	12/8	9/6	202-6961
	LSMSCS- 24	18	24	1.620	0.915	0.630	11.5	STD-1	1,000	12/6	9/5	12/6	9/5	202-6962
	LSMSCS- 30	18	30	1.620	0.961	0.727	12.0	STD-1	1,000	12/6	9/5	12/6	9/5	202-6963
	LSMSCS- 37	18	37	1.620	1.065	0.810	13.0	STD-1	1,000	12/6	9/5	12/6	9/5	202-6964
	LSMSCS- 44	18	44	1.620	1.174	0.991	14.0	STD-1	1,000	12/5	9/4	12/5	9/4	201-9510
	LSMSCS- 61	18	61	1.620	1.310	1.227	16.0	STD-1	1,000	12/4	9/3	12/4	9/3	202-6965
	LSMSCS- 91	18	91	1.620	1.540	1.887	18.5	STD-1	1,000	12/4	9/3	12/4	9/3	202-6966
24643/18	LSMSCU- 7	18	7	1.620	0.484	0.148	3.5	STD-1	1,000	12/8	9/6	12/8	9/6	202-2041
	LSMSCU- 10	18	10	1.620	0.622	0.227	4.0	STD-1	1,000	12/8	9/6	12/8	9/6	201-9503
	LSMSCU- 14	18	14	1.620	0.668	0.280	4.5	STD-1	1,000	12/8	9/6	12/8	9/6	201-9504
	LSMSCU- 19	18	19	1.620	0.738	0.354	5.0	STD-1	1,000	12/8	9/6	12/8	9/6	201-9505
	LSMSCU- 24	18	24	1.620	0.855	0.467	5.5	STD-1	1,000	12/6	9/5	12/6	9/5	202-6958
	LSMSCU- 30	18	30	1.620	0.901	0.539	5.5	STD-1	1,000	12/6	9/5	12/6	9/5	202-6959
	LSMSCU- 37	18	37	1.620	1.002	0.648	6.0	STD-1	1,000	12/6	9/5	12/6	9/5	202-2042
	LSMSCU- 44	18	44	1.620	1.114	0.793	7.0	STD-1	1,000	12/5	9/4	12/5	9/4	202-2043

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TABLE VI (a)(2). MIL-DTL-24643 cable ratings and characteristics, watertight (with circuit integrity), non-flexing service (control) – Continued.

MIL-DTL-	Cable type designation	Conductor size: AWG or MCM	Number of conductors, pairs, or triads in cable	Area of each conductor (MCM)	Cable overall diameter max. (inches)	Cable weight per ft approx. (lbs)	Radius of bend min. (inches)	CDR ID	Rated voltage, max. (RMS)	Ampacity, each conductor (amps, max)				NSN 6145-01
										DC or 60 Hz		400 Hz		
										40 °C Ambient	50 °C Ambient	40 °C Ambient	50 °C Ambient	
24643/18	LSMSCU- 61	18	61	1.620	1.250	0.982	8.0	STD-1	1,000	12/4	9/3	12/4	9/3	202-2044
	LSMSCU- 91	18	91	1.620	1.480	1.510	9.0	STD-1	1,000	12/4	9/3	12/4	9/3	202-2045
24643/85	LSE7CIW- 2	18	7	1.62	0.564	---	3.4	STD-1 or 3, or LTR-5	1,000	13	12	13	12	---
	LSE7CIW- 3	16	7	2.58	0.606	---	3.6	STD-1 or 3, or LTR-5	1,000	16	13	16	13	---
	LSE7CIW- 4	14	7	4.11	0.666	---	4	STD-1 or 3, or LTR-5	1,000	23	21	23	21	---
	LSE10CIW-2	18	10	1.62	0.679	---	4.1	STD-1 or 3 or LTR-5	1,000	13	12	13	12	---
	LSE14CIW-2	18	14	1.62	0.72	---	4.3	STD-1 or 3 or LTR-5	1,000	13	12	13	12	---
	LSE19CIW-2	18	19	1.62	0.794	---	4.8	STD-1 or 3 or LTR-5	1,000	13	12	13	12	---
	LSE30CIW-2	18	30	1.62	0.945	---	5.7	STD-1 or 3 or LTR-5	1,000	13	12	13	12	---
	LSE37CIW-2	18	37	1.62	1.004	---	6	STD-1 or 3 or LTR-5	1,000	13	12	13	12	---
	LSE44CIW-2	18	44	1.62	1.109	---	6.7	STD-1 or 3 or LTR-5	1,000	13	12	13	12	---
	LSE61CIW-2	18	61	1.62	1.219	---	7.3	STD-1 or 3 or LTR-5	1,000	13	12	13	12	---
	LSE91CIW-2	18	91	1.62	1.419	---	8.5	STD-1 or 3 or LTR-5	1,000	13	12	13	12	---

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TABLE VI (a)(3). MIL-DTL-24643 cable ratings and characteristics, watertight (with circuit integrity), non-flexing service (electronic, communication, and instrumentation).

MIL-DTL-	Cable type designation	Conductor size: AWG or MCM	Number of conductors, pairs, or triads in cable	Area of each conductor (MCM)	Cable overall diameter max. (inches)	Cable weight per ft approx. (lbs)	Radius of bend min. (inches)	CDR ID	Rated voltage, max. (RMS)	Ampacity, each conductor (amps, max.)				NSN 6145-01
										DC or 60 Hz		400 Hz		
										40 °C Ambient	50 °C Ambient	40 °C Ambient	50 °C Ambient	
24643/21 [4/]	LSTCJA- 4	14	2	4.110	0.480	0.136	3.0	---	---	---	---	---	---	203-6565
24643/21	LSTCJU- 4	14	2	4.110	0.430	0.117	3.0	---	300	---	---	---	---	201-9520
24643/24	LSTCJX- 3	16	3 pairs	2.582	0.742	0.231	4.0	LTR-4	300	---	---	---	---	977-6160 [3/]
	LSTCJX- 7	16	7 pairs	2.582	0.983	0.515	5.5	LTR-4	300	---	---	---	---	157-2037
	LSTCJX- 12	16	12 pairs	2.582	1.269	0.844	6.5	LTR-4	300	---	---	---	---	931-8420 [3/]
24643/24	LSTCKX- 1	16	1 pair	2.582	0.456	0.108	2.5	LTR-4	300	---	---	---	---	598-9397 [3/]
	LSTCKX- 3	16	3 pairs	2.582	0.742	0.257	4.0	LTR-4	300	---	---	---	---	948-4135 [3/]
	LSTCKX- 7	16	7 pairs	2.582	0.983	0.520	5.5	LTR-4	300	---	---	---	---	901-6242 [3/]
	LSTCKX- 12	16	12 pairs	2.582	1.269	0.844	6.5	LTR-4	300	---	---	---	---	542-6703 [3/]
24643/21 [4/]	LSTCTA- 4	14	2	4.110	0.480	0.136	3.0	---	---	---	---	---	---	201-8818
24643/21	LSTCTU- 4	14	2	4.110	0.430	0.117	3.0	---	300	---	---	---	---	201-9521
24643/24	LSTCTX- 1	21	1 pair	0.810	0.350	0.108	2.0	LTR-4	300	---	---	---	---	156-8817
	LSTCTX- 3	21	3 pairs	0.810	0.552	0.262	3.5	LTR-4	300	---	---	---	---	542-6704 [3/]
	LSTCTX- 7	21	7 pairs	0.810	0.731	0.515	4.0	LTR-4	300	---	---	---	---	577-3358 [3/]
	LSTCTX- 12	21	12 pairs	0.810	0.964	0.844	5.0	LTR-4	300	---	---	---	---	542-6705 [3/]
24643/23 [4/]	LSTTSA- 1.5	22	3	0.640	0.380	0.083	2.5	TEL-6	300	---	---	---	---	201-9540
	LSTTSA- 3	22	6	0.640	0.500	0.137	3.0	TEL-6	300	---	---	---	---	202-0692
	LSTTSA- 5	22	10	0.640	0.590	0.195	3.5	TEL-6	300	---	---	---	---	202-0693
	LSTTSA- 10	22	20	0.640	0.725	0.296	4.0	TEL-6	300	---	---	---	---	201-9541
	LSTTSA- 15	22	30	0.640	0.850	0.415	5.0	TEL-6	300	---	---	---	---	201-9542
	LSTTSA- 20	22	40	0.640	0.920	0.504	5.5	TEL-6	300	---	---	---	---	201-9543
	LSTTSA- 30	22	60	0.640	1.130	0.742	6.0	TEL-6	300	---	---	---	---	202-9537
	LSTTSA- 40	22	80	0.640	1.250	0.930	7.0	TEL-6	300	---	---	---	---	201-9544
	LSTTSA- 50	22	100	0.640	1.450	1.193	7.5	TEL-6	300	---	---	---	---	202-9538
	LSTTSA- 60	22	120	0.640	1.500	1.317	8.0	TEL-6	300	---	---	---	---	201-9545

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TABLE VI (a)(3). MIL-DTL-24643 cable ratings and characteristics, watertight (with circuit integrity), non-flexing service (electronic, communication, and instrumentation) – Continued.

MIL-DTL-	Cable type designation	Conductor size: AWG or MCM	Number of conductors, pairs, or triads in cable	Area of each conductor (MCM)	Cable overall diameter max. (inches)	Cable weight per ft approx. (lbs)	Radius of bend min. (inches)	CDR ID	Rated voltage, max. (RMS)	Ampacity, each conductor (amps, max.)				NSN 6145-01
										DC or 60 Hz		400 Hz		
										40 °C Ambient	50 °C Ambient	40 °C Ambient	50 °C Ambient	
24643/23	LSTTSU- 1.5	22	3	0.640	0.330	0.066	2.5	TEL-6	300	---	---	---	---	202-9528
	LSTTSU- 3	22	3 pairs	0.640	0.450	0.113	3.0	TEL-6	300	---	---	---	---	202-9529
	LSTTSU- 5	22	5 pairs	0.640	0.540	0.167	3.5	TEL-6	300	---	---	---	---	202-9530
	LSTTSU- 10	22	10 pairs	0.640	0.675	0.262	4.0	TEL-6	300	---	---	---	---	202-9531
	LSTTSU- 15	22	15 pairs	0.640	0.800	0.379	5.0	TEL-6	300	---	---	---	---	202-9532
	LSTTSU- 20	22	20 pairs	0.640	0.870	0.461	5.5	TEL-6	300	---	---	---	---	202-9533
	LSTTSU- 30	22	30 pairs	0.640	1.080	0.697	6.0	TEL-6	300	---	---	---	---	202-9534
	LSTTSU- 40	22	40 pairs	0.640	1.200	0.874	7.0	TEL-6	300	---	---	---	---	202-9535
	LSTTSU- 50	22	50 pairs	0.640	1.400	1.148	7.5	TEL-6	300	---	---	---	---	202-9536
	LSTTSU- 60	22	60 pairs	0.640	1.450	1.267	8.0	TEL-6	300	---	---	---	---	204-4264

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TABLE VI (b)(1). MIL-DTL-24643 cable ratings and characteristics, watertight, non-flexing service (power and lighting).

MIL-DTL-	Cable type designation	Conductor size: AWG or MCM	Number of conductors, pairs, or triads in cable	Area of each conductor (MCM)	Cable overall diameter max. (inches)	Cable weight per ft approx. (lbs)	Radius of bend min. (inches)	CDR ID	Rated voltage, max. (RMS)	Ampacity, each conductor (amps, max.)				NSN 6145-01
										DC or 60 Hz		400 Hz		
										40 °C Ambient	50 °C Ambient	40 °C Ambient	50 °C Ambient	
24643/53	LSDRW	14	2	4.110	0.670	0.298	3.0	STD-1 or 3	3,000	26	24	---	---	201-9057
24643/53 [4/]	LSDRWA	14	2	4.110	0.720	0.336	6.0	STD-1	3,000	26	24	---	---	201-9058
24643/5	LSMDU- 6	12	19	6.530	1.000	1.143	8.0	STD-1 or 3	600	---	---	---	---	202-2780
	LSMDU- 14	9	19	13.090	1.395	1.783	9.5	STD-1 or 3	600	---	---	---	---	202-2781
	LSMDU- 23	7	19	20.820	1.765	2.566	10.5	STD-1 or 3	600	---	---	---	---	202-2782
	LSMDU- 40	4	19	41.740	2.040	4.191	12.5	STD-1 or 3	600	---	---	---	---	203-0366
	LSMDU- 60	2	19	66.360	2.330	5.843	14.0	STD-1 or 3	600	---	---	---	---	202-2783
24643/6 [5/]	LSMDY- 6	12	19	6.530	1.190	1.360	7.0	STD-1	600	---	---	---	---	202-2784
	LSMDY- 14	9	19	13.090	1.570	2.041	9.5	STD-1	600	---	---	---	---	203-5387
	LSMDY- 23	7	19	20.820	1.960	2.978	12.0	STD-1	600	---	---	---	---	202-2785
	LSMDY- 40	4	19	41.740	2.240	4.480	13.5	STD-1	600	---	---	---	---	202-2786
	LSMDY- 60	2	19	66.360	2.525	6.471	15.0	STD-1	600	---	---	---	---	202-2787
24643/53	LSTRW	14	3	4.110	0.710	0.323	4.5	STD-1 or 3	3,000	24	22	---	---	201-9523
24643/53 [4/]	LSTRWA	14	3	4.110	0.760	0.353	4.5	STD-1	3,000	24	22	---	---	201-9524
24643/53	LSSRW	14	1	4.110	0.400	0.106	3.0	STD-1 or 4	3,000	32	30	---	---	202-2054
24643/53 [4/]	LSSRWA	14	1	4.110	0.450	0.134	3.0	STD-1	3,000	32	30	---	---	201-9056
24643/80	LSTW- 2	18	1	1.624	0.11	---	0.7	See spec sheet	1,000	13	12	13	12	---
	LSTW- 3	16	1	2.582	0.12	---	0.7	See spec sheet	1,000	16	15	16	15	---
	LSTW- 4	14	1	4.106	0.135	---	0.8	See spec sheet	1,000	23	21	23	21	---
	LSTW- 6	12	1	6.526	0.155	---	0.9	See spec sheet	1,000	27	25	27	25	---
	LSTW- 9	10	1	10.382	0.19	---	1.1	See spec sheet	1,000	36	34	36	34	---
	LSTW- 14	9	1	13.08	0.23	---	1.4	See spec sheet	1,000	44	40	44	40	---
	LSTW- 23	7	1	20.92	0.265	---	1.6	See spec sheet	1,000	59	55	59	55	---
	LSTW- 40	4	1	41.84	0.335	---	2	See spec sheet	1,000	86	80	86	80	---
	LSTW- 50	3	1	52.69	0.365	---	2.2	See spec sheet	1,000	105	97	105	97	---
	LSTW- 60	2	1	66.31	0.405	---	2.4	See spec sheet	1,000	118	109	118	109	---
	LSTW- 75	1	1	83.68	0.46	---	2.8	See spec sheet	1,000	132	122	132	122	---
	LSTW- 100	0	1	105.58	0.495	---	3	See spec sheet	1,000	155	143	155	143	---
	LSTW- 150	000	1	167.75	0.55	---	3.3	See spec sheet	1,000	205	189	205	189	---

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TABLE VI (b)(1). MIL-DTL-24643 cable ratings and characteristics, watertight, non-flexing service (power and lighting) – Continued.

MIL-DTL-	Cable type designation		Conductor size: AWG or MCM	Number of conductors, pairs, or triads in cable	Area of each conductor (MCM)	Cable overall diameter max. (inches)	Cable weight per ft approx. (lbs)	Radius of bend min. (inches)	CDR ID	Rated voltage, max. (RMS)	Ampacity, each conductor (amps, max.)				NSN 6145-01
											DC or 60 Hz		400 Hz		
											40 °C Ambient	50 °C Ambient	40 °C Ambient	50 °C Ambient	
24643/80	LSTW-	300	300	1	300	0.775	---	4.7	See spec sheet	1,000	291	269	291	269	---
	LSTW-	400	400 (Navy STD)	1	405.4	0.91	13	5.5	See spec sheet	1,000	346	319	346	319	---
24643/81	LSE19EW-	2	18	19	1.624	0.82	---	4.9	STD-1 or 3	1,000	13	12	13	12	---
	LSE2EW-	3	16	2	2.582	0.47	---	2.8	STD-1 or 3	1,000	16	15	16	15	---
	LSE3EW-	3	16	3	2.582	0.49	---	2.9	STD-1 or 3	1,000	16	15	16	15	---
	LSE4EW-	3	16	4	2.582	0.52	---	3.1	STD-1 or 3	1,000	16	15	16	15	---
	LSE19EW-	3	16	19	2.582	0.88	---	5.3	STD-1 or 3	1,000	16	15	16	15	---
	LSE2EW-	4	14	2	4.106	0.5	---	3	STD-1 or 3	1,000	23	21	23	21	---
	LSE3EW-	4	14	3	4.106	0.52	---	3.1	STD-1 or 3	1,000	23	21	23	21	---
	LSE4EW-	4	14	4	4.106	0.555	---	3.3	STD-1 or 3	1,000	23	21	23	21	---
	LSE19EW-	6	12	19	6.526	1.095	---	6.6	STD-1 or 3	1,000	27	25	27	25	---
	LSE2EW-	9	10	2	10.382	0.505	---	3	STD-1 or 3	1,000	36	34	36	34	---
	LSE3EW-	9	10	3	10.382	0.635	---	3.8	STD-1 or 3	1,000	36	34	36	34	---
	LSE4EW-	9	10	4	10.382	0.695	---	4.2	STD-1 or 3	1,000	36	34	36	34	---
	LSE2EW-	14	9	2	13.08	0.695	---	4.2	STD-1 or 3	1,000	44	40	44	40	---
	LSE3EW-	14	9	3	13.08	0.735	---	4.4	STD-1 or 3	1,000	44	40	44	40	---
	LSE4EW-	14	9	4	13.08	0.81	---	4.9	STD-1 or 3	1,000	44	40	44	40	---
	LSE19EW-	14	9	19	13.08	1.51	---	9.1	STD-1 or 3	1,000	44	40	44	40	---
	LSE2EW-	23	7	2	20.92	0.715	---	4.3	STD-1 or 3	1,000	59	55	59	55	---
	LSE3EW-	23	7	3	20.92	0.825	---	5	STD-1 or 3	1,000	59	55	59	55	---
	LSE4EW-	23	7	4	20.92	0.91	---	5.5	STD-1 or 3	1,000	59	55	59	55	---
	LSE19EW-	23	7	19	20.92	1.725	---	10.4	STD-1 or 3	1,000	59	55	59	55	---
	LSE19EW-	40	4	19	41.84	2.16	---	13	STD-1 or 3	1,000	86	80	86	80	---
	LSE2EW-	50	3	2	52.69	1.03	---	6.2	STD-1 or 3	1,000	105	97	105	97	---
	LSE3EW-	50	3	3	52.69	1.19	---	7.1	STD-1 or 3	1,000	105	97	105	97	---
	LSE19EW-	60	2	19	66.31	2.56	---	15.4	STD-1 or 3	1,000	118	109	118	109	---
	LSE2EW-	75	1	2	83.68	1.255	---	7.5	STD-1 or 3	1,000	132	122	132	122	---

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TABLE VI (b)(1). MIL-DTL-24643 cable ratings and characteristics, watertight, non-flexing service (power and lighting) – Continued.

MIL-DTL-	Cable type designation		Conductor size: AWG or MCM	Number of conductors, pairs, or triads in cable	Area of each conductor (MCM)	Cable overall diameter max. (inches)	Cable weight per ft approx. (lbs)	Radius of bend min. (inches)	CDR ID	Rated voltage, max. (RMS)	Ampacity, each conductor (amps, max.)				NSN 6145-01
											DC or 60 Hz		400 Hz		
											40 °C Ambient	50 °C Ambient	40 °C Ambient	50 °C Ambient	
24643/81	LSE3EW-	75	1	3	83.68	1.335	---	8	STD-1 or 3	1,000	132	122	132	122	---
	LSE2EW-	100	0	2	105.58	1.34	---	8	STD-1 or 3	1,000	155	143	155	143	---
	LSE3EW-	100	0	3	105.58	1.425	---	8.6	STD-1 or 3	1,000	155	143	155	143	---
	LSE3EW-	150	000	2	167.75	1.525	---	9.2	STD-1 or 3	1,000	205	189	205	189	---
	LSE1EW-	300	300	1	300	1.09	---	6.5	White or natural	1,000	291	269	291	269	---

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TABLE VI (b)(2). MIL-DTL-24643 cable ratings and characteristics, watertight, non-flexing service (electronic, communication, and instrumentation).

MIL-DTL-	Cable type designation	Conductor size: AWG or MCM	Number of conductors, pairs, or triads in cable	Area of each conductor (MCM)	Cable overall diameter max. (inches)	Cable weight per ft approx. (lbs)	Radius of bend min. (inches)	CDR ID	Rated voltage, max. (RMS)	Ampacity, each conductor (amps, max.)				NSN 6145-01
										DC or 60 Hz		400 Hz		
										40 °C Ambient	50 °C Ambient	40 °C Ambient	50 °C Ambient	
24643/47 [4/]	LS1SMWA- 70	22	70	0.700	1.605	2.052	10.0	STD-1	600	---	---	---	---	202-3535
24643/47	LS1SMWU- 70	22	70	0.700	1.555	1.565	10.0	STD-2	600	---	---	---	---	202-3534
24643/30 [4/]	LS1SWA- 2	22	2	0.700	0.505	0.120	3.5	STD-2	600	---	---	---	---	205-3661
	LS1SWA- 14	22	14	0.700	0.920	0.506	5.5	STD-2	600	---	---	---	---	205-3662
	LS1SWA- 20	22	20	0.700	1.080	0.658	6.5	STD-2	600	---	---	---	---	205-3663
	LS1SWA- 30	22	30	0.700	1.250	0.942	7.5	STD-2	600	---	---	---	---	203-0381
24643/30	LS1SWU- 2	22	2	0.700	0.455	0.103	3.5	STD-2	600	---	---	---	---	203-5406
	LS1SWU- 14	22	14	0.700	0.870	0.463	5.5	STD-2	600	---	---	---	---	203-5407
	LS1SWU- 20	22	20	0.700	1.030	0.618	6.5	STD-2	600	---	---	---	---	203-5408
	LS1SWU- 30	22	30	0.700	1.200	0.885	7.5	STD-2	600	---	---	---	---	205-3660
24643/32 [4/]	LS2SWA- 3	22	6	0.700	0.520	0.206	4.5	STD-2	600	---	---	---	---	202-9554
	LS2SWA- 7	22	14	0.700	0.710	0.298	4.5	STD-2	600	---	---	---	---	202-9555
	LS2SWA- 10	22	20	0.700	0.880	0.422	5.0	STD-2	600	---	---	---	---	202-7767
	LS2SWA- 14	22	28	0.700	0.980	0.618	6.0	STD-2	600	---	---	---	---	205-5182
	LS2SWA- 19	22	38	0.700	1.090	0.793	6.5	STD-2	600	---	---	---	---	202-7768
	LS2SWA- 24	22	48	0.700	1.260	1.030	7.5	STD-2	600	---	---	---	---	204-4873
	LS2SWA- 30	22	60	0.700	1.330	1.256	8.0	STD-2	600	---	---	---	---	203-7624
	LS2SWA- 37	22	74	0.700	1.430	1.545	8.5	STD-2	600	---	---	---	---	202-3529
	LS2SWA- 44	22	88	0.700	1.600	1.905	9.0	STD-2	600	---	---	---	---	202-3530
	LS2SWA- 61	22	122	0.700	1.790	2.317	10.0	STD-2	600	---	---	---	---	203-4068
24643/32	LS2SWAU- 3	22	3 pairs	0.700	0.520	0.175	3.5	STD-2	600	---	---	---	---	204-5540
	LS2SWAU- 7	22	7 pairs	0.700	0.660	0.257	4.5	STD-2	600	---	---	---	---	202-9552
	LS2SWAU- 10	22	10 pairs	0.700	0.830	0.370	5.0	STD-2	600	---	---	---	---	204-4872
	LS2SWAU- 14	22	14 pairs	0.700	0.930	0.556	6.0	STD-2	600	---	---	---	---	202-3537
	LS2SWAU- 19	22	19 pairs	0.700	1.040	0.731	6.5	STD-2	600	---	---	---	---	202-9553
	LS2SWAU- 24	22	24 pairs	0.700	1.210	0.957	7.5	STD-2	600	---	---	---	---	202-3538
	LS2SWAU- 30	22	30 pairs	0.700	1.280	1.174	8.0	STD-2	600	---	---	---	---	202-3539

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TABLE VI (b)(2). MIL-DTL-24643 cable ratings and characteristics, watertight, non-flexing service (electronic, communication, and instrumentation) – Continued.

MIL-DTL-	Cable type designation	Conductor size: AWG or MCM	Number of conductors, pairs, or triads in cable	Area of each conductor (MCM)	Cable overall diameter max. (inches)	Cable weight per ft approx. (lbs)	Radius of bend min. (inches)	CDR ID	Rated voltage, max. (RMS)	Ampacity, each conductor (amps, max.)				NSN 6145-01
										DC or 60 Hz		400 Hz		
										40 °C Ambient	50 °C Ambient	40 °C Ambient	50 °C Ambient	
24643/32	LS2SWAU- 37	22	37 pairs	0.700	1.380	1.462	8.5	STD-2	600	---	---	---	---	203-0391
	LS2SWAU- 44	22	44 pairs	0.700	1.550	1.812	9.0	STD-2	600	---	---	---	---	202-7766
	LS2SWAU- 61	22	61 pairs	0.700	1.740	2.214	10.0	STD-2	600	---	---	---	---	205-5181
24643/56	LS2SWL- 7	16	7 pairs	2.580	0.910	0.427	7.3	STD-2	600	---	---	---	---	209-7450
24643/56 [4/]	LS2SWLA- 7	16	14	2.580	0.960	0.467	7.3	STD-2	600	---	---	---	---	205-3645
24643/33	LS2SWU- 1	18	1 pair	1.620	0.255	0.103	2.0	STD-2	600	---	---	---	---	205-3632
	LS2SWU- 3	18	3 pairs	1.620	0.710	0.207	4.0	STD-2	600	---	---	---	---	203-0371
	LS2SWU- 7	18	7 pairs	1.620	0.910	0.358	5.0	STD-2	600	---	---	---	---	204-0543
	LS2SWU- 12	18	12 pairs	1.620	1.200	0.700	6.5	STD-2	600	---	---	---	---	204-6993
	LS2SWU- 19	18	19 pairs	1.620	1.380	0.810	8.0	STD-2	600	---	---	---	---	204-0544
	LS2SWU- 24	18	24 pairs	1.620	1.590	1.042	9.0	STD-2	600	---	---	---	---	210-9472
	LS2SWU- 30	18	30 pairs	1.620	1.760	1.256	9.5	STD-2	600	---	---	---	---	205-3633
	LS2SWU- 37	18	37 pairs	1.620	1.870	1.512	10.5	STD-2	600	---	---	---	---	205-3634
	LS2SWU- 61	18	61 pairs	1.620	2.300	2.321	13.5	STD-2	600	---	---	---	---	204-6994
24643/33 [5/]	LS2SWUA- 1	18	1 pair	1.620	0.305	0.130	2.0	STD-2	600	---	---	---	---	202-9507
	LS2SWUA- 3	18	3 pairs	1.620	0.760	0.228	4.0	STD-2	600	---	---	---	---	203-3393
	LS2SWUA- 7	18	7 pairs	1.620	0.960	0.392	5.0	STD-2	600	---	---	---	---	204-8927
	LS2SWUA- 12	18	12 pairs	1.620	1.250	0.745	6.5	STD-2	600	---	---	---	---	204-8928
	LS2SWUA- 19	18	19 pairs	1.620	1.420	0.842	8.0	STD-2	600	---	---	---	---	205-9307
	LS2SWUA- 24	18	24 pairs	1.620	1.640	1.070	9.0	STD-2	600	---	---	---	---	204-0545
	LS2SWUA- 30	18	30 pairs	1.620	1.810	1.289	9.5	STD-2	600	---	---	---	---	205-5935
	LS2SWUA- 37	18	37 pairs	1.620	1.920	1.552	10.5	STD-2	600	---	---	---	---	205-9308
	LS2SWUA- 61	18	61 pairs	1.620	2.350	2.367	13.5	STD-2	600	---	---	---	---	204-6995
24643/57	LS2UW- 42	26	42 pairs	0.278	0.790	0.493	6.5	TEL-3	300	---	---	---	---	202-7000
24643/57 [4/]	LS2UWA- 42	26	84	0.278	0.840	0.559	6.5	TEL-3	300	---	---	---	---	202-7001
24643/57	LS2UWS- 42	26	42 pairs	0.278	0.825	0.665	10.0	TEL-3	300	---	---	---	---	202-7002
24643/46 [4/]	LS2WA- 40	22	40 pairs	0.700	1.420	1.030	4.0	TEL-3	600	---	---	---	---	202-3533

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TABLE VI (b)(2). MIL-DTL-24643 cable ratings and characteristics, watertight, non-flexing service (electronic, communication, and instrumentation) – Continued.

MIL-DTL-	Cable type designation	Conductor size: AWG or MCM	Number of conductors, pairs, or triads in cable	Area of each conductor (MCM)	Cable overall diameter max. (inches)	Cable weight per ft approx. (lbs)	Radius of bend min. (inches)	CDR ID	Rated voltage, max. (RMS)	Ampacity, each conductor (amps, max.)				NSN 6145-01
										DC or 60 Hz		400 Hz		
										40 °C Ambient	50 °C Ambient	40 °C Ambient	50 °C Ambient	
24643/46	LS2WAU- 40	22	40 pairs	0.700	1.370	0.741	5.0	TEL-3	600	---	---	---	---	202-3532
24643/36 [4/]	LS3SWA- 3	18	9	1.620	0.705	0.309	4.5	STD-2	600	---	---	---	---	203-0373
	LS3SWA- 7	18	21	1.620	0.990	0.587	6.0	STD-2	600	---	---	---	---	202-9018
	LS3SWA- 10	18	30	1.620	1.230	0.865	7.5	STD-2	600	---	---	---	---	204-4260
	LS3SWA- 14	18	42	1.620	1.330	1.133	8.5	STD-2	600	---	---	---	---	203-7602
	LS3SWA- 19	18	57	1.620	1.500	1.442	9.0	STD-2	600	---	---	---	---	203-7603
	LS3SWA- 24	18	72	1.620	1.790	1.905	10.5	STD-2	600	---	---	---	---	203-7604
	LS3SWA- 30	18	90	1.620	1.910	2.266	11.0	STD-2	600	---	---	---	---	203-7605
	LS3SWA- 37	18	111	1.620	2.010	2.781	12.0	STD-2	600	---	---	---	---	203-7606
	LS3SWA- 44	18	132	1.620	2.290	3.399	13.0	STD-2	600	---	---	---	---	203-6567
24643/36	LS3SWU- 3	18	3 triads	1.620	0.655	0.298	4.5	STD-2	600	---	---	---	---	204-4862
	LS3SWU- 7	18	7 triads	1.620	0.940	0.528	6.0	STD-2	600	---	---	---	---	202-9011
	LS3SWU- 10	18	10 triads	1.620	1.180	0.791	7.5	STD-2	600	---	---	---	---	202-9012
	LS3SWU- 14	18	14 triads	1.620	1.280	1.052	8.5	STD-2	600	---	---	---	---	202-9013
	LS3SWU- 19	18	19 triads	1.620	1.450	1.353	9.0	STD-2	600	---	---	---	---	202-9014
	LS3SWU- 24	18	24 triads	1.620	1.760	1.799	10.5	STD-2	600	---	---	---	---	202-9015
	LS3SWU- 30	18	30 triads	1.620	1.860	2.156	11.0	STD-2	600	---	---	---	---	205-5175
	LS3SWU- 37	18	37 triads	1.620	1.940	2.664	12.0	STD-2	600	---	---	---	---	202-9016
	LS3SWU- 44	18	44 triads	1.620	2.240	3.278	13.0	STD-2	600	---	---	---	---	202-9017
24643/38	LSECM [6/]	20	8 pairs	1.111	1.370	1.545	11.0	STD-2	600	---	---	---	---	202-3527
		18	56	1.620	---	---	---	---	---	---	---	---	---	---
24643/38 [4/]	LSECMA	20	16	1.111	1.420	1.600	11.5	STD-2	600	---	---	---	---	202-3528
		18	56	1.620	---	---	---		---	---	---	---	---	---
24643/10 [4/]	LSPBTM- 5	22	10	0.640	0.590	0.167	3.5	TEL-3	600	---	---	---	---	205-3654
	LSPBTM- 15	22	30	0.640	0.800	0.331	5.0	TEL-3	600	---	---	---	---	205-3655
	LSPBTM- 30	22	60	0.640	1.030	0.544	6.5	TEL-3	600	---	---	---	---	203-6581
24643/10	LSPBTMU- 5	22	5 pairs	0.672 to 0.728	0.540	0.131	3.5	TEL-3	600	---	---	---	---	202-7765

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TABLE VI (b)(2). MIL-DTL-24643 cable ratings and characteristics, watertight, non-flexing service (electronic, communication, and instrumentation) – Continued.

MIL-DTL-	Cable type designation	Conductor size: AWG or MCM	Number of conductors, pairs, or triads in cable	Area of each conductor (MCM)	Cable overall diameter max. (inches)	Cable weight per ft approx. (lbs)	Radius of bend min. (inches)	CDR ID	Rated voltage, max. (RMS)	Ampacity, each conductor (amps, max.)				NSN 6145-01
										DC or 60 Hz		400 Hz		
										40 °C Ambient	50 °C Ambient	40 °C Ambient	50 °C Ambient	
24643/10	LSPBTMU- 15	22	15 pairs	0.672 to 0.728	0.750	0.283	5.0	TEL-3	600	---	---	---	---	203-6580
	LSPBTMU- 30	22	30 pairs	0.672 to 0.728	0.980	0.484	6.5	TEL-3	600	---	---	---	---	202-2830
24643/36	LS3SWUS- 3	18	3 triads	1.620	0.710	0.298	4.9	STD-2	600	---	---	---	---	---
	LS3SWUS- 7	18	7 triads	1.620	1.000	0.528	6.3	STD-2	600	---	---	---	---	---
	LS3SWUS- 10	18	10 triads	1.620	1.240	0.791	7.9	STD-2	600	---	---	---	---	---
	LS3SWUS- 14	18	14 triads	1.620	1.340	1.052	9.0	STD-2	600	---	---	---	---	---
	LS3SWUS- 19	18	19 triads	1.620	1.510	1.353	9.3	STD-2	600	---	---	---	---	---
	LS3SWUS- 24	18	24 triads	1.620	1.820	1.799	10.9	STD-2	600	---	---	---	---	---
	LS3SWUS- 30	18	30 triads	1.620	1.920	2.156	11.3	STD-2	600	---	---	---	---	---
	LS3SWUS- 37	18	37 triads	1.620	2.050	2.664	12.4	STD-2	600	---	---	---	---	---
	LS3SWUS- 44	18	44 triads	1.620	2.300	3.278	13.3	STD-2	600	---	---	---	---	---
24643/45	LS2WU- 10	26	10 pairs	0.254	0.490	0.118	4.0	TEL-3	300	---	---	---	---	---
	LS2WU- 15	26	15 pairs	0.254	0.570	0.164	5.0	TEL-3	300	---	---	---	---	---
	LS2WU- 19	26	19 pairs	0.254	0.590	0.185	6.0	TEL-3	300	---	---	---	---	---
	LS2WU- 30	26	30 pairs	0.254	0.720	0.257	7.0	TEL-3	300	---	---	---	---	---
	LS2WU- 45	26	45 pairs	0.254	0.890	0.360	8.5	TEL-3	300	---	---	---	---	---
	LS2WU- 60	26	60 pairs	0.254	0.980	0.442	10.0	TEL-3	300	---	---	---	---	---
24643/59	LSC5FSW- 4	24	4 pairs	0.404	0.38	0.074	2.3	TIA 568.2	150	---	---	---	---	---
	LSC5OSW- 4	24	4 pairs	0.404	0.42	0.086	2.5	TIA 568.2	150	---	---	---	---	525-5216
24643/62	LSPB2SDW- 1	22C/24D	1 pair/ 1 drain	0.642C/ 0.404D	0.405	0.067	2.4	Red/Green	150	100	100	---	---	---
	LSPB2SDOSW- 1	22C/24D	1 pair/ 1 drain	0.642C/ 0.404D	0.53	0.129	3.2	Red/Green	150	100	100	---	---	---
24643/69	LSC264- 4	31	4	4.11	0.765	0.409	4.6	STD-2	300	---	---	---	---	---
24643/77	LSC6OSW- 4	23	4 pairs	0.509	0.44	0.096	2.6	TIA 568.2	150	---	---	---	---	---
24643/78	LSCB1PSJW- 22	22	1 pair	0.7	0.273	---	1.6	Green/ Yellow	300	---	---	---	---	---

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TABLE VI (b)(2). MIL-DTL-24643 cable ratings and characteristics, watertight, non-flexing service (electronic, communication, and instrumentation) – Continued.

MIL-DTL-	Cable type designation	Conductor size: AWG or MCM	Number of conductors, pairs, or triads in cable	Area of each conductor (MCM)	Cable overall diameter max. (inches)	Cable weight per ft approx. (lbs)	Radius of bend min. (inches)	CDR ID	Rated voltage, max. (RMS)	Ampacity, each conductor (amps, max.)				NSN 6145-01
										DC or 60 Hz		400 Hz		
										40 °C Ambient	50 °C Ambient	40 °C Ambient	50 °C Ambient	
24643/78	LSCB2PSJW- 22	22	2 pairs	0.7	0.392	---	2.4	P1-Green/ Yellow, P2-Brown/ White	300	---	---	---	---	---
	LSCB1PSJW- 20	20	1 pair	1.111	0.309	---	1.9	Green/ Yellow	300	---	---	---	---	---
	LSCB2PSJW- 20	20	2 pairs	1.111	0.449	---	2.7	P1-Green/ Yellow, P2-Brown/ White	300	---	---	---	---	---
	LSCB1PSJW- 18	18	1 pair	1.9	0.367	---	2.2	Green/ Yellow	300	---	---	---	---	---
	LSCB2PSJW- 18	18	2 pairs	1.9	0.542	---	3.3	P1-Green/ Yellow, P2-Brown/ White	300	---	---	---	---	---
24643/82	LSE7UW- 26	26	7 pairs	0.254	0.495	---	3	STD-3	300	---	---	---	---	---
	LSE10UW- 26	26	10 pairs	0.254	0.57	---	3.4	STD-3	300	---	---	---	---	---
	LSE19UW- 26	26	19 pairs	0.254	0.65	---	3.9	STD-3	300	---	---	---	---	---
	LSE37UW- 26	26	37 pairs	0.254	0.8	---	4.8	STD-3	300	---	---	---	---	---
	LSE42UW- 26	26	42 pairs	0.254	0.85	---	5.1	STD-3	300	---	---	---	---	---
	LSE7AUW- 22	22	7 pairs	0.642	0.665	---	4	STD-3	300	---	---	---	---	---
	LSE10AUW- 22	22	10 pairs	0.642	0.795	---	4.8	STD-3	300	---	---	---	---	---
	LSE19AUW- 22	22	19 pairs	0.642	0.95	---	5.7	STD-3	300	---	---	---	---	---
	LSE37AUW- 22	22	37 pairs	0.642	1.255	---	7.5	STD-3	300	---	---	---	---	---
	LSE40AUW- 22	22	40 pairs	0.642	1.305	---	7.8	STD-3	300	---	---	---	---	---
24643/83	LSE3SPEW- 22	22	3 pairs	0.754	0.5	---	3	STD-2	300	---	---	---	---	---
	LSE7SPEW- 22	22	7 pairs	0.754	0.6	---	3.6	STD-2	300	---	---	---	---	---
	LSE10SPEW- 22	22	10 pairs	0.754	0.76	---	4.6	STD-2	300	---	---	---	---	---
	LSE14SPEW- 22	22	14 pairs	0.754	0.83	---	5	STD-2	300	---	---	---	---	---
	LSE19SPEW- 22	22	19 pairs	0.754	0.93	---	5.6	STD-2	300	---	---	---	---	---

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TABLE VI (b)(2). MIL-DTL-24643 cable ratings and characteristics, watertight, non-flexing service (electronic, communication, and instrumentation) – Continued.

MIL-DTL-	Cable type designation	Conductor size: AWG or MCM	Number of conductors, pairs, or triads in cable	Area of each conductor (MCM)	Cable overall diameter max. (inches)	Cable weight per ft approx. (lbs)	Radius of bend min. (inches)	CDR ID	Rated voltage, max. (RMS)	Ampacity, each conductor (amps, max.)				NSN 6145-01
										DC or 60 Hz		400 Hz		
										40 °C Ambient	50 °C Ambient	40 °C Ambient	50 °C Ambient	
24643/83	LSE24SPEW- 22	22	24 pairs	0.754	1.09	---	6.5	STD-2	300	---	---	---	---	---
	LSE30SPEW- 22	22	30 pairs	0.754	1.15	---	6.9	STD-2	300	---	---	---	---	---
	LSE37SPEW- 22	22	37 pairs	0.754	1.245	---	7.5	STD-2	300	---	---	---	---	---
	LSE44SPEW- 22	22	44 pairs	0.754	1.395	---	8.4	STD-2	300	---	---	---	---	---
	LSE61SPEW- 22	22	61 pairs	0.754	1.56	---	9.4	STD-2	300	---	---	---	---	---
	LSE1SPEW- 20	20	1 pair	1	0.37	---	2.2	STD-2	300	---	---	---	---	---
	LSE2SPEW- 20	20	2 pairs	1	0.52	---	3.1	STD-2	300	---	---	---	---	---
	LSE4SPEW- 20	20	4 pairs	1	0.565	---	3.4	STD-2	300	---	---	---	---	---
	LSE6SPEW- 20	20	6 pairs	1	0.675	---	4.1	STD-2	300	---	---	---	---	---
	LSE8SPEW- 20	20	8 pairs	1	0.73	---	4.4	STD-2	300	---	---	---	---	---
	LSE10SPEW- 20	20	10 pairs	1	0.855	---	5.1	STD-2	300	---	---	---	---	---
	LSE12SPEW- 20	20	12 pairs	1	0.88	---	5.3	STD-2	300	---	---	---	---	---
	LSE16SPEW- 20	20	16 pairs	1	0.985	---	5.9	STD-2	300	---	---	---	---	---
	LSE1SPEW- 18	18	1 pair	1.62	0.39	---	2.3	STD-2	300	---	---	---	---	---
	LSE3SPEW- 18	18	3 pairs	1.62	0.57	---	3.4	STD-2	300	---	---	---	---	---
	LSE7SPEW- 18	18	7 pairs	1.62	0.745	---	4.5	STD-2	300	---	---	---	---	---
	LSE10SPEW- 18	18	10 pairs	1.62	0.96	---	5.8	STD-2	300	---	---	---	---	---
	LSE12SPEW- 18	18	12 pairs	1.62	0.99	---	5.9	STD-2	300	---	---	---	---	---
	LSE14SPEW- 18	18	14 pairs	1.62	1.045	---	6.3	STD-2	300	---	---	---	---	---
	LSE19SPEW- 18	18	19 pairs	1.62	1.165	---	7	STD-2	300	---	---	---	---	---
	LSE24SPEW- 18	18	24 pairs	1.62	1.365	---	8.2	STD-2	300	---	---	---	---	---
	LSE30SPEW- 18	18	30 pairs	1.62	1.445	---	8.7	STD-2	300	---	---	---	---	---
	LSE37SPEW- 18	18	37 pairs	1.62	1.57	---	9.4	STD-2	300	---	---	---	---	---
	LSE61SPEW- 18	18	61 pairs	1.62	1.97	---	11.8	STD-2	300	---	---	---	---	---
	LSE3SPEW- 16	16	3 pairs	2.58	0.605	---	3.6	STD-2	300	---	---	---	---	---
	LSE7SPEW- 16	16	7 pairs	2.58	0.81	---	4.9	STD-2	300	---	---	---	---	---
24643/84	LSEC6OSW- 4	23	4 pairs	0.509	0.645	0.28	5	TIA 568.2	150	---	---	---	---	---

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TABLE VI (b)(3). MIL-DTL-24643 cable ratings and characteristics, watertight, non-flexing service (control).

MIL-DTL-	Cable type designation		Conductor size: AWG or MCM	Number of conductors, pairs, or triads in cable	Area of each conductor (MCM)	Cable overall diameter max. (inches)	Cable weight per ft approx. (lbs)	Radius of bend min. (inches)	CDR ID	Rated voltage, max. (RMS)	Ampacity, each conductor (amps, max.)				NSN 6145-01
											DC or 60 Hz		400 Hz		
											40 °C Ambient	50 °C Ambient	40 °C Ambient	50 °C Ambient	
24643/80	LSTW-	2	18	1	1.624	0.11	---	0.7	See spec sheet	1,000	13	12	13	12	---
	LSTW-	3	16	1	2.582	0.12	---	0.7	See spec sheet	1,000	16	15	16	15	---
	LSTW-	4	14	1	4.106	0.135	---	0.8	See spec sheet	1,000	23	21	23	21	---
	LSTW-	6	12	1	6.526	0.155	---	0.9	See spec sheet	1,000	27	25	27	25	---
	LSTW-	9	10	1	10.382	0.19	---	1.1	See spec sheet	1,000	36	34	36	34	---
	LSTW-	14	9	1	13.08	0.23	---	1.4	See spec sheet	1,000	44	40	44	40	---
	LSTW-	23	7	1	20.92	0.265	---	1.6	See spec sheet	1,000	59	55	59	55	---
	LSTW-	40	4	1	41.84	0.335	---	2	See spec sheet	1,000	86	80	86	80	---
	LSTW-	50	3	1	52.69	0.365	---	2.2	See spec sheet	1,000	105	97	105	97	---
	LSTW-	60	2	1	66.31	0.405	---	2.4	See spec sheet	1,000	118	109	118	109	---
	LSTW-	75	1	1	83.68	0.46	---	2.8	See spec sheet	1,000	132	122	132	122	---
	LSTW-	100	0	1	105.58	0.495	---	3	See spec sheet	1,000	155	143	155	143	---
	LSTW-	150	000	1	167.75	0.55	---	3.3	See spec sheet	1,000	205	189	205	189	---
	LSTW-	300	300	1	300	0.775	---	4.7	See spec sheet	1,000	291	269	291	269	---
	LSTW-	400	400 (Navy STD)	1	405.4	0.91	13	5.5	See spec sheet	1,000	346	319	346	319	---
24643/81	LSE7EW-	2	18	7	1.624	0.56	---	3.4	STD-1 or 3	1,000	13	12	13	12	---
	LSE10EW-	2	18	10	1.624	0.685	---	4.1	STD-1 or 3	1,000	13	12	13	12	---
	LSE14EW-	2	18	14	1.624	0.74	---	4.4	STD-1 or 3	1,000	13	12	13	12	---
	LSE19EW-	2	18	19	1.624	0.82	---	4.9	STD-1 or 3	1,000	13	12	13	12	---
	LSE24EW-	2	18	24	1.624	0.955	---	5.7	STD-1 or 3	1,000	13	12	13	12	---
	LSE30EW-	2	18	30	1.624	1.005	---	6	STD-1 or 3	1,000	13	12	13	12	---
	LSE37EW-	2	18	37	1.624	1.09	---	6.5	STD-1 or 3	1,000	13	12	13	12	---
	LSE44EW-	2	18	44	1.624	1.22	---	7.3	STD-1 or 3	1,000	13	12	13	12	---
	LSE7EW-	3	16	7	2.582	0.59	---	3.5	STD-1 or 3	1,000	16	15	16	15	---
	LSE10EW-	3	16	10	2.582	0.735	---	4.4	STD-1 or 3	1,000	16	15	16	15	---
	LSE14EW-	3	16	14	2.582	0.795	---	4.8	STD-1 or 3	1,000	16	15	16	15	---
	LSE19EW-	3	16	19	2.582	0.88	---	5.3	STD-1 or 3	1,000	16	15	16	15	---
	LSE24EW-	3	16	24	2.582	1.025	---	6.2	STD-1 or 3	1,000	16	15	16	15	---

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TABLE VI (b)(3). MIL-DTL-24643 cable ratings and characteristics, watertight, non-flexing service (control) – Continued.

MIL-DTL-	Cable type designation		Conductor size: AWG or MCM	Number of conductors, pairs, or triads in cable	Area of each conductor (MCM)	Cable overall diameter max. (inches)	Cable weight per ft approx. (lbs)	Radius of bend min. (inches)	CDR ID	Rated voltage, max. (RMS)	Ampacity, each conductor (amps, max.)				NSN 6145-01
											DC or 60 Hz		400 Hz		
											40 °C Ambient	50 °C Ambient	40 °C Ambient	50 °C Ambient	
24643/81	LSE30EW-	3	16	30	2.582	1.09	---	6.5	STD-1 or 3	1,000	16	15	16	15	---
	LSE33EW-	3	16	33	2.582	1.13	---	6.8	STD-1 or 3	1,000	16	15	16	15	---
	LSE37EW-	3	16	37	2.582	1.175	---	7.1	STD-1 or 3	1,000	16	15	16	15	---
	LSE44EW-	3	16	44	2.582	1.315	---	7.9	STD-1 or 3	1,000	16	15	16	15	---
	LSE61EW-	3	16	61	2.582	1.46	---	8.8	STD-1 or 3	1,000	16	15	16	15	---
	LSE7EW-	4	14	7	4.106	0.64	---	3.8	STD-1 or 3	1,000	23	21	23	21	---
	LSE19EW-	6	12	19	6.526	1.095	---	6.6	STD-1 or 3	1,000	27	25	27	25	---
	LSE19EW-	14	9	19	13.08	1.51	---	9.1	STD-1 or 3	1,000	44	40	44	40	---
	LSE19EW-	23	7	19	20.92	1.725	---	10.4	STD-1 or 3	1,000	59	55	59	55	---
	LSE19EW-	40	4	19	41.84	2.16	---	13	STD-1 or 3	1,000	86	80	86	80	---
	LSE19EW-	60	2	19	66.31	2.56	---	15.4	STD-1 or 3	1,000	118	109	118	109	---

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TABLE VI (c)(1). MIL-DTL-24643 cable ratings and characteristics, non-watertight (with circuit integrity), non-flexing service (electronic, communication, and instrumentation).

MIL-DTL-	Cable type designation	Conductor size: AWG or MCM	Number of conductors, pairs, or triads in cable	Area of each conductor (MCM)	Cable overall diameter max. (inches)	Cable weight per ft approx. (lbs)	Radius of bend min. (inches)	CDR ID	Rated voltage, max. (RMS)	Ampacity, each conductor (amps, max.)				NSN 6145-01
										DC or 60 Hz		400 Hz		
										40 °C Ambient	50 °C Ambient	40 °C Ambient	50 °C Ambient	
24643/25	LSPI- 3	18	3 pairs	1.620	0.685	0.285	4.5	STD-4	300	---	---	---	---	161-150
	LSPI- 7	18	7 pairs	1.620	0.900	0.505	5.5	STD-4	300	---	---	---	---	495-3384
	LSPI- 12	18	12 pairs	1.620	1.155	1.372	7.0	STD-4	300	---	---	---	---	163-7375
24643/24	LSTCJXN- 3	16	3 pairs	2.582	0.742	0.231	4.0	LTR-4	300	---	---	---	---	---
	LSTCJXN- 7	16	7 pairs	2.582	0.983	0.515	5.5	LTR-4	300	---	---	---	---	---
	LSTCJXN- 12	16	12 pairs	2.582	1.269	0.844	6.5	LTR-4	300	---	---	---	---	---
24643/24	LSTCKXN- 1	16	1 pair	2.582	0.456	0.108	2.5	LTR-4	300	---	---	---	---	---
	LSTCKXN- 3	16	3 pairs	2.582	0.742	0.257	4.0	LTR-4	300	---	---	---	---	---
	LSTCKXN- 7	16	7 pairs	2.582	0.983	0.520	5.5	LTR-4	300	---	---	---	---	---
	LSTCKXN- 12	16	12 pairs	2.582	1.269	0.844	6.5	LTR-4	300	---	---	---	---	---
24643/24	LSTCTXN- 1	21	1 pair	0.810	0.350	0.108	2.0	LTR-4	300	---	---	---	---	---
	LSTCTXN- 3	21	3 pairs	0.810	0.552	0.262	3.5	LTR-4	300	---	---	---	---	---
	LSTCTXN- 7	21	7 pairs	0.810	0.731	0.515	4.0	LTR-4	300	---	---	---	---	---
	LSTCTXN- 12	21	12 pairs	0.810	0.964	0.844	5.0	LTR-4	300	---	---	---	---	---

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TABLE VI (c)(2). MIL-DTL-24643 cable ratings and characteristics, non-watertight (with circuit integrity), non-flexing service (power and lighting).

MIL-DTL-	Cable type designation	Conductor size: AWG or MCM	Number of conductors, pairs, or triads in cable	Area of each conductor (MCM)	Cable overall diameter max. (inches)	Cable weight per ft approx. (lbs)	Radius of bend min. (inches)	CDR ID	Rated voltage, max. (RMS)	Ampacity, each conductor (amps, max.)				NSN 6145-01
										DC or 60 Hz		400 Hz		
										40 °C Ambient	50 °C Ambient	40 °C Ambient	50 °C Ambient	
24643/26	LSDPSN- 3	16	2	2.580	0.455	0.121	3.0	STD-4	600	10	---	10	---	---
	LSDPSN- 4	14	2	4.110	0.489	0.141	3.0	STD-4	600	20	---	20	---	---
	LSDPSN- 6	12	2	6.530	0.585	0.194	3.5	STD-4	600	28	---	28	---	---
	LSDPSN- 9	10	2	10.380	0.628	0.232	4.0	STD-4	600	41	---	41	---	---
	LSDPSN- 14	9	2	13.090	0.730	0.323	4.5	STD-4	600	54	---	54	---	---
24643/26	LSTPSN- 3	16	3	2.580	0.475	0.129	3.0	STD-4	600	10	---	10	---	---
	LSTPSN- 4	14	3	4.110	0.553	0.160	3.5	STD-4	600	17	---	17	---	---
	LSTPSN- 6	12	3	6.530	0.620	0.229	4.0	STD-4	600	23	---	23	---	---
	LSTPSN- 9	10	3	10.380	0.657	0.299	4.0	STD-4	600	36	---	36	---	---
	LSTPSN- 14	9	3	13.090	0.751	0.411	4.5	STD-4	600	47	---	47	---	---
	LSTPSN- 23	7	3	20.815	0.866	0.579	5.5	STD-4	600	64	---	64	---	---
	LSTPSN- 30	5	3	33.098	0.989	0.738	5.0	STD-4	600	77	---	77	---	---
24643/26	LSFPSN- 14	9	4	13.090	0.815	0.452	5.0	STD-4	600	42	---	---	---	---
24643/26	LS7PSN- 6	12	7	6.530	0.775	0.361	4.5	STD-4	600	35/20 1/	---	---	---	---
	LS7PSN- 14	9	7	13.090	0.986	---	5.7	STD-4	600	---	---	---	---	---
24643/85	LSE1CI- 2	18	1	1.62	0.335	---	2	STD-1 or 3 or LTR-5	1,000	13	12	13	12	---
	LSE2CI- 2	18	2	1.62	0.44	---	2.6	STD-1 or 3 or LTR-5	1,000	13	12	13	12	---
	LSE3CI- 2	18	3	1.62	0.455	---	2.7	STD-1 or 3 or LTR-5	1,000	13	12	13	12	---
	LSE4CI- 2	18	4	1.62	0.48	---	2.9	STD-1 or 3 or LTR-5	1,000	13	12	13	12	---
	LSE7CI- 2	18	7	1.62	0.54	---	3.2	STD-1 or 3 or LTR-5	1,000	13	12	13	12	---
	LSE2CI- 3	16	2	2.58	0.46	---	2.8	STD-1 or 3 or LTR-5	1,000	16	13	16	13	---
	LSE3CI- 3	16	3	2.58	0.48	---	2.9	STD-1 or 3 or LTR-5	1,000	16	13	16	13	---
	LSE4CI- 3	16	4	2.58	0.51	---	3.1	STD-1 or 3 or LTR-5	1,000	16	13	16	13	---

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TABLE VI (c)(2). MIL-DTL-24643 cable ratings and characteristics, non-watertight (with circuit integrity), non-flexing service (power and lighting) – Continued.

MIL-DTL-	Cable type designation	Conductor size: AWG or MCM	Number of conductors, pairs, or triads in cable	Area of each conductor (MCM)	Cable overall diameter max. (inches)	Cable weight per ft approx. (lbs)	Radius of bend min. (inches)	CDR ID	Rated voltage, max. (RMS)	Ampacity, each conductor (amps, max.)				NSN 6145-01
										DC or 60 Hz		400 Hz		
										40 °C Ambient	50 °C Ambient	40 °C Ambient	50 °C Ambient	
24643/85	LSE7CI- 3	16	7	2.58	0.575	---	3.5	STD-1 or 3 or LTR-5	1,000	16	13	16	13	---
	LSE2CI- 4	14	2	4.11	0.49	---	2.9	STD-1 or 3 or LTR-5	1,000	23	21	23	21	---
	LSE3CI- 4	14	3	4.11	0.51	---	3.1	STD-1 or 3 or LTR-5	1,000	23	21	23	21	---
	LSE4CI- 4	14	4	4.11	0.54	---	3.2	STD-1 or 3 or LTR-5	1,000	23	21	23	21	---
	LSE7CI- 4	14	7	4.11	0.615	---	3.7	STD-1 or 3 or LTR-5	1,000	23	21	23	21	---
	LSE2CI- 9	10	2	10.38	0.57	---	3.4	STD-1 or 3 or LTR-5	1,000	32	29	32	29	---
	LSE3CI- 9	10	3	10.38	0.595	---	3.6	STD-1 or 3 or LTR-5	1,000	32	29	32	29	---
	LSE4CI- 9	10	4	10.38	0.665	---	4	STD-1 or 3 or LTR-5	1,000	32	29	32	29	---
	LSE2CI- 14	9	2	13.09	0.615	---	3.7	STD-1 or 3 or LTR-5	1,000	38	35	38	35	---
	LSE3CI- 14	9	3	13.09	0.655	---	3.9	STD-1 or 3 or LTR-5	1,000	38	35	38	35	---
	LSE1CI- 23	7	1	20.82	0.455	---	2.7	STD-1 or 3 or LTR-5	1,000	53	49	53	49	---
	LSE2CI- 23	7	2	20.82	0.69	---	4.1	STD-1 or 3 or LTR-5	1,000	53	49	53	49	---
	LSE3CI- 23	7	3	20.82	0.725	---	4.4	STD-1 or 3 or LTR-5	1,000	53	49	53	49	---
	LSE4CI- 23	7	4	20.82	0.805	---	4.8	STD-1 or 3 or LTR-5	1,000	53	49	53	49	---
	LSE1CI- 50	3	1	52.62	0.59	---	3.5	STD-1 or 3 or LTR-5	1,000	90	84	90	84	---
	LSE2CI- 50	3	2	52.62	0.985	---	5.9	STD-1 or 3 or LTR-5	1,000	90	84	90	84	---

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TABLE VI (c)(2). MIL-DTL-24643 cable ratings and characteristics, non-watertight (with circuit integrity), non-flexing service (power and lighting) – Continued.

MIL-DTL-	Cable type designation	Conductor size: AWG or MCM	Number of conductors, pairs, or triads in cable	Area of each conductor (MCM)	Cable overall diameter max. (inches)	Cable weight per ft approx. (lbs)	Radius of bend min. (inches)	CDR ID	Rated voltage, max. (RMS)	Ampacity, each conductor (amps, max.)				NSN 6145-01
										DC or 60 Hz		400 Hz		
										40 °C Ambient	50 °C Ambient	40 °C Ambient	50 °C Ambient	
24643/85	LSE3CI- 50	3	3	52.62	1.045	---	6.3	STD-1 or 3 or LTR-5	1,000	90	84	90	84	---
	LSE4CI- 50	3	4	52.62	1.15	---	6.9	STD-1 or 3 or LTR-5	1,000	90	84	90	84	---
	LSE2CI- 60	2	2	66.36	1.06	---	6.4	STD-1 or 3 or LTR-5	1,000	105	88	105	88	---
	LSE3CI- 60	2	3	66.36	1.12	---	6.7	STD-1 or 3 or LTR-5	1,000	105	88	105	88	---
	LSE1CI- 75	1	1	83.69	0.675	---	4.1	STD-1 or 3 or LTR-5	1,000	132	122	132	122	---
	LSE2CI- 75	1	2	83.69	1.13	---	6.8	STD-1 or 3 or LTR-5	1,000	132	122	132	122	---
	LSE3CI- 75	1	3	83.69	1.205	---	7.2	STD-1 or 3 or LTR-5	1,000	132	122	132	122	---
	LSE4CI- 75	1	4	83.69	1.33	---	8	STD-1 or 3 or LTR-5	1,000	132	122	132	122	---
	LSE1CI- 100	0	1	105.6	0.715	---	4.3	STD-1 or 3 or LTR-5	1,000	155	143	155	143	---
	LSE2CI- 100	0	2	105.6	1.22	---	7.3	STD-1 or 3 or LTR-5	1,000	155	143	155	143	---
	LSE3CI- 100	0	3	105.6	1.29	---	7.7	STD-1 or 3 or LTR-5	1,000	155	143	155	143	---
	LSE4CI- 100	0	4	105.6	1.43	---	8.6	STD-1 or 3 or LTR-5	1,000	155	143	155	143	---
	LSE6CI- 100	0	6	105.6	1.685	---	10.1	STD-1 or 3 or LTR-5	1,000	155	143	155	143	---
	LSE1CI- 125	00	1	133.1	0.795	---	4.8	STD-1 or 3 or LTR-5	1,000	177	164	177	164	---
	LSE2CI- 125	00	2	133.1	1.35	---	8.1	STD-1 or 3 or LTR-5	1,000	177	164	177	164	---
	LSE3CI- 125	00	3	133.1	1.43	---	8.6	STD-1 or 3 or LTR-5	1,000	177	164	177	164	---

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TABLE VI (c)(2). MIL-DTL-24643 cable ratings and characteristics, non-watertight (with circuit integrity), non-flexing service (power and lighting) – Continued.

MIL-DTL-	Cable type designation	Conductor size: AWG or MCM	Number of conductors, pairs, or triads in cable	Area of each conductor (MCM)	Cable overall diameter max. (inches)	Cable weight per ft approx. (lbs)	Radius of bend min. (inches)	CDR ID	Rated voltage, max. (RMS)	Ampacity, each conductor (amps, max.)				NSN 6145-01
										DC or 60 Hz		400 Hz		
										40 °C Ambient	50 °C Ambient	40 °C Ambient	50 °C Ambient	
24643/85	LSE1CI- 150	000	1	167.8	0.84	---	5	STD-1 or 3 or LTR-5	1,000	205	189	205	189	---
	LSE2CI- 150	000	2	167.8	1.44	---	8.6	STD-1 or 3 or LTR-5	1,000	205	189	205	189	---
	LSE3CI- 150	000	3	167.8	1.525	---	9.2	STD-1 or 3 or LTR-5	1,000	205	189	205	189	---
	LSE4CI- 150	000	4	167.8	1.72	---	10.3	STD-1 or 3 or LTR-5	1,000	205	189	205	189	---
	LSE6CI- 150	000	6	167.8	2.035	---	12.2	STD-1 or 3 or LTR-5	1,000	205	189	205	189	---
	LSE1CI- 200	0000	1	211.6	0.935	---	5.6	STD-1 or 3 or LTR-5	1,000	237	218	237	218	---
	LSE2CI- 200	0000	2	211.6	1.61	---	9.7	STD-1 or 3 or LTR-5	1,000	237	218	237	218	---
	LSE3CI- 200	0000	3	211.6	1.73	---	10.4	STD-1 or 3 or LTR-5	1,000	237	218	237	218	---
	LSE4CI- 200	0000	4	211.6	1.93	---	11.6	STD-1 or 3 or LTR-5	1,000	237	218	237	218	---
	LSE6CI- 200	0000	6	211.6	2.29	---	13.7	STD-1 or 3 or LTR-5	1,000	237	218	237	218	---
	LSE1CI- 300	300	1	300	1.03	---	6.2	STD-1 or 3 or LTR-5	1,000	291	269	291	269	---
	LSE2CI- 300	300	2	300	1.82	---	10.9	STD-1 or 3 or LTR-5	1,000	291	269	291	269	---
	LSE3CI- 300	300	3	300	1.935	---	11.6	STD-1 or 3 or LTR-5	1,000	291	269	291	269	---
	LSE1CI- 400	400 (Navy STD)	1	405.4	1.15	13	6.9	STD-1 or 3 or LTR-5	1,000	346	319	346	319	---
	LSE2CI- 400	400 (Navy STD)	2	405.4	2.05	13	12.3	STD-1 or 3 or LTR-5	1,000	346	319	346	319	---
	LSE3CI- 400	400 (Navy STD)	3	405.4	2.18	13	13.1	STD-1 or 3 or LTR-5	1,000	346	319	346	319	---

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TABLE VI (c)(3). MIL-DTL-24643 cable ratings and characteristics, non-watertight (with circuit integrity), non-flexing service (control).

MIL-DTL-	Cable type designation	Conductor size: AWG or MCM	Number of conductors, pairs, or triads in cable	Area of each conductor (MCM)	Cable overall diameter max. (inches)	Cable weight per ft approx. (lbs)	Radius of bend min. (inches)	CDR ID	Rated voltage, max. (RMS)	Ampacity, each conductor (amps, max.)				NSN 6145-01
										DC or 60 Hz		400 Hz		
										40 °C Ambient	50 °C Ambient	40 °C Ambient	50 °C Ambient	
24643/85	LSE7CI- 2	18	7	1.62	0.54	---	3.2	STD-1 or 3 or LTR-5	1,000	13	12	13	12	---
	LSE7CI- 3	16	7	2.58	0.575	---	3.5	STD-1 or 3 or LTR-5	1,000	16	13	16	13	---
	LSE7CI- 4	14	7	4.11	0.615	---	3.7	STD-1 or 3 or LTR-5	1,000	23	21	23	21	---
	LSE10CI- 2	18	10	1.62	0.66	---	4	STD-1 or 3 or LTR-5	1,000	13	12	13	12	---
	LSE14CI- 2	18	14	1.62	0.7	---	4.2	STD-1 or 3 or LTR-5	1,000	13	12	13	12	---
	LSE19CI- 2	18	19	1.62	0.77	---	4.6	STD-1 or 3 or LTR-5	1,000	13	12	13	12	---
	LSE30CI- 2	18	30	1.62	0.93	---	5.6	STD-1 or 3 or LTR-5	1,000	13	12	13	12	---
	LSE37CI- 2	18	37	1.62	0.99	---	5.9	STD-1 or 3 or LTR-5	1,000	13	12	13	12	---
	LSE44CI- 2	18	44	1.62	1.095	---	6.6	STD-1 or 3 or LTR-5	1,000	13	12	13	12	---
	LSE61CI- 2	18	61	1.62	1.21	---	7.3	STD-1 or 3 or LTR-5	1,000	13	12	13	12	---
	LSE91CI- 2	18	91	1.62	1.415	---	8.5	STD-1 or 3 or LTR-5	1,000	13	12	13	12	---

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TABLE VI (d)(1). MIL-DTL-24643 cable ratings and characteristics, non-watertight, non-flexing service (power and lighting).

MIL-DTL-	Cable type designation	Conductor size: AWG or MCM	Number of conductors, pairs, or triads in cable	Area of each conductor (MCM)	Cable overall diameter max. (inches)	Cable weight per ft approx. (lbs)	Radius of bend min. (inches)	CDR ID	Rated voltage, max. (RMS)	Ampacity, each conductor (amps, max.)				NSN 6145-01
										DC or 60 Hz		400 Hz		
										40 °C Ambient	50 °C Ambient	40 °C Ambient	50 °C Ambient	
24643/48	LSDNW- 3	16	2	2.580	0.390	0.071	2.0	STD-1 or 3	1,000	13	12	13	12	202-3513
	LSDNW- 4	14	2	4.110	0.430	0.092	2.0	STD-1 or 3	1,000	22	20	22	20	203-0376
	LSDNW- 9	10	2	10.380	0.545	0.179	2.5	STD-1 or 3	1,000	44	41	44	41	202-3514
	LSDNW- 14	9	2	13.090	0.610	0.204	3.0	STD-1 or 3	1,000	60	55	60	55	203-4032
	LSDNW- 23	7	2	20.820	0.690	0.292	3.5	STD-1 or 3	1,000	78	72	78	72	202-3515
	LSDNW- 50	3	2	52.620	0.910	0.603	4.5	STD-1 or 3	1,000	126	116	126	116	202-3516
	LSDNW- 75	1	2	83.690	1.080	0.882	5.0	STD-1 or 3	1,000	168	155	168	155	202-3517
	LSDNW- 100	0	2	105.600	1.170	1.058	5.5	STD-1 or 3	1,000	199	183	199	183	202-3492
	LSDNW- 200	0000	2	211.600	1.583	---	7.4	STD-1 or 3	1,000	312	287	312	287	---
	LSDNW- 300	300	2	300	1.841	---	8.7	STD-1 or 3	1,000	416	383	416	383	---
	LSDNW- 400	400 (Navy STD	2	413.5	2.069	---	9.7	STD-1 or 3	1,000	496	456	496	456	---
24643/48 [4/]	LSDNWA- 3	16	2	2.580	0.440	0.090	2.0	STD-1	1,000	13	12	13	12	202-3493
	LSDNWA- 4	14	2	4.410	0.480	0.112	2.0	STD-1	1,000	22	20	22	20	202-3494
	LSDNWA- 9	10	2	10.380	0.595	0.209	2.5	STD-1	1,000	44	41	44	41	202-3495
	LSDNWA- 14	9	2	13.090	0.660	0.230	3.0	STD-1	1,000	60	55	60	55	202-3496
	LSDNWA- 23	7	2	20.820	0.740	0.329	3.5	STD-1	1,000	78	72	78	72	202-3497
	LSDNWA- 50	3	2	52.620	0.960	0.660	4.0	STD-1	1,000	126	116	126	116	202-3498
	LSDNWA- 75	1	2	83.690	1.130	0.939	5.0	STD-1	1,000	168	155	168	155	202-3499
	LSDNWA- 100	0	2	105.600	1.220	1.126	5.5	STD-1	1,000	199	183	199	183	202-3500
24643/50	LSFNW- 3	16	4	2.580	0.447	0.104	2.0	STD-1 or 3	1,000	11	10	11	10	203-6571
	LSFNW- 4	14	4	4.110	0.513	0.141	2.0	STD-1 or 3	1,000	18	17	18	17	203-6572
	LSFNW- 9	10	4	10.380	0.630	0.268	2.5	STD-1 or 3	1,000	39	36	39	36	203-6573
	LSFNW- 23	7	4	20.820	0.830	0.482	4.0	STD-1 or 3	1,000	69	64	69	64	203-6574
	LSFNW- 50	3	4	52.627	1.074	---	5.2	STD-1 or 3	1,000	111	103	111	103	---
	LSFNW- 75	1	4	83.680	1.332	---	6.4	STD-1 or 3	1,000	150	139	150	139	---
	LSFNW- 100	0	4	105.518	1.446	---	7.0	STD-1 or 3	1,000	177	164	177	164	---
	LSFNW- 150	000	4	167.780	1.692	---	8.2	STD-1 or 3	1,000	240	223	240	223	---
	LSFNW- 200	0000	4	211.566	1.836	---	8.8	STD-1 or 3	1,000	279	259	279	259	---

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TABLE VI (d)(1). MIL-DTL-24643 cable ratings and characteristics, non-watertight, non-flexing service (power and lighting) – Continued.

MIL-DTL-	Cable type designation	Conductor size: AWG or MCM	Number of conductors, pairs, or triads in cable	Area of each conductor (MCM)	Cable overall diameter max. (inches)	Cable weight per ft approx. (lbs)	Radius of bend min. (inches)	CDR ID	Rated voltage, max. (RMS)	Ampacity, each conductor (amps, max.)				NSN 6145-01
										DC or 60 Hz		400 Hz		
										40 °C Ambient	50 °C Ambient	40 °C Ambient	50 °C Ambient	
24643/50 [4/]	LSFNWA- 3	16	4	2.580	0.497	0.126	2.0	STD-1	1,000	11	10	11	10	203-6575
	LSFNWA- 4	14	4	4.110	0.563	0.164	2.0	STD-1	1,000	18	17	18	17	203-6482
	LSFNWA- 9	10	4	10.380	0.680	0.302	2.5	STD-1	1,000	39	36	39	36	206-7007
	LSFNWA- 23	7	4	20.820	0.880	0.527	4.0	STD-1	1,000	69	64	69	64	203-6483
24643/49	LSTNW- 3	16	3	2.580	0.411	0.085	1.5	STD-1 or 3	1,000	11	10	11	10	202-3501
	LSTNW- 4	14	3	4.110	0.449	0.107	2.0	STD-1 or 3	1,000	18	17	18	17	203-4033
	LSTNW- 9	10	3	10.380	0.625	0.240	2.5	STD-1 or 3	1,000	39	36	39	36	202-3502
	LSTNW- 14	9	3	13.090	0.670	0.271	2.5	STD-1 or 3	1,000	51	47	51	47	202-3503
	LSTNW- 23	7	3	20.820	0.760	0.390	3.5	STD-1 or 3	1,000	69	64	69	64	202-3504
	LSTNW- 30	5	3	33.098	0.865	---	4.0	STD-1 or 3	1,000	70	65	70	65	---
	LSTNW- 50	3	3	52.620	0.969	0.793	4.5	STD-1 or 3	1,000	110	101	110	101	202-3505
	LSTNW- 75	1	3	83.690	1.134	1.200	5.0	STD-1 or 3	1,000	148	136	148	136	202-3506
	LSTNW- 100	0	3	105.600	1.266	1.452	6.0	STD-1 or 3	1,000	174	160	174	160	202-3507
	LSTNW- 150	000	3	167.800	1.515	2.218	7.0	STD-1 or 3	1,000	235	216	235	216	202-3508
	LSTNW- 200	0000	3	211.566	1.669	---	7.7	STD-1 or 3	1,000	270	248	270	248	---
	LSTNW- 300	300	3	300	1.957	---	9.0	STD-1 or 3	1,000	347	319	347	319	---
	LSTNW- 400	400	3	400	2.203	---	10.2	STD-1 or 3	1,000	441	405	441	405	---
24643/49 [4/]	LSTNWA- 3	16	3	2.580	0.461	1.107	1.5	STD-1	1,000	11	10	11	10	203-4034
	LSTNWA- 4	14	3	4.110	0.499	0.130	2.0	STD-1	1,000	18	17	18	17	203-3518
	LSTNWA- 9	10	3	10.380	0.675	0.271	2.5	STD-1	1,000	39	36	39	36	202-3519
	LSTNWA- 14	9	3	13.090	0.720	0.306	2.5	STD-1	1,000	51	47	51	47	202-3520
	LSTNWA- 23	7	3	20.820	0.810	0.427	3.5	STD-1	1,000	69	64	69	64	202-3521
	LSTNWA- 50	3	3	52.620	1.019	0.868	4.5	STD-1	1,000	110	101	110	101	202-3522
	LSTNWA- 75	1	3	83.690	1.184	1.276	5.0	STD-1	1,000	148	136	148	136	202-3523
	LSTNWA- 100	0	3	105.600	1.316	1.510	6.0	STD-1	1,000	174	160	174	160	202-3524
	LSTNWA- 150	000	3	167.800	1.565	2.277	7.0	STD-1	1,000	235	216	235	216	202-3525
24643/43	LS2SJ- 7	7	2	20.920	0.615	0.288	2.5	STD-3	600	56	49	56	49	202-3636
	LS2SJ- 9	9	2	13.080	0.545	0.232	2.5	STD-3	600	42	35	42	35	202-7749
	LS2SJ- 11	10	2	10.380	0.460	0.164	2.5	STD-3	600	31	25	16	14	202-6974

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TABLE VI (d)(1). MIL-DTL-24643 cable ratings and characteristics, non-watertight, non-flexing service (power and lighting) – Continued.

MIL-DTL-	Cable type designation	Conductor size: AWG or MCM	Number of conductors, pairs, or triads in cable	Area of each conductor (MCM)	Cable overall diameter max. (inches)	Cable weight per ft approx. (lbs)	Radius of bend min. (inches)	CDR ID	Rated voltage, max. (RMS)	Ampacity, each conductor (amps, max.)				NSN 6145-01
										DC or 60 Hz		400 Hz		
										40 °C Ambient	50 °C Ambient	40 °C Ambient	50 °C Ambient	
24643/43	LS2SJ- 12	12	2	6.530	0.430	0.134	2.0	STD-3	600	23	17	16	14	202-7748
	LS2SJ- 14	14	2	3.831	0.350	0.083	2.0	STD-3	600	16	14	16	14	202-7747
	LS2SJ- 16	16	2	2.426	0.325	0.074	2.0	STD-3	600	13	11	---	---	202-7746
	LS2SJ- 18	18	2	1.900	0.310	0.060	1.5	STD-3	600	10	8	---	---	202-6973
	LS2SJ- 20	20	2	1.216	0.290	0.056	1.5	STD-3	600	6	5	---	---	202-7769
	LS2SJ- 22	22	2	0.754	0.275	0.050	1.0	STD-3	600	3	2	---	---	205-3665
24643/43 [4/]	LS2SJA- 7	7	2	20.820	0.665	0.325	2.5	STD-3	600	56	49	56	49	205-3640
	LS2SJA- 9	9	2	13.090	0.595	0.271	2.5	STD-3	600	42	35	42	35	205-5176
	LS2SJA- 11	10	2	10.380	0.510	0.200	2.5	STD-3	600	31	25	16	14	205-3639
	LS2SJA- 12	12	2	6.530	0.480	0.163	2.0	STD-3	600	23	17	16	14	208-6876
	LS2SJA- 14	14	2	3.831	0.400	0.105	2.0	STD-3	600	16	14	16	14	203-0375
	LS2SJA- 16	16	2	2.426	0.397	0.093	2.0	STD-3	600	13	11	---	---	205-3638
	LS2SJA- 18	18	2	1.900	0.360	0.076	1.5	STD-3	600	10	8	---	---	203-0374
	LS2SJA- 20	20	2	1.216	0.340	0.071	1.5	STD-3	600	6	5	---	---	205-3667
	LS2SJA- 22	22	2	0.754	0.325	0.063	1.5	STD-3	600	3	2	---	---	205-3666
24643/43	LS3SJ- 9	9	3	13.090	0.620	0.310	3.5	STD-3	600	33	27	33	27	202-7753
	LS3SJ- 12	12	3	6.530	0.455	0.170	2.5	STD-3	600	21	15	21	15	202-7752
	LS3SJ- 14	14	3	3.831	0.370	0.101	2.0	STD-3	600	14	12	14	12	202-7751
	LS3SJ- 16	16	3	2.426	0.340	0.085	2.0	STD-3	600	11	10	11	10	205-3637
	LS3SJ- 18	18	3	1.900	0.325	0.076	2.0	STD-3	600	9	7	9	7	202-7750
	LS3SJ- 20	20	3	1.216	0.300	0.064	1.5	STD-3	600	6	5	6	5	202-7771
	LS3SJ- 22	22	3	0.754	0.285	0.054	1.5	STD-3	600	3	2	3	2	202-7770
24643/43 [4/]	LS3SJA- 9	9	3	13.090	0.670	0.350	3.5	STD-3	600	33	27	33	27	202-7754
	LS3SJA- 12	12	3	6.530	0.505	0.207	2.5	STD-3	600	21	15	21	15	202-6976
	LS3SJA- 14	14	3	3.831	0.420	0.128	2.0	STD-3	600	14	12	14	12	206-9745
	LS3SJA- 16	16	3	2.426	0.390	0.107	2.0	STD-3	600	11	10	11	10	205-3642
	LS3SJA- 18	18	3	1.900	0.375	0.096	2.0	STD-3	600	9	7	9	7	205-3641
	LS3SJA- 20	20	3	1.216	0.350	0.081	1.5	STD-3	600	6	5	6	5	205-3669

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TABLE VI (d)(1). MIL-DTL-24643 cable ratings and characteristics, non-watertight, non-flexing service (power and lighting) – Continued.

MIL-DTL-	Cable type designation	Conductor size: AWG or MCM	Number of conductors, pairs, or triads in cable	Area of each conductor (MCM)	Cable overall diameter max. (inches)	Cable weight per ft approx. (lbs)	Radius of bend min. (inches)	CDR ID	Rated voltage, max. (RMS)	Ampacity, each conductor (amps, max.)				NSN 6145-01
										DC or 60 Hz		400 Hz		
										40 °C Ambient	50 °C Ambient	40 °C Ambient	50 °C Ambient	
24643/43 [4/]	LS3SJA- 22	22	3	0.754	0.335	0.058	1.5	STD-3	600	3	2	3	2	205-3668
24643/43	LS4SJ- 14	14	4	3.831	0.395	0.128	2.5	STD-3	600	11	9	---	---	202-6975
	LS4SJ- 16	16	4	2.426	0.360	0.079	2.5	STD-3	600	9	7	---	---	204-4867
	LS4SJ- 20	20	4	1.216	0.320	0.048	2.0	STD-3	600	6	5	---	---	202-7772
24643/43 [4/]	LS4SJA- 14	14	4	3.831	0.445	0.158	2.5	STD-3	600	11	9	---	---	205-3643
	LS4SJA- 16	16	4	2.426	0.410	0.133	2.5	STD-3	600	9	7	---	---	202-6977
	LS4SJA- 20	20	4	1.216	0.370	0.091	2.0	STD-3	600	6	5	---	---	202-6999
24643/80	LST- 2	18	1	1.624	0.110	---	0.7	See spec sheet	1,000	13	12	13	12	---
	LST- 3	16	1	2.582	0.120	---	0.7	See spec sheet	1,000	16	15	16	15	---
	LST- 4	14	1	4.106	0.135	---	0.8	See spec sheet	1,000	23	21	23	21	---
	LST- 6	12	1	6.526	0.155	---	0.9	See spec sheet	1,000	27	25	27	25	---
	LST- 9	10	1	10.382	0.19	---	1.1	See spec sheet	1,000	36	34	36	34	---
	LST- 14	9	1	13.08	0.23	---	1.4	See spec sheet	1,000	44	40	44	40	---
	LST- 23	7	1	20.92	0.265	---	1.6	See spec sheet	1,000	59	55	59	55	---
	LST- 40	4	1	41.84	0.335	---	2	See spec sheet	1,000	86	80	86	80	---
	LST- 50	3	1	52.69	0.365	---	2.2	See spec sheet	1,000	105	97	105	97	---
	LST- 60	2	1	66.31	0.405	---	2.4	See spec sheet	1,000	118	109	118	109	---
	LST- 75	1	1	83.68	0.46	---	2.8	See spec sheet	1,000	132	122	132	122	---
	LST- 100	0	1	105.58	0.495	---	3	See spec sheet	1,000	155	143	155	143	---
	LST- 150	000	1	167.75	0.55	---	3.3	See spec sheet	1,000	205	189	205	189	---
	LST- 300	300	1	300	0.775	---	4.7	See spec sheet	1,000	291	269	291	269	---
	LST- 400	400 (Navy STD)	1	405.4	0.91	---	5.5	See spec sheet	1,000	346	319	346	319	---
24643/81	LSE2E- 3	16	2	2.582	0.44	---	2.6	STD-1 or 3	1,000	16	15	16	15	---
	LSE3E- 3	16	3	2.582	0.46	---	2.8	STD-1 or 3	1,000	16	15	16	15	---
	LSE4E- 3	16	4	2.582	0.49	---	2.9	STD-1 or 3	1,000	16	15	16	15	---
	LSE2E- 4	14	2	4.106	0.47	---	2.8	STD-1 or 3	1,000	23	21	23	21	---
	LSE3E- 4	14	3	4.106	0.49	---	2.9	STD-1 or 3	1,000	23	21	23	21	---
	LSE4E- 4	14	4	4.106	0.525	---	3.2	STD-1 or 3	1,000	23	21	23	21	---

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TABLE VI (d)(1). MIL-DTL-24643 cable ratings and characteristics, non-watertight, non-flexing service (power and lighting) – Continued.

MIL-DTL-	Cable type designation	Conductor size: AWG or MCM	Number of conductors, pairs, or triads in cable	Area of each conductor (MCM)	Cable overall diameter max. (inches)	Cable weight per ft approx. (lbs)	Radius of bend min. (inches)	CDR ID	Rated voltage, max. (RMS)	Ampacity, each conductor (amps, max.)				NSN 6145-01
										DC or 60 Hz		400 Hz		
										40 °C Ambient	50 °C Ambient	40 °C Ambient	50 °C Ambient	
24643/81	LSE2E- 9	10	2	10.382	0.575	---	3.5	STD-1 or 3	1,000	36	34	36	34	---
	LSE3E- 9	10	3	10.382	0.6	---	3.6	STD-1 or 3	1,000	36	34	36	34	---
	LSE4E- 9	10	4	10.382	0.66	---	4	STD-1 or 3	1,000	36	34	36	34	---
	LSE2E- 14	9	2	13.08	0.66	---	4	STD-1 or 3	1,000	44	40	44	40	---
	LSE3E- 14	9	3	13.08	0.7	---	4.2	STD-1 or 3	1,000	44	40	44	40	---
	LSE4E- 14	9	4	13.08	0.77	---	4.6	STD-1 or 3	1,000	44	40	44	40	---
	LSE2E2- 23	7	2	20.92	0.74	---	4.4	STD-1 or 3	1,000	59	55	59	55	---
	LSE3E- 23	7	3	20.92	0.795	---	4.8	STD-1 or 3	1,000	59	55	59	55	---
	LSE4E- 23	7	4	20.92	0.875	---	5.3	STD-1 or 3	1,000	59	55	59	55	---
	LSE2E- 50	3	2	52.69	0.99	---	5.9	STD-1 or 3	1,000	105	97	105	97	---
	LSE3E- 50	3	3	52.69	1.06	---	6.4	STD-1 or 3	1,000	105	97	105	97	---
	LSE2E- 75	1	2	83.68	1.22	---	7.3	STD-1 or 3	1,000	132	122	132	122	---
	LSE3E- 75	1	3	83.68	1.3	---	7.8	STD-1 or 3	1,000	132	122	132	122	---
	LSE2E- 100	0	2	105.58	1.305	---	7.8	STD-1 or 3	1,000	155	143	155	143	---
	LSE3E- 100	0	3	105.58	1.39	---	8.3	STD-1 or 3	1,000	155	143	155	143	---
	LSE2E- 150	000	2	167.75	1.43	---	8.6	STD-1 or 3	1,000	205	189	205	189	---
	LSE3E- 150	000	3	167.75	1.525	---	9.2	STD-1 or 3	1,000	205	189	205	189	---
	LSE1E- 300	300	1	300	1.05	---	6.3	White or natural	1,000	291	269	291	269	---
	LSE1E- 400	400 (Navy STD)	1	405.4	1.25	13	7.5	White or natural	1,000	346	319	346	319	---
	LSE2E- 400	400 (Navy STD)	2	405.4	2.22	13	13.3	STD-1 or 3	1,000	346	319	346	319	---
	LSE3E- 400	400 (Navy STD)	3	400	2.35	13	14.1	STD-1 or 3	1,000	346	319	346	319	---

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TABLE VI (d)(1). MIL-DTL-24643 cable ratings and characteristics, non-watertight, non-flexing service (power and lighting) – Continued.

MIL-DTL-	Cable type designation	Conductor size: AWG or MCM	Number of conductors, pairs, or triads in cable	Area of each conductor (MCM)	Cable overall diameter max. (inches)	Cable weight per ft approx. (lbs)	Radius of bend min. (inches)	CDR ID	Rated voltage, max. (RMS)	Ampacity, each conductor (amps, max.)				NSN 6145-01
										DC or 60 Hz		400 Hz		
										40 °C Ambient	50 °C Ambient	40 °C Ambient	50 °C Ambient	
24643/81	LSE19E- 2	18	19	1.624	0.78	---	4.7	STD-1 or 3	1,000	13	12	13	12	---
	LSE19E- 3	16	19	2.582	0.845	---	5.1	STD-1 or 3	1,000	16	15	16	15	---
	LSE19E- 6	12	19	6.526	1.055	---	6.3	STD-1 or 3	1,000	27	25	27	25	---
	LSE19E- 14	9	19	13.08	1.475	---	8.9	STD-1 or 3	1,000	44	40	44	40	---
	LSE19E- 23	7	19	20.92	1.69	---	10.1	STD-1 or 3	1,000	59	55	59	55	---
	LSE19E- 40	4	19	41.84	2.125	---	12.8	STD-1 or 3	1,000	86	80	86	80	---
	LSE19E- 60	2	19	66.31	2.525	---	15.2	STD-1 or 3	1,000	118	109	118	109	---

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TABLE VI (d)(2). MIL-DTL-24643 cable ratings and characteristics, non-watertight, non-flexing service (control) – Continued.

MIL-DTL-	Cable type designation	Conductor size: AWG or MCM	Number of conductors, pairs, or triads in cable	Area of each conductor (MCM)	Cable overall diameter max. (inches)	Cable weight per ft approx. (lbs)	Radius of bend min. (inches)	CDR ID	Rated voltage, max. (RMS)	Ampacity, each conductor (amps, max.)				NSN 6145-01
										DC or 60 Hz		400 Hz		
										40 °C Ambient	50 °C Ambient	40 °C Ambient	50 °C Ambient	
24643/55	LS4NW- 8	8	4	16.510	0.740	0.612	6.0	STD-1 or 3	1,000	---	---	---	---	210-2345
24643/55 [4/]	LS4NWA- 8	8	4	16.510	0.790	0.691	6.0	STD-1	1,000	---	---	---	---	205-9306
24643/54	LS8NW- 6	12	8	6.530	0.670	0.366	5.5	STD-1or 3	1,000	---	---	---	---	201-9059
24643/54 [4/]	LS8NWA- 6	12	8	6.530	0.720	0.413	5.5	STD-1	1,000	---	---	---	---	201-9060
24643/51	LSMNW- 7	18	7	1.620	0.400	0.088	1.5	STD-1 or 3	1,000	12/8 [1/]	9/6 [1/]	12/8 [1/]	9/6 [1/]	202-9019
	LSMNW- 10	18	10	1.620	0.495	0.127	1.5	STD-1 or 3	1,000	12/8	9/6	12/8	9/6	204-6997
	LSMNW- 14	18	14	1.620	0.535	0.155	2.0	STD-1 or 3	1,000	12/8	9/6	12/8	9/6	205-3644
	LSMNW- 19	18	19	1.620	0.590	0.198	2.0	STD-1 or 3	1,000	12/8	9/6	12/8	9/6	203-8503
	LSMNW- 24	18	24	1.620	0.685	0.259	2.5	STD-1 or 3	1,000	12/6	9/5	12/6	9/5	202-9020
	LSMNW- 30	18	30	1.620	0.725	0.300	2.5	STD-1 or 3	1,000	12/6	9/5	12/6	9/5	202-9021
	LSMNW- 37	18	37	1.620	0.785	0.361	3.0	STD-1 or 3	1,000	12/6	9/5	12/6	9/5	202-9022
	LSMNW- 44	18	44	1.620	0.890	0.447	3.5	STD-1 or 3	1,000	12/5	9/4	12/5	9/4	202-9023
24643/51 [4/]	LSMNWA- 7	18	7	1.620	0.450	0.111	1.5	STD-1	1,000	12/8	9/6	12/8	9/6	202-9024
	LSMNWA- 10	18	10	1.620	0.545	0.154	1.5	STD-1	1,000	12/8	9/6	12/8	9/6	202-9025
	LSMNWA- 14	18	14	1.620	0.585	0.181	2.0	STD-1	1,000	12/8	9/6	12/8	9/6	202-9026
	LSMNWA- 19	18	19	1.620	0.640	0.231	2.0	STD-1	1,000	12/8	9/6	12/8	9/6	203-9430
	LSMNWA- 24	18	24	1.620	0.735	0.292	2.5	STD-1	1,000	12/6	9/5	12/6	9/5	203-9431
	LSMNWA- 30	18	30	1.620	0.775	0.339	2.5	STD-1	1,000	12/6	9/5	12/6	9/5	202-9027
	LSMNWA- 37	18	37	1.620	0.835	0.395	3.0	STD-1	1,000	12/6	9/5	12/6	9/5	203-9432
	LSMNWA- 44	18	44	1.620	0.940	0.489	3.5	STD-1	1,000	12/5	9/4	12/5	9/4	202-6978
24643/34	LSMS- 37	16	37	2.426	0.800	0.618	5.0	STD-1	300	11/5	9/3	11/5	9/3	201-9522
24643/34 [4/]	LSMSA- 37	16	37	2.426	0.850	0.676	5.0	STD-1	300	11/5	9/3	11/5	9/3	202-6967
24643/29	LSMU- 14	20	14	1.111	0.400	0.132	3.0	STD-1	300	---	---	---	---	205-9320
24643/29	LSMUS- 14	20	14	1.111	0.460	0.217	5.5	STD-1	300	---	---	---	---	204-0570
24643/29 [4/]	LSMA- 14	20	14	1.111	0.450	0.160	3.0	STD-1	300	---	---	---	---	204-0669
24643/80	LST- 2	18	1	1.624	0.110	---	0.7	See spec sheet	1,000	13	12	13	12	---
	LST- 3	16	1	2.582	0.120	---	0.7	See spec sheet	1,000	16	15	16	15	---
	LST- 4	14	1	4.106	0.135	---	0.8	See spec sheet	1,000	23	21	23	21	---

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TABLE VI (d)(2). MIL-DTL-24643 cable ratings and characteristics, non-watertight, non-flexing service (control) – Continued.

MIL-DTL-	Cable type designation	Conductor size: AWG or MCM	Number of conductors, pairs, or triads in cable	Area of each conductor (MCM)	Cable overall diameter max. (inches)	Cable weight per ft approx. (lbs)	Radius of bend min. (inches)	CDR ID	Rated voltage, max. (RMS)	Ampacity, each conductor (amps, max.)				NSN 6145-01
										DC or 60 Hz		400 Hz		
										40 °C Ambient	50 °C Ambient	40 °C Ambient	50 °C Ambient	
24643/80	LST- 6	12	1	6.526	0.155	---	0.9	See spec sheet	1,000	27	25	27	25	---
	LST- 9	10	1	10.382	0.19	---	1.1	See spec sheet	1,000	36	34	36	34	---
	LST- 14	9	1	13.08	0.23	---	1.4	See spec sheet	1,000	44	40	44	40	---
	LST- 23	7	1	20.92	0.265	---	1.6	See spec sheet	1,000	59	55	59	55	---
	LST- 40	4	1	41.84	0.335	---	2	See spec sheet	1,000	86	80	86	80	---
	LST- 50	3	1	52.69	0.365	---	2.2	See spec sheet	1,000	105	97	105	97	---
	LST- 60	2	1	66.31	0.405	---	2.4	See spec sheet	1,000	118	109	118	109	---
	LST- 75	1	1	83.68	0.46	---	2.8	See spec sheet	1,000	132	122	132	122	---
	LST- 100	0	1	105.58	0.495	---	3	See spec sheet	1,000	155	143	155	143	---
	LST- 150	000	1	167.75	0.55	---	3.3	See spec sheet	1,000	205	189	205	189	---
	LST- 300	300	1	300	0.775	---	4.7	See spec sheet	1,000	291	269	291	269	---
	LST- 400	400 (Navy STD)	1	405.4	0.91	---	5.5	See spec sheet	1,000	346	319	346	319	---
24643/81	LSE14E- 2	18	14	1.624	0.705	---	4.2	STD-1 or 3	1,000	13	12	13	12	---
	LSE19E- 2	18	19	1.624	0.78	---	4.7	STD-1 or 3	1,000	13	12	13	12	---
	LSE24E- 2	18	24	1.624	0.915	---	5.5	STD-1 or 3	1,000	13	12	13	12	---
	LSE30E- 2	18	30	1.624	0.97	---	5.8	STD-1 or 3	1,000	13	12	13	12	---
	LSE37E- 2	18	37	1.624	1.055	---	6.3	STD-1 or 3	1,000	13	12	13	12	---
	LSE44E- 2	18	44	1.624	1.185	---	7.1	STD-1 or 3	1,000	13	12	13	12	---
	LSE7E- 2	18	7	1.624	0.53	---	3.2	STD-1 or 3	1,000	13	12	13	12	---
	LSE7E- 3	16	7	2.582	0.56	---	3.4	STD-1 or 3	1,000	16	15	16	15	---
	LSE10E- 3	16	10	2.582	0.7	---	4.2	STD-1 or 3	1,000	16	15	16	15	---
	LSE14E- 3	16	14	2.582	0.755	---	4.5	STD-1 or 3	1,000	16	15	16	15	---
	LSE19E- 3	16	19	2.582	0.845	---	5.1	STD-1 or 3	1,000	16	15	16	15	---
	LSE24E- 3	16	24	2.582	0.99	---	5.9	STD-1 or 3	1,000	16	15	16	15	---
	LSE30E- 3	16	30	2.582	1.05	---	6.3	STD-1 or 3	1,000	16	15	16	15	---
	LSE33E- 3	16	33	2.582	1.095	---	6.6	STD-1 or 3	1,000	16	15	16	15	---
	LSE37E- 3	16	37	2.582	1.135	---	6.8	STD-1 or 3	1,000	16	15	16	15	---

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TABLE VI (d)(2). MIL-DTL-24643 cable ratings and characteristics, non-watertight, non-flexing service (control) – Continued.

MIL-DTL-	Cable type designation	Conductor size: AWG or MCM	Number of conductors, pairs, or triads in cable	Area of each conductor (MCM)	Cable overall diameter max. (inches)	Cable weight per ft approx. (lbs)	Radius of bend min. (inches)	CDR ID	Rated voltage, max. (RMS)	Ampacity, each conductor (amps, max.)				NSN 6145-01
										DC or 60 Hz		400 Hz		
										40 °C Ambient	50 °C Ambient	40 °C Ambient	50 °C Ambient	
	LSE44E- 3	16	44	2.582	1.28	---	7.7	STD-1 or 3	1,000	16	15	16	15	---
	LSE61E- 3	16	61	2.582	1.425	---	8.6	STD-1 or 3	1,000	16	15	16	15	---
	LSE7E- 4	14	7	4.106	0.605	---	3.6	STD-1 or 3	1,000	23	21	23	21	---
	LSE19E- 6	12	19	6.526	1.055	---	6.3	STD-1 or 3	1,000	27	25	27	25	---
	LSE19E- 14	9	19	13.08	1.475	---	8.9	STD-1 or 3	1,000	44	40	44	40	---
	LSE19E- 23	7	19	20.92	1.69	---	10.1	STD-1 or 3	1,000	59	55	59	55	---
	LSE4E- 40	4	4	41.84	1.15	---	6.9	STD-1 or 3	1,000	86	80	86	80	---
	LSE19E- 40	4	19	41.84	2.125	---	12.8	STD-1 or 3	1,000	86	80	86	80	---
	LSE19E- 60	2	19	66.31	2.525	---	15.2	STD-1 or 3	1,000	118	109	118	109	---

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TABLE VI (d)(3). MIL-DTL-24643 cable ratings and characteristics, non-watertight, non-flexing service (electronic, communication, and instrumentation).

MIL-DTL-	Cable type designation	Conductor size: AWG or MCM	Number of conductors, pairs, or triads in cable	Area of each conductor (MCM)	Cable overall diameter max. (inches)	Cable weight per ft approx. (lbs)	Radius of bend min. (inches)	CDR ID	Rated voltage, max. (RMS)	Ampacity, each conductor (amps, max.)				NSN 6145-01
										DC or 60 Hz		400 Hz		
										40 °C Ambient	50 °C Ambient	40 °C Ambient	50 °C Ambient	
24643/28 [4/]	LS1S5OMA- 16	22	16	0.700	0.875	0.360	5.5	STD-1	300	---	---	---	---	205-3656
	LS1S5OMA- 20	22	20	0.700	0.955	0.417	6.0	STD-1	300	---	---	---	---	205-3657
	LS1S5OMA- 40	22	40	0.700	1.235	0.756	7.5	STD-1	300	---	---	---	---	205-3658
	LS1S5OMA- 70	22	70	0.700	1.605	1.798	10.0	STD-1	300	---	---	---	---	205-3659
24643/28	LS1S5OMU- 16	22	16	0.700	0.825	0.329	5.5	STD-2	300	---	---	---	---	202-9539
	LS1S5OMU- 20	22	20	0.700	0.905	0.381	6.0	STD-2	300	---	---	---	---	203-5465
	LS1S5OMU- 40	22	40	0.700	1.185	0.710	7.5	STD-2	300	---	---	---	---	203-6582
	LS1S5OMU- 70	22	70	0.700	1.555	1.751	10.5	STD-2	300	---	---	---	---	203-3308
24643/28	LS1S5OMUS- 16	22	16	0.700	0.885	0.444	10.5	STD-2	300	---	---	---	---	202-9540
	LS1S5OMUS- 20	22	20	0.700	0.965	0.514	11.5	STD-2	300	---	---	---	---	202-9541
	LS1S5OMUS- 40	22	40	0.700	1.245	0.887	15.0	STD-2	300	---	---	---	---	202-9542
	LS1S5OMUS- 70	22	70	0.700	1.615	2.188	19.5	STD-2	300	---	---	---	---	203-6583
24643/39 [4/]	LS1S75MA- 8	22	8	0.700	1.080	0.612	6.5	STD-2	300	---	---	---	---	202-3531
24643/39	LS1S75MU- 8	22	8	0.700	1.030	0.552	6.5	STD-2	300	---	---	---	---	202-8476
24643/41 [4/]	LS1SA- 44	20	44	1.111	1.040	0.698	6.5	STD-2	600	---	---	---	---	203-4069
24643/41	LS1SAU- 44	20	44	1.111	0.990	0.638	6.5	STD-2	600	---	---	---	---	205-5183
24643/40 [4/]	LS1SMA- 5	22	5	0.700	0.550	0.156	4.5	STD-2	600	---	---	---	---	202-0695
24643/40	LS1SMU- 5	22	5	0.700	0.500	0.128	3.0	STD-2	600	---	---	---	---	202-0694
24643/42 [6/]	LS1SU- 36	20	32	1.111	0.985	0.642	6.0	STD-2	600	---	---	---	---	203-4065
		18	4	1.620	---	---	---	---	---	---	---	---	---	---
	LS1SU- 60	20	60	1.111	1.310	1.060	10.0	STD-2	600	---	---	---	---	202-9047
24643/42 [4/]	LS1SUA- 36	20	32	1.111	1.035	0.702	6.0	STD-2	600	---	---	---	---	203-4066
		18	4	1.620	---	---	---	---	---	---	---	---	---	---
	LS1SUA- 60	20	60	1.111	1.360	1.102	10.0	STD-2	600	---	---	---	---	205-3664
24643/27 [4/]	LS2A- 40	22	40 pairs	0.700	1.420	0.741	4.0	TEL-3	600	---	---	---	---	202-8474
24643/27	LS2AU- 40	22	40 pairs	0.700	1.370	0.721	4.0	TEL-3	600	---	---	---	---	202-8473
	LS2AUS- 40	22	40 pairs	0.700	1.430	0.901	17.0	TEL-3	600	---	---	---	---	202-8475
24643/58	LS2CS- 6	26	6 pairs	0.278	0.430	0.126	5.5	STD-6	300	---	---	---	---	203-9466
	LS2CS- 18	26	18 pairs	0.278	0.590	0.235	7.0	STD-6	300	---	---	---	---	202-9058

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TABLE VI (d)(3). MIL-DTL-24643 cable ratings and characteristics, non-watertight, non-flexing service (electronic, communication, and instrumentation) – Continued.

MIL-DTL-	Cable type designation		Conductor size: AWG or MCM	Number of conductors, pairs, or triads in cable	Area of each conductor (MCM)	Cable overall diameter max. (inches)	Cable weight per ft approx. (lbs)	Radius of bend min. (inches)	CDR ID	Rated voltage, max. (RMS)	Ampacity, each conductor (amps, max.)				NSN 6145-01
											DC or 60 Hz		400 Hz		
											40 °C Ambient	50 °C Ambient	40 °C Ambient	50 °C Ambient	
24643/58	LS2CS-	42	26	42 pairs	0.278	0.800	0.419	9.5	STD-6	300	---	---	---	---	203-9467
	LS2CS-	60	26	60 pairs	0.278	0.930	0.562	11.5	STD-6	300	---	---	---	---	202-9059
	LS2CS-	77	26	77 pairs	0.278	1.070	0.651	13.0	STD-6	300	---	---	---	---	203-9468
24643/31 [4/]	LS2SA-	3	22	3 pairs	0.700	0.570	0.206	3.5	STD-2	600	---	---	---	---	202-9546
	LS2SA-	7	22	7 pairs	0.700	0.710	0.298	4.5	STD-2	600	---	---	---	---	203-7621
	LS2SA-	10	22	10 pairs	0.700	0.880	0.422	5.0	STD-2	600	---	---	---	---	203-7622
	LS2SA-	14	22	14 pairs	0.700	0.980	0.618	6.0	STD-2	600	---	---	---	---	204-7020
	LS2SA-	19	22	19 pairs	0.700	1.090	0.793	6.5	STD-2	600	---	---	---	---	203-7623
	LS2SA-	24	22	24 pairs	0.700	1.260	1.030	7.5	STD-2	600	---	---	---	---	203-0383
	LS2SA-	30	22	30 pairs	0.700	1.330	1.256	7.5	STD-2	600	---	---	---	---	203-0384
	LS2SA-	37	22	37 pairs	0.700	1.430	1.545	11.5	STD-2	600	---	---	---	---	203-0385
	LS2SA-	44	22	44 pairs	0.700	1.600	1.905	13.0	STD-2	600	---	---	---	---	203-0386
	LS2SA-	61	22	61 pairs	0.700	1.790	2.317	14.5	STD-2	600	---	---	---	---	203-4067
24643/31	LS2SU-	3	22	3 pairs	0.700	0.520	0.170	3.5	STD-2	600	---	---	---	---	203-0382
	LS2SU-	7	22	7 pairs	0.700	0.660	0.255	4.5	STD-2	600	---	---	---	---	202-9543
	LS2SU-	10	22	10 pairs	0.700	0.830	0.368	5.0	STD-2	600	---	---	---	---	202-9544
	LS2SU-	14	22	14 pairs	0.700	0.930	0.558	6.0	STD-2	600	---	---	---	---	204-8939
	LS2SU-	19	22	19 pairs	0.700	1.040	0.732	6.5	STD-2	600	---	---	---	---	203-7618
	LS2SU-	24	22	24 pairs	0.700	1.210	0.953	7.5	STD-2	600	---	---	---	---	204-2281
	LS2SU-	30	22	30 pairs	0.700	1.280	1.176	8.0	STD-2	600	---	---	---	---	203-7619
	LS2SU-	37	22	37 pairs	0.700	1.380	1.459	8.5	STD-2	600	---	---	---	---	202-9545
	LS2SU-	44	22	44 pairs	0.700	1.550	1.809	9.0	STD-2	600	---	---	---	---	204-4871
	LS2SU-	61	22	61 pairs	0.700	1.740	2.209	10.0	STD-2	600	---	---	---	---	203-7620
	LS2SUS-	3	22	3 pairs	0.700	0.580	0.261	7.0	STD-2	600	---	---	---	---	203-0387
	LS2SUS-	7	22	7 pairs	0.700	0.720	0.367	8.5	STD-2	600	---	---	---	---	204-4265
	LS2SUS-	10	22	10 pairs	0.700	0.890	0.496	11.0	STD-2	600	---	---	---	---	203-0388
	LS2SUS-	14	22	14 pairs	0.700	0.990	0.753	12.0	STD-2	600	---	---	---	---	202-9547

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TABLE VI (d)(3). MIL-DTL-24643 cable ratings and characteristics, non-watertight, non-flexing service (electronic, communication, and instrumentation) – Continued.

MIL-DTL-	Cable type designation		Conductor size: AWG or MCM	Number of conductors, pairs, or triads in cable	Area of each conductor (MCM)	Cable overall diameter max. (inches)	Cable weight per ft approx. (lbs)	Radius of bend min. (inches)	CDR ID	Rated voltage, max. (RMS)	Ampacity, each conductor (amps, max.)				NSN 6145-01
											DC or 60 Hz		400 Hz		
											40 °C Ambient	50 °C Ambient	40 °C Ambient	50 °C Ambient	
24643/31	LS2SUS-	19	22	19 pairs	0.700	1.100	0.915	13.5	STD-2	600	---	---	---	---	203-0389
	LS2SUS-	24	22	24 pairs	0.700	1.270	1.191	15.5	STD-2	600	---	---	---	---	202-9548
	LS2SUS-	30	22	30 pairs	0.700	1.340	1.470	16.0	STD-2	600	---	---	---	---	202-9549
	LS2SUS-	37	22	37 pairs	0.700	1.440	1.823	17.5	STD-2	600	---	---	---	---	202-9550
	LS2SUS-	44	22	44 pairs	0.700	1.610	2.080	19.5	STD-2	600	---	---	---	---	202-9551
	LS2SUS-	61	22	61 pairs	0.700	1.800	2.540	22.0	STD-2	600	---	---	---	---	203-0390
24643/45	LS2U-	10	26	10 pairs	0.278	0.480	0.118	4.0	TEL-3	300	---	---	---	---	202-8477
	LS2U-	15	26	15 pairs	0.278	0.560	0.164	5.0	TEL-3	300	---	---	---	---	202-8478
	LS2U-	19	26	19 pairs	0.278	0.580	0.185	6.0	TEL-3	300	---	---	---	---	202-9048
	LS2U-	30	26	30 pairs	0.278	0.700	0.257	7.0	TEL-3	300	---	---	---	---	202-9049
	LS2U-	45	26	45 pairs	0.278	0.870	0.360	8.5	TEL-3	300	---	---	---	---	202-9050
	LS2U-	60	26	60 pairs	0.278	0.960	0.442	10.0	TEL-3	300	---	---	---	---	202-9051
24643/45 [4/]	LS2UA-	10	26	10 pairs	0.278	0.530	0.143	4.0	TEL-3	300	---	---	---	---	202-9052
	LS2UA-	15	26	15 pairs	0.278	0.610	0.191	5.0	TEL-3	300	---	---	---	---	202-9053
	LS2UA-	19	26	19 pairs	0.278	0.630	0.216	6.0	TEL-3	300	---	---	---	---	202-9054
	LS2UA-	30	26	30 pairs	0.278	0.750	0.290	7.0	TEL-3	300	---	---	---	---	202-9055
	LS2UA-	45	26	45 pairs	0.278	0.920	0.394	8.5	TEL-3	300	---	---	---	---	202-9056
	LS2UA-	60	26	60 pairs	0.278	1.010	0.483	10.0	TEL-3	300	---	---	---	---	202-9057
24643/35 [4/]	LS3SA-	3	18	3 triads	1.620	0.750	0.309	4.5	STD-2	600	---	---	---	---	202-9509
	LS3SA-	7	18	7 triads	1.620	0.960	0.587	6.0	STD-2	600	---	---	---	---	203-5394
	LS3SA-	10	18	10 triads	1.620	1.240	0.865	7.5	STD-2	600	---	---	---	---	202-3489
	LS3SA-	14	18	14 triads	1.620	1.340	1.133	8.5	STD-2	600	---	---	---	---	202-3512
	LS3SA-	19	18	19 triads	1.620	1.480	1.442	9.0	STD-2	600	---	---	---	---	202-0689
	LS3SA-	24	18	24 triads	1.620	1.720	1.905	10.5	STD-2	600	---	---	---	---	202-6968
	LS3SA-	30	18	30 triads	1.620	1.820	2.266	11.0	STD-2	600	---	---	---	---	202-6969
	LS3SA-	37	18	37 triads	1.620	1.980	2.781	12.0	STD-2	600	---	---	---	---	205-5174
	LS3SA-	44	18	44 triads	1.620	2.200	3.399	13.0	STD-2	600	---	---	---	---	202-9510

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TABLE VI (d)(3). MIL-DTL-24643 cable ratings and characteristics, non-watertight, non-flexing service (electronic, communication, and instrumentation) – Continued.

MIL-DTL-	Cable type designation		Conductor size: AWG or MCM	Number of conductors, pairs, or triads in cable	Area of each conductor (MCM)	Cable overall diameter max. (inches)	Cable weight per ft approx. (lbs)	Radius of bend min. (inches)	CDR ID	Rated voltage, max. (RMS)	Ampacity, each conductor (amps, max.)				NSN 6145-01
											DC or 60 Hz		400 Hz		
											40 °C Ambient	50 °C Ambient	40 °C Ambient	50 °C Ambient	
24643/35	LS3SU-	3	18	3 triads	1.620	0.700	0.262	4.5	STD-2	600	---	---	---	---	202-3487
	LS3SU-	7	18	7 triads	1.620	0.910	0.528	6.5	STD-2	600	---	---	---	---	203-330
	LS3SU-	10	18	10 triads	1.620	1.190	0.791	7.5	STD-2	600	---	---	---	---	202-8467
	LS3SU-	14	18	14 triads	1.620	1.290	1.054	8.5	STD-2	600	---	---	---	---	202-8468
	LS3SU-	19	18	19 triads	1.620	1.430	1.353	9.0	STD-2	600	---	---	---	---	202-3488
	LS3SU-	24	18	24 triads	1.620	1.670	1.799	10.5	STD-2	600	---	---	---	---	202-8469
	LS3SU-	30	18	30 triads	1.620	1.770	2.266	11.0	STD-2	600	---	---	---	---	202-9508
	LS3SU-	37	18	37 triads	1.620	1.930	2.664	12.0	STD-2	600	---	---	---	---	202-8470
	LS3SU-	44	18	44 triads	1.620	2.150	3.273	13.0	STD-2	600	---	---	---	---	203-0372
	LS3SUS-	3	18	3 triads	1.620	0.760	0.444	9.5	STD-2	600	---	---	---	---	202-6970
	LS3SUS-	7	18	7 triads	1.620	0.970	0.792	12.0	STD-2	600	---	---	---	---	202-0690
	LS3SUS-	10	18	10 triads	1.620	1.250	1.081	15.0	STD-2	600	---	---	---	---	202-0691
	LS3SUS-	14	18	14 triads	1.620	1.350	1.416	16.5	STD-2	600	---	---	---	---	202-3490
	LS3SUS-	19	18	19 triads	1.620	1.490	1.802	18.0	STD-2	600	---	---	---	---	202-6971
	LS3SUS-	24	18	24 triads	1.620	1.730	2.190	21.0	STD-2	600	---	---	---	---	204-4859
	LS3SUS-	30	18	30 triads	1.620	1.830	2.605	22.0	STD-2	600	---	---	---	---	204-4860
	LS3SUS-	37	18	37 triads	1.620	1.990	3.301	24.0	STD-2	600	---	---	---	---	202-6972
	LS3SUS-	44	18	44 triads	1.620	2.210	3.908	26.5	STD-2	600	---	---	---	---	204-4861
24643/37	LS3U-	3	18	3 triads	1.620	0.620	0.123	4.5	STD-4	300	---	---	---	---	203-9429
	LS3U-	7	18	7 triads	1.620	0.810	0.288	5.5	STD-4	300	---	---	---	---	205-3635
	LS3U-	12	18	12 triads	1.620	1.090	0.491	7.5	STD-4	600	---	---	---	---	204-4864
24643/37 [4/]	LS3UA-	3	18	3 triads	1.620	0.670	0.138	5.5	STD-4	300	---	---	---	---	204-4865
	LS3UA-	7	18	7 triads	1.620	0.860	0.315	5.5	STD-4	300	---	---	---	---	204-4866
	LS3UA-	12	18	12 triads	1.620	1.140	0.522	7.5	STD-4	300	---	---	---	---	204-6996
24643/9	LSMRI-	D1	20	2	1.020	0.075	0.010	---	STD-3	600	3	3	---	---	202-7764
	LSMRI-	D2.5	16	2	2.580	0.102	0.022	---	STD-3	600	8	6	---	---	208-5455
	LSMRI-	T2.5	16	3	2.580	0.102	0.033	---	STD-3	600	7	6	---	---	205-3627

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TABLE VI (d)(3). MIL-DTL-24643 cable ratings and characteristics, non-watertight, non-flexing service (electronic, communication, and instrumentation) – Continued.

MIL-DTL-	Cable type designation	Conductor size: AWG or MCM	Number of conductors, pairs, or triads in cable	Area of each conductor (MCM)	Cable overall diameter max. (inches)	Cable weight per ft approx. (lbs)	Radius of bend min. (inches)	CDR ID	Rated voltage, max. (RMS)	Ampacity, each conductor (amps, max.)				NSN 6145-01
										DC or 60 Hz		400 Hz		
										40 °C Ambient	50 °C Ambient	40 °C Ambient	50 °C Ambient	
24643/52	LSTPNW- 1.5	22	1 triad	0.640	0.235	0.024	1.0	TEL-3	300	---	---	---	---	205-3670
	LSTPNW- 3	22	3 pairs	0.640	0.310	0.039	1.0	TEL-3	300	---	---	---	---	205-3671
	LSTPNW- 5	22	5 pairs	0.640	0.365	0.059	1.5	TEL-3	300	---	---	---	---	205-3672
	LSTPNW- 10	22	10 pairs	0.640	0.470	0.104	1.5	TEL-3	300	---	---	---	---	206-7518
	LSTPNW- 15	22	15 pairs	0.640	0.530	0.138	2.0	TEL-3	300	---	---	---	---	205-3673
	LSTPNW- 20	22	20 pairs	0.640	0.575	0.172	2.0	TEL-3	300	---	---	---	---	205-3674
	LSTPNW- 30	22	30 pairs	0.640	0.680	0.255	2.5	TEL-3	300	---	---	---	---	201-8819
	LSTPNW- 40	22	40 pairs	0.640	0.765	0.328	3.0	TEL-3	300	---	---	---	---	201-8820
24643/52 [4/]	LSTPNWA- 1.5	22	3	0.640	0.285	0.032	1.0	TEL-3	300	---	---	---	---	201-8821
	LSTPNWA- 3	22	6	0.640	0.360	0.062	1.0	TEL-3	300	---	---	---	---	201-9546
	LSTPNWA- 5	22	10	0.640	0.415	0.074	1.5	TEL-3	300	---	---	---	---	201-8822
	LSTPNWA- 10	22	20	0.640	0.520	0.126	1.5	TEL-3	300	---	---	---	---	201-9547
	LSTPNWA- 15	22	30	0.640	0.580	0.161	2.0	TEL-3	300	---	---	---	---	201-9548
	LSTPNWA- 20	22	40	0.640	0.625	0.201	2.0	TEL-3	300	---	---	---	---	201-9549
	LSTPNWA- 30	22	60	0.640	0.730	0.288	2.5	TEL-3	300	---	---	---	---	201-9550
	LSTPNWA- 40	22	80	0.640	0.815	0.370	3.0	TEL-3	300	---	---	---	---	205-9321
24643/43	LS2SJ- 7	7	2	20.920	0.615	0.288	2.5	STD-3	600	56	49	56	49	202-3636
	LS2SJ- 9	9	2	13.080	0.545	0.232	2.5	STD-3	600	42	35	42	35	202-7749
	LS2SJ- 11	10	2	10.380	0.460	0.164	2.5	STD-3	600	31	25	16	14	202-6974
	LS2SJ- 12	12	2	6.530	0.430	0.134	2.0	STD-3	600	23	17	16	14	202-7748
	LS2SJ- 14	14	2	3.831	0.350	0.083	2.0	STD-3	600	16	14	16	14	202-7747
	LS2SJ- 16	16	2	2.426	0.325	0.074	2.0	STD-3	600	13	11	---	---	202-7746
	LS2SJ- 18	18	2	1.900	0.310	0.060	1.5	STD-3	600	10	8	---	---	202-6973
	LS2SJ- 20	20	2	1.216	0.290	0.056	1.5	STD-3	600	6	5	---	---	202-7769
	LS2SJ- 22	22	2	0.754	0.275	0.050	1.0	STD-3	600	3	2	---	---	205-3665
	LS3SJ- 9	9	3	13.090	0.620	0.310	3.5	STD-3	600	33	27	33	27	202-7753
	LS3SJ- 12	12	3	6.530	0.455	0.170	2.5	STD-3	600	21	15	21	15	202-7752

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TABLE VI (d)(3). MIL-DTL-24643 cable ratings and characteristics, non-watertight, non-flexing service (electronic, communication, and instrumentation) – Continued.

MIL-DTL-	Cable type designation		Conductor size: AWG or MCM	Number of conductors, pairs, or triads in cable	Area of each conductor (MCM)	Cable overall diameter max. (inches)	Cable weight per ft approx. (lbs)	Radius of bend min. (inches)	CDR ID	Rated voltage, max. (RMS)	Ampacity, each conductor (amps, max.)				NSN 6145-01
											DC or 60 Hz		400 Hz		
											40 °C Ambient	50 °C Ambient	40 °C Ambient	50 °C Ambient	
24643/43	LS3SJ-	14	14	3	3.831	0.370	0.101	2.0	STD-3	600	14	12	14	12	202-7751
	LS3SJ-	16	16	3	2.426	0.340	0.085	2.0	STD-3	600	11	10	11	10	205-3637
	LS3SJ-	18	18	3	1.900	0.325	0.076	2.0	STD-3	600	9	7	9	7	202-7750
	LS3SJ-	20	20	3	1.216	0.300	0.064	1.5	STD-3	600	6	5	6	5	202-7771
	LS3SJ-	22	22	3	0.754	0.285	0.054	1.5	STD-3	600	3	2	3	2	202-7770
	LS4SJ-	14	14	4	3.831	0.395	0.128	2.5	STD-3	600	11	9	---	---	202-6975
	LS4SJ-	16	16	4	2.426	0.360	0.079	2.5	STD-3	600	9	7	---	---	204-4867
	LS4SJ-	20	20	4	1.216	0.320	0.048	2.0	STD-3	600	6	5	---	---	202-7772
24643/43 [4/]	LS2SJA-	7	7	2	20.820	0.665	0.325	2.5	STD-3	600	56	49	56	49	205-3640
	LS2SJA-	9	9	2	13.090	0.595	0.271	2.5	STD-3	600	42	35	42	35	205-5176
	LS2SJA-	11	10	2	10.380	0.510	0.200	2.5	STD-3	600	31	25	16	14	205-3639
	LS2SJA-	12	12	2	6.530	0.480	0.163	2.0	STD-3	600	23	17	16	14	208-6876
	LS2SJA-	14	14	2	3.831	0.400	0.105	2.0	STD-3	600	16	14	16	14	203-0375
	LS2SJA-	16	16	2	2.426	0.397	0.093	2.0	STD-3	600	13	11	---	---	205-3638
	LS2SJA-	18	18	2	1.900	0.360	0.076	1.5	STD-3	600	10	8	---	---	203-0374
	LS2SJA-	20	20	2	1.216	0.340	0.071	1.5	STD-3	600	6	5	---	---	205-3667
	LS2SJA-	22	22	2	0.754	0.325	0.063	1.5	STD-3	600	3	2	---	---	205-3666
	LS3SJA-	9	9	3	13.090	0.670	0.350	3.5	STD-3	600	33	27	33	27	202-7754
	LS3SJA-	12	12	3	6.530	0.505	0.207	2.5	STD-3	600	21	15	21	15	202-6976
	LS3SJA-	14	14	3	3.831	0.420	0.128	2.0	STD-3	600	14	12	14	12	206-9745
	LS3SJA-	16	16	3	2.426	0.390	0.107	2.0	STD-3	600	11	10	11	10	205-3642
	LS3SJA-	18	18	3	1.900	0.375	0.096	2.0	STD-3	600	9	7	9	7	205-3641
	LS3SJA-	20	20	3	1.216	0.350	0.081	1.5	STD-3	600	6	5	6	5	205-3669
	LS3SJA-	22	22	3	0.754	0.335	0.058	1.5	STD-3	600	3	2	3	2	205-3668
	LS4SJA-	14	14	4	3.831	0.445	0.158	2.5	STD-3	600	11	9	---	---	205-3643
	LS4SJA-	16	16	4	2.426	0.410	0.133	2.5	STD-3	600	9	7	---	---	202-6977
	LS4SJA-	20	20	4	1.216	0.370	0.091	2.0	STD-3	600	6	5	---	---	202-6999

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TABLE VI (d)(3). MIL-DTL-24643 cable ratings and characteristics, non-watertight, non-flexing service (electronic, communication, and instrumentation) – Continued.

MIL-DTL-	Cable type designation	Conductor size: AWG or MCM	Number of conductors, pairs, or triads in cable	Area of each conductor (MCM)	Cable overall diameter max. (inches)	Cable weight per ft approx. (lbs)	Radius of bend min. (inches)	CDR ID	Rated voltage, max. (RMS)	Ampacity, each conductor (amps, max.)				NSN 6145-01
										DC or 60 Hz		400 Hz		
										40 °C Ambient	50 °C Ambient	40 °C Ambient	50 °C Ambient	
24643/58	LS2CS- 6	26	6 pairs	0.278	0.43	0.126	2.6	STD-6	300	---	---	---	---	203-9466
	LS2CS- 18	26	18 pairs	0.278	0.59	0.235	3.5	STD-6	300	---	---	---	---	202-9058
	LS2CS- 42	26	42 pairs	0.278	0.8	0.419	4.8	STD-6	300	---	---	---	---	203-9467
	LS2CS- 60	26	60 pairs	0.278	0.93	0.562	5.6	STD-6	300	---	---	---	---	202-9059
	LS2CS- 77	26	77 pairs	0.278	1.07	0.651	6.4	STD-6	300	---	---	---	---	203-9468
24643/59	LSC5OS- 4	24	4 pairs	0.404	0.4	0.068	2.4	TIA 568.2	150	---	---	---	---	---
	LSC5FS- 4	24	4 pairs	0.404	0.36	0.055	2.2	TIA 568.2	150	---	---	---	---	546-5154
24643/61	LSC5P- 4	24	4 pairs	0.448	0.325	0.059	2	TIA 568.2	150	---	---	---	---	---
	LSC5POS- 4	24	4 pairs	0.448	0.37	0.059	2.2	TIA 568.2	150	---	---	---	---	575-8783
	LSC5POSR- 4	26	4 pairs	0.278	0.325	0.059	2	TIA 568.2	150	---	---	---	---	---
24643/62	LSPB2SDOS- 1	22C/24D	1 pair/ 1 drain	0.642C/ 0.404D	0.5	---	3	Red/Green	150	100	100	---	---	---
	LSPB2SD- 1	22C/24D	1 pair/ 1 drain	0.642C/ 0.404D	0.39	0.059	2.3	Red/Green	150	---	---	---	---	---
24643/64	LS1TPSJ- 24	24	1 pair	0.475	0.17	0.022	1	STD-6	600	---	---	---	---	---
	LS1TPSJ- 22	22	1 pair	0.754	0.182	0.037	1.1	STD-6	600	---	---	---	---	595-7155
	LS1TPSJ- 20	20	1 pair	1	0.2	0.042	1.2	STD-6	600	---	---	---	---	---
	LS1TPSJ- 18	18	1 pair	1.9	0.22	---	1.3	STD-6	600	13	12	13	12	---
	LS1TPSJ- 16	16	1 pair	2.426	0.234	0.055	1.4	STD-6	600	16	15	16	15	---
	LS1TPSJ- 14	14	1 pair	3.831	0.262	---	1.6	STD-6	600	23	21	23	21	---
	LS2TPSJ- 24	24	2 pairs	0.475	0.26	0.05	1.6	STD-6	600	---	---	---	---	---
	LS2TPSJ- 22	22	2 pairs	0.754	0.285	---	1.7	STD-6	600	---	---	---	---	---
	LS2TPSJ- 20	20	2 pairs	1	0.325	---	2	STD-6	600	---	---	---	---	---
	LS2TPSJ- 18	18	2 pairs	1.9	0.39	---	2.3	STD-6	600	13	12	13	12	---
	LS2TPSJ- 16	16	2 pairs	2.426	0.42	---	2.5	STD-6	600	16	15	16	15	---
	LS2TPSJ- 14	14	2 pairs	3.831	0.465	---	2.8	STD-6	600	23	21	23	21	---
	LS3TPSJ- 24	24	3 pairs	0.475	0.275	---	1.7	STD-6	600	---	---	---	---	---
	LS3TPSJ- 22	22	3 pairs	0.754	0.285	---	1.7	STD-6	600	---	---	---	---	---

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TABLE VI (d)(3). MIL-DTL-24643 cable ratings and characteristics, non-watertight, non-flexing service (electronic, communication, and instrumentation) – Continued.

MIL-DTL-	Cable type designation	Conductor size: AWG or MCM	Number of conductors, pairs, or triads in cable	Area of each conductor (MCM)	Cable overall diameter max. (inches)	Cable weight per ft approx. (lbs)	Radius of bend min. (inches)	CDR ID	Rated voltage, max. (RMS)	Ampacity, each conductor (amps, max.)				NSN 6145-01
										DC or 60 Hz		400 Hz		
										40 °C Ambient	50 °C Ambient	40 °C Ambient	50 °C Ambient	
24643/64	LS3TPSJ- 20	20	3 pairs	1	0.325	0.085	2	STD-6	600	---	---	---	---	---
	LS3TPSJ- 18	18	3 pairs	1.9	0.39	---	2.3	STD-6	600	13	12	13	12	---
	LS3TPSJ- 16	16	3 pairs	2.426	0.45	0.15	2.7	STD-6	600	16	15	16	15	---
	LS3TPSJ- 14	14	3 pairs	3.831	0.49	---	2.9	STD-6	600	23	21	23	21	---
	LS4TPSJ- 24	24	4 pairs	0.475	0.295	---	1.8	STD-6	600	---	---	---	---	---
	LS4TPSJ- 22	22	4 pairs	0.754	0.325	---	2	STD-6	600	---	---	---	---	---
	LS4TPSJ- 20	20	4 pairs	1	0.395	0.087	2.4	STD-6	600	---	---	---	---	---
	LS4TPSJ- 18	18	4 pairs	1.9	0.45	---	2.7	STD-6	600	13	12	13	12	---
	LS4TPSJ- 16	16	4 pairs	2.426	0.48	---	2.9	STD-6	600	16	15	16	15	---
	LS4TPSJ- 14	14	4 pairs	3.831	0.54	---	3.2	STD-6	600	23	21	23	21	---
24643/65	LS3C179DT- 3	28	3 coaxes	0.16	0.365	0.085	2.2	STD-1 or 3	150	---	---	---	---	---
	LS6C179DT- 6	28	6 coaxes	0.16	0.455	0.128	2.7	STD-1 or 3	150	---	---	---	---	---
24643/75	LSLCFS- 25	24	12 pairs/1 single	0.4	0.53	0.122	3.2	STD-3	300	---	---	---	---	---
	LSLCF- 50	24	25 pairs	0.4	0.57	---	3.4	STD-3	300	---	---	---	---	---
24643/77	LSC6FS- 4	23	4 pairs	0.509	0.38	---	2.3	ANSI/ TIA 568.2	150	---	---	---	---	---
	LSC6OS- 4	23	4 pairs	0.509	0.4	0.074	2.4	ANSI/ TIA 568.2	150	---	---	---	---	---
24643/78	LSCB1PSJ- 22	22	1 pair	0.7	0.257	---	1.5	Green/Yellow	300	---	---	---	---	---
	LSCB2PSJ- 22	22	2 pairs	0.7	0.356	---	2.1	P1- Green/Yellow P2- Brown/White	300	---	---	---	---	---
	LSCB1PSJ- 20	20	1 pair	1.111	0.293	---	1.8	Green/Yellow	300	---	---	---	---	---
	LSCB2PSJ- 20	20	2 pairs	1.111	0.43	---	2.6	P1- Green/Yellow P2- Brown/White	300	---	---	---	---	---

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TABLE VI (d)(3). MIL-DTL-24643 cable ratings and characteristics, non-watertight, non-flexing service (electronic, communication, and instrumentation) – Continued.

MIL-DTL-	Cable type designation	Conductor size: AWG or MCM	Number of conductors, pairs, or triads in cable	Area of each conductor (MCM)	Cable overall diameter max. (inches)	Cable weight per ft approx. (lbs)	Radius of bend min. (inches)	CDR ID	Rated voltage, max. (RMS)	Ampacity, each conductor (amps, max.)				NSN 6145-01
										DC or 60 Hz		400 Hz		
										40 °C Ambient	50 °C Ambient	40 °C Ambient	50 °C Ambient	
24643/78	LSCB1PSJ- 18	18	1 pair	1.9	0.351	---	2.1	Green/Yellow	300	---	---	---	---	---
	LSCB2PSJ- 18	18	2 pairs	1.9	0.526	---	3.2	P1- Green/Yellow P2- Brown/White	300	---	---	---	---	---
24643/82	LSE6U- 26	26	6 pairs	0.278	0.495	---	3	STD-3	300	---	---	---	---	---
	LSE10U- 26	26	10 pairs	0.278	0.57	---	3.4	STD-3	300	---	---	---	---	---
	LSE15U- 26	26	15 pairs	0.278	0.615	---	3.7	STD-3	300	---	---	---	---	---
	LSE19U- 26	26	19 pairs	0.278	0.65	---	3.9	STD-3	300	---	---	---	---	---
	LSE30U- 26	26	30 pairs	0.278	0.755	---	4.5	STD-3	300	---	---	---	---	---
	LSE45U- 26	26	45 pairs	0.278	0.9	---	5.4	STD-3	300	---	---	---	---	---
	LSE60U- 26	26	60 pairs	0.278	0.98	---	5.9	STD-3	300	---	---	---	---	---
	LSE7AU- 22	22	7 pairs	0.642	0.665	---	4	STD-3	300	---	---	---	---	---
	LSE19AU- 22	22	19 pairs	0.642	0.95	---	5.7	STD-3	300	---	---	---	---	---
	LSE40AU- 22	22	40 pairs	0.642	1.305	---	7.8	STD-3	300	---	---	---	---	---
	LSE3TTOP- 20	20	3 pairs	1.021	0.57	---	3.4	STD-3	300	---	---	---	---	---
	LSE5TTOP- 20	20	5 pairs	1.021	0.66	---	4	STD-3	300	---	---	---	---	---
	LSE10TTOP- 20	20	10 pairs	1.021	0.81	---	4.9	STD-3	300	---	---	---	---	---
	LSE15TTOP- 20	20	15 pairs	1.021	0.905	---	5.4	STD-3	300	---	---	---	---	---
24643/83	LSE3SPE- 22	22	3 pairs	0.754	0.47	---	2.8	STD-2	300	---	---	---	---	---
	LSE7SPE- 22	22	7 pairs	0.754	0.555	---	3.3	STD-2	300	---	---	---	---	---
	LSE10SPE- 22	22	10 pairs	0.754	0.7	---	4.2	STD-2	300	---	---	---	---	---
	LSE14SPE- 22	22	14 pairs	0.754	0.755	---	4.5	STD-2	300	---	---	---	---	---
	LSE19SPE- 22	22	19 pairs	0.754	0.845	---	5.1	STD-2	300	---	---	---	---	---
	LSE24SPE- 22	22	24 pairs	0.754	0.995	---	6	STD-2	300	---	---	---	---	---
	LSE30SPE- 22	22	30 pairs	0.754	1.06	---	6.4	STD-2	300	---	---	---	---	---
	LSE37SPE- 22	22	37 pairs	0.754	1.145	---	6.9	STD-2	300	---	---	---	---	---
	LSE44SPE- 22	22	44 pairs	0.754	1.285	---	7.7	STD-2	300	---	---	---	---	---
	LSE61SPE- 22	22	61 pairs	0.754	1.425	---	8.6	STD-2	300	---	---	---	---	---

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TABLE VI (d)(3). MIL-DTL-24643 cable ratings and characteristics, non-watertight, non-flexing service (electronic, communication, and instrumentation) – Continued.

MIL-DTL-	Cable type designation	Conductor size: AWG or MCM	Number of conductors, pairs, or triads in cable	Area of each conductor (MCM)	Cable overall diameter max. (inches)	Cable weight per ft approx. (lbs)	Radius of bend min. (inches)	CDR ID	Rated voltage, max. (RMS)	Ampacity, each conductor (amps, max.)				NSN 6145-01
										DC or 60 Hz		400 Hz		
										40 °C Ambient	50 °C Ambient	40 °C Ambient	50 °C Ambient	
24643/83	LSE1SPE- 20	20	1 pair	1	0.345	---	2.1	STD-2	300	---	---	---	---	---
	LSE2SPE- 20	20	2 pairs	1	0.485	---	2.9	STD-2	300	---	---	---	---	---
	LSE4SPE- 20	20	4 pairs	1	0.53	---	3.2	STD-2	300	---	---	---	---	---
	LSE6SPE- 20	20	6 pairs	1	0.61	---	3.7	STD-2	300	---	---	---	---	---
	LSE8SPE- 20	20	8 pairs	1	0.67	---	4	STD-2	300	---	---	---	---	---
	LSE10SPE- 20	20	10 pairs	1	0.78	---	4.7	STD-2	300	---	---	---	---	---
	LSE12SPE- 20	20	12 pairs	1	0.81	---	4.9	STD-2	300	---	---	---	---	---
	LSE16SPE- 20	20	16 pairs	1	0.9	---	5.4	STD-2	300	---	---	---	---	---
	LSE1SPE- 18	18	1 pair	1.62	0.355	---	2.1	STD-2	300	---	---	---	---	---
	LSE3SPE- 18	18	3 pairs	1.62	0.51	---	3.1	STD-2	300	---	---	---	---	---
	LSE7SPE- 18	18	7 pairs	1.62	0.645	---	3.9	STD-2	300	---	---	---	---	---
	LSE10SPE- 18	18	10 pairs	1.62	0.82	---	4.9	STD-2	300	---	---	---	---	---
	LSE12SPE- 18	18	12 pairs	1.62	0.85	---	5.1	STD-2	300	---	---	---	---	---
	LSE14SPE- 18	18	14 pairs	1.62	0.89	---	5.3	STD-2	300	---	---	---	---	---
	LSE19SPE- 18	18	19 pairs	1.62	1.005	---	6	STD-2	300	---	---	---	---	---
	LSE24SPE- 18	18	24 pairs	1.62	1.18	---	7.1	STD-2	300	---	---	---	---	---
	LSE30SPE- 18	18	30 pairs	1.62	1.25	---	7.5	STD-2	300	---	---	---	---	---
	LSE37SPE- 18	18	37 pairs	1.62	1.35	---	8.1	STD-2	300	---	---	---	---	---
	LSE61SPE- 18	18	61 pairs	1.62	1.7	---	10.2	STD-2	300	---	---	---	---	---
	LSE3SPE- 16	16	3 pairs	2.58	0.565	---	3.4	STD-2	300	---	---	---	---	---
	LSE7SPE- 16	16	7 pairs	2.58	0.74	---	4.4	STD-2	300	---	---	---	---	---
24643/84	LSEC6FS- 4	23	4 pairs	0.509	0.595	0.225	2.8	TIA 568.2	150	---	---	---	---	---
	LSEC6OS- 4	23	4 pairs	0.509	0.605	0.24	2.9	TIA 568.2	150	---	---	---	---	---

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TABLE VI (e)(1). MIL-DTL-24643 cable ratings and characteristics, non-watertight, flexing service (power and lighting).

MIL-DTL-	Cable type designation		Conductor size: AWG or MCM	Number of conductors, pairs, or triads in cable	Area of each conductor (MCM)	Cable overall diameter max. (inches)	Cable weight per ft approx. (lbs)	Radius of bend min. (inches)	CDR ID	Rated voltage, max (RMS)	Ampacity, each conductor (amps, max.)				NSN 6145-01
											DC or 60 Hz		400 Hz		
											40 °C Ambient	50 °C Ambient	40 °C Ambient	50 °C Ambient	
24643/1 [5/]	LSCVSF-	4	3	3	52.620	1.450	1.328	9.0	STD-3	600	100	75	100	75	202-9490
			5	1	33.090	---	---	---	---	---	---	---	---	---	---
24643/3	LSDHOF-	3	16	2	2.580	0.425	0.101	3.0	STD-1 or 3	600	23	21	23	21	201-9494
	LSDHOF-	4	14	2	4.110	0.460	0.117	3.5	STD-1or 3	600	30	28	30	28	202-7745
	LSDHOF-	6	12	2	6.529	0.510	0.150	4.0	STD-1 or 3	600	41	37	41	37	202-2036
	LSDHOF-	9	9 (Navy STD)	2	9.045	0.570	0.172	4.5	STD-1 or 3	600	50	45	50	45	202-2037
	LSDHOF-	14	14 (Navy STD)	2	14.070	0.705	0.293	5.0	STD-1 or 3	600	60	54	60	54	203-0365
	LSDHOF-	23	7	2	20.815	0.860	0.395	6.5	STD-1 or 3	600	80	72	80	72	202-2038
	LSDHOF-	30	5	2	33.090	0.960	0.690	7.5	STD-1 or 3	600	90	83	90	83	202-2039
	LSDHOF-	83	83 (Navy STD)	2	84.230	1.450	1.359	11.0	STD-1 or3	600	169	152	169	152	202-0661
	LSDHOF-	250	250 (Navy STD)	2	252.700	2.100	2.811	17.0	STD-1 or 3	600	322	287	285	254	202-0662
	LSDHOF-	400	400 (Navy STD)	2	413.500	2.500	4.532	20.0	STD-1 or 3	600	422	382	290	262	202-0663
24643/3	LSFHOF-	3	16	4	2.580	0.480	0.130	4.0	STD-1 or 3	600	17	16	17	16	201-9495
	LSFHOF-	4	14	4	4.110	0.550	0.165	4.5	STD-1 or3	600	23	21	23	21	202-0672
	LSFHOF-	9	9 (Navy STD)	4	9.045	0.660	0.281	5.5	STD-1 or3	600	36	34	36	34	202-0673
	LSFHOF-	42	42 (Navy STD)	4	42.110	1.380	1.210	11.0	STD-1 or 3	600	79	73	79	73	202-0674
	LSFHOF-	60	60 (Navy STD)	4	61.200	1.510	1.550	12.0	STD-1 or 3	600	95	80	95	80	202-9491
	LSFHOF-	133	133 (Navy STD)	4	137.800	2.000	2.863	16.0	STD-1 or 3	600	163	148	155	140	202-0675
24643/3	LSSHOF-	3	16	1	2.580	0.210	0.027	1.5	STD-1 or 3	600	20	18	18	18	201-9530
	LSSHOF-	23	7	1	20.815	0.460	0.143	3.5	STD-1 or 3	600	88	80	80	80	201-9531
	LSSHOF-	60	60 (Navy STD)	1	61.200	0.600	0.341	5.0	STD-1 or 3	600	162	153	163	153	201-9532
	LSSHOF-	150	150 (Navy STD)	1	153.100	0.870	0.769	7.0	STD-1 or 3	600	285	263	263	263	210-4000

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TABLE VI (e)(1). MIL-DTL-24643 cable ratings and characteristics, non-watertight, flexing service (power and lighting) – Continued.

MIL-DTL-	Cable type designation	Conductor size: AWG or MCM	Number of conductors, pairs, or triads in cable	Area of each conductor (MCM)	Cable overall diameter max. (inches)	Cable weight per ft approx. (lbs)	Radius of bend min. (inches)	CDR ID	Rated voltage, max (RMS)	Ampacity, each conductor (amps, max.)				NSN 6145-01
										DC or 60 Hz		400 Hz		
										40 °C Ambient	50 °C Ambient	40 °C Ambient	50 °C Ambient	
24643/3	LSSHOF- 200	200 (Navy STD)	1	199.100	0.980	0.966	8.0	STD-1 or 3	600	323	306	---	---	201-9533
	LSSHOF- 250	250 (Navy STD)	1	252.700	1.085	1.318	8.5	STD-1 or 3	600	397	362	---	---	202-2051
	LSSHOF- 500	500	1	500.000	1.450	2.585	11.5	STD-1 or 3	600	602	578	---	---	202-2052
	LSSHOF- 650	650	1	650.000	1.610	3.090	13.0	STD-1 or 3	600	698	658	---	---	202-2053
	LSSHOF- 800	800 Navy STD)	1	812.700	1.670	3.306	13.5	STD-1 or 3	600	803	732	---	---	201-9534
24643/11	LSSSF- 300	300	1	300.000	1.100	1.287	6.6	---	600	---	---	---	---	204-5529
24643/3	LSTHOF- 3	16	3	2.580	0.450	0.094	3.5	STD-1 or 3	600	19	17	19	17	202-0664
	LSTHOF- 4	14	3	4.110	0.480	0.136	4.0	STD-1 or 3	600	25	23	25	23	205-3626
	LSTHOF- 6	12	3	6.530	0.550	0.179	4.5	STD-1 or 3	600	33	31	33	31	202-0665
	LSTHOF- 9	9 (Navy STD)	3	9.045	0.600	0.201	5.0	STD-1 or 3	600	388	34	38	34	202-0666
	LSTHOF- 14	14 (Navy STD)	3	14.070	0.750	0.346	6.0	STD-1 or 3	600	50	46	50	46	202-0667
	LSTHOF- 23	7	3	20.815	0.900	0.506	7.0	STD-1 or 3	600	70	64	70	64	202-0668
	LSTHOF- 42	42 (Navy STD)	3	42.110	1.250	0.986	10.0	STD-1 or 3	600	93	86	93	86	202-0669
	LSTHOF- 150	150 (Navy STD)	3	153.100	1.820	2.470	13.5	STD-1 or 3	600	197	180	197	180	203-5384
	LSTHOF- 250	250 (Navy STD)	3	252.700	2.240	3.872	18.0	STD-1 or 3	600	287	264	---	---	202-0670
	LSTHOF- 400	400 (Navy STD)	3	413.500	2.800	6.128	22.0	STD-1 or 3	600	400	365	315	265	202-0671
	LSTHOF- 500	500	3	500.000	3.100	7.313	25.0	STD-1 or 3	600	500	450	350	275	202-2040
	LSTHOF- 600	600	3	600.000	3.150	7.983	25.0	STD-1 or 3	600	600	550	---	---	203-5385
24643/71	LSSCF- 3	16	1	2.58	0.15	---	0.9	---	1,000	16	15	16	15	---
	LSSCF- 4	14	1	4.11	0.195	---	1.2	---	1,000	23	21	23	21	---
	LSSCF- 10	10	1	10.38	0.245	---	1.5	---	1,000	36	33	36	33	---
	LSSCF- 16	8	1	16.51	0.34	---	2	---	1,000	50	45	50	45	---
	LSSCF- 26	6	1	26.24	0.385	---	2.3	---	1,000	68	62	68	62	---

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TABLE VI (e)(1). MIL-DTL-24643 cable ratings and characteristics, non-watertight, flexing service (power and lighting) – Continued.

MIL-DTL-	Cable type designation		Conductor size: AWG or MCM	Number of conductors, pairs, or triads in cable	Area of each conductor (MCM)	Cable overall diameter max. (inches)	Cable weight per ft approx. (lbs)	Radius of bend min. (inches)	CDR ID	Rated voltage, max (RMS)	Ampacity, each conductor (amps, max.)				NSN 6145-01
											DC or 60 Hz		400 Hz		
											40 °C Ambient	50 °C Ambient	40 °C Ambient	50 °C Ambient	
24643/71	LSSCF-	42	4	1	41.74	0.46	---	2.8	---	1,000	86	78	86	78	---
	LSSCF-	66	2	1	66.36	0.555	---	3.3	---	1,000	118	197	118	197	---
	LSSCF-	105	0	1	105.6	0.685	---	4.1	---	1,000	155	139	155	139	---
	LSSCF-	123	00	1	133.1	0.735	---	4.4	---	1,000	177	160	177	160	---
	LSSCF-	165	000	1	167.8	0.82	---	4.9	---	1,000	205	185	205	185	---
	LSSCF-	210	0000	1	211.6	0.87	---	5.2	---	1,000	237	213	237	213	---
	LSSCF-	250	250 MCM	1	250	0.97	---	5.8	---	1,000	264	238	264	238	---
	LSSCF-	300	300 MCM	1	300	1.01	---	6.1	---	1,000	291	262	291	262	---
	LSSCF-	350	350 MCM	1	400	1.04	---	6.2	---	1,000	319	287	319	287	---
	LSSCF-	400	400 MCM	1	500	1.115	---	6.7	---	1,000	346	312	346	312	---
24643/72	LSFCF-	42	4AWG, 16AWG	4 4	41.740 2.426	1.35	---	7.5	STD-1	1,000	86 16	80 15	86 16	80 15	---
24643/73	LSTCF-	200	0000	3	211.6	1.8	---	9	STD-1 or 3	1,000	237	213	237	213	---

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TABLE VI (e)(2). MIL-DTL-24643 cable ratings and characteristics, non-watertight, flexing service (control).

MIL-DTL-	Cable type designation		Conductor size: AWG or MCM	Number of conductors, pairs, or triads in cable	Area of each conductor (MCM)	Cable overall diameter max. (inches)	Cable weight per ft approx. (lbs)	Radius of bend min. (inches)	CDR ID	Rated voltage, max. (RMS)	Ampacity, each conductor (amps, max.)				NSN 6145-01
											DC or 60 Hz		400 Hz		
											40 °C Ambient	50 °C Ambient	40 °C Ambient	50 °C Ambient	
24643/7	LSMHOF-	7	16	7	2.580	0.500	0.164	4.0	STD-1 or 3	600	11/7 [1/]	9/6 [1/]	11/7 [1/]	9/6 [1/]	202-2788
	LSMHOF-	10	16	10	2.580	0.585	0.224	4.5	STD-1 or 3	600	11/7	9/6	11/7	9/6	203-5388
	LSMHOF-	14	16	14	2.580	0.635	0.290	5.0	STD-1 or 3	600	11/7	9/6	11/7	9/6	202-2789
	LSMHOF-	19	16	19	2.580	0.705	0.360	5.5	STD-1 or 3	600	11/7	9/6	11/7	9/6	202-2790
	LSMHOF-	24	16	24	2.580	0.795	0.457	6.5	STD-1 or 3	600	11/7	9/6	11/7	9/6	202-2791
	LSMHOF-	30	16	30	2.580	0.835	0.541	6.5	STD-1 or 3	600	11/7	9/6	11/7	9/6	202-2792
	LSMHOF-	37	16	37	2.580	0.925	0.674	7.5	STD-1 or 3	600	11/5	9/4	11/5	9/4	202-2793
	LSMHOF-	44	16	44	2.580	1.000	0.771	8.0	STD-1 or 3	600	11/4	9/3	11/4	9/3	203-5389
	LSMHOF-	61	16	61	2.580	1.175	1.1125	9.5	STD-1 or 3	600	11/3	9/2	11/3	9/2	202-2794

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TABLE VI (e)(3). MIL-DTL-24643 cable ratings and characteristics, non-watertight, flexing service (electronic, communication, and instrumentation).

MIL-DTL-	Cable type designation		Conductor size: AWG or MCM	Number of conductors, pairs, or triads in cable	Area of each conductor (MCM)	Cable overall diameter max. (inches)	Cable weight per ft approx. (lbs)	Radius of bend min. (inches)	CDR ID	Rated voltage, max. (RMS)	Ampacity, each conductor (amps, max.)				NSN 6145-01
											DC or 60 Hz		400 Hz		
											40 °C Ambient	50 °C Ambient	40 °C Ambient	50 °C Ambient	
24643/44	LS3SF-	7	18	7 triads	1.620	1.040	0.618	2.0	STD-2	600	---	---	---	---	202-3491
24643/2	LSDCOP-	1	20	2	1.020	0.250	0.031	2.0	STD-3	300	---	---	---	---	202-9527
	LSDCOP-	1.5	18	2	1.620	0.315	0.050	2.52	STD-3	300	---	---	---	---	202-2776
	LSDCOP-	2	18	2	1.620	0.330	0.052	2.64	STD-3	300	---	---	---	---	202-2777
24643/4	LSMCOS-	2	18	2	1.620	0.460	0.130	3.68	SPL-3	600	5/5 [1/]	4/4 [1/]	5/5 [1/]	4/4 [1/]	203-5386
	LSMCOS-	4	18	4	1.620	0.510	0.167	4.08	SPL-3	600	5/3	4/2	5/3	4/2	202-2778
	LSMCOS-	5	20	5	1.020	0.390	0.091	3.12	SPL-3	600	2.5/1	2/1	2.5/1	2/1	203-5400
	LSMCOS-	6	20	6	1.020	0.480	0.102	3.84	SPL-3	600	2.5/1	2/1	2.5/1	2/1	202-2828
	LSMCOS-	7	18	7	1.620	0.595	0.237	4.76	SPL-3	600	5/2.5	4/2.5	5/2.5	4/2.5	202-2779
24643/8	LSMMOP-	5	24	5	0.404	0.305	0.059	2.44	STD-1 or 3	300	3/1	2/1	---	---	202-2829
24643/2	LSTCOP-	2	18	3	1.620	0.345	0.067	2.76	STD-3	300	6	5	---	---	203-5383
24643/12	LSTTOP-	3	20	3 pairs	1.020	0.480	0.113	4.0	TEL-3	300	5/4	4/3	5/4	4/3	201-9538
	LSTTOP-	5	20	5 pairs	1.020	0.590	0.154	5.0	TEL-3	300	5/3	4/2	5/3	4/2	202-6998
	LSTTOP-	10	20	10 pairs	1.020	0.700	0.259	5.5	TEL-3	300	5/2	4/1	5/2	4/1	201-9539
	LSTTOP-	15	20	15 pairs	1.020	0.830	0.383	6.5	TEL-3	300	5/1	4/0.5	5/1	4/0.5	202-3536
24643/13	LSTTRS-	2	20	2 pairs	1.020	0.680	0.233	1.0	STD-2	300	---	---	---	---	202-2831
	LSTTRS-	4	20	4 pairs	1.020	0.740	0.281	1.1	STD-2	300	---	---	---	---	203-5401
	LSTTRS-	6	20	6 pairs	1.020	0.880	0.388	1.5	STD-2	300	---	---	---	---	202-2832
	LSTTRS-	8	20	8 pairs	1.020	0.990	0.474	1.7	STD-2	300	---	---	---	---	202-2833
	LSTTRS-	10	20	10 pairs	1.020	1.080	0.561	2.0	STD-2	300	---	---	---	---	202-2834
	LSTTRS-	12	20	12 pairs	1.020	1.100	0.611	2.0	STD-2	300	---	---	---	---	202-2835
	LSTTRS-	16	20	16 pairs	1.020	1.190	0.723	2.2	STD-2	300	---	---	---	---	203-5402
24643/13 [4/]	LSTTRSA-	2	20	4	1.020	0.730	0.297	4.5	STD-2	300	---	---	---	---	202-2836
	LSTTRSA-	4	20	8	1.020	0.790	0.394	5.0	STD-2	300	---	---	---	---	203-5403
	LSTTRSA-	6	20	12	1.020	0.930	0.470	5.5	STD-2	300	---	---	---	---	202-2837
	LSTTRSA-	8	20	16	1.020	1.040	0.521	6.5	STD-2	300	---	---	---	---	203-5404
	LSTTRSA-	10	20	20	1.020	1.130	0.557	7.0	STD-2	300	---	---	---	---	203-5405

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TABLE VI (e)(3). MIL-DTL-24643 cable ratings and characteristics, non-watertight, flexing service (electronic, communication, and instrumentation).

MIL-DTL-	Cable type designation	Conductor size: AWG or MCM	Number of conductors, pairs, or triads in cable	Area of each conductor (MCM)	Cable overall diameter max. (inches)	Cable weight per ft approx. (lbs)	Radius of bend min. (inches)	CDR ID	Rated voltage, max. (RMS)	Ampacity, each conductor (amps, max.)				NSN 6145-01
										DC or 60 Hz		400 Hz		
										40 °C Ambient	50 °C Ambient	40 °C Ambient	50 °C Ambient	
24643/13 [4/]	LSTTRSA- 12	20	24	1.020	1.150	0.608	7.0	STD-2	300	---	---	---	---	210-2346
	LSTTRSA- 16	20	32	1.020	1.240	0.764	7.5	STD-2	300	---	---	---	---	202-8472

NOTES:

1/ Ind/Avg indicates the maximum current for each conductor (Ind), and the maximum current (Avg) for each conductor when all conductors in the cable are used.

2/ May be STD-3 or STD-4, manufacturer's option.

3/ 6145-00

4/ The armored cable versions are not in the most current versions of the respective specification sheets.

5/ Canceled and replaced, see 5.1.

6/ See 5.1 for detailed description of LSECM and LS1SU cable.

7/ Cable information for 15 kV rated cable. Note that, at the time of this document's publication, the MIL-DTL-24643 specification sheet for this cable type has not yet been issued.

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5.7 Cable classification (MIL-DTL-24640). Cables specified in MIL-DTL-24640 are listed in table VII under the following general classifications:

a. Watertight (with circuit integrity), non-flexing service:
   (1) power
   (2) control
b. Watertight, non-flexing service:
   (1) electronic, communication, and instrumentation
   (2) power
c. Non-watertight, non-flexing service:
   (1) power
   (2) electronic, communication, and instrumentation

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TABLE VII (a)(1). MIL-DTL-24640 cable ratings and characteristics, watertight (with circuit integrity), non-flexing service (power).

MIL-DTL-	Cable type designation	Conductor size: AWG or MCM	Number of conductors in cable	Area of each conductor (MCM)	Cable overall diameter max. (inches)	Cable weight per ft max (lbs)	Radius of bend min. (inches)	CDR ID	Rated voltage, max. (RMS)	Ampacity, each conductor (amps, max.)				NSN 6145-01
										DC or 60 Hz		400 Hz		
										40 °C Ambient	50 °C Ambient	40 °C Ambient	50 °C Ambient	
24640/22	7XW- 3	16	7	2.580	0.339	0.110	3.0	STD-1 or 3	600	15/11 [1/]	14/10 [1/]	15/11 [1/]	14/10 [1/]	224-8321
	7XW- 4	14	7	4.110	0.404	0.162	3.5	STD-1 or 3	600	26/14	24/13	26/14	24/13	224-8322
24640/22 [5/]	7XWA- 3	16	7	2.580	0.389	0.138	3.0	STD-1 or 3	600	15/11	14/10	15/11	14/10	225-1379
	7XWA- 4	14	7	4.110	0.454	0.203	3.5	STD-1 or 3	600	26/14	24/13	26/14	24/13	225-1380
24640/19	DXOW- 3	16	2	2.580	0.316	0.099	4.0	STD-1 or 3	600	13	12	13	12	225-2664
	DXOW- 4	14	2	4.110	0.354	0.125	4.5	STD-1 or 3	600	22	20	22	20	225-6382
24640/19	DXW- 3	16	2	2.580	0.257	0.050	2.0	STD-1 o r3	600	13	12	13	12	224-9185
	DXW- 4	14	2	4.110	0.303	0.071	2.5	STD-1or 3	600	22	20	22	20	224-8317
24640/19 [5/]	DXWA- 3	16	2	2.580	0.307	0.067	2.0	STD-1 or 3	600	13	12	13	12	224-8318
	DXWA- 4	14	2	4.110	0.353	0.089	2.5	STD-1 or 3	600	22	20	22	20	226-5011
24640/21	FXOW- 3	16	4	2.580	0.350	0.132	4.2	STD-1 or 3	600	11	10	11	10	---
	FXOW- 4	14	4	4.110	0.394	0.165	5.0	STD-1 or 3	600	11	10	11	10	---
24640/21	FXW- 3	16	4	2.580	0.286	0.078	3.0	STD-1 or 3	600	11	10	11	10	225-2153
	FXW- 4	14	4	4.110	0.339	0.115	3.0	STD-1 or 3	600	18	17	18	17	225-2154
24640/21 [5/]	FXWA- 3	16	4	2.580	0.336	0.095	3.0	STD-1 or 3	600	11	10	11	10	224-8319
	FXWA- 4	14	4	4.110	0.389	0.128	3.0	STD-1 or 3	600	18	17	18	17	224-8320
24640/20	TXOW- 3	16	3	2.580	0.329	0.114	4.0	STD-1 or 3	600	11	10	11	10	---
	TXOW- 4	14	3	4.110	0.369	0.148	4.5	STD-1 or 3	600	18	17	18	17	---
24640/20	TXW- 3	16	3	2.580	0.266	0.065	3.0	STD-1 or 3	600	11	10	11	10	225-2665
	TXW- 4	14	3	4.110	0.314	0.099	3.0	STD-1 or 3	600	18	17	18	17	225-2151
24640/20 [5/]	TXWA- 3	16	3	2.580	0.316	0.084	3.0	STD-1 or 3	600	11	10	11	10	225-2152
	TXWA- 4	14	3	4.110	0.364	0.125	3.0	STD-1 or 3	600	18	17	18	17	225-2151

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TABLE VII (a)(2). MIL-DTL-24640 cable ratings and characteristics, watertight (with circuit integrity), non-flexing service (control).

MIL-DTL-	Cable type designation		Conductor size: AWG or MCM	Number of conductors in cable	Area of each conductor (MCM)	Cable overall diameter max. (inches)	Cable weight per ft max. (lbs)	Radius of bend min. (inches)	CDR ID	Rated voltage, max. (RMS)	Ampacity, each conductor (amps, max.)				NSN 6145-01
											DC or 60 Hz		400 Hz		
											40 °C Ambient	50 °C Ambient	40 °C Ambient	50 °C Ambient	
24640/23	MXCOW-	7	18	7	1.620	0.366	0.140	4.5	STD-1 or 3	600	12/8 [1/]	9/6 [1/]	---	---	227-9946
	MXCOW-	10	18	10	1.620	0.447	0.185	5.5	STD-1 or 3	600	12/8	9/6	---	---	229-0031
	MXCOW-	14	18	14	1.620	0.474	0.225	6.0	STD-1 or 3	600	12/8	9/6	---	---	225-3450
	MXCOW-	19	18	19	1.620	0.515	0.280	6.5	STD-1 or 3	600	12/8	9/6	---	---	225-3451
	MXCOW-	24	18	24	1.620	0.601	0.362	7.5	STD-1 or 3	600	12/6	9/5	---	---	225-2161
	MXCOW-	30	18	30	1.620	0.630	0.399	7.5	STD-1 or 3	600	12/6	9/5	---	---	225-2162
	MXCOW-	37	18	37	1.620	0.670	0.466	8.0	STD-1 or 3	600	12/6	9/5	---	---	225-2163
	MXCOW-	44	18	44	1.620	0.751	0.574	9.0	STD-1 or 3	600	12/5	9/4	---	---	225-2164
	MXCOW-	61	18	61	1.620	0.817	0.715	10.0	STD-1 or 3	600	12/4.5	9/3.5	---	---	225-2165
24640/23	MXCW-	7	18	7	1.620	0.319	0.009	2.5	STD-1 or 3	600	12/8	9/6	---	---	225-3448
	MXCW-	10	18	10	1.620	0.405	0.139	3.5	STD-1 or 3	600	12/8	9/6	---	---	224-8233
	MXCW-	14	18	14	1.620	0.434	0.173	3.5	STD-1 or 3	600	12/8	9/6	---	---	225-1381
	MXCW-	19	18	19	1.620	0.474	0.215	4.0	STD-1 or 3	600	12/8	9/6	---	---	226-3691
	MXCW-	24	18	24	1.620	0.560	0.295	4.5	STD-1 or 3	600	12/6	9/5	---	---	224-8323
	MXCW-	30	18	30	1.620	0.589	0.340	4.5	STD-1 or 3	600	12/6	9/5	---	---	225-2155
	MXCW-	37	18	37	1.620	0.630	0.398	5.0	STD-1 or 3	600	12/6	9/5	---	---	225-2156
	MXCW-	44	18	44	1.620	0.708	0.492	5.5	STD-1 or 3	600	12/5	9/4	---	---	225-2157
	MXCW-	61	18	61	1.620	0.785	0.640	6.5	STD-1 or 3	600	12/4.5	9/3.5	---	---	225-2158
24640/23 [5/]	MXCWA-	7	18	7	1.620	0.369	0.110	2.5	STD-1	600	12/8	9/6	---	---	226-2052
	MXCWA-	10	18	10	1.620	0.455	0.177	3.5	STD-1	600	12/8	9/6	---	---	225-2666
	MXCWA-	14	18	14	1.620	0.485	0.216	3.5	STD-1	600	12/6	9/5	---	---	225-2667
	MXCWA-	19	18	19	1.620	0.524	0.264	4.0	STD-1	600	12/6	9/5	---	---	225-2668
	MXCWA-	24	18	24	1.620	0.610	0.337	4.5	STD-1	600	12/6	9/5	---	---	225-2669
	MXCWA-	30	18	30	1.620	0.639	0.385	4.5	STD-1	600	12/6	9/5	---	---	225-2159
	MXCWA-	37	18	37	1.620	0.680	0.453	5.0	STD-1	600	12/6	9/5	---	---	225-0961
	MXCWA-	44	18	44	1.620	0.758	0.548	5.5	STD-1	600	12/5	9/4	---	---	225-2160
	MXCWA-	61	18	61	1.620	0.835	0.564	6.5	STD-1	600	12/4.5	9/3.5	---	---	225-3449

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TABLE VII (a)(2). MIL-DTL-24640 cable ratings and characteristics, watertight (with circuit integrity), non-flexing service (control) – Continued.

MIL-DTL-	Cable type designation		Conductor size: AWG or MCM	Number of conductors in cable	Area of each conductor (MCM)	Cable overall diameter max. (inches)	Cable weight per ft max. (lbs)	Radius of bend min. (inches)	CDR ID	Rated voltage, max. (RMS)	Ampacity, each conductor (amps, max.)				NSN 6145-01
											DC or 60 Hz		400 Hz		
											40 °C Ambient	50 °C Ambient	40 °C Ambient	50 °C Ambient	
24640/24	TTXOW-	1.5	22	1.5 pairs	0.640	0.273	0.068	3.5	TEL-3 or 6	600	---	---	---	---	226-8808
	TTXOW-	3	22	3 pairs	0.640	0.359	0.108	4.5	TEL-3 or 6	600	---	---	---	---	228-4262
	TTXOW-	5	22	5 pairs	0.640	0.406	0.134	5.0	TEL-3 or 6	600	---	---	---	---	237-1290
	TTXOW-	15	22	15 pairs	0.640	0.600	0.299	7.5	TEL-3 or 6	600	---	---	---	---	231-3926
	TTXOW-	20	22	20 pairs	0.640	0.662	0.371	8.0	TEL-3 or 6	600	---	---	---	---	226-8809
	TTXOW-	30	22	30 pairs	0.640	0.772	0.497	9.5	TEL-3 or 6	600	---	---	---	---	229-0032
	TTXOW-	40	22	40 pairs	0.640	0.887	0.631	11.0	TEL-3 or 6	600	---	---	---	---	226-4565
24640/24	TTXW-	1.5	22	1.5 pairs	0.640	0.195	0.026	1.5	TEL-3 or 6	600	---	---	---	---	225-2205
	TTXW-	3	22	3 pairs	0.640	0.307	0.062	2.5	TEL-3 or 6	600	---	---	---	---	225-2206
	TTXW-	5	22	5 pairs	0.640	0.357	0.091	3.0	TEL-3 or 6	600	---	---	---	---	225-2207
	TTXW-	10	22	10 pairs	0.640	0.492	0.172	4.0	TEL-3 or 6	600	---	---	---	---	225-2208
	TTXW-	15	22	15 pairs	0.640	0.569	0.235	4.5	TEL-3 or 6	600	---	---	---	---	225-2209
	TTXW-	20	22	20 pairs	0.640	0.621	0.285	5.0	TEL-3 or 6	600	---	---	---	---	225-9664
	TTXW-	30	22	30 pairs	0.640	0.738	0.417	6.0	TEL-3 or 6	600	---	---	---	---	237-1289
	TTXW-	40	22	40 pairs	0.640	0.852	0.544	7.0	TEL-3 or 6	600	---	---	---	---	226-6028
24640/24 [5/]	TTXWA-	1.5	22	3	0.640	0.245	0.041	1.5	TEL-6	600	---	---	---	---	226-6029
	TTXWA-	3	22	6	0.640	0.357	0.084	2.5	TEL-6	600	---	---	---	---	225-3453
	TTXWA-	5	22	10	0.640	0.407	0.119	3.5	TEL-6	600	---	---	---	---	226-6030
	TTXWA-	10	22	20	0.640	0.542	0.209	4.5	TEL-6	600	---	---	---	---	226-6031
	TTXWA-	15	22	30	0.640	0.619	0.270	5.0	TEL-6	600	---	---	---	---	226-6032
	TTXWA-	20	22	40	0.640	0.671	0.328	5.5	TEL-6	600	---	---	---	---	226-7410
	TTXWA-	30	22	60	0.640	0.788	0.463	6.0	TEL-6	600	---	---	---	---	226-6033
	TTXWA-	40	22	80	0.640	0.902	0.593	7.0	TEL-6	600	---	---	---	---	228-0701

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TABLE VII (b)(1). MIL-DTL-24640 cable ratings and characteristics, watertight, non-flexing service (electronic, communication, and instrumentation).

MIL-DTL-	Cable type designation		Conductor size: AWG or MCM	Number of conductors in cable	Area of each conductor (MCM)	Cable overall diameter max. (inches)	Cable weight per ft max. (lbs)	Radius of bend min. (inches)	CDR ID	Rated voltage, max. (RMS)	Ampacity, each conductor (amps, max.)				NSN 6145-01
											DC or 60 Hz		400 Hz		
											40 °C Ambient	50 °C Ambient	40 °C Ambient	50 °C Ambient	
24640/7	1XMSOW-	7	22	7	0.640	0.385	0.144	4.5	STD-2	600	---	---	---	---	---
	1XMSOW-	16	22	16	0.640	0.510	0.255	6.0	STD-2	600	---	---	---	---	---
	1XMSOW-	70	22	70	0.640	0.925	0.900	11.0	STD-2	600	---	---	---	---	---
24640/14	1XSOW-	2	22	2	0.640	0.314	0.085	4.0	STD-2	600	---	---	---	---	225-2199
	1XSOW-	14	22	14	0.640	0.506	0.280	6.0	STD-2	600	---	---	---	---	225-2200
	1XSOW-	20	22	20	0.640	0.584	0.330	7.0	STD-2	600	---	---	---	---	231-4589
	1XSOW-	30	22	30	0.640	0.662	0.429	8.0	STD-2	600	---	---	---	---	225-2201
24640/6	2XAOW-	2	22	2 pairs	0.640	0.335	0.115	4.0	Method 6	600	---	---	---	---	---
	2XAOW-	7	22	7 pairs	0.640	0.425	0.135	5.0	Method 6	600	---	---	---	---	---
	2XAOW-	10	22	10 pairs	0.640	0.515	0.184	6.0	Method 6	600	---	---	---	---	---
	2XAOW-	18	22	18 pairs	0.640	0.615	0.263	7.5	Method 6	600	---	---	---	---	---
	2XAOW-	40	22	40 pairs	0.640	0.86	0.495	10.5	Method 6	600	---	---	---	---	---
24640/12	2XOW-	6	26	6 pairs	0.250	0.340	0.078	4.0	Method 6 or TEL-3	600	---	---	---	---	---
	2XOW-	18	26	18 pairs	0.250	0.460	0.129	5.5	Method 6 or TEL-3	600	---	---	---	---	---
	2XOW-	24	26	24 pairs	0.250	0.525	0.178	6.0	Method 6 or TEL-3	600	---	---	---	---	---
	2XOW-	42	26	42 pairs	0.250	0.630	0.243	7.5	Method 6 or TEL-3	600	---	---	---	---	---
	2XOW-	60	26	60 pairs	0.250	0.710	0.315	8.5	Method 6 or TEL-3	600	---	---	---	---	---
	2XOW-	77	26	77 pairs	0.250	0.800	0.400	9.5	Method 6 or TEL-3	600	---	---	---	---	---
24640/17	2XOW-	6	26	6 pairs	0.278	0.363	0.182	4.5	STD-6 or TEL-3	600	---	---	---	---	224-8341
	2XOW-	18	26	18 pairs	0.278	0.504	0.201	6.0	STD-6 or TEL-3	600	---	---	---	---	224-9205
	2XOW-	24	26	24 pairs	0.278	0.588	0.266	7.0	STD-6 or TEL-3	600	---	---	---	---	225-8952
	2XOW-	42	26	42 pairs	0.278	0.686	0.380	8.5	STD-6 or TEL-3	600	---	---	---	---	225-8953

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TABLE VII (b)(1). MIL-DTL-24640 cable ratings and characteristics, watertight, non-flexing service (electronic, communication, and instrumentation) – Continued.

MIL-DTL-	Cable type designation		Conductor size: AWG or MCM	Number of conductors in cable	Area of each conductor (MCM)	Cable overall diameter max. (inches)	Cable weight per ft max. (lbs)	Radius of bend min. (inches)	CDR ID	Rated voltage, max. (RMS)	Ampacity, each conductor (amps, max.)				NSN 6145-01
											DC or 60 Hz		400 Hz		
											40 °C Ambient	50 °C Ambient	40 °C Ambient	50 °C Ambient	
24640/17	2XOW-	60	26	60 pairs	0.278	0.802	0.485	10.5	STD-6 or TEL-3	600	---	---	---	---	225-8954
	2XOW-	77	26	77 pairs	0.278	0.906	0.615	11.0	STD-6 or TEL-3	600	---	---	---	---	225-8955
24640/15	2XSAOW-	3	22	3 pairs	0.640	0.437	0.160	5.5	STD-2	600	---	---	---	---	225-1396
	2XSAOW-	7	22	7 pairs	0.640	0.550	0.260	7.5	STD-2	600	---	---	---	---	225-1397
	2XSAOW-	10	22	10 pairs	0.640	0.681	0.385	8.5	STD-2	600	---	---	---	---	224-9204
	2XSAOW-	14	22	14 pairs	0.640	0.743	0.465	9.0	STD-2	600	---	---	---	---	225-1398
	2XSAOW-	19	22	19 pairs	0.640	0.817	0.627	10.0	STD-2	600	---	---	---	---	225-1399
	2XSAOW-	24	22	24 pairs	0.640	0.952	0.800	11.5	STD-2	600	---	---	---	---	225-2204
	2XSAOW-	30	22	30 pairs	0.640	1.020	0.910	12.5	STD-2	600	---	---	---	---	229-8306
	2XSAOW-	37	22	37 pairs	0.640	1.090	1.050	13.0	STD-2	600	---	---	---	---	224-8340
24640/15	2XSAW-	3	22	3 pairs	0.640	0.396	0.115	3.5	STD-2	600	---	---	---	---	225-2202
	2XSAW-	7	22	7 pairs	0.640	0.497	0.195	4.0	STD-2	600	---	---	---	---	225-2203
	2XSAW-	14	22	14 pairs	0.640	0.691	0.390	5.5	STD-2	600	---	---	---	---	225-1394
24640/15	2XSAWA-	3	22	3 pairs	0.640	0.456	0.148	3.5	STD-2	600	---	---	---	---	224-9202
	2XSAWA-	7	22	7 pairs	0.640	0.557	0.244	4.0	STD-2	600	---	---	---	---	224-9203
	2XSAWA-	14	22	14 pairs	0.640	0.751	0.445	5.0	STD-2	600	---	---	---	---	225-1395
24640/16	2XSOW-	3	18	3 pairs	1.620	0.525	0.227	6.5	STD-2	600	---	---	---	---	224-8311
	2XSOW-	7	18	7 pairs	1.620	0.656	0.401	8.0	STD-2	600	---	---	---	---	224-8312
	2XSOW-	12	18	12 pairs	1.620	0.864	0.665	10.5	STD-2	600	---	---	---	---	224-8313
	2XSOW-	19	18	19 pairs	1.620	1.010	0.950	12.5	STD-2	600	---	---	---	---	224-9181
	2XSOW-	30	18	30 pairs	1.620	1.270	1.442	15.5	STD-2	600	---	---	---	---	225-1376
24640/9	2XSOW-	3	22	3 pairs	0.640	0.430	0.153	5.0	STD-2	600	---	---	---	---	---
	2XSOW-	7	22	7 pairs	0.640	0.540	0.267	6.0	STD-2	600	---	---	---	---	---
	2XSOW-	10	22	10 pairs	0.640	0.640	0.350	8.0	STD-2	600	---	---	---	---	---
	2XSOW-	14	22	14 pairs	0.640	0.720	0.460	8.5	STD-2	600	---	---	---	---	---

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TABLE VII (b)(1). MIL-DTL-24640 cable ratings and characteristics, watertight, non-flexing service (electronic, communication, and instrumentation) – Continued.

MIL-DTL-	Cable type designation		Conductor size: AWG or MCM	Number of conductors in cable	Area of each conductor (MCM)	Cable overall diameter max. (inches)	Cable weight per ft max. (lbs)	Radius of bend min. (inches)	CDR ID	Rated voltage, max. (RMS)	Ampacity, each conductor (amps, max.)				NSN 6145-01
											DC or 60 Hz		400 Hz		
											40 °C Ambient	50 °C Ambient	40 °C Ambient	50 °C Ambient	
24640/9	2XSOW-	19	22	19 pairs	0.640	0.805	0.600	9.5	STD-2	600	---	---	---	---	---
	2XSOW-	30	22	30 pairs	0.640	0.975	0.895	11.5	STD-2	600	---	---	---	---	---
24640/9	2XSW-	2	22	2 pairs	0.640	0.335	0.079	3.0	STD-2	600	---	---	---	---	---
	2XSW-	3	22	3 pairs	0.640	0.38	0.109	3.0	STD-2	600	---	---	---	---	---
	2XSW-	7	22	7 pairs	0.640	0.505	0.204	4.0	STD-2	600	---	---	---	---	---
	2XSW-	10	22	10 pairs	0.640	0.61	0.295	5.0	STD-2	600	---	---	---	---	---
	2XSW-	14	22	14 pairs	0.640	0.675	0.387	5.0	STD-2	600	---	---	---	---	---
	2XSW-	19	22	19 pairs	0.640	0.765	0.513	5.5	STD-2	600	---	---	---	---	---
	2XSW-	24	22	24 pairs	0.640	0.855	0.642	6.5	STD-2	600	---	---	---	---	---
	2XSW-	30	22	30 pairs	0.640	0.955	0.805	7.0	STD-2	600	---	---	---	---	---
24640/16	2XSW-	1	18	1 pair	1.620	0.258	0.053	2.0	STD-2	600	---	---	---	---	224-8308
	2XSW-	3	18	3 pairs	1.620	0.470	0.175	4.0	STD-2	600	---	---	---	---	224-8309
	2XSW-	7	18	7 pairs	1.620	0.617	0.330	5.0	STD-2	600	---	---	---	---	225-8947
24640/16 [5/]	2XSWA-	1	18	2	1.620	0.308	0.069	2.0	STD-2	600	---	---	---	---	229-8294
24640/16	2XSWA-	3	18	6	1.620	0.520	0.210	4.0	STD-2	600	---	---	---	---	225-8948
	2XSWA-	7	18	14	1.620	0.667	0.375	5.0	STD-2	600	---	---	---	---	224-8310
24640/13	2XSXOW-	4	26	4 pairs	0.278	0.390	0.147	4.5	STD-2	600	---	---	---	---	---
24640/18	3XSOW-	3	18	3 triads	1.620	0.559	0.271	7.0	STD-2	600	---	---	---	---	224-9182
	3XSOW-	7	18	7 triads	1.620	0.711	0.475	8.5	STD-2	600	---	---	---	---	224-9183
	3XSOW-	10	18	10 triads	1.620	0.901	0.730	11.0	STD-2	600	---	---	---	---	224-8314
	3XSOW-	14	18	14 triads	1.620	0.968	0.880	12.0	STD-2	600	---	---	---	---	224-8315
	3XSOW-	19	18	19 triads	1.620	1.090	1.115	13.0	STD-2	600	---	---	---	---	224-8316
	3XSOW-	24	18	24 triads	1.620	1.300	1.607	16.0	STD-2	600	---	---	---	---	224-9184
24640/11	3XSW-	7	18	3 triads	1.620	0.645	0.350	8.0	STD-2	600	---	---	---	---	---
24640/18	3XSW-	3	18	3 triads	1.620	0.508	0.205	4.0	STD-2	600	---	---	---	---	225-8949
	3XSW-	7	18	7 triads	1.620	0.668	0.390	5.5	STD-2	600	---	---	---	---	225-8950
	3XSW-	10	18	10 triads	1.620	0.865	0.625	7.0	STD-2	600	---	---	---	---	225-4894
	3XSW-	14	18	14 triads	1.620	0.941	0.775	7.5	STD-2	600	---	---	---	---	225-2660

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TABLE VII (b)(1). MIL-DTL-24640 cable ratings and characteristics, watertight, non-flexing service (electronic, communication, and instrumentation) – Continued.

MIL-DTL-	Cable type designation		Conductor size: AWG or MCM	Number of conductors in cable	Area of each conductor (MCM)	Cable overall diameter max. (inches)	Cable weight per ft max. (lbs)	Radius of bend min. (inches)	CDR ID	Rated voltage, max. (RMS)	Ampacity, each conductor (amps, max.)				NSN 6145-01
											DC or 60 Hz		400 Hz		
											40 °C Ambient	50 °C Ambient	40 °C Ambient	50 °C Ambient	
24640/18 [5/]	3XSWA-	3	18	9	1.620	0.558	0.255	4.0	STD-2	600	---	---	---	---	225-2661
	3XSWA-	7	18	21	1.620	0.718	0.458	5.5	STD-2	600	---	---	---	---	228-4261
	3XSWA-	10	18	30	1.620	0.915	0.701	7.0	STD-2	600	---	---	---	---	225-2662
	3XSWA-	14	18	42	1.620	0.991	0.873	7.5	STD-2	600	---	---	---	---	225-2663
24640/8	MXOW-	10	20	10	1.020	0.380	0.119	4.5	STD-1 or 3	600	---	---	---	---	---
	MXOW-	14	20	14	1.020	0.415	0.147	5.0	STD-1 or 3	600	---	---	---	---	---
24640/10	MXSOW-	2	16	2	2.580	0.304	0.083	4.0	STD-1	600	---	---	---	---	225-2148
	MXSOW-	9	16	9	2.580	0.458	0.200	5.5	STD-1	600	---	---	---	---	225-2149
	MXSOW-	21	16	21	2.580	0.596	0.365	7.5	STD-1	600	---	---	---	---	225-1377
	MXSOW-	37	16	37	2.580	0.694	0.550	8.5	STD-1	600	---	---	---	---	225-1378
24640/5	TTXSW-	2	20	2 pairs	1.020	0.400	0.095	3.2	STD-2	600	---	---	---	---	---
	TTXSW-	4	20	4 pairs	1.020	0.455	0.135	3.6	STD-2	600	---	---	---	---	---
24640/4	TTXWB-	3	20	3 pairs	1.020	0.000	0.070	2.6	Method 6	600	5/4.0 [1/]	4/3.0 [1/]	5/4.0 [1/]	4/3.0 [1/]	---
	TTXWB-	15	20	15 pairs	1.020	0.591	0.244	5.0	Method 6	600	5/0.6	4/0.5	5/0.6	4/0.5	---
24640/5	TTXSOW-	2	20	2 pairs	1.020	0.430	0.138	3.4	STD-2	600	---	---	---	---	---
	TTXSOW-	6	20	6 pairs	1.020	0.585	0.250	4.7	STD-2	600	---	---	---	---	---
	TTXSOW-	8	20	8 pairs	1.020	0.635	0.302	5.1	STD-2	600	---	---	---	---	---
	TTXSOW-	10	20	10 pairs	1.020	0.725	0.385	5.8	STD-2	600	---	---	---	---	---

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TABLE VII (b)(2). MIL-DTL-24640 cable ratings and characteristics, watertight, non-flexing service (power).

MIL-DTL-	Cable type designation		Conductor size: AWG or MCM	Number of conductors in cable	Area of each conductor (MCM)	Cable overall diameter max. (inches)	Cable weight per ft max. (lbs)	Radius of bend min. (inches)	CDR ID	Rated voltage, max. (RMS)	Ampacity, each conductor (amps, max.)				NSN 6145-01
											DC or 60 Hz		400 Hz		
											40 °C Ambient	50 °C Ambient	40 °C Ambient	50 °C Ambient	
24640/1	DXWB-	3	16	2	2.582	0.251	0.054	2.0	STD 1 or 3	600	13	12	13	12	---
	DXWB-	4	14	2	4.106	0.296	0.068	2.4	STD 1 or 3	600	22	20	22	20	---
24640/2	TXWB-	3	16	3	2.582	0.265	0.602	2.1	STD 1 or 3	600	13	12	13	12	---
	TXWB-	4	14	3	4.106	0.320	0.084	2.6	STD 1 or 3	600	22	20	22	20	---
24640/3	FXWB-	3	16	4	2.582	0.300	0.602	2.4	STD 1 or 3	600	11	10	11	10	---
	FXWB-	4	14	4	4.106	0.350	0.084	2.8	STD 1 or 3	600	18	17	18	17	---

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TABLE VII (c)(1). MIL-DTL-24640 cable ratings and characteristics, non-watertight, non-flexing service (power).

MIL-DTL-	Cable type designation		Conductor size: AWG or MCM	Number of conductors in cable	Area of each conductor (MCM)	Cable overall diameter max. (inches)	Cable weight per ft max. (lbs)	Radius of bend min. (inches)	CDR ID	Rated voltage, max. (RMS)	Ampacity, each conductor (amps, max.)				NSN 6145-01
											DC or 60 Hz		400 Hz		
											40 °C Ambient	50 °C Ambient	40 °C Ambient	50 °C Ambient	
24640/1	DX-	3	16	2	2.580	0.241	0.039	2.0	STD-1 or 3	600	13	12	13	12	224-6905
	DX-	4	14	2	4.110	0.286	0.057	2.5	STD-1 or 3	600	22	20	22	20	224-6906
24640/1 [5/]	DXA-	3	16	2	2.580	0.291	0.054	2.0	STD-1	600	13	12	13	12	224-9180
	DXA-	4	14	2	4.110	0.336	0.074	2.5	STD-1	600	22	20	22	20	224-6907
24640/3	FX-	3	16	4	2.580	0.282	0.063	2.5	STD-1 or 3	600	11	10	11	10	224-6912
	FX-	4	14	4	4.110	0.335	0.093	3.0	STD-1 or 3	600	18	17	18	17	224-6913
24640/3 [5/]	FXA-	3	16	4	2.580	0.332	0.080	2.5	STD-1	600	13	12	13	12	224-6914
	FXA-	4	14	4	4.110	0.383	0.113	3.0	STD-1	600	22	20	22	20	225-2147
24640/2	TX-	3	16	3	2.580	0.261	0.052	2.0	STD-1 or 3	600	11	10	11	10	224-6908
	TX-	4	14	3	4.110	0.310	0.076	2.5	STD-1 or 3	600	18	17	18	17	224-6909
24640/2 [5/]	TXA-	3	16	3	2.580	0.311	0.068	2.0	STD-1	600	11	10	11	10	224-6910
	TXA-	4	14	3	4.110	0.360	0.095	2.5	STD-1	600	18	17	18	17	224-6911
24643/25	9XS-	12	12	9	6.529	0.675	0.22	5	---	600	27	25	27	25	---
	9XS-	10	10	9	10.382	0.905	0.36	7	---	600	36	33	36	33	---
	9XS-	8	8	9	16.507	1.3	0.94	10	---	600	46	41	46	41	---
24643/26	5XO-	3	16	5	2.582	0.33	0.09	2	---	600	16	15	16	15	---
	5XO-	4	14	5	4.106	0.37	0.14	2.5	---	600	23	21	23	21	---
	5XO-	6	12	5	6.526	0.41	0.199	3	---	600	27	25	27	25	---

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TABLE VII (c)(2). MIL-DTL-24640 cable ratings and characteristics, non-watertight, non-flexing service (electronic, communication, and instrumentation).

MIL-DTL-	Cable type designation		Conductor size: AWG or MCM	Number of conductors in cable	Area of each conductor (MCM)	Cable overall diameter max. (inches)	Cable weight per ft max. (lbs)	Radius of bend min. (inches)	CDR ID	Rated voltage, max. (RMS)	Ampacity, each conductor (amps, max.)				NSN 6145-01
											DC or 60 Hz		400 Hz		
											40 °C Ambient	50 °C Ambient	40 °C Ambient	50 °C Ambient	
24640/7	1XMSO-	7	22	7	0.640	0.370	0.116	4.5	STD-2	600	---	---	---	---	226-2650
	1XMSO-	16	22	16	0.640	0.507	0.212	6.0	STD-2	600	---	---	---	---	226-2651
	1XMSO-	70	22	70	0.640	0.925	0.735	11.0	STD-2	600	---	---	---	---	226-7409
24640/6	2XAO-	2	22	2 pairs	0.640	0.331	0.079	4.0	Method 6	600	---	---	---	---	225-0964
	2XAO-	7	22	7 pairs	0.640	0.422	0.131	5.0	Method 6	600	---	---	---	---	225-0965
	2XAO-	10	22	10 pairs	0.640	0.511	0.175	6.0	Method 6	600	---	---	---	---	224-8337
	2XAO-	18	22	18 pairs	0.640	0.611	0.263	7.5	Method 6	600	---	---	---	---	224-8338
	2XAO-	40	22	40 pairs	0.640	0.858	0.513	10.5	Method 6	600	---	---	---	---	225-0966
24640/12	2XO-	6	26	6 pairs	0.250	0.329	0.082	4.0	Method 6 or TEL-3	600	---	---	---	---	225-2196
	2XO-	18	26	18 pairs	0.250	0.449	0.140	5.5	Method 6 or TEL-3	600	---	---	---	---	225-4904
	2XO-	24	26	24 pairs	0.250	0.509	0.175	6.0	Method 6 or TEL-3	600	---	---	---	---	224-9199
	2XO-	42	26	42 pairs	0.250	0.609	0.275	7.5	Method 6 or TEL-3	600	---	---	---	---	224-9200
	2XO-	60	26	60 pairs	0.250	0.691	0.334	8.5	Method 6 or TEL-3	600	---	---	---	---	225-2197
	2XO-	77	26	77 pairs	0.250	0.785	0.395	9.5	Method 6 or TEL-3	600	---	---	---	---	225-2198
24640/9	2XS-	2	22	2 pairs	0.640	0.332	0.061	3.0	STD-2	600	---	---	---	---	225-0967
	2XS-	3	22	3 pairs	0.640	0.350	0.074	3.0	STD-2	600	---	---	---	---	225-2193
	2XS-	7	22	7 pairs	0.640	0.455	0.135	4.0	STD-2	600	---	---	---	---	225-0968
	2XS-	10	22	10 pairs	0.640	0.579	0.194	5.0	STD-2	600	---	---	---	---	225-0969
	2XS-	14	22	14 pairs	0.640	0.627	0.245	5.0	STD-2	600	---	---	---	---	225-1387
	2XS-	19	22	19 pairs	0.640	0.694	0.319	5.5	STD-2	600	---	---	---	---	225-3452
	2XS-	24	22	24 pairs	0.640	0.818	0.431	6.5	STD-2	600	---	---	---	---	224-9191
	2XS-	30	22	30 pairs	0.640	0.866	0.500	7.0	STD-2	600	---	---	---	---	224-9192
24640/9 [5/]	2XSA-	2	22	4	0.640	0.382	0.081	3.0	STD-2	600	---	---	---	---	225-1388
	2XSA-	3	22	6	0.640	0.400	0.102	3.0	STD-2	600	---	---	---	---	225-1389

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TABLE VII (c)(2). MIL-DTL-24640 cable ratings and characteristics, non-watertight, non-flexing service (electronic, communication, and instrumentation) – Continued.

MIL-DTL-	Cable type designation		Conductor size: AWG or MCM	Number of conductors in cable	Area of each conductor (MCM)	Cable overall diameter max. (inches)	Cable weight per ft max. (lbs)	Radius of bend min. (inches)	CDR ID	Rated voltage, max. (RMS)	Ampacity, each conductor (amps, max.)				NSN 6145-01
											DC or 60 Hz		400 Hz		
											40 °C Ambient	50 °C Ambient	40 °C Ambient	50 °C Ambient	
24640/9 [5/]	2XSA-	7	22	14	0.640	0.505	0.176	4.0	STD-2	600	---	---	---	---	225-1390
	2XSA-	10	22	20	0.640	0.629	0.229	5.0	STD-2	600	---	---	---	---	225-1391
	2XSA-	14	22	28	0.640	0.677	0.280	5.0	STD-2	600	---	---	---	---	225-1392
	2XSA-	19	22	38	0.640	0.744	0.360	5.5	STD-2	600	---	---	---	---	225-6388
	2XSA-	24	22	48	0.640	0.868	0.443	6.5	STD-2	600	---	---	---	---	225-2194
	2XSA-	30	22	60	0.640	0.916	0.531	7.0	STD-2	600	---	---	---	---	224-9193
24640/9	2XSO-	3	22	3 pairs	0.640	0.410	0.117	5.0	STD-2	600	---	---	---	---	224-9194
	2XSO-	7	22	7 pairs	0.640	0.510	0.190	6.0	STD-2	600	---	---	---	---	224-9195
	2XSO-	10	22	10 pairs	0.640	0.640	0.287	8.0	STD-2	600	---	---	---	---	224-9196
	2XSO-	14	22	14 pairs	0.640	0.686	0.323	8.5	STD-2	600	---	---	---	---	224-9197
	2XSO-	19	22	19 pairs	0.640	0.765	0.423	9.5	STD-2	600	---	---	---	---	224-9198
	2XSO-	30	22	30 pairs	0.640	0.937	0.644	11.5	STD-2	600	---	---	---	---	225-2195
24640/13	2XSXO-	4	26	4 pairs	0.278	0.359	0.101	4.5	STD-2	600	---	---	---	---	224-9201
24640/11	3XS-	7	18	3 triads	1.620	0.647	.315	8.0	STD-2	600	---	---	---	---	225-2150
24640/11 [5/]	3XSA-	7	18	7 triads	1.620	0.697	0.342	8.5	STD-2	600	---	---	---	---	225-8946
24640/8	MXO-	10	20	10	1.020	0.377	0.118	4.5	STD-1 or 3	600	---	---	---	---	227-5743
	MXO-	14	20	14	1.020	0.410	0.147	5.0	STD-1 or 3	600	---	---	---	---	224-8339
24640/10	MXSO-	2	16	2	2.580	0.304	0.076	4.0	STD-1	600	---	---	---	---	225-2148
	MXSO-	9	16	9	2.580	0.458	0.180	5.5	STD-1	600	---	---	---	---	225-2149
	MXSO-	21	16	21	2.580	0.596	0.329	7.5	STD-1	600	---	---	---	---	225-1377
	MXSO-	37	16	37	2.580	0.694	0.497	8.5	STD-1	600	---	---	---	---	225-1378
24640/4	TTX-	3	20	3 pairs	1.020	0.320	0.062	2.5	Method 6	600	5/4.0 [1/]	4/3.0 [1/]	5/4.0 [1/]	4/3.0 [1/]	225-2190
	TTX-	15	20	15 pairs	1.020	0.591	0.221	5.0	Method 6	600	5/0.6	4/0.5	5/0.6	4/0.5	225-1393
24640/4 [5/]	TTXA-	3	20	3 pairs	1.020	0.370	0.082	2.5	TEL-3	600	5/4.0	4/3.0	5/4.0	4/3.0	225-2191
	TTXA-	15	20	15 pairs	1.020	0.641	0.275	5.0	TEL-3	600	5/0.6	4/0.5	5/0.6	4/0.5	224-6916
24640/5	TTXS-	2	20	2 pairs	1.020	0.361	0.074	3.0	STD-2	600	---	---	---	---	224-6917
	TTXS-	4	20	4 pairs	1.020	0.424	0.115	3.5	STD-2	600	---	---	---	---	226-5038

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TABLE VII (c)(2). MIL-DTL-24640 cable ratings and characteristics, non-watertight, non-flexing service (electronic, communication, and instrumentation) – Continued.

MIL-DTL-	Cable type designation		Conductor size: AWG or MCM	Number of conductors in cable	Area of each conductor (MCM)	Cable overall diameter max. (inches)	Cable weight per ft max. (lbs)	Radius of bend min. (inches)	CDR ID	Rated voltage, max. (RMS)	Ampacity, each conductor (amps, max.)				NSN 6145-01
											DC or 60 Hz		400 Hz		
											40 °C Ambient	50 °C Ambient	40 °C Ambient	50 °C Ambient	
24640/5 [5/]	TTXSA-	2	20	2 pairs	1.020	0.411	0.100	3.5	STD-2	600	---	---	---	---	224-8334
	TTXSA-	4	20	4 pairs	1.020	0.474	0.150	4.0	STD-2	600	---	---	---	---	224-8335
24640/5	TTXSO-	2	20	2 pairs	1.020	0.416	0.126	5.0	STD-2	600	---	---	---	---	224-8336
	TTXSO-	6	20	6 pairs	1.020	0.546	0.219	6.5	STD-2	600	---	---	---	---	225-2192
	TTXSO-	8	20	8 pairs	1.020	0.633	0.284	8.0	STD-2	600	---	---	---	---	225-0963
	TTXSO-	10	20	10 pairs	1.020	0.675	0.322	8.0	STD-2	600	---	---	---	---	226-7408

NOTES:

1/ Ind/Avg indicates the maximum current for each conductor (Ind), and the maximum current (Avg) for each conductor when all conductors in the cable are used.

2/ May be STD-1 or LTR-5, manufacturer's option.

3/ May be STD-3 or STD-4, manufacturer's option.

4/ 6145-00.

5/ The armored cable versions are not in the most current versions of the respective specification sheets.

6/ Canceled and replaced, see 5.1.

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5.8 Cable classification (MIL-DTL-915). Cables specified in MIL-DTL-915 are listed in table VIII under the following general classifications:

a. Watertight, flexing service:
   (1) power
   (2) control
   (3) electronic, communication, and instrumentation
b. Non-watertight, flexing service:
   (1) power and lighting
   (2) electronic, communication, and instrumentation
c. Non-watertight, non-flexing service:
   (1) electronic, communication, and instrumentation
d. Watertight, non-flexing service:
   (1) electronic, communication, and instrumentation

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TABLE VIII (a)(1). MIL-DTL-915 cable ratings and characteristics watertight, flexing service (power). [6/]

MIL-DTL-	Cable type designation	Conductor size: AWG or MCM	Number of conductors or pairs in cable	Area of each conductor (MCM)	Cable overall diameter max. (inches)	Cable weight per ft max. (lbs)	Radius of bend min. (inches)	CDR ID	Rated voltage, max. (RMS)	Ampacity, each conductor (amps, max.)				NSN 6145-01
										DC or 60 Hz		400 Hz		
										40 °C Ambient	50 °C Ambient	40 °C Ambient	50 °C Ambient	
915/7	DSWS- 4	14	2	4.110	0.800	0.050	3.0	STD-3	600	---	---	---	---	776-8513
915/10	MCSF- 4	6	2	26.240	1.500	1.625	12.0	STD-3	600	---	---	---	---	802-2059
		1	2	83.690	---	---	---	---	---	---	---	---	---	---

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TABLE VIII (a)(2). MIL-DTL-915 cable ratings and characteristics, watertight, flexing service (control).

MIL-DTL-	Cable type designation		Conductor size: AWG or MCM	Number of conductors or pairs in cable	Area of each conductor (MCM)	Cable overall diameter max. (inches)	Cable weight per ft max. (lbs)	Radius of bend min. (inches)	CDR ID	Rated voltage, max. (RMS)	Ampacity, each conductor (amps, max.)				NSN 6145-01
											DC or 60 Hz		400 Hz		
											40 °C Ambient	50 °C Ambient	40 °C Ambient	50 °C Ambient	
915/58	MWF-	7	18	7	1.620	0.500	0.145	1.5	STD-1	600	1.5	1.5	1.5	1.5	995-2543
	MWF-	10	18	10	1.620	0.635	0.230	2.0	STD-1	600	2.0	2.0	2.0	2.0	162-4794
	MWF-	14	18	14	1.620	0.635	0.250	2.0	STD-1	600	2.0	2.0	2.0	2.0	985-8320
	MWF-	19	18	19	1.620	0.745	0.350	2.5	STD-1	600	2.5	2.5	2.5	2.5	983-8639
	MWF-	24	18	24	1.620	0.836	0.430	2.5	STD-1	600	2.5	2.5	2.5	2.5	985-8318
	MWF-	30	18	30	1.620	0.945	0.550	3.0	STD-1	600	3.0	3.0	3.0	3.0	084-1342
	MWF-	37	18	37	1.620	1.045	0.680	3.5	STD-1	600	3.5	3.5	3.5	3.5	154-4373

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TABLE VIII (a)(3). MIL-DTL-915 cable ratings and characteristics, watertight, flexing service (electronic, communication, and instrumentation).

MIL-DTL- or MIL-C-	Cable type designation		Conductor size: AWG or MCM	Number of conductors in cable	Area of each conductor (MCM)	Cable overall diameter max. (inches)	Cable weight per ft max. (lbs)	Radius of bend min. (inches)	CDR ID	Rated voltage, max. (RMS)	Ampacity, each conductor (amps, max.)				NSN 6145-01
											DC or 60 Hz		400 Hz		
											40 °C Ambient	50 °C Ambient	40 °C Ambient	50 °C Ambient	
915/47	1SWF-	2	22	2	0.700	0.625	0.188	2.0	STD-2	300	---	---	---	---	051-4825
915/48	2SWF-	3	22	3 pairs	0.700	0.625	0.200	2.0	STD-2	300	---	---	---	---	158-6329
	2SWF-	4	22	4 pairs	0.700	0.625	0.211	2.0	STD-2	300	---	---	---	---	985-7926
	2SWF-	7	22	7 pairs	0.700	0.815	0.366	2.5	STD-2	300	---	---	---	---	964-4662
	2SWF-	24	22	24 pairs	0.700	1.250	0.620	7.5	STD-2	300	---	---	---	---	
915/8	7SS-	2	18	7	1.620	0.625	0.185	1.5	SPL-3	600	---	---	---	---	905-6797
915/8	DSS-	2	18	2	1.620	0.390	0.120	1.0	SPL-3	600	6	5	---	---	914-9973
	DSS-	3	16	2	2.580	0.500	0.160	1.5	SPL-3	600	9	8	---	---	912-2616
	DSS-	4	14	2	4.110	0.500	0.180	1.5	SPL-3	600	13	11	---	---	913-2401
915/8	FSS-	2	18	4	1.620	0.500	0.180	1.5	SPL-3	600	5	4	---	---	913-2402
	FSS-	3	16	4	2.580	0.500	0.210	1.5	SPL-3	600	8	7	---	---	159-6655
	FSS-	4	14	4	4.110	0.625	0.240	2.0	SPL-3	600	13	11	---	---	985-7843
915/61 [4/]	S2S	---	18	1	1.620	0.500	1.000	3.0	SPL-3	600	---	---	---	---	155-3526
915/79 [7/]	TPUM-	6	16	6 pairs	1.700	0.925	0.125	7.5	TEL-1 or 2	600	---	---	---	---	156-7864
915/8	TSS-	2	18	3	1.620	0.400	0.140	1.0	SPL-3	600	6	5	---	---	162-2600
	TSS-	3	16	3	2.580	0.500	0.180	1.5	SPL-3	600	9	8	---	---	159-8619
	TSS-	4	14	3	4.110	0.500	0.200	1.5	SPL-3	600	13	11	---	---	905-6795

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TABLE VIII (b)(1). MIL-DTL-915 cable ratings and characteristics, non-watertight, flexing service (power and lighting).

MIL-DTL- or MIL-C-	Cable type designation		Conductor size: AWG or MCM	Number of conductors in cable	Area of each conductor (MCM)	Cable overall diameter max. (inches)	Cable weight per ft max. (lbs)	Radius of bend min. (inches)	CDR ID	Rated voltage, max. (RMS)	Ampacity, each conductor (amps, max.)				NSN 6145-01
											DC or 60 Hz		400 Hz		
											40 °C Ambient	50 °C Ambient	40 °C Ambient	50 °C Ambient	
915/1 [5/]	CVSF-	4	5	1	33.090	1.450	1.300	9.0	STD-3 [1/]	600	100	75	100	75	191-2037
	---	---	3	3	52.620	---	---	---	---	---	---	---	---	---	---
915/9 [3/]	JAS-	250	12	2	6.530	1.260	0.917	10.0	STD-4	600	---	---	---	---	753-2264
	---	---	250	2	250.000	2.480	---	20.0	STD-3	600	---	---	---	---	---
915/6	THOF-	400	400 (Navy STD)	3	413.500	2.800	3.950	22.0	STD-3 [1/]	600	400	365	315	265	202-0671
	THOF-	500	500 MCM	3	500.000	3.100	7.100	25.0	STD-3 [1/]	600	500	450	350	275	202-2040
915/20 [4/]	TRF-	105	0	1	105.600	0.760	0.510	6.0	---	600	143	121	---	---	943-7407
	TRF-	133	00	1	133.100	0.810	0.618	6.5	---	600	167	141	---	---	956-3451
	TRF-	168	000	1	167.800	0.860	0.704	7.0	---	600	201	170	---	---	928-9103
915/21 [4/]	TRXF-	84	1	1	83.700	0.600	0.360	5.0	---	125	130	110	---	---	914-9010
	TRXF-	105	0	1	105.600	0.680	0.460	5.5	---	125	143	121	---	---	913-2065
	TRXF-	133	00	1	133.100	0.750	0.567	6.0	---	125	167	141	---	---	914-0517
915/5 [4/]	DLT	---	18	4	1.620	0.710	0.050	6.0	STD-3	300	---	---	---	---	574-2040

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TABLE VIII (b)(2). MIL-DTL-915 cable ratings and characteristics, non-watertight, flexing service (electronic, communication, and instrumentation).

MIL-DTL-	Cable type designation	Conductor size: AWG or MCM	Number of conductors in cable	Area of each conductor (MCM)	Cable overall diameter max. (inches)	Cable weight per ft max. (lbs)	Radius of bend min. (inches)	CDR ID	Rated voltage, max. (RMS)	Ampacity, each conductor (amps, max.)				NSN 6145-01
										DC or 60 Hz		400 Hz		
										40 °C Ambient	50 °C Ambient	40 °C Ambient	50 °C Ambient	
915/74 [4/]	5SS ---	18	5	1.620	0.560	0.240	2.0	STD-3	600	---	---	---	---	155-4488

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TABLE VIII (c)(1). MIL-DTL-915 cable ratings and characteristics, non-watertight, non-flexing service (electronic, communication, and instrumentation).

MIL-DTL- or MIL-C-	Cable type designation		Conductor size: AWG or MCM	Number of conductors in cable	Area of each conductor (MCM)	Cable overall diameter max. (inches)	Cable weight per ft max. (lbs)	Radius of bend min. (inches)	CDR ID	Rated voltage, max. (RMS)	Ampacity, each conductor (amps, max.)				NSN 6145-01
											DC or 60 Hz		400 Hz		
											40 °C Ambient	50 °C Ambient	40 °C Ambient	50 °C Ambient	
915/67	MSP	---	22	16	0.700	1.635	1.500	10.5	---	---	---	---	---	---	791-3234
	---	---	20	18	1.111	---	---	---	---	---	---	---	---	---	---
	---	---	16	25	2.580	---	---	---	---	---	---	---	---	---	---
915/81	1PR-	A20E	20	2	1.021	0.29	---	2.3	---	300	---	---	---	---	6145-21-882-4970
915/82 [7/]	1PR-	16	16	1 pair	2.582	0.4	0.158	3.2	---	600	---	---	---	---	---
915/83 [7/]	7PR-	16	16	7 pairs	2.582	0.82	---	8.56	---	600	---	---	---	---	---
915/84 [7/]	2SPR-	16	16	2 pairs	2.582	0.65	---	5.2	---	600	---	---	---	---	---
915/85 [7/]	3PR-	16	16	3 pairs	2.582	0.55	---	4.4	---	600	---	---	---	---	---
915/86 [7/]	1Q-	16	16	4	2.582	0.4	---	3.2	---	600	---	---	---	---	---
915/87 [7/]	1TR-	16	16	3	2.582	0.4	0.175	3.2	---	600	---	---	---	---	---
915/88 [7/]	7SPR-	16S	16	7 pairs	2.582	0.95	---	7.6	---	600	---	---	---	---	---
915/89 [7/]	1SPR-	16	16	1 pair	2.582	0.4	---	3.2	---	600	---	---	---	---	---

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TABLE VIII (d)(1). MIL-DTL-915 cable ratings and characteristics, watertight, non-flexing service (electronic, communication, and instrumentation).

MIL-DTL-	Cable type designation		Conductor size: AWG or MCM	Number of conductors in cable	Area of each conductor (MCM)	Cable overall diameter max. (inches)	Cable weight per ft max. (lbs)	Radius of bend min. (inches)	CDR ID	Rated voltage, max. (RMS)	Ampacity, each conductor (amps, max.)				NSN 6145-01
											DC or 60 Hz		400 Hz		
											40 °C Ambient	50 °C Ambient	40 °C Ambient	50 °C Ambient	
915/66	MSPW	---	22	16	0.700	1.635	1.500	10.5	STD-2	---	---	---	---	---	791-3233
	---	---	20	18	1.111	---	---	---	---	---	---	---	---	---	---
	---	---	16	25	2.580	---	---	---	---	---	---	---	---	---	---
915/22	TSP-	11	22	11 pairs	0.640	0.730	0.275	4.5	TEL-3	300	4/1 2/ 2/	3/0.5 2/	---	---	916-7217
	TSP-	31	22	31 pairs	0.640	1.062	0.611	6.5	TEL-3	300	---	---	---	---	940-8711
915/22 8/	TSPA-	11	22	11 pairs	0.640	0.780	0.310	4.0	TEL-3	300	4/1	3/0.5	---	---	156-9581
	TSPA-	31	22	31 pairs	0.640	1.112	0.650	6.5	TEL-3	300	---	---	---	---	155-8741

NOTES:

1/ Identification may be STD-3 or STD-4, manufacturer's option.

2/ Ind/Avg indicates the maximum current for each conductor (Ind), and the maximum current (Avg) for each conductor when all conductors in the cable are used.

3/ Oval shape, minor and major diameter.

4/ Specification sheets have been cancelled and replaced by A-A-59608 (915/20), A-A-59609 (915/21), A-A-59615 (915/5), A-A-59612 (915/74), and A-A-59610 (915/61).

5/ Specification sheet has been cancelled and replaced by MIL-DTL-32180.

6/ Cables in this table are for outboard use or portable use consistent with MIL-STD-2003.

7/ Inactive for new design but in use.

8/ Cable type TSPA no longer covered in MIL-DTL-915/22.

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5.9 Ampacity rating. See 4.1.10 and 4.1.11 for guidance to determine the ampacity of electrical cables installed in power and control systems. In general, to obtain the ampacity rating for an ambient temperature of either 60 or 70 °C, first determine the ampacity rating for 40 °C ambient from tables VI, VII, or VIII. Then multiply that value by the applicable derating factor. For specific cable types, the applicable derating is shown in tables IX and X. Cables listed in this table should not be used in ambient temperatures above 70 °C. Cables not listed in this table should not be used in ambient temperatures above 60 °C. The ampacity derating factor does not change for the limited cable types with armored variants.

TABLE IX. Ampacity derating factors for ambient temperatures above 50 °C.

Cable type	Ambient temperature (°C)	
	60	70
LSDHOF	0.78	0.67
LSDNW	0.78	0.67
LSDSGU	0.80	0.69
DSS	0.63	---
LSFSGU	0.80	0.69
FSS	0.63	---
LSMDU	0.72	---
LSMHOF	0.80	0.69
LSMNW	0.78	0.67
LSMSCU	0.80	0.69
LSSHOF	0.78	0.67
LSTHOF	0.78	0.67
LSTNW	0.78	0.67
TRF	0.63	---
TRXF	0.63	---
LSTSGU	0.80	0.69
TXW		
TSS	0.63	---
LS5KVTSGU	0.80	0.69
LS6SGU	0.80	0.69
LS7SGU	0.80	0.69
7SS	0.63	---

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TABLE X. Ampacities of degaussing cables.

(Maximum amperes for each conductor at 40 and 50 °C ambient temperatures)											
Cable type and size		**1 cable in air**		**2 cables in air**		**3 or 4 cables in air**		**5 to 7 cables in air**		**1 cable in conduit or 8 or more cables in air**	
		40°	**50°**	**40°**	**50°**	**40°**	**50°**	**40°**	**50°**	**40°**	**50°**
LSDSGU/A-	14	66	61	60	56	55	51	49	46	46	43
	23	86	79	79	73	72	67	64	60	61	56
	50	139	128	128	119	117	109	104	97	98	90
	75	185	171	170	159	156	146	139	130	132	122
	100	219	201	202	187	184	171	164	153	155	143
	200	339	312	312	290	285	265	254	237	241	222
LSTSGU/A-	9	43	40	39	36	34	30	29	27	27	26
	14	56	52	51	48	47	44	42	39	40	37
	23	76	70	70	65	64	59	57	53	54	50
	50	121	111	111	103	102	94	91	84	86	79
	75	163	150	150	140	137	128	122	114	116	106
	100	191	176	176	164	160	150	143	134	136	125
	150	259	238	238	221	218	202	194	181	184	169
	200	298	275	274	256	250	234	224	209	212	195
	300	383	352	352	324	322	296	287	264	276	253
	400	478	440	440	405	401	370	358	330	344	317
LSFSGU/A-	9	43	40	41	38	38	36	36	34	34	32
	23	76	70	72	67	68	63	64	60	61	56
	50	121	112	115	107	109	101	103	95	97	90
	75	163	151	154	144	146	136	138	129	130	121
	100	191	178	182	169	172	160	162	151	153	142
	150	258	240	245	228	232	216	220	204	207	192
	200	298	277	283	263	268	249	253	235	238	221
LS6SGU/A-	100	163	153	150	140	136	127	122	114	116	108
	125	192	173	175	163	160	147	144	133	136	125
	150	225	204	207	190	188	173	169	156	160	147
	200	250	243	235	222	219	202	188	182	177	172
LS7SGU-	6	41	38	39	35	36	32	31	28	29	27

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TABLE X. Ampacities of degaussing cables – Continued.

(Maximum amperes for each conductor at 40 and 50 °C ambient temperatures)											
Cable type and size		**1 cable in air**		**2 cables in air**		**3 or 4 cables in air**		**5 to 7 cables in air**		**1 cable in conduit or 8 or more cables in air**	
		40°	**50°**	**40°**	**50°**	**40°**	**50°**	**40°**	**50°**	**40°**	**50°**
LSMDU-	6	15	14	14	13	14	12	13	12	12	11
	14	25	22	24	21	22	20	21	19	20	18
	23	34	30	32	28	30	27	29	25	27	24
	40	49	43	46	41	44	37	41	37	39	36
	60	65	59	62	57	58	52	55	52	51	50
LSMDY-	6	15	14	14	13	14	12	13	12	12	11
	14	25	22	24	21	22	20	21	19	20	18
	23	34	30	32	28	30	27	29	25	27	24
	40	49	43	46	41	44	39	41	37	39	36
	60	65	59	62	57	58	54	55	52	51	50
LSMSCU/A/S-	7	9	8	8.5	7.5	8	7	7.5	7	7	6.5
	10	9	8	8.5	7.5	8	7	7.5	7	7	6.5
	14	9	8	8.5	7.5	8	7	7.5	7	7	6.5
	19	9	8	8.5	7.5	8	7	7.5	7	7	6.5
	24	9	8	8.5	7.5	8	7	7.5	7	7	6.5
	30	9	8	8.5	7.5	8	7		7	7	6.5
	37	9	8	8.5	7.5	8	7	7.5	7	7	6.5
	44	5.5	5	5.0	4.5	5	4.5	4.5	4	4.5	4.0
NOTE: Cables should be tested at the highest current that will be encountered after installation to ensure that overheating does not take place.											

6. NOTES

6.1 Intended use. This handbook is intended to aid ship design, supply, installation, and maintenance activities in the utilization of electric shipboard cable, particularly in selection of the appropriate cables for an electrical system and for the selection of alternate or substitute cables for use in lieu of specified types and sizes that might not be immediately available. This handbook is also intended to aid in selecting currently available cables for replacement of obsolete cables. This handbook does not cover shore-use cable, magnet wire, fiberoptic cable, coaxial cables, or radio frequency special cables used in connection with minesweeping and harbor defense.

6.2 Subject term (key word) listing.

Ampacity
Cable, degaussing
Cable, flexing service
Cable, non-flexing service
Codes, identification
Conductor, multiple

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Insulation
Jacket, armored
Jacket, unarmored
Medium voltage
Non-watertight
Shield, braided
Watertight

6.3 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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ELECTRICAL CABLE VOLTAGE DROP CALCULATIONS

A.1. SCOPE

A.1.1 Scope. This appendix is intended for use as a guide to determining cable voltage drops for alternating current (AC) and direct current (DC) power, lighting, electronic, interior communication, and weapon control systems.

A.2. APPLICABLE PUBLICATIONS

A.2.1 Government publications. This paragraph is not applicable to this appendix.

A.2.2 Non-Government publications. The following publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of publications are those cited in the solicitation or contract.

ASTM INTERNATIONAL

ASTM B8 - Standard Specification for Concentric-Lay-Stranded Copper Conductors, Hard, Medium-Hard, or Soft

ASTM B258 - Standard Specification for Standard Nominal Diameters and Cross-Sectional Areas of AWG Sizes of Solid Round Wires Used as Electrical Conductors

(Copies of these publications are available online at www.astm.org.)

A.3. DEFINITIONS

A.3.1 Symbols and abbreviations. The symbols and abbreviations used in this appendix are as follows:

Symbols		Parameter	Units
A	-	Cross-sectional area of a conductor	Cmil
AWG	-	American Wire Guage	---
D%	-	Cable voltage drop expressed in percent with respect to system or source voltage	---
DF	-	Cable drop factor for equation using current	$(Amp.ft)^{-1}$
DF′	-	Cable drop factor for equation using power	$(Watt.ft)^{-1}$
D or d	-	Conductor diameter	Mils (thousandth of an inch)
E	-	Line-to-neutral rated voltage at the switchboard of a three-phase system or rated voltage of a single single-phase or DC system	Volt
E_X	-	Line-to-line rate voltage of a three-phase system (E_{AB}, E_{BC}, E_{CA})	Volt
I	-	Line current (I_A, I_B, or I_C)	Ampere
I_{LO}	-	Resultant load currents for each phase (leg) of a three-phase, delta circuit (I_{AB}, I_{BC}, or I_{CA})	Ampere
I_X	-	Difference in two line currents of a three-phase system ($I_A - I_B$, $I_B - I_C$, $I_C - I_A$)	Ampere
L	-	Cable length	Feet
P	-	Real power for each phase (leg) load of a three-phase delta circuit (P_{AB}, P_{BC}, or P_{CA})	Watt
P_X	-	Net real power in two lines of a three-phase delta system ($P_A - P_B$, $P_B - P_C$, $P_C - P_A$)	Watt
pf	-	Load power factor	---
Q	-	Reactive power for each phase (leg) load of a three-phase delta circuit (Q_{AB}, Q_{BC}, or Q_{CA})	Var

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Q_X	-	Net reactive power in two lines of a three-phase delta system ($Q_A - Q_B$, $Q_B - Q_C$, $Q_C - Q_A$)	Var
R	-	Total cable resistance per phase	Ohm
R_{DC}	-	Conductor DC resistance	Ohm
S	-	Apparent power for each phase (leg) load of a three-phase delta circuit (S_{AB}, S_{BC}, or S_{CA})	Volt-amp
V	-	Terminal voltage or load voltage	Volt
V_{AN}	-	Line-to-neutral terminal load voltage for phase A of a three-phase, four-wire system	Volt
X	-	Total cable reactance per phase	Ohm
Z	-	Total cable impedance per phase	Ohm
z	-	Cable impedance per phase per foot	Ohm/ft
Z_{AN}	-	Load impedance in phase A of a three-phase, four-wire (wye) circuit	Ohm
Z_{LO}	-	Load impedance in each phase (leg) of a three-phase delta circuit (Z_{AB}, Z_{BC}, or Z_{CA})	Ohm
α	-	Angle between terminal load voltage and cable voltage drop (IZ)	Degree
α_x	-	Angle between terminal load voltage and cable voltage drop (I_xZ)	Degree
β	-	Cable impedance angle	Degree
δ	-	Mass density of a selected material	g/cm^3
Θ	-	Load power factor angle or angle between load voltage and line current	Degree
Θ_x	-	Angle between terminal load voltage and current I_x	Degree
ρ	-	Resistivity of a selected material at desired operating temperature	Ohm.ft
ρ_o	-	Resistivity of a selected material at 20 °C	Ohm.ft

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A.4. GENERAL EQUATIONS FOR CABLE VOLTAGE DROP CALCULATIONS

A.4.1 Voltage drop calculations for DC systems. The voltage drop for DC circuits is calculated from the following equations as derived in A.8.

A.4.1.1 Two-wire circuits. For power systems, use a resistivity based on an ambient temperature of 65 °C based on the likely ambient for power cables. For non-power systems, use an ambient temperature of 45 °C to determine resistivity.

- For all systems except power

$$D\% = 22.78 \left(\frac{IL}{AE}\right) * 100 \tag{8-5}$$

- For power systems only

$$D\% = 24.42 \left(\frac{IL}{AE}\right) * 100 \tag{8-6}$$

A.4.2 Voltage drop calculations for AC systems. The voltage drops for AC circuits are calculated from the following equations as derived in A.9, A.10, A.11, and A.12.

A.4.2.1 Single-phase circuits.

- For all systems

$$D\% = 2IL \left(\frac{zcos(\alpha)}{E}\right) * 100 \tag{9-9}$$

A.4.2.2 Three-phase circuits.

- Voltage drop in each line for balanced systems, such as, electronic, interior communication, or weapon control systems

$$D\% = \sqrt{3} I_{LO} L \left(\frac{zcos(\alpha)}{E}\right) * 100 \tag{9-10}$$

- Voltage drop in each line for all systems

$$D\% = IL \left(\frac{zcos(\alpha)}{E}\right) * 100 \tag{10-10}$$

- Voltage drop in each phase for all systems

$$D\% = I_X L \left(\frac{zcos(\alpha_X)}{E_X}\right) * 100 \tag{10-17}$$

A.4.3 Voltage drop calculations using drop factors. The following simplified equations are used for percent drop calculations in lighting and power systems in conjunction with the cable drop factors listed in tables XIII through XVII.

A.4.3.1 Lighting systems.

A.4.3.1.1 Single-phase circuits.

$$D\% = 2IL\ (DF) \qquad \text{Table XVI}$$

$$D\% = 2PL\ (DF') \qquad \text{Table XVII}$$

A.4.3.1.2 Three-phase circuits.

$$D\% = I_X L\ (DF) \qquad \text{Table XVI}$$

$$D\% = P_X L\ (DF') \qquad \text{Table XVII}$$

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A.4.3.2 Power systems.

A.4.3.2.1 Three-phase circuits. The drop in each line is:

$$D\% = IL\ (DF)$$ Tables XIII, XIV, and XV.

Where:

- I, I_{LO}, I_X, Θ, α, Θ_X, and α_X are calculated in A.5
- P and P_X are calculated in A.11
- DF and DF′ are calculated and tabulated in A.7.1

A.5. CURRENT CALCULATIONS FOR AC SYSTEMS

A.5.1 Single-phase circuits. The line current (I) used in the voltage drop equations is the scalar magnitude of current (I_{LO}) obtained by adding vectorially all load currents in the branch:

$$I = I_{LO}$$

The angles Θ, α, and β are calculated from the following equations as derived in A.9:

$$\theta = cos^{-1}\ (pf),\ \beta = tan^{-1}(X/R)$$

$$\alpha = -\theta + \beta \qquad (9\text{-}2)$$

Θ is positive if the load power factor (pf) is lagging

Θ is negative if the load power factor (pf) is leading

A.5.2 Three-phase-delta circuits.

A.5.2.1 Balanced systems. If the system is balanced, the magnitude of the line current is equal to:

$$I = \sqrt{3}I_{LO} = \sqrt{3}I_{AB} = \sqrt{3}I_{BC} = \sqrt{3}I_{CA}$$

The total load currents $\bar{I}_{AB}$, $\bar{I}_{BC}$, and $\bar{I}_{CA}$ are the vectorial sum of all the currents in their respective phase (leg) loads. If the load power factor angles in legs AB, BC, and CA are θ_{AB}, θ_{BC}, and θ_{CA}, respectively, the phase load currents can be written as:

$$\bar{I}_{AB} = I_{AB}\underline{/(0° - \theta_{AB})},\ \bar{I}_{BC} = I_{BC}\underline{/(\text{-}120° - \theta_{BC})},\ \bar{I}_{CA} = I_{CA}\underline{/(120° - \theta_{CA})}$$

Similarly, the phase voltages $\bar{V}_{AB}$, $\bar{V}_{BC}$, and $\bar{V}_{CA}$ are defined as:

$$\bar{V}_{AB} = V_{AB}\underline{/0°},\ \bar{V}_{BC} = V_{BC}\underline{/\text{-}120°},\ \text{and}\ \bar{V}_{CA} = V_{CA}\underline{/\text{-}120°}$$

A.5.2.2 Balanced/unbalanced systems. If the system is unbalanced or balanced, the magnitude of line currents I and the associated angles Θ and α are calculated from the following equations as derived in A.10:

$$\bar{I}_A = \bar{I}_{AB} - \bar{I}_{CA} \qquad (10\text{-}7)$$

$$\theta_A = 0° - \underline{/I_A}$$

$$\alpha_A = \text{-}\theta_A + \beta$$

$$\bar{I}_B = \bar{I}_{BC} - \bar{I}_{AB} \qquad (10\text{-}8)$$

$$\theta_B = \text{-}120° - \underline{/I_B}$$

$$\alpha_B = \text{-}\theta_B + \beta$$

$$\bar{I}_C = \bar{I}_{CA} - \bar{I}_{BC} \qquad (10\text{-}9)$$

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$$\theta_C = 120° - \underline{/I_C}$$

$$\alpha_C = -\theta_C + \beta$$

The currents I_X and the associated angles Θ_X and α_X are calculated as follows:

$$\bar{I}_{X(AB)} = \bar{I}_A - \bar{I}_B \qquad (10\text{-}12)$$

$$\theta_{X(AB)} = 0° - \underline{/I_{X(AB)}}$$

$$\alpha_{X(AB)} = -\theta_{X(AB)} + \beta$$

$$\bar{I}_{X(BC)} = \bar{I}_B - \bar{I}_C \qquad (10\text{-}13)$$

$$\theta_{X(BC)} = -120° - \underline{/I_{X(BC)}}$$

$$\alpha_{X(BC)} = -\theta_{X(BC)} + \beta$$

$$\bar{I}_{X(CA)} = \bar{I}_C - \bar{I}_A \qquad (10\text{-}14)$$

$$\theta_{X(CA)} = 120° - \underline{/I_{X(CA)}}$$

$$\alpha_{X(CA)} = -\theta_{X(CA)} + \beta$$

A.5.3 Three-phase four-wire circuits. The individual line currents (I) are calculated from the following equation as derived in A.12:

$$I = I_{LO} = \frac{V_{AN}}{Z_{AN}} \qquad (12\text{-}3)$$

A.5.4 Example of voltage drop calculations. The following example is a sample calculation of percent voltage drop for a three-phase lighting system (see figures A-3 and A-4).

Step 1.

Determine the load current (I_{LO}) in each phase by adding vectorially all the connected load currents in that phase. Let assume the total load current for each phase as follows:

$$I_{AB} = 6.16A$$

$$I_{BC} = 5.7A$$

$$I_{CA} = 9.5A$$

Also, the following parameters are given in table XVI:

Cable type: LSTSGU-9
Length: L = 45ft
Impedance: z = $1.120(10^{-3})$ Ω/ft
Angle: β = 1.82°

Step 2.

Calculate phase load currents. Let the terminal load voltages be:

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$$\bar{V}_{AB} = V_{AB}\underline{/0°}$$

$$\bar{V}_{BC} = V_{BC}\underline{/\text{-}120°}$$

$$\bar{V}_{CA} = V_{CA}\underline{/120°}$$

Assume load power factor for each phase of 0.80 lagging, the phase currents with respect to their respective terminal voltages can be written as follows:

$$\bar{I}_{AB} = I_{AB}\underline{/0° - \theta_{AB}} = 6.16\underline{/\text{-}37°}$$

$$\bar{I}_{BC} = I_{BC}\underline{/\text{-}120° - \theta_{BC}} = 5.7\underline{/\text{-}157°}$$

$$\bar{I}_{CA} = I_{CA}\underline{/120° - \theta_{CA}} = 9.5\underline{/83°}$$

Where $\theta_{AB} = \theta_{BC} = \theta_{CA} = \cos^{-1}(0.80) = 37°$

Step 3.

Determine line current I, angles Θ, and angle α from equations (10-7), (10-8), and (10-9), respectively, after converting the phase currents from their polar forms to rectangular forms.

$$\bar{I}_{AB} = 6.16\underline{/\text{-}37°} = 4.92 - j3.71$$

$$\bar{I}_{BC} = 5.7\underline{/\text{-}157°} = \text{-}5.25 - j2.23$$

$$\bar{I}_{CA} = 9.5\underline{/83°} = 1.16 + j9.43$$

The line currents are calculated as follows:

$$\bar{I}_A = \bar{I}_{AB} - \bar{I}_{CA} \qquad (10\text{-}7)$$
$$= (4.92 - j3.71) - (1.16 + j9.43)$$

$$\bar{I}_A = 3.76 - j13.14$$
$$= 13.67\underline{/285.97°}$$

and,

$$\theta_A = 360° - 285.97°$$
$$= 74.03°$$

$$\alpha_A = -74.03° + 1.82°$$
$$= -72.21°$$

$$\bar{I}_B = \bar{I}_{BC} - \bar{I}_{AB} \qquad (10\text{-}8)$$
$$= (\text{-}5.25 - j2.23) - (4.92 - j3.71)$$
$$= \text{-}10.17 + j1.48$$

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$= 10.28\underline{/171.72°}$

and,

$\theta_B = \text{-}120° - 171.72°$

$= \text{-}291.72°$

$\alpha_B = \text{-}(\text{-}291.72°) + 1.82°$

$= 293.54°,$

$$\bar{I}_C = \bar{I}_{CA} - \bar{I}_{BC} \quad (10\text{-}9)$$

$= (1.16 + j9.43) - (\text{-}5.25 - j2.23)$

$= 6.41 + j11.66$

$= 13.31\underline{/61.20°}$

$\theta_C = 120° - 61.20°$

$= 58.8°$

$\alpha_C = \text{-}58.80° + 1.82°$

$= \text{-}56.98°$

Step 4.

The drop in each line from the switchboard to the load is calculated as follows:

$$D_A\% = I_A L\left(\frac{z\cos(\alpha_A)}{E}\right) * 100 \quad (10\text{-}10)$$

$$D_A\% = (13.67)(45)\left(\frac{(1.120*10^{-3})\cos(\text{-}72.21°)}{(120/\sqrt{3})}\right) * 100$$

$= 0.30\%$

$$D_B\% = I_B L\left(\frac{z\cos(\alpha_B)}{E}\right) * 100 \quad (10\text{-}10)$$

$$= (10.28)(45)\left(\frac{(1.120*10^{-3})\cos(293.54°)}{(120/\sqrt{3})}\right) * 100$$

$= 0.30\%$

$$D_C\% = I_C L\left(\frac{z\cos_{(\alpha_C)}}{E}\right) * 100 \quad (10\text{-}10)$$

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$$= (13.30)(45)\left(\frac{(1.120*10^{-3})\cos(-57.98°)}{(120/\sqrt{3})}\right) * 100$$

$$= 0.53\%$$

The total drop ($D_T\%$) in this portion of cable is the combination of the drops in the lines:

$$D_T\% = D_A\% + D_B\% + D_C\%$$

$$= 0.30\% + 0.30\% + 0.53\%$$

$$= 1.13\%$$

Step 5.

If the voltage drop in each phase is desired, the currents I_X should be determined from the following equations: The current loop E_{AB} is:

$$\bar{I}_{X(AB)} = \bar{I}_A - \bar{I}_B \qquad (10\text{-}12)$$

$$= (3.76 - j13.14) - (-10.17 + j1.48)$$

$$= 13.93 - j14.62$$

$$= 20.19/313.62°$$

and,

$$\theta_{X(AB)} = 360° - 313.62°$$

$$= 46.38°$$

$$\alpha_{X(AB)} = -46.38° + 1.82°$$

$$= -44.54°$$

The current in loop E_{BC} is:

$$\bar{I}_{X(BC)} = \bar{I}_B - \bar{I}_C \qquad (10\text{-}13)$$

$$= (-10.17 + j1.48) - (6.41 + j11.66)$$

$$= -16.58 - j10.18$$

$$= 19.46/238.45°$$

and,

$$\theta_{X(BC)} = -120° - 238.45°$$

$$= -358.45°$$

$$\alpha_{X(BC)} = -(-358.45°) + 1.82°$$

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$= 360.27°$

$= 0.27°$

The current in loop E_{CA} is:

$$\bar{I}_{X(CA)} = \bar{I}_C - \bar{I}_A \quad (10\text{-}14)$$

$= (6.41 + j11.66) - (3.76 - j13.14)$

$= 2.65 + j24.80$

$= 24.94\underline{/83.9°}$

and,

$\theta_{X(CA)} = 120° - 83.90°$

$= 36.1°$

$\alpha_{X(CA)} = -36.21° + 1.82°$

$= -34.28°$

Step 6.

The percent drop in each phase is calculated as follows:

$$D_{AB}\% = (20.19)(45)\left(\frac{(1.120)(10^{-3})\cos(-44.54°)}{(120)}\right) * 100$$

$= 0.60\%$

$$D_{BC}\% = I_{X(BC)}L\left(\frac{z\cos(\alpha_{X(BC)})}{E_{BC}}\right) * 100 \quad (10\text{-}17)$$

$= (19.46)(45)\left(\frac{(1.120)(10^{-3})\cos(360.27°)}{120}\right) * 100$

$= 0.82\%$

$$D_{CA}\% = I_{X(CA)}L\left(\frac{z\cos(\alpha_{X(CA)})}{E_{CA}}\right) * 100 \quad (10\text{-}17)$$

$= (24.94)(45)\left(\frac{(1.120)(10^{-3})\cos(-34.28°)}{120}\right) * 100$

$= 0.87\%$

The drop in this portion of the cable is the combination of the results in step 6 as follows:

$D_T(\%) = 1/2(D_{AB}\% + D_{BC}\% + D_{CA}\%)$

$= 1/2(0.60\% + 0.82\% + 0.87\%)$

$= 1.14\%$

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A.6. VOLTAGE DROP CALCULATIONS FOR UNBALANCED SYSTEMS BY SYMMETRICAL COMPONENT METHOD

To calculate cable voltage drop for unbalanced system, the line currents supplying the loads should be calculated. Assume all three phase loads are unbalanced. Therefore, the phase load currents are different and should be calculated individually. From the phase load currents, the individual line currents will be calculated. From the line currents, the voltage drop in each line will be determined. All necessary equations are derived in A.13.

A.6.1 Calculation procedure. The step-by-step procedure for the percent drop determination is as follows:

Step 1.

Locate or determine the following necessary parameters:

- Vectorial sum of all load currents for each phase
- Real power for each phase load
- Load power factor for each phase
- Load impedance for each phase
- Rated load voltage for each phase

Step 2.

Calculate the load current for each phase. Two methods will be used. If real power, power factor, and rated voltage of the loads are given, use the following equations:

$$\bar{I}_{AB} = [P_{AB}/V_{AB}\cos(\theta_{AB})]\underline{/(0-\theta_{AB})^\circ} \quad (13\text{-}31)$$

$$\bar{I}_{BC} = [P_{BC}/V_{BC}\cos(\theta_{BC})]\underline{/(-120-\theta_{BC})^\circ} \quad (13\text{-}32)$$

$$\bar{I}_{CA} = [P_{CA}/V_{CA}\cos(\theta_{CA})]\underline{/(120-\theta_{CA})^\circ} \quad (13\text{-}33)$$

If the impedances and rated voltages of the loads are given, use the following equations:

$$\bar{I}_{AB} = \bar{V}_{AB}/\bar{Z}_{AB} \quad (13\text{-}34)$$

$$\bar{I}_{BC} = \bar{V}_{BC}/\bar{Z}_{BC} \quad (13\text{-}35)$$

$$\bar{I}_{CA} = \bar{V}_{CA}/\bar{Z}_{CA} \quad (13\text{-}36)$$

Step 3.

Calculate the positive and negative sequence currents. Taking phase AB as reference, the positive and negative sequence currents are calculated as follows:

$$\bar{I}_{AB1} = (1/3)(\bar{I}_{AB} + a\bar{I}_{BC} + a^2\bar{I}_{CA}) \quad (13\text{-}29)$$

$$\bar{I}_{AB2} = (1/3)(\bar{I}_{AB} + a^2\bar{I}_{BC} + a\bar{I}_{CA}) \quad (13\text{-}30)$$

Where:

$$a = \underline{1/120^\circ},\ a^2 = \underline{1/240^\circ}$$

Note: In a three-phase and three-wire system, the zero sequence currents are zero as shown in A.13.

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Step 4.

Calculate the sequence components of line currents from the following equations:

$$\bar{I}_{A1} = \bar{I}_{AB1}\left(\sqrt{3}\underline{/\text{-}30°}\right) \quad (13\text{-}8)$$

$$\bar{I}_{A2} = \bar{I}_{AB2}\left(\sqrt{3}\underline{/30°}\right) \quad (13\text{-}11)$$

$$\bar{I}_{B1} = \bar{I}_{AB1}\left(\sqrt{3}\underline{/\text{-}150°}\right) \quad (13\text{-}14)$$

$$\bar{I}_{B2} = \bar{I}_{AB2}\left(\sqrt{3}\underline{/150°}\right) \quad (13\text{-}17)$$

$$\bar{I}_{C1} = \bar{I}_{AB1}\left(\sqrt{3}\underline{/90°}\right) \quad (13\text{-}21)$$

$$\bar{I}_{C2} = \bar{I}_{AB2}\left(\sqrt{3}\underline{/\text{-}90°}\right) \quad (13\text{-}25)$$

Step 5.

The line currents flowing into each node of the delta connected loads as shown in figure A-7 are calculated as follows:

$$\bar{I}_A = \bar{I}_{AB1}\left(\sqrt{3}\underline{/\text{-}30°}\right) + \bar{I}_{AB2}\left(\sqrt{3}\underline{/30°}\right) \quad (13\text{-}26)$$

$$\bar{I}_B = \bar{I}_{AB1}\left(\sqrt{3}\underline{/\text{-}150°}\right) + \bar{I}_{AB2}\left(\sqrt{3}\underline{/150°}\right) \quad (13\text{-}27)$$

$$\bar{I}_C = \bar{I}_{AB1}\left(\sqrt{3}\underline{/90°}\right) + \bar{I}_{AB2}\left(\sqrt{3}\underline{/\text{-}90°}\right) \quad (13\text{-}28)$$

The corresponding angles Θ and α are calculated from equations derived in A.10.:

$$\theta_A = 0° - \underline{/I_A}\,,\ \alpha_A = \text{-}\theta_A + \beta \quad (10\text{-}7)$$

$$\theta_B = \text{-}120° - \underline{/I_B}\,,\ \alpha_B = \text{-}\theta_B + \beta \quad (10\text{-}8)$$

$$\theta_C = 120° - \underline{/I_C}\,,\ \alpha_C = \text{-}\theta_C + \beta \quad (10\text{-}9)$$

Step 6.

If the cable length and cable impedance are known, the voltage drop in each line is:

$$D\% = IL\left(\frac{z\cos(\alpha)}{E}\right) * 100 \quad (10\text{-}10)$$

A.6.2 Example of voltage drop calculations.

Step 1.

The following parameters are given:

$$\bar{V}_{AB} = 118\underline{/0°}, \bar{V}_{BC} = 118\underline{/\text{-}120°}, \bar{V}_{CA} = 118\underline{/120°}$$

$$P_{AB} = 3400W, P_{BC} = 2500W, P_{CA} = 2900W$$

$$E_{AB} = E_{BC} = E_{CA} = 120V$$

All load power factors (pf) = 0.80 lagging. From table XVI, use LSTSGU-50 cable with z = $0.223(10^{-3})\Omega$/ft, β = 7.8°.

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Step 2.

Calculate the phase currents from the following equations:

$$\bar{I}_{AB} = [P_{AB}/V_{AB}\cos(\theta_{AB})]\underline{/(0 - \theta_{AB})^\circ} \quad (13\text{-}31)$$

$$= [3400/(118)(0.8)]\underline{/(0^\circ - 37^\circ)}$$

$$= 36\underline{/\text{-}37^\circ}$$

$$\bar{I}_{BC} = [P_{BC}/V_{BC}\cos(\theta_{BC})]\underline{/(\text{-}120 - \theta_{BC})^\circ} \quad (13\text{-}32)$$

$$= [2500/(118)(0.8)]\underline{/(\text{-}120^\circ - 37^\circ)}$$

$$= 26.5\underline{/\text{-}157^\circ}$$

$$\bar{I}_{CA} = [P_{CA}/V_{CA}\cos(\theta_{CA})]\underline{/(120 - \theta_{CA})^\circ} \quad (13\text{-}33)$$

$$= [2900/(118)(0.8)]\underline{/(120^\circ - 37^\circ)}$$

$$= 30.7\underline{/83^\circ}$$

Step 3.

Calculate the positive and negative sequences of $\bar{I}_{AB}$ from the following equations:

$$\bar{I}_{AB1} = (1/3)(\bar{I}_{AB} + a\bar{I}_{BC} + a^2\bar{I}_{CA}) \quad (13\text{-}29)$$

$$= (1/3)\left[36.0\underline{/\text{-}37^\circ} + \left(1\underline{/120^\circ}\right)\left(26.5\underline{/\text{-}157^\circ}\right) + \left(1\underline{/240^\circ}\right)\left(30.7\underline{/83^\circ}\right)\right]$$

$$= 31\underline{/\text{-}37^\circ}$$

$$\bar{I}_{AB2} = (1/3)(\bar{I}_{AB} + a^2\bar{I}_{BC} + a\bar{I}_{CA}) \quad (13\text{-}30)$$

$$= (1/3)\left[36.0\underline{/\text{-}37^\circ} + (1/240^\circ)\left(26.5\underline{/\text{-}157^\circ}\right) + \left(1\underline{/120^\circ}\right)\left(30.7\underline{/83^\circ}\right)\right]$$

$$= 2.6\underline{/\text{-}62.5^\circ}$$

Step 4.

Calculate the sequence components of line currents from the following equations:

$$\bar{I}_{A1} = \bar{I}_{AB1}\left(\sqrt{3}\underline{/\text{-}30^\circ}\right) \quad (13\text{-}8)$$

$$= \left(31.0\underline{/\text{-}37^\circ}\right)\left(\sqrt{3}\underline{/\text{-}30^\circ}\right)$$

$$= 53.7\underline{/\text{-}67^\circ}$$

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$$\bar{I}_{A2} = \bar{I}_{AB2}\left(\sqrt{3}/\underline{30°}\right) \tag{13-11}$$
$$= \left(2.6/\underline{-62.5°}\right)\left(\sqrt{3}/\underline{30°}\right)$$
$$= 4.5/\underline{-32.5°}$$

$$\bar{I}_{B1} = \bar{I}_{AB1}\left(\sqrt{3}/\underline{-150°}\right) \tag{13-14}$$
$$= \left(31.0/\underline{-37°}\right)\left(\sqrt{3}/\underline{-150°}\right)$$
$$= 53.7/\underline{-187°}$$

$$\bar{I}_{B2} = \bar{I}_{AB2}\left(\sqrt{3}/\underline{150°}\right) \tag{13-17}$$
$$= \left(2.6/\underline{-62.5°}\right)\left(\sqrt{3}/\underline{150°}\right)$$
$$= 4.5/\underline{87.5°}$$

$$\bar{I}_{C1} = \bar{I}_{AB1}\left(\sqrt{3}/\underline{90°}\right) \tag{13-21}$$
$$= \left(31.0/\underline{-37°}\right)\left(\sqrt{3}/\underline{90°}\right)$$
$$= 53.7/\underline{53°}$$

$$\bar{I}_{C2} = \bar{I}_{AB2}\left(\sqrt{3}/\underline{-90°}\right) \tag{13-25}$$
$$= \left(2.6/\underline{-62.5°}\right)\left(\sqrt{3}/\underline{-90°}\right)$$
$$= 4.5/\underline{-152.5°}$$

Step 5.

Calculate the line currents $\bar{I}_A$, $\bar{I}_B$, and $\bar{I}_C$ from the sequence currents as follows:

$$\bar{I}_A = \bar{I}_{A0} + \bar{I}_{A1} + \bar{I}_{A2} \tag{13-26}$$
$$= 0 + 53.7/\underline{-67°} + 4.5/\underline{-32.5°}$$
$$= 24.78 - j51.85$$
$$= 54.47/\underline{-64.46°}$$

Then,

$$\theta_A = 0° - (-64.46°)$$
$$= 64.46°$$

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$$\alpha_A = -\theta_A + \beta$$
$$= -64.46° + 7.8°$$
$$= -56.66°$$

$$\bar{I}_B = \bar{I}_{B0} + \bar{I}_{B1} + \bar{I}_{B2} \quad (13\text{-}27)$$
$$= 0 + 53.7\underline{/-187°} + 4.5\underline{/87.5°}$$
$$= -53.10 + j11.04$$
$$= 54.24\underline{/168.26°}$$

Then,

$$\theta_B = -120° - 168.26°$$
$$= -288.26°$$

$$\alpha_B = -\theta_B + \beta$$
$$= 288.26° + 7.8°$$
$$= 296.06°$$

$$\bar{I}_C = \bar{I}_{C0} + \bar{I}_{C1} + \bar{I}_{C2} \quad (13\text{-}28)$$
$$= 0 + 53.7\underline{/53°} + 4.5\underline{/-152.5°}$$
$$= 28.33 + j40.81$$
$$= 49.68\underline{/55.24°}$$

Then,

$$\theta_C = 120° - 55.24°$$
$$= 64.76°$$

$$\alpha_C = -\theta_C + \beta$$
$$= -64.76° + 7.8°$$
$$= -56.96°$$

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Step 6.

Determine the percent voltage drops in the lines and compare the results.

$$D_A\% = I_A L\left(\frac{z\cos(\alpha_A)}{E}\right) * 100 \quad (10\text{-}10)$$

$$= (57.47)(65)\left(\frac{(0.223)(10^{-3})cos(\text{-}56.66°)}{120/\sqrt{3}}\right) * 100$$

$$= 0.66\%$$

$$D_B\% = I_B L\left(\frac{z\cos(\alpha_B)}{E}\right) * 100 \quad (10\text{-}10)$$

$$= (54.24)(65)\left(\frac{(0.223)(10^{-3})cos(296.06°)}{120/\sqrt{3}}\right) * 100$$

$$= 0.50\%$$

$$D_C\% = I_C L\left(\frac{z\cos(\alpha_C)}{E}\right) * 100 \quad (10\text{-}10)$$

$$= (49.68)(65)\left(\frac{(0.223)(10^{-3})cos(\text{-}56.96°)}{120/\sqrt{3}}\right) * 100$$

$$= 0.57\%$$

The drop in this portion of cable is obtained by combining the drop in the individual lines (phase):

$$D_T\% = D_A\% + D_B\% + D_C\%$$

$$= 0.66\% + 0.50\% + 0.57\%$$

$$= 1.73\%$$

A.7. CABLE IMPEDANCES AND DROP FACTORS

The drop factors and impedances are calculated based on the characteristics of LSTSGU and LS6SGU cables in accordance with MIL-DTL-24643. The tabulated drop factors for these cables may be used for other types of cables with similar impedance characteristics including lightweight cables in accordance with MIL-DTL-24640.

A.7.1 Drop factor calculations. The drop factors in tables XIII through XVII for power and lighting systems are calculated from the following equations derived in A.9, A.10, and A.11:

A.7.1.1 Power systems.

$$DF = \left(\frac{z\cos(\alpha)}{E} + \frac{(2)z\tan(\alpha)\sin(\alpha)}{200E}\right) * 100 \quad (9\text{-}13)$$

A.7.1.2 Lighting systems.

A.7.1.2.1 Single-phase circuits.

- For equations using Current I.

$$DF = \left(\frac{z\cos(\alpha)}{E}\right) * 100 \quad (10\text{-}20)$$

- For equations using Power P.

$$DF' = \left(\frac{1}{V\cos(\theta)}\right)\left(\frac{z\cos(\alpha)}{E}\right) * 100 \quad (11\text{-}26)$$

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A.7.1.2.2 Three-phase circuits.

- For equations using Current I_X.

$$DF = \left(\frac{z\cos(\alpha_X)}{E_X}\right) * 100 \quad (10\text{-}19)$$

TABLE XI. Impedances for LSTSGU cable, 60 Hz electronics and communications.

Size	Cable characteristics				
	A 1/ Cmil	R 2/, 3/	X 3/	Z	β Degrees
		Ohms per 100 feet			
3	2,580	4.503	0.046	4.503	0.55
4	4,110	2.825	0.040	2.825	0.81
9	10,380	1.119	0.036	1.120	1.82
14	13,090	0.887	0.041	0.888	2.63
23	20,820	0.558	0.039	0.559	4.02
50	52,620	0.221	0.030	0.223	7.81
75	83,690	0.138	0.030	0.141	12.07
100	105,600	0.110	0.029	0.114	14.92
150	167,800	0.069	0.029	0.075	22.58
200	211,600	0.055	0.029	0.062	27.80
300	300,000	0.040	0.029	0.049	35.75
400	413,600	0.029	0.029	0.041	44.80

NOTES:

1/ Conductor cross-sectional areas are based on MIL-DTL-24643.

2/ Resistances are derived at a temperature of 45 °C. For other copper temperatures, calculate the DC resistance using equation (14-5), convert to AC resistance using table XXIII, and adjust impedance Z and drop factors accordingly.

3/ Resistances and reactances per phase are calculated in A.14.

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TABLE XII. Impedances for LSTSGU cable, 400 Hz electronics and communications.

Size	Cable characteristics				
	A [1/] Cmil	R [2/, 3/]	X [3/]	Z	β Degrees
		Ohms per 100 feet			
3	2,580	4.503	0.288	4.512	3.66
4	4,110	2.825	0.267	2.838	5.40
9	10,380	1.119	0.237	1.144	11.98
14	13,090	0.887	0.371	0.928	17.01
23	20,820	0.558	0.362	0.616	25.12
50	52,620	0.221	0.202	0.299	42.44
75	83,690	0.143	0.197	0.243	53.98
100	105,600	0.116	0.195	0.227	59.29
150	167,800	0.077	0.191	0.206	68.10
200	211,600	0.065	0.194	0.204	71.44
300	300,000	0.051	0.192	0.199	75.14
400	413,600	0.042	0.192	0.196	77.65

NOTES:

1/ Conductor cross-sectional areas are based on MIL-DTL-24643.

2/ Resistances are derived at a temperature of 45 °C. For other copper temperatures, calculate the DC resistance using equation (14-5), convert to AC resistance using table XXIII, and adjust impedance Z and drop factors accordingly.

3/ Resistances and reactances per phase are calculated inA.14.

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TABLE XIII. Drop factors for LSTSGU cable, 450V, three-phase, 60 Hz power systems. [1/]

Size	Cable characteristics					Drop factors at cos (Θ) [2/] below (multiply by 10^{-5})													
	A [1/] Cmil	R [4/, 5/]	X [4/]	Z	β Degrees	1.0	0.95	0.90	0.85	0.80	0.75	0.70	0.65	0.60	0.55	0.50	0.45	0.40	0.35	0.30
		Ohms per 1000 feet																		
3	2,580	4.827	0.043	4.827	0.51	185.8	177.2	168.4	159.4	150.4	141.5	132.5	123.5	114.8	106.0	97.1	88.2	79.7	71.1	62.0
4	4,110	3.028	0.040	3.028	0.76	116.5	111.3	105.8	100.2	94.6	89.1	83.5	77.9	72.4	66.9	61.3	55.7	50.4	45.1	39.9
9	10,380	1.199	0.036	1.200	1.70	46.1	44.3	42.2	40.1	37.9	35.8	33.6	31.4	29.3	27.1	24.9	22.7	20.6	18.5	16.4
14	13,090	0.951	0.041	0.952	2.45	36.6	35.3	33.7	32.0	30.4	28.7	27.0	25.3	23.6	21.9	20.1	18.4	16.8	15.1	13.4
23	20,820	0.598	0.039	0.599	3.75	23.0	22.4	21.4	20.4	19.4	18.4	17.3	16.3	15.2	14.2	13.1	12.0	11.0	9.9	8.9
50	52,620	0.237	0.030	0.239	7.29	9.1	9.0	8.9	8.4	8.0	7.6	7.3	6.9	6.5	6.1	5.7	5.3	4.9	4.4	4.0
75	83,690	0.148	0.030	0.151	11.27	5.7	5.8	5.6	5.4	5.2	5.0	4.8	4.6	4.4	4.1	3.9	3.6	3.4	3.1	2.9
100	105,600	0.118	0.029	0.122	13.94	4.5	4.7	4.6	4.5	4.3	4.2	4.0	3.8	3.7	3.5	3.3	3.1	2.9	2.7	2.9
150	167,800	0.074	0.029	0.079	21.20	2.9	3.1	3.0	3.0	2.9	2.9	2.8	2.7	2.6	2.5	2.4	2.3	2.2	2.1	1.9
200	211,600	0.059	0.029	0.066	26.18	2.3	2.5	2.5	2.5	2.5	2.4	2.4	2.3	2.3	2.2	2.1	2.0	1.9	1.9	1.8
300	300,000	0.042	0.029	0.051	34.44	1.6	1.9	1.9	2.0	2.0	1.9	1.9	1.9	1.9	1.8	1.8	1.7	1.7	1.6	1.6
400	413,600	0.031	0.029	0.042	42.89	1.2	1.6	1.6	1.6	1.6	1.6	1.6	1.6	1.6	1.6	1.6	1.5	1.5	1.5	1.4

NOTES:

1/ Drop factor equations are derived in A.9.

2/ Θ is the load power factor angle or angle between load terminal voltage and line current.

3/ Conductor cross-sectional areas are based on MIL-DTL-24643.

4/ Resistances and reactances per phase are calculated in A.14.

5/ Resistances are derived at a temperature of 65 °C. For other copper temperatures, calculate the DC resistance using equation (14-5), convert to AC resistance using table XXIII, and adjust impedance Z and drop factors accordingly.

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TABLE XIV. Drop factors for LSTSGU cable, 450V, three-phase, 400 Hz power systems. [1/]

Size	Cable characteristics					Drop factors at cos (Θ) [2/] below (multiply by 10^{-5})														
	A [1/] Cmil	R [4/, 5/]	X [4/]	Z	β Degrees	1.0	0.95	0.90	0.85	0.80	0.75	0.70	0.65	0.60	0.55	0.50	0.45	0.40	0.35	0.30
		Ohms per 1000 feet																		
3	2,580	4.827	0.288	4.836	3.42	185.8	180.1	172.4	164.2	155.9	147.6	139.1	130.5	122.1	113.5	104.9	96.2	87.8	79.3	71.0
4	4,110	3.028	0.267	3.040	5.04	116.6	114.0	109.6	104.7	99.8	94.7	89.5	84.3	79.2	73.9	68.5	63.2	58.0	52.7	47.0
9	10,380	1.199	0.237	1.222	11.20	46.2	46.7	45.6	44.1	42.5	40.8	39.0	37.2	35.3	33.4	31.4	29.4	24.4	25.4	23.0
14	13,090	0.951	0.271	0.989	15.92	36.6	38.0	37.5	36.6	35.6	34.4	33.2	31.9	30.5	29.1	27.6	26.1	24.6	23.0	21.0
23	20,820	0.598	0.262	0.653	23.63	23.1	25.0	25.1	24.9	24.5	23.9	23.3	22.7	21.9	21.2	20.3	19.5	18.6	17.7	16.7
50	52,620	0.237	0.202	0.311	40.46	9.2	11.1	11.6	11.9	12.0	12.0	11.9	11.8	11.7	11.5	11.3	11.1	10.8	10.5	10.2
75	83,690	0.153	0.197	0.249	51.12	6.0	8.0	8.6	9.0	9.3	9.4	9.5	9.6	9.6	9.6	9.5	9.4	9.3	9.2	9.0
100	105,600	0.124	0.195	0.231	57.59	4.9	6.9	7.6	8.0	8.3	8.6	8.7	8.8	8.9	8.9	8.9	8.9	8.8	8.7	8.6
150	167,800	0.082	0.191	0.208	66.82	3.3	5.4	6.1	6.6	7.0	7.3	7.5	7.7	7.8	7.9	8.0	8.0	8.0	8.0	8.0
200	211,600	0.069	0.194	0.206	70.38	2.9	4.9	5.7	6.2	6.6	6.9	7.2	7.4	7.6	7.7	7.8	7.9	7.9	7.9	7.9
300	300,000	0.054	0.192	0.200	74.31	2.3	4.4	5.2	5.7	6.1	6.5	6.8	7.0	7.2	7.3	7.5	7.5	7.6	7.7	7.7
400	413,600	0.045	0.192	0.197	76.80	2.0	4.1	4.8	5.4	5.9	6.2	6.5	6.8	7.0	7.1	7.3	7.4	7.5	7.5	7.6

NOTES:

1/ Drop factor equations are derived in A.9.

2/ Θ is the load power factor angle or angle between load terminal voltage and line current.

3/ Conductor cross-sectional areas are based on MIL-DTL-24643.

4/ Resistances and reactances per phase are calculated in A.14.

5/ Resistances are derived at a temperature of 65 °C. For other copper temperatures, calculate the DC resistance using equation (14-5), convert to AC resistance using table XXIII, and adjust impedance Z and drop factors accordingly.

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TABLE XV. Drop factors for LS6SGU cable, 450V, three-phase, 400 Hz power systems. [1/]

Size	Cable characteristics					Drop factors at cos (Θ) [2/] below (multiply by 10^{-5})														
	A [1/] Cmil	R [4/, 5/]	X [4/]	Z	β Degrees	1.0	0.95	0.90	0.85	0.80	0.75	0.70	0.65	0.60	0.55	0.50	0.45	0.40	0.35	0.30
		Ohms per 1000 feet																		
100	211,200	0.062	0.098	0.116	57.60	2.4	3.5	3.8	4.0	4.2	4.3	4.4	4.4	4.4	4.5	4.4	4.4	4.4	4.4	4.3
125	266,200	0.049	0.097	0.109	63.29	2.0	3.0	3.3	3.6	3.8	3.9	4.0	4.1	4.1	4.2	4.2	4.2	4.2	4.2	4.1
150	335,600	0.041	0.096	0.104	66.83	1.7	2.7	3.0	3.3	3.5	3.6	3.7	3.8	3.9	3.9	4.0	4.0	4.0	4.0	4.0
200	423,200	0.035	0.097	0.103	70.12	1.5	2.5	2.9	3.1	3.3	3.5	3.6	3.7	3.8	3.9	3.9	3.9	4.0	4.0	4.0

NOTES:

1/ Drop factor equations are derived in A.9.

2/ Θ is the load power factor angle or angle between load terminal voltage and line current.

3/ Conductor cross-sectional areas are based on MIL-DTL-24643.

4/ Resistances and reactances per phase are calculated in A.14.

5/ Resistances are derived at a temperature of 65 °C. For other copper temperatures, calculate the DC resistance using equation (14-5), convert to AC resistance using table XXIII, and adjust impedance Z and drop factors accordingly.

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TABLE XVI. Drop factors for LSTSGU cable, 120V, three-phase/single-phase, 60 Hz lighting systems using I or I_X. [1/]

Size	Cable characteristics					Drop factors at angle (Θ) [2/] for single-phase loads or Θ [3/] for three-phase loads (multiply by 10^{-5})													
	A [4/] Cmil	R [5/, 6/]	X [5/]	Z	B Degrees	-70	-65	-60	-55	-50	-45	-40	-30	-20	-10	0	10	20	30	40
		Ohms per 1000 feet																		
3	2,580	4.503	0.043	4.503	0.55	125.0	155.3	184.5	212.3	238.4	262.8	285.1	323.2	351.4	368.9	375.2	370.2	353.9	326.8	289.8
4	4,110	2.825	0.040	2.825	0.81	77.4	96.5	114.8	132.3	148.8	164.1	178.2	202.2	220.1	231.3	235.4	232.4	222.4	205.5	182.5
9	10,380	1.119	0.036	1.120	1.82	29.1	36.7	44.1	51.1	57.7	63.8	69.5	79.3	86.6	91.3	93.2	92.3	88.6	82.2	73.6
14	13,090	0.887	0.041	0.888	2.63	22.1	28.2	34.0	39.6	44.9	49.9	54.4	62.3	68.3	72.2	73.9	73.4	70.6	65.7	58.8
23	20,820	0.558	0.039	0.559	4.02	12.8	16.7	20.4	24.0	27.4	30.6	33.5	38.6	42.6	45.2	46.5	46.4	44.8	41.9	37.7
50	52,620	0.221	0.030	0.223	7.81	3.9	5.5	7.0	8.5	9.9	11.2	12.5	14.7	16.4	17.7	18.4	18.6	18.2	17.2	15.7
75	83,690	0.138	0.030	0.141	12.07	1.6	2.6	3.6	4.6	5.5	6.4	7.2	8.7	10.0	10.9	11.5	11.8	11.6	11.2	10.4
100	105,600	0.110	0.029	0.114	14.92	0.8	1.7	2.5	3.3	4.0	4.8	5.5	6.7	7.8	8.6	9.2	9.5	9.4	9.2	8.6
150	167,800	0.069	0.029	0.075	22.58	0.3	0.3	0.8	1.3	1.9	2.4	2.9	3.8	4.6	5.2	5.8	6.1	6.2	6.2	5.9
200	211,600	0.055	0.029	0.062	27.80	0.7	0.3	0.2	0.6	1.1	1.5	2.0	2.8	3.5	4.1	4.6	4.9	5.1	5.2	5.1
300	300,000	0.040	0.029	0.049	35.75	1.1	0.8	0.4	0.1	0.3	0.7	1.0	1.7	2.3	2.9	3.3	3.7	4.0	4.1	4.1
400	413,600	0.029	0.029	0.041	44.80	1.4	1.2	0.9	0.6	0.3	0.0	0.3	0.9	1.5	2.0	2.4	2.8	3.1	3.3	3.4

NOTES:

1/ Drop factor equations are derived in A.10.

2/ Θ is the load power factor angle or angle between load terminal voltage and line current I.

3/ Θ_X is the angle between load terminal voltage and current I_X.

4/ Conductor cross-sectional areas are based on MIL-DTL-24643.

5/ Resistances and reactances per phase are calculated in A.14.

6/ Resistances are derived at a temperature of 45 °C. For other copper temperatures, calculate the DC resistance using equation (14-5), convert to AC resistance using table XXIII, and adjust impedance Z and drop factors accordingly.

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TABLE XVII. Drop factors for LSTSGU cable, 120V, three-phase/single-phase, 60 Hz lighting systems using P or P_X. [1/]

Size	Cable characteristics					Drop factors at angle (Θ) [2/] for single-phase loads or Θ[3/] for three-phase loads (multiply by 10^{-5})														
	A [4/] Cmil	R [5/, 6/]	X [5/]	Z	β Degrees	-70	-65	-60	-55	-50	-45	-40	-30	-20	-10	0	10	20	30	40
		Ohms per 1000 feet																		
3	2,580	4.503	0.043	4.503	0.55	317.7	319.6	320.9	321.8	322.6	323.2	323.7	324.5	325.5	325.8	326.3	326.9	327.4	328.1	328.9
4	4,110	2.825	0.040	2.825	0.81	196.7	198.5	199.7	200.6	201.0	201.8	202.3	203.0	203.7	204.2	204.7	205.2	205.8	206.4	207.1
9	10,380	1.119	0.036	1.120	1.82	74.0	75.6	76.6	77.4	78.0	78.5	78.9	79.6	80.1	80.6	81.1	81.5	82.0	82.6	83.3
14	13,090	0.887	0.041	0.888	2.63	56.2	58.0	59.2	60.1	60.8	61.3	61.8	62.6	63.2	63.8	64.3	64.8	65.3	66.0	66.8
23	20,820	0.558	0.039	0.559	4.02	32.6	34.3	35.5	36.4	37.0	37.6	38.1	38.8	39.4	39.9	40.4	40.9	41.5	42.1	42.8
50	52,620	0.221	0.030	0.223	7.81	10.0	11.3	12.2	12.9	13.4	13.8	14.2	14.7	15.2	15.6	16.0	16.4	16.8	17.3	17.9
75	83,690	0.138	0.030	0.141	12.07	4.1	5.4	6.3	6.9	7.5	7.9	8.2	8.8	9.2	9.6	10.0	10.4	10.8	11.2	11.8
100	105,600	0.110	0.029	0.114	14.92	2.1	3.4	4.3	4.9	5.4	5.8	6.2	6.7	7.2	7.6	8.0	8.3	8.7	9.2	9.8
150	167,800	0.069	0.029	0.075	22.58	0.7	0.5	1.4	2.0	2.5	2.9	3.3	3.8	4.2	4.6	5.0	5.4	5.8	6.2	6.7
200	211,600	0.055	0.029	0.062	27.80	1.8	0.5	0.3	1.0	1.5	1.9	2.2	2..8	3.2	3.6	4.0	4.4	4.8	5.2	5.7
300	300,000	0.040	0.029	0.049	35.75	2.8	1.6	0.7	0.1	0.4	0.8	1.1	1.7	2.1	2.5	2.9	3.3	3.7	4.1	4.6
400	413,600	0.029	0.029	0.041	44.80	3.6	2.4	1.5	0.9	0.4	0.0	0.4	0.9	1.3	1.7	2.1	2.5	2.9	3.3	3.9

NOTES:

1/ Drop factor equations are derived in A.10.

2/ Θ is the load power factor angle or angle between P and S vectors.

3/ Θ_X is the angle between P_X and apparent power S_X vectors.

4/ Conductor cross-sectional areas are based on MIL-DTL-24643.

5/ Resistances and reactances per phase are calculated in A.14.

6/ Resistances are derived at a temperature of 45 °C. For other copper temperatures, calculate the DC resistance using equation (14-5), convert to AC resistance using table XXIII, and adjust impedance Z and drop factors accordingly.

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- For equations using Power P_X.

$$DF' = \left(\frac{1}{Vcos(\theta_X)}\right)\left(\frac{zcos(\alpha_X)}{E_X}\right) * 100, \quad (11\text{-}23)$$

The drop factors for other cables not listed in the tables can be also calculated from the above equations if the cable characteristic are given. Data may be obtained from the cable manufacturer or applicable design organization or can be derived from cable geometry. Alternatively, voltage drop calculations may be performed by engineering software such as those produced by ETAP, Paladin, SKM, EasyPower, or equal, if the proper cable characteristics are used.

A.8. DERIVATION OF VOLTAGE DROP EQUATIONS FOR DC SYSTEMS

A.8.1 Single-wire circuits. The R_{DC} resistance for a single wire in DC systems is:

$$R_{DC} = \left(\frac{\rho L}{A}\right) \quad (8\text{-}1)$$

Where:

ρ = Conductor resistivity (copper) of a cable at desired operating temperature (t).

L = Cable length.

A = Conductor cross-sectional area in circular mil (cmil).

The standard nominal cross-sectional area of a conductor in circular mils is calculated in accordance with the following equation:

$$A = d^2 \quad (8\text{-}2)$$

Where:

d = conductor diameter in mils.

The conductor resistivity (copper) at operating temperature (t) is given by:

$$\rho = \rho_O[1 + 0.00393(t - t_O)], \quad (8\text{-}3)$$

From A.14, the conductor resistivity (copper) at 20 °C is:

$$\rho_O = 10.371\ \Omega\ cmil/ft$$

At t = 65 °C,

$$\rho = 10.371[1 + 0.00393(65 - 20)]$$
$$= 12.21\ \Omega\ cmil/ft$$

At t = 45 °C,

$$\rho = 10.371[1 + 0.00393(45 - 20)]$$
$$= 11.39\ \Omega\ cmil/ft$$

With respect to the system voltage, the percent voltage drop in a single line is:

$$D\% = \left(\frac{IR_{dc}}{E}\right) * 100 \quad (8\text{-}4)$$

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A.8.2 Two-wire circuits. In a two-wire circuit, the percent drop in equation (8-4) should be multiplied by 2 to include the drop in the return path:

For all systems except power.

$$D\% = 2\left(\frac{IR_{dc}}{E}\right) * 100$$

At 45 °C, the resistance R_{DC} is:

$$R_{dc} = 11.39\left(\frac{L}{A}\right)$$

Then,

$$D\% = 22.78\left(\frac{IL}{AE}\right) * 100 \quad (8\text{-}5)$$

For power system only.

$$D\% = 2\left(\frac{IR_{dc}}{E}\right) * 100$$

$$D\% = 2\left(\frac{12.21IL}{AE}\right) * 100$$

$$= 24.42\left(\frac{IL}{AE}\right) * 100 \quad (8\text{-}6)$$

A.9. DERIVATION OF VOLTAGE DROP EQUATIONS FOR SINGLE-PHASE/POWER SYSTEMS

The derivation of equations is based on Navy shipboard power systems and following assumptions:

Cable impedance is purely resistive and inductive.

The terminal load voltage is used as reference at $V/0°$.

Let derive the voltage drop equations based on figures A-1 and A-2:

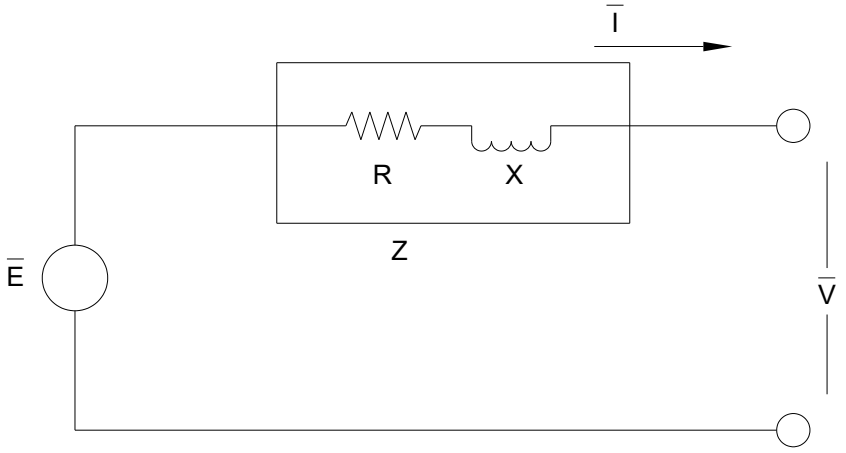


FIGURE A-1. Single-phase circuit representation.

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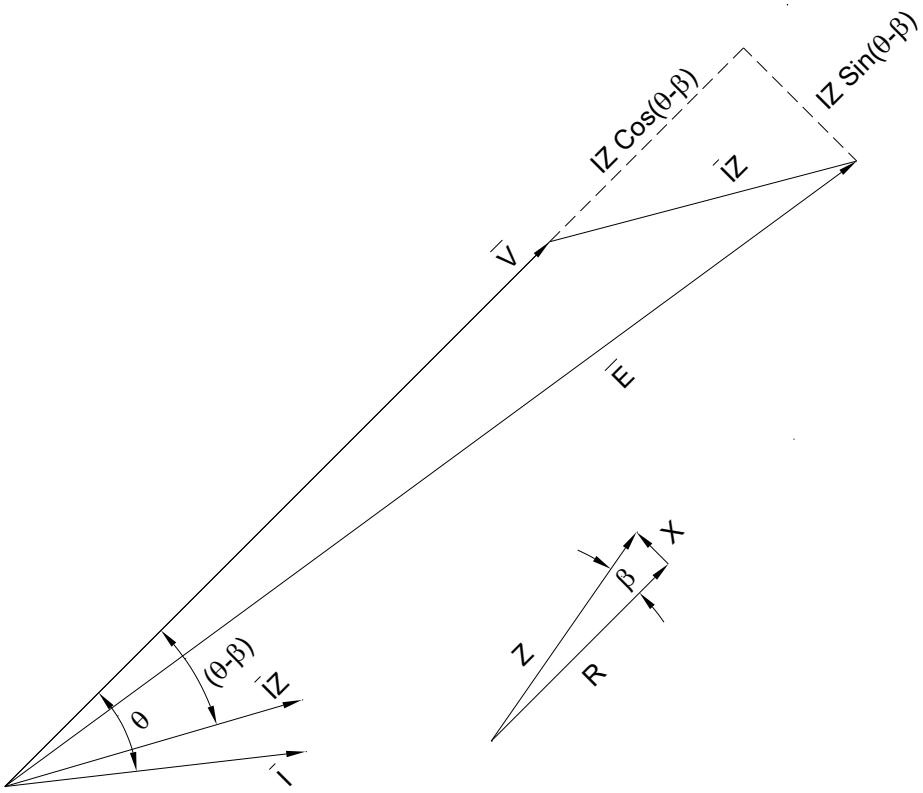


FIGURE A-2. Single-phase voltage and current phasor diagram.

From the above figures, define:

Θ = Load power factor angle or angle between load voltage and line current

β = Cable impedance angle

= $\tan^{-1}(X/R)$

The relationship between the sending bus and the terminal load voltage is as follows:

$$\bar{E} = \overline{IZ} + \bar{V} \tag{9-1}$$

$$\bar{E} = \left(I\underline{/-\theta}\right)\left(Z\underline{/\beta}\right) + V\underline{/0°}$$

$$\bar{E} = IZ\underline{/(-\theta + \beta)} + V\underline{/0°}$$

Let:

$$\alpha = -\theta + \beta \tag{9-2}$$

$$\bar{E} = i(IZcos(\alpha)) + j(IZsin(\alpha)) + iV$$

$$\bar{E} = i(IZcos(\alpha) + V) + jIZsin(\alpha)$$

$$E = \sqrt{(IZcos(\alpha) + V)^2 + I^2Z^2sin^2(\alpha)} \tag{9-3}$$

The cable voltage drop in percent is calculated as follows:

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$$D\% = \left(\frac{E-V}{E}\right) * 100$$

$$= \left(1 - \frac{V}{E}\right) * 100 \quad (9\text{-}4)$$

From equation (9-3), the terminal load voltage is:

$$V = \sqrt{E^2 - I^2Z^2sin^2(\alpha)} - IZcos(\alpha)$$

Substitute V into equation (9-4) and simplify:

$$D\% = \left[1 - \frac{\sqrt{E^2 - I^2Z^2sin^2(\alpha)}}{E} + \frac{IZcos(\alpha)}{E}\right] * 100$$

$$D\% = \left[1 - \sqrt{1 - \frac{I^2Z^2sin^2(\alpha)}{E^2}} + \frac{IZcos(\alpha)}{E}\right] * 100 \quad (9\text{-}5)$$

Set Z=zL, the exact equation for voltage drop is:

$$D\% = \left[1 - \sqrt{1 - \frac{I^2L^2z^2sin^2(\alpha)}{E^2}} + \frac{ILzcos(\alpha)}{E}\right] * 100 \quad (9\text{-}6)$$

The rule for approximation by binomial expansion can be applied to simplify equation (9-6):

$$(1 \pm x)^n = 1 \pm nx + \frac{n(n-1)x^2}{2!} \pm \frac{n(n-1)(n-2)x^3}{3!} + .., (x^2 < 1)$$

In the above expression, let:

$x^2 = \left(\frac{I^2L^2z^2 \quad (\alpha)}{E^2}\right)^2$ which is approximately equal to zero.

Therefore, the term in equation (9-6) can be written:

$$\sqrt{1 - \frac{I^2L^2z^2sin^2(\alpha)}{E^2}} = 1 - \frac{1}{2}\left(\frac{I^2L^2z^2sin^2(\alpha)}{E^2}\right) \quad (9\text{-}7)$$

Substitute equation (9-7) into equation (9-6):

$$D\% = \left[1 - \left(1 - \frac{1}{2}\left(\frac{I^2L^2z^2sin^2(\alpha)}{E^2}\right)\right) + \frac{ILzcos(\alpha)}{E}\right] * 100$$

The voltage drop equation reduces to:

$$D\% = IL\left(\frac{zcos(\alpha)}{E} + \frac{IL \quad z^2sin^2(\alpha)}{2E^2}\right) * 100 \quad (9\text{-}8)$$

Since the quantity $\sin^2(\alpha)/2E^2$ is very small with respect to $\cos(\alpha)/E$, it can be neglected and the acceptable equation for the voltage drop is obtained:

$$D\% = IL\left(\frac{zcos(\alpha)}{E}\right) * 100 \quad (9\text{-}9)$$

Since it is a single-phase circuit, equation (9-9) should be multiplied by 2 to include the drop in the return path.

Equation (9-9) can also be used to determine the percent drop in the individual lines for three-phase balanced systems such as for electronic, internal communication, or weapon control systems with the substitution of line current by $\sqrt{3}I_{LO}$:

$$D\% = \sqrt{3}I_{LO}L\left(\frac{zcos(\alpha)}{E}\right) * 100 \quad (9\text{-}10)$$

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Where I_{LO} is the resultant load current in each phase.

In equation (9-8), since $(z/E)\cos(\alpha) >> (z^2/2E^2)\sin^2(\alpha)$, and by setting D% equal to the assumed drop (AD), an estimate of (IL) may be determined as follows:

$$(AD) = IL[(z/E)cos(\alpha)] * 100$$

Solve for IL:

$$IL = [(AD)E/100zcos(\alpha)] \quad (9\text{-}11)$$

Substitute equation (9-11) into equation (9-8) for IL in the bracket:

$$D\% = IL\left(\frac{zcos(\alpha)}{E} + \frac{(AD)Ez^2sin^2(\alpha)}{200zE^2cos(\alpha)}\right) * 100$$

Simplification of the above equation gives the voltage drop equation which includes the assumed drop (AD) in the systems:

$$D\% = IL\left(\frac{zcos(\alpha)}{E} + \frac{(AD)ztan(\alpha)sin(\alpha)}{200E}\right) * 100 \quad (9\text{-}12)$$

The assumed drop (AD) for a 60 Hz or 400 Hz power system at normal operation is 2 percent.

In equation (9-12), let define the drop factor as:

$$DF = \left(\frac{zcos(\alpha)}{E} + \frac{(AD)ztan(\alpha)sin(\alpha)}{200E}\right) * 100 \quad (9\text{-}13)$$

Equation (9-12) can be rewritten in the simple form:

$$D\% = IL(DF) \quad (9\text{-}14)$$

As before, the percent drop for a single-phase circuit in equation (9-14) should be multiplied by 2 to include the drop in the return path:

$$D\% = 2IL(DF) \quad (9\text{-}15)$$

A.10. DERIVATION OF VOLTAGE DROP EQUATIONS FOR THREE-PHASE/LIGHTING SYSTEMS

In order to determine the best cable selection for three-phase systems, voltage drop calculations for all phases should be performed. The combination of the individual drops will be the drop of the cable. Figures A-3 and A-4 represent three-phase circuits used to calculate voltage drop in a three-phase delta configuration.

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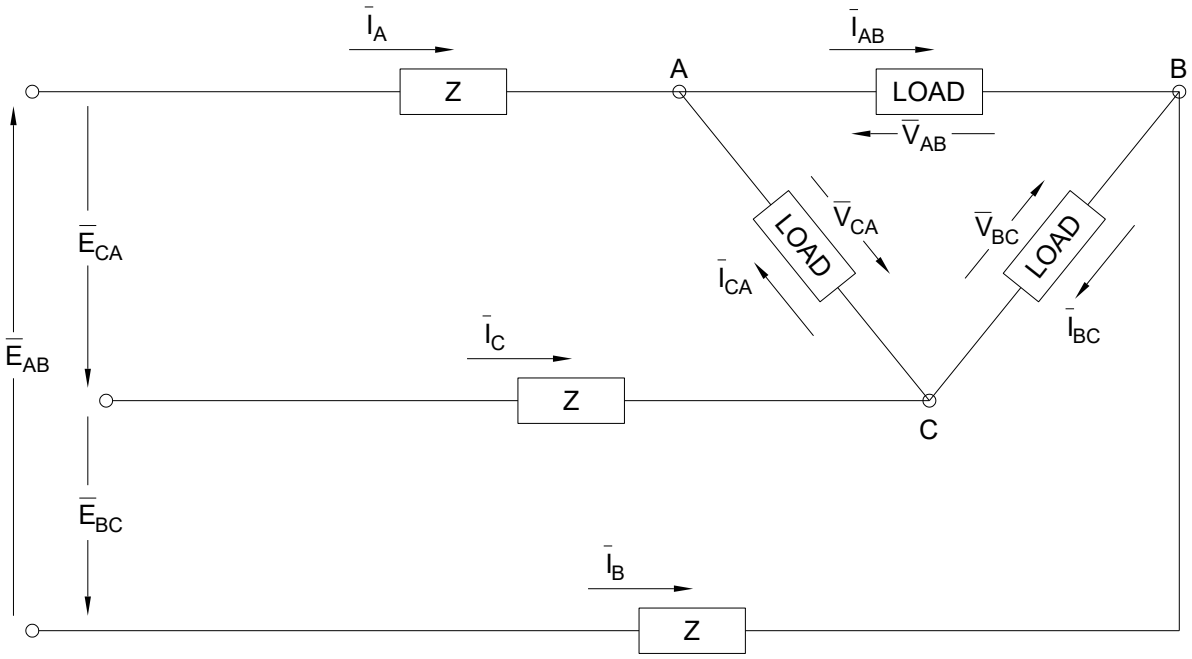


FIGURE A-3. Three-phase circuit representation.

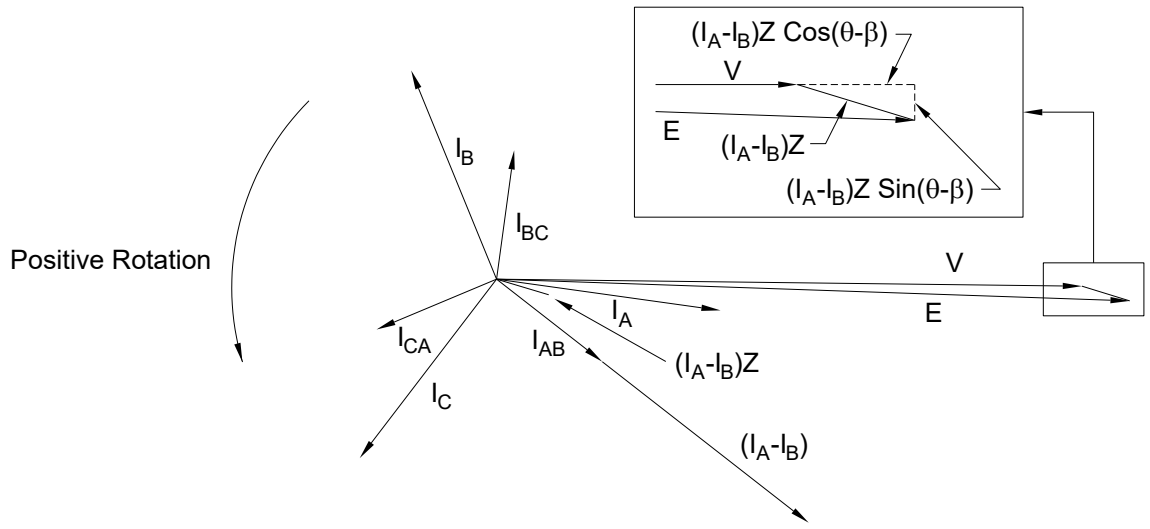


FIGURE A-4. Three-phase voltage and current phasor diagram.

From figure A-3, the voltage loop method gives:

$$\bar{E}_{AB} - \bar{I}_A\bar{Z} - \bar{V}_{AB} + \bar{I}_B\bar{Z} = 0 \tag{10-1}$$

$$\bar{E}_{BC} - \bar{I}_B\bar{Z} - \bar{V}_{BC} + \bar{I}_C\bar{Z} = 0 \tag{10-2}$$

$$\bar{E}_{CA} - \bar{I}_C\bar{Z} - \bar{V}_{CA} + \bar{I}_A\bar{Z} = 0 \tag{10-3}$$

Regroup the above equations as follows:

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$$\bar{E}_{AB} = \bar{V}_{AB} + \bar{Z}(\bar{I}_A - \bar{I}_B) \quad (10\text{-}4)$$

$$\bar{E}_{BC} = \bar{V}_{BC} + \bar{Z}(\bar{I}_B - \bar{I}_C) \quad (10\text{-}5)$$

$$\bar{E}_{CA} = \bar{V}_{CA} + \bar{Z}(\bar{I}_C - \bar{I}_A) \quad (10\text{-}6)$$

Next, the difference in line current terms in equations (10-4) through (10-6) should be solved. From figures A-3 and A-4:

$$\bar{I}_A = \bar{I}_{AB} - \bar{I}_{CA} \quad (10\text{-}7)$$

$$\theta_A = 0° - \underline{/I_A}$$

$$\alpha_A = \text{-}\theta_A + \beta$$

$$\bar{I}_B = \bar{I}_{BC} - \bar{I}_{AB} \quad (10\text{-}8)$$

$$\theta_B = \text{-}120° - \underline{/I_B}$$

$$\alpha_B = \text{-}\theta_B + \beta$$

$$\bar{I}_C = \bar{I}_{CA} - \bar{I}_{BC} \quad (10\text{-}9)$$

$$\theta_C = 120° - \underline{/I_C}$$

$$\alpha_C = \text{-}\theta_C + \beta$$

Where:

$\underline{/I_A}$, $\underline{/I_B}$, $\underline{/I_C}$: angle of $\bar{I}_A$, $\bar{I}_B$, and $\bar{I}_C$

θ_A: angle between $\bar{V}_{AB}$ and $\bar{I}_A$

θ_B: angle between $\bar{V}_{BC}$ and $\bar{I}_B$

θ_C: angle between $\bar{V}_{CA}$ and $\bar{I}_C$

The phase load currents are expressed as:

$$\bar{I}_{AB} = I_{AB}\underline{/(0° - \theta_{AB})}$$

$$\bar{I}_{BC} = I_{BC}\underline{/(\text{-}120° - \theta_{BC})}$$

$$\bar{I}_{CA} = I_{CA}\underline{/(120° - \theta_{CA})}$$

From equations (10-7), (10-8), and (10-9), the percent voltage drop in each line is computed as follows:

$$D\% = IL\left(\frac{z\cos(\alpha)}{E}\right) * 100 \quad (10\text{-}10)$$

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Let the drop factor as follows:

$$DF = \left(\frac{zcos(\alpha)}{E}\right) * 100$$

Equation (10-10) reduces to:

$$D\% = IL(DF) \tag{10-11}$$

The difference in line currents $\bar{I}_X$ are computed as follows:

$$\begin{aligned} \bar{I}_{X(AB)} &= \bar{I}_A - \bar{I}_B \\ &= \bar{I}_{AB} - \bar{I}_{BC} - \bar{I}_{CA} + \bar{I}_{AB} \\ &= 2\bar{I}_{AB} - \bar{I}_{BC} - \bar{I}_{CA} \end{aligned} \tag{10-12}$$

and

$$\theta_{X(AB)} = 0° - \underline{/I_{X(AB)}}$$

$$\alpha_{X(AB)} = \text{-}\theta_{X(AB)} + \beta$$

$$\begin{aligned} \bar{I}_{X(BC)} &= \bar{I}_B - \bar{I}_C \\ &= \bar{I}_{BC} - \bar{I}_{CA} - \bar{I}_{AB} + \bar{I}_{BC} \\ &= 2\bar{I}_{BC} - \bar{I}_{CA} - \bar{I}_{AB} \end{aligned} \tag{10-13}$$

and

$$\theta_{X(BC)} = \text{-}120° - \underline{/I_{X(BC)}}$$

$$\alpha_{X(BC)} = \text{-}\theta_{BC} + \beta$$

$$\begin{aligned} \bar{I}_{X(CA)} &= \bar{I}_C - \bar{I}_A \\ &= \bar{I}_{CA} - \bar{I}_{AB} - \bar{I}_{BC} + \bar{I}_{CA} \\ &= 2\bar{I}_{CA} - \bar{I}_{AB} - \bar{I}_{BC} \end{aligned} \tag{10-14}$$

and

$$\theta_{X(CA)} = 120° - \underline{/I_{X(CA)}}$$

$$\alpha_{X(CA)} = \text{-}\theta_{X(CA)} + \beta$$

Where:

$\theta_{X(AB)}$: Angle between $\bar{V}_{AB}$ and $\bar{I}_{X(AB)}$

$\theta_{X(BC)}$: Angle between $\bar{V}_{BC}$ and $\bar{I}_{X(BC)}$

$\theta_{X(CA)}$: Angle between $\bar{V}_{CA}$ and $\bar{I}_{X(CA)}$

From equations (104), (10-5), and (10-6), the general equation for the percent drop in the individual phase is:

$$D\% = \frac{|\bar{E}_X| - |\bar{V}|}{E_X} * 100 \tag{10-15}$$

The magnitude of $\bar{V}$ may be approximated as:

$$V = E_X - I_X Z cos(\alpha_X)$$

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Substitution of V in equation (10-15) gives:

$$D\% = I_X Z \left(\frac{cos(\alpha_X)}{E_X}\right) * 100 \quad (10\text{-}16)$$

Let Z=zL, equation (10-16) becomes:

$$D\% = I_X L \left(\frac{zcos(\alpha_X)}{E_X}\right) * 100 \quad (10\text{-}17)$$

For example, the percent drop in phase AB is:

$$D_{AB}\% = I_{X(AB)} L \left(\frac{zcos(\alpha_{X(AB)})}{E_{AB}}\right) * 100$$

Equation (10-17) can also be written as:

$$D\% = I_X L(DF) \quad (10\text{-}18)$$

With,

$$DF = \left(\frac{zcos(\alpha_X)}{E_X}\right) * 100 \quad (10\text{-}19)$$

For a single-phase system, equation (10-19) becomes:

$$DF = \left(\frac{zcos(\alpha)}{E}\right) * 100 \quad (10\text{-}20)$$

Then, the percent drop equation is as follows:

$$D\% = 2IL(DF) \quad (10\text{-}21)$$

A.11. DERIVATION OF VOLTAGE DROP EQUATIONS FOR THREE-PHASE/LIGHTING SYSTEMS USING WATTS AND VARS

In lighting systems, the use of watts (P) and vars (Q) instead of currents (I) in voltage drop equations avoids necessity for calculating all phase and line currents. This is an advantage when real and reactive power or the apparent power (S) and power factor of connected loads are known.

Consider a balanced three-phase voltage and current vector diagram as shown in figure A-5:

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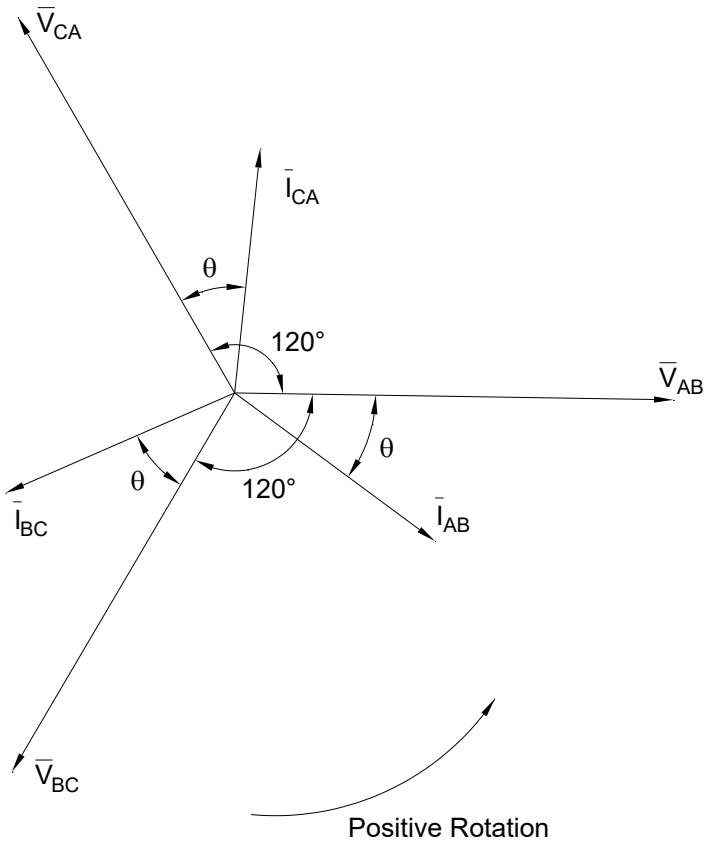


FIGURE A-5. Three-phase voltage and current phasor diagram.

Assume all phase loads are inductive. The total complex power for each phase can be determined as follows:

$$\bar{S} = \bar{V}\bar{I}_L^{\,*}$$

Where $\bar{I}_L^{\,*}$ is the complex conjugate of $\bar{I}_L$

The apparent power (S) for each phase (leg) is defined as:

$$S = |S| = \bar{V}I_L$$

For phase AB:

$$S_{AB} = V_{AB}I_{AB}$$

In rectangular form:

$$S_{AB} = V_{AB}(1 + j0) * I_{AB}[cos(\theta_{AB}) + jsin(\theta_{AB})]$$

$$S_{AB} = V_{AB}I_{AB}cos(\theta_{AB}) + JV_{AB}I_{AB}sin(\theta_{AB}) \quad (11\text{-}1)$$

Since,

$$P_{AB} = V_{AB}I_{AB}cos(\theta_{AB}),\ Q_{AB} = V_{AB}I_{AB}sin(\theta_{AB})$$

Equation (11-1) becomes:

$$S_{AB} = P_{AB} + jQ_{AB} \quad (11\text{-}2)$$

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The apparent powers S_{BC} for phase BC and S_{CA} for phase CA can be referenced to the same axis as S_{AB}. For inductive loads, their horizontal projections are:

$$S_{BC}\cos(\text{-}120° - \theta_{BC}) \text{ and } S_{CA}\cos(120° - \theta_{CA})$$

For the trigonometric identity:

$$cos(\theta_1 - \theta_2) = cos(\theta_1)cos(\theta_2) + sin(\theta_1)sin(\theta_2)$$

$$S_{BC}cos(\text{-}120° - \theta_{BC}) = S_{BC}[cos(\text{-}120°)cos(\theta_{BC}) + sin(\text{-}120°)sin(\theta_{BC})]$$

$$= S_{BC}\left[\text{-}(1/2)cos(\theta_{BC}) - \left(\sqrt{3}/2\right)sin(\theta_{BC})\right]$$

For phase CA:

$$S_{CA}cos(120° - \theta_{CA}) = S_{CA}[cos(120°)cos(\theta_{CA}) + sin(120°)sin(\theta_{CA})]$$

$$= S_{CA}\left[\text{-}(1/2)cos(\theta_{CA}) + \left(\sqrt{3}/2\right)sin(\theta_{CA})\right]$$

As indicated above, the two quantities can be rewritten:

$$S_{BC}cos(\text{-}120° - \theta_{BC}) = \text{-}(1/2)P_{BC} - \left(\sqrt{3}/2\right)Q_{BC}$$

$$S_{CA}cos(120° - \theta_{CA}) = \text{-}(1/2)P_{CA} + \left(\sqrt{3}/2\right)Q_{CA}$$

The apparent power S_A and real power P_A in line A (or at node A) can be calculated as follows:

$$S_A = S_{AB} - S_{CA} \quad P_A = P_{AB} - P_{CA}$$

Substitute P_{CA} in the above equation:

$$\begin{aligned} P_A &= P_{AB} - S_{CA}cos(120° - \theta_{CA}) \\ &= P_{AB} - (-(1/2)P_{CA} + \left(\sqrt{3}\right)Q_{CA}) \\ &= P_{AB} + (1/2)P_{CA} - \left(\sqrt{3}\right)Q_{CA} \end{aligned} \tag{11-3}$$

For lines B and C:

$$\begin{aligned} P_B &= P_{BC} - P_{AB} \\ &= P_{BC} + (1/2)P_{AB} - \left(\sqrt{3}\right)Q_{AB} \end{aligned} \tag{11-4}$$

$$\begin{aligned} P_C &= P_{CA} - P_{BC} \\ &= P_{CA} + (1/2)P_{BC} - \left(\sqrt{3}\right)Q_{BC} \end{aligned} \tag{11-5}$$

Similarly,

$$\begin{aligned} Q_A &= Q_{AB} - Q_{CA} \\ &= Q_{AB} + (1/2)Q_{CA} - \left(\sqrt{3}/2\right)P_{CA} \end{aligned} \tag{11-6}$$

$$\begin{aligned} Q_B &= Q_{BC} - Q_{AB}{}' \\ &= Q_{BC} + (1/2)Q_{AB} - \left(\sqrt{3}/2\right)P_{AB} \end{aligned} \tag{11-7}$$

$$Q_C = Q_{CA} - Q_{BC}$$

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$$= Q_{CA} + (1/2)Q_{BC} - (\sqrt{3}/2)P_{BC} \quad (11\text{-}8)$$

The net power P_X at two nodes of a three-phase delta connected load is as follows:

Phase AB:

$$P_{X(AB)} = P_A - P_B \quad (11\text{-}9)$$

In equation (11-4) for line B, P_B can be expressed as:

$$P_B = P_{BC} - P_{AB}$$
$$= S_{BC}[-(1/2)cos(\theta_{BC}) - (\sqrt{3}/2)sin(\theta_{BC})] - P_{AB}$$
$$= -P_{AB} - (1/2)P_{BC} - (\sqrt{3}/2)Q_{BC} \quad (11\text{-}10)$$

Substitute P_A and P_B in equation (11-9), equation for P_X in phase AB is obtained:

$$P_{X(AB)} = 2P_{AB} + (1/2)(P_{BC} + P_{CA}) + (\sqrt{3}/2)(Q_{BC} - Q_{CA}) \quad (11\text{-}11)$$

For phases BC and CA:

$$P_{X(BC)} = 2P_{CB} + (1/2)(P_{CA} + P_{AB}) + (\sqrt{3}/2)(Q_{CA} - Q_{AB}) \quad (11\text{-}12)$$

$$P_{X(CA)} = 2P_{CA} + (1/2)(P_{AB} + P_{BC}) + (\sqrt{3}/2)(Q_{AB} - Q_{BC}) \quad (11\text{-}13)$$

Similarly, Q_X can be shown as:

$$Q_{X(AB)} = -[2Q_{AB} + (1/2)(Q_{BC} + Q_{CA}) - (\sqrt{3}/2)(P_{BC} - P_{CA})] \quad (11\text{-}14)$$

$$Q_{X(BC)} = -[2Q_{BC} + (1/2)(Q_{CA} + Q_{AB}) - (\sqrt{3}/2)(P_{CA} - P_{AB})] \quad (11\text{-}15)$$

$$Q_{X(CA)} = -[2Q_{CA} + (1/2)(Q_{AB} + Q_{BC}) - (\sqrt{3}/2)(P_{AB} - P_{BC})] \quad (11\text{-}16)$$

Equations for capacitive loads, or mixed inductive and capacitive loads, can also be derived by assigning the proper sign to each phase angle (negative for inductive loads and positive for capacitive loads) in determining the cosine and sine of (±120° ±Θ). Therefore, the capacitive loads, the signs of the last terms in equations (11-11) through (11-16) should be reversed.

Note that if these equations are divided by load voltage V, similar expressions involving resistive and reactive components of current will result. In some cases, the use of these equations may be more convenient than the vector rotation method when currents are used in the calculations.

For checking current carrying capacity of conductors, only the largest phase apparent power (S) or (S_X) needs to be considered. The general equations are:

$S = \sqrt{P^2 + Q^2}$	$S_X = \sqrt{{P_X}^2 + {Q_X}^2}$	(11-17)
$I = \frac{S}{V} = \left(\frac{P}{V}\right)\left(\frac{1}{cos(\Theta)}\right)$	$I_X = \frac{S_X}{V} = \left(\frac{P_X}{V}\right)\left(\frac{1}{cos(\Theta_X)}\right)$	(11-18)
$\theta = cos^{-1}\left(\frac{P}{S}\right)$	$\theta_X = cos^{-1}\left(\frac{P_X}{S_X}\right)$	(11-19)
or	or	
$\theta = tan^{-1}\left(\frac{Q}{P}\right)$	$\theta_X = tan^{-1}\left(\frac{Q_X}{P_X}\right)$	(11-20)

Θ_X is positive if Q_X is positive.

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Θ_X is greater than 90° if P_X is negative.

Recall the general voltage drop equation for three-phase lighting systems derived in A.10:

$$D\% = I_X L \left(\frac{z\cos(\alpha_X)}{E_X}\right) * 100 \quad (10\text{-}17)$$

Since,

$$P_X = I_X V \cos(\theta_X)$$

The current I_X is then:

$$I_X = \left(\frac{P_X}{V\cos(\theta_X)}\right) \quad (11\text{-}21)$$

Substitution of I_X in equation (10-17) gives:

$$D\% = P_X L \left(\frac{1}{V\cos(\theta_X)}\right)\left(\frac{z\cos(\alpha_X)}{E_X}\right) * 100 \quad (11\text{-}22)$$

From equation (11-22), a new drop factor is defined as:

$$DF' = \left(\frac{1}{V\cos(\theta_X)}\right)\left(\frac{z\cos(\alpha_X)}{E_X}\right) * 100 \quad (11\text{-}23)$$

Finally,

$$D\% = P_X L(DF) \quad (11\text{-}24)$$

For single-phase circuits, in equation (11-22), the parameters E_X, P_X, α_X, and Θ_X become E, P, α, and Θ, respectively. The percent voltage drop for a single-phase circuit is then:

$$D\% = 2PL(DF) \quad (11\text{-}25)$$

With,

$$DF' = \left(\frac{1}{V\cos(\theta)}\right)\left(\frac{z\cos(\alpha)}{E}\right) * 100 \quad (11\text{-}26)$$

The computed drop factors for LSTSGU cables from equations (11-23) or (11-26) are shown in table XVII. Now, let derive equation for D% in function of watts (P) and vars (Q). Equation (11-18) can be written as:

$$P_X = S_X \cos(\theta_X), \quad (11\text{-}27)$$

Substitution of P_X in equation (11-22) gives:

$$D\% = \left(\frac{S_X L z\cos(\theta_X)\cos(\alpha_X)}{E_X V\cos(\theta_X)}\right) * 100 \quad (11\text{-}28)$$

$$D\% = \left(\frac{S_X L z\cos(\alpha_X)}{E_X V}\right) * 100 \quad (11\text{-}28)$$

Rewrite equation (11-28) as:

$$D\% = \left(\frac{S_X L z\cos(\beta \pm \theta_X)}{E_X V}\right) * 100 \quad (11\text{-}29)$$

$$D\% = \left(\frac{S_X L z[\cos(\beta)\cos(\theta_X) \pm \sin(\beta)\sin(\theta_X)]}{E_X V}\right) * 100 \quad (11\text{-}30)$$

Recall:

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$$P_X = S_X cos(\theta_X) \;\; Q_X = S_X sin(\theta_X)$$

$$R_X = zcos(\beta) \qquad\qquad X = zsin(\beta)$$

After the substitution, equation (11-30) becomes:

$$D\% = \left(\frac{L(RP_X \pm XQ_X)}{E_X V}\right) * 100 \tag{11-31}$$

For lighting systems, let assume:

$$E_X = 120V, \qquad V = 115V$$

Equation (11-31) can be written as:

$$D\% = (7.25)(10^{-5})L(RP_X \pm XQ_X) * 100 \tag{11-32}$$

For a single-phase system, equation (11-31) becomes:

$$D\% = 2\left(\frac{L(RP \pm XQ)}{EV}\right) * 100 \tag{11-33}$$

The signs "+" and "-" are for inductive and capacitive loads, respectively. The values of R and X for different types of cables can be found in tables XI through XVII.

A.12. DERIVATION OF VOLTAGE DROP EQUATIONS FOR THREE-PHASE FOUR-WIRE SYSTEMS

Consider a three-phase four-wire (Wye) system as shown in :

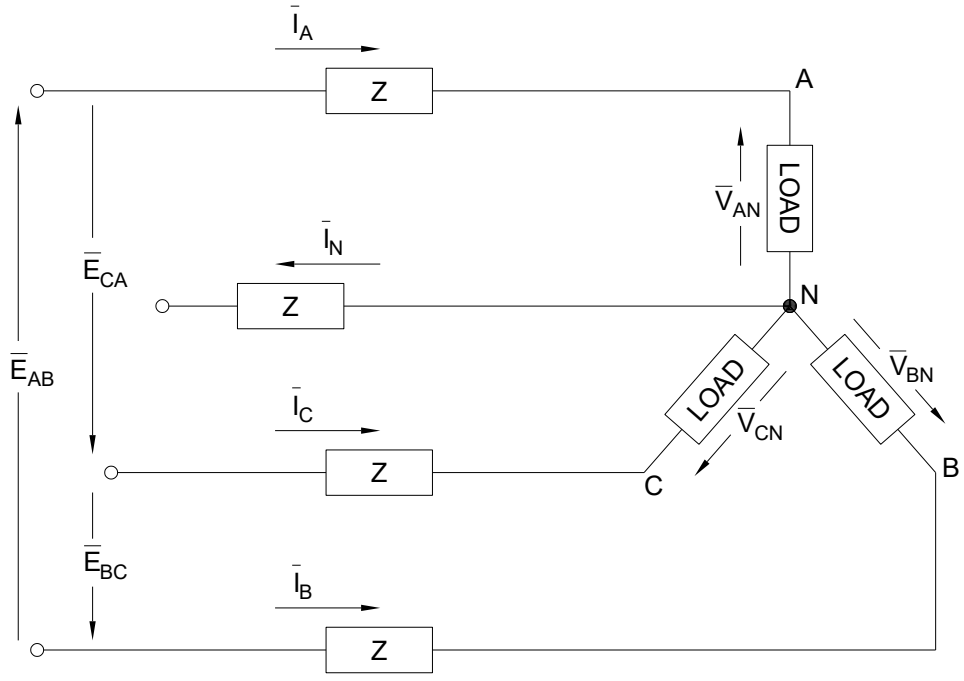


FIGURE A-6. Three-phase four-wire system.

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If the system is balanced, the voltage drops for all lines are the same. For example, in line A, the precent drop is:

$$D_A\% = \left(\frac{E_{AN}-V_{AN}}{E_{AN}}\right) * 100 \quad (12\text{-}1)$$

or

$$D_A\% = IL\left(\frac{z\cos(\alpha_A)}{E_{AN}}\right) * 100 \quad (12\text{-}2)$$

Where:

$$I = I_A = \frac{V_{AN}}{Z_{AN}} \quad (12\text{-}3)$$

$$\alpha = \text{-}\theta_A + \beta \quad (10\text{-}7)$$

With, $E_{AN} = \left(E_{AB}/\sqrt{3}\right) = E$, equation (12-2) becomes:

$$D_A\% = IL\left(\frac{z\cos(\alpha_A)}{E}\right) * 100 \quad (12\text{-}4)$$

Or

$$D_A\% = IL(DF) \quad (12\text{-}5)$$

With,

$$DF = \left(\frac{z\cos(\alpha_A)}{E}\right) * 100$$

Now, let derive equations for the drop in each phase (the drop in phase A plus the drop in phase B). From figure A-6, the source voltage in line A to line B is:

$$\bar{E}_{AB} = \bar{V}_{AN} - \bar{V}_{BN} + \bar{Z}(\bar{I}_A - \bar{I}_B) \quad (12\text{-}5)$$

For $\bar{V}_{BN} = \bar{V}_{AN}\left(\underline{1/\text{-}120°}\right)$

$$\bar{E}_{AB} = \bar{V}_{AN} - \bar{V}_{AN}\left(\underline{1/\text{-}120°}\right) + \bar{Z}(\bar{I}_A - \bar{I}_B)$$

$$= \bar{V}_{AN}(1.5 + j0.866) + \bar{Z}(\bar{I}_A - \bar{I}_B)$$

$$= \bar{V}_{AN}\left(\underline{\sqrt{3}/30°}\right) + \bar{Z}(\bar{I}_A - \bar{I}_B)$$

Since $\bar{V}_{AB} = \bar{V}_{AN}\left(\underline{\sqrt{3}/30°}\right)$, equation (12-5) becomes:

$$\bar{E}_{AB} = \bar{V}_{AB} + \bar{Z}(\bar{I}_A - \bar{I}_B) \quad (12\text{-}6)$$

Similarly:

$$\bar{E}_{BC} = \bar{V}_{BC} + \bar{Z}(\bar{I}_B - \bar{I}_C) \quad (12\text{-}7)$$

and

$$\bar{E}_{CA} = \bar{V}_{CA} + \bar{Z}(\bar{I}_C - \bar{I}_A) \quad (12\text{-}8)$$

Note that equations (12-6), (12-7), and (12-8) are similar to equations (10-4), (10-5), (10-6) in A.10. After the line currents I are computed from equation (12-3), the currents I_X and the associated angles Θ_X and α_X can be determined in accordance with the following equations as derived in A.10:

$$\bar{I}_{X(AB)} = \bar{I}_A - \bar{I}_B \quad (10\text{-}12)$$

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$$\theta_{X(AB)} = 0° - \underline{/I_{X(AB)}}$$

$$\alpha_{X(AB)} = \text{-}\theta_{x(AB)} + \beta$$

$$\bar{I}_{X(BC)} = \bar{I}_B - \bar{I}_C \tag{10-13}$$

$$\theta_{X(BC)} = \text{-}120° - \underline{/I_{X(BC)}}$$

$$\alpha_{X(BC)} = \text{-}\theta_{x(BC)} + \beta$$

$$\bar{I}_{X(CA)} = \bar{I}_C - \bar{I}_A \tag{10-14}$$

$$\theta_{X(CA)} = 120° - \underline{/I_{X(CA)}}$$

$$\alpha_{X(CA)} = \text{-}\theta_{x(CA)} + \beta$$

The percent drop in each phase is calculated from equation (10-17) as derived in A.10:

$$D\% = I_X L \left(\frac{z\cos(\alpha_X)}{E_X}\right) * 100 \tag{10-17}$$

With,

$$DF = \left(\frac{z\cos(\alpha_X)}{E_X}\right) * 100$$

Equation (10-17) reduces to:

$$D\% = I_X L(DF)$$

A.13. DERIVATION OF VOLTAGE DROP EQUATIONS FOR UNBALANCED SYSTEMS BY SYMMETRICAL COMPONENT METHOD

For analyzing unbalanced circuit, all loads are assumed unequal. Therefore, the currents in all phase loads are different. The symmetrical component method is preferred here to analyze the network. To use this method, it is convenient to keep the voltages, currents, and impedances in phasor forms. Figures A-7 and A-8 represent a three-phase unbalanced system and a three-phase unbalanced currents.

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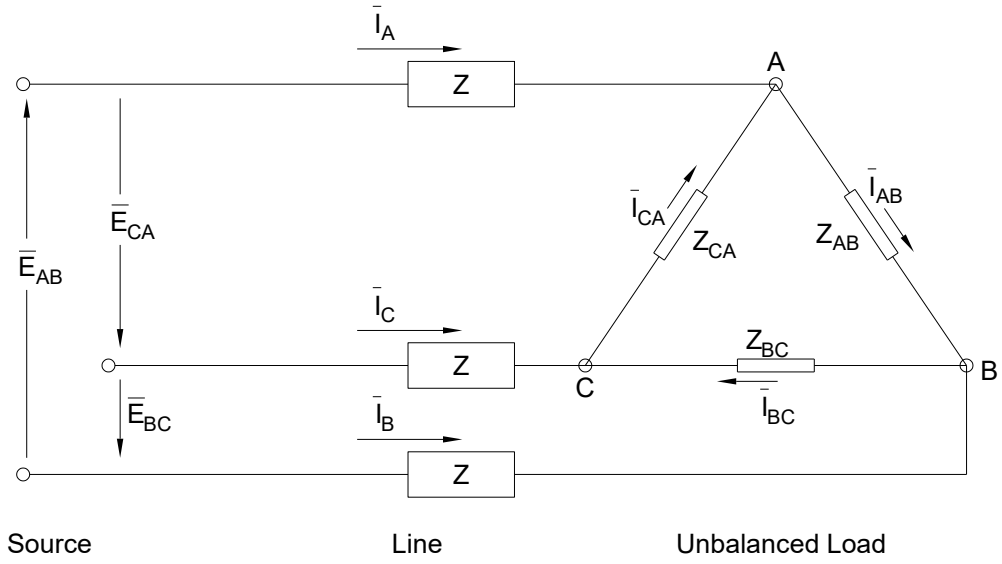


FIGURE A-7. Three-phase unbalanced system.

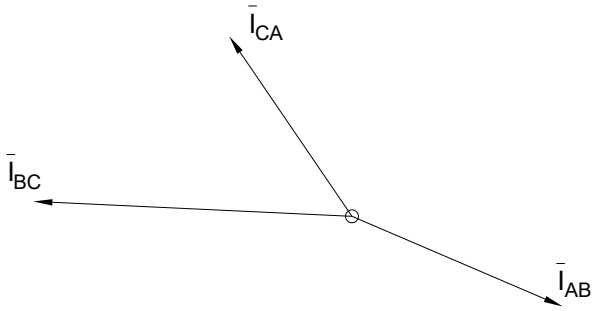


FIGURE A-8. Three-phase unbalanced currents.

Assume all three loads are unbalanced. The load currents are:

$$|\bar{I}_{AB}| \neq |\bar{I}_{BC}| \neq |\bar{I}_{CA}|$$

and

$$\Theta_{AB} \neq \Theta_{BC} \neq \Theta_{CA}$$

The line currents, as functions of load currents at nodes A, B, and C of figure A-7, are determined from equations derived in A.10 as follows:

$$\bar{I}_A = \bar{I}_{AB} - \bar{I}_{CA} \quad (10\text{-}7)$$

$$\bar{I}_B = \bar{I}_{BC} - \bar{I}_{AB} \quad (10\text{-}8)$$

$$\bar{I}_C = \bar{I}_{CA} - \bar{I}_{BC} \quad (10\text{-}9)$$

Rearranging equation (10-7) as follows:

$$\bar{I}_A + \bar{I}_{CA} - \bar{I}_{AB} = 0 \quad (13\text{-}1)$$

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Substitution of equation (10-9) into equation (13-1) yields:

$$\bar{I}_A + \bar{I}_C + \bar{I}_{BC} - \bar{I}_{AB} = 0 \qquad (13\text{-}2)$$

Substitution of equation (10-8) into equation (13-2) yields:

$$\bar{I}_A + \bar{I}_B + \bar{I}_C = 0 \qquad (13\text{-}3)$$

Consider the zero sequence component of $\bar{I}_A$:

$$\bar{I}_{A0} = (1/3)(\bar{I}_A + \bar{I}_B + \bar{I}_C) \qquad (13\text{-}4)$$

Substitution from equation (13-3) gives:

$$\bar{I}_{A0} = (1/3)(0) = 0$$

By similar derivation:

$$\bar{I}_{A0} = \bar{I}_{B0} = \bar{I}_{C0} = 0 \qquad (13\text{-}5)$$

Thus, in a three-phase delta system the zero sequence currents are zero.

Now, let calculate the positive and negative sequence components of the line currents in term of current in phase AB. Figures A-9 and A-10 show the rotation sequences of positive and negative sequence currents.

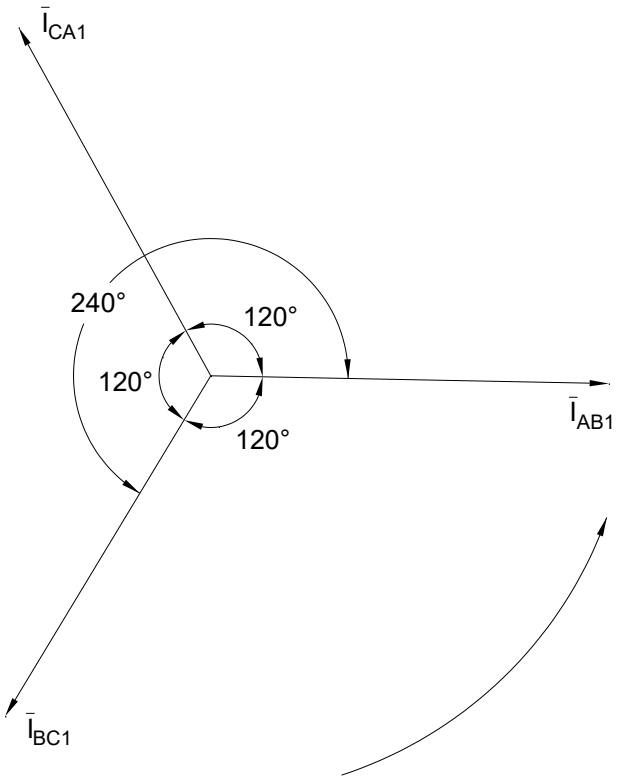


FIGURE A-9. Positive sequence currents.

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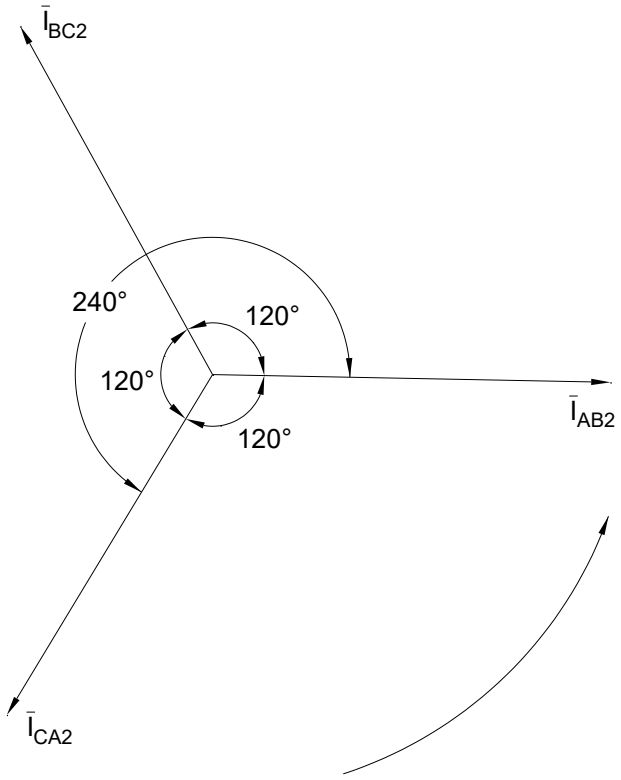


FIGURE A-10. Negative sequence currents.

Components of $\bar{I}_A$

From equation (10-7), the positive sequence current is:

$$\bar{I}_{A1} = \bar{I}_{AB1} - \bar{I}_{CA1} \tag{13-6}$$

From figure A-9:

$$\bar{I}_{CA1} = \bar{I}_{AB1}\underline{/120°} \tag{13-7}$$

Substitution equation (13-7) into equation (13-6) yields:

$$\begin{aligned}\bar{I}_{A1} &= \bar{I}_{AB1} - \bar{I}_{AB1}\underline{/120°}, \\ &= \bar{I}_{AB1}(1\left(1\underline{/0°} - 1\underline{/120°}\right)) \\ &= \bar{I}_{AB1}(1 + 0.5 - j0.866) \\ &= \bar{I}_{AB1}\left(\sqrt{3}\underline{/\text{-}30°}\right)\end{aligned} \tag{13-8}$$

From equation (13-7), the negative sequence current is:

$$\bar{I}_{A2} = \bar{I}_{AB2} - \bar{I}_{CA2} \tag{13-9}$$

From figure A-10:

$$\bar{I}_{CA2} = \bar{I}_{AB2}\underline{/240°} \tag{13-10}$$

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Substitution of equation (13-10) into equation (13-9) gives:

$$\bar{I}_{A2} = \bar{I}_{AB2} - \bar{I}_{AB2}\underline{/240°}$$

$$= \bar{I}_{AB2}(1 - (\underline{1/0°} - \underline{1/240°}))$$

$$= \bar{I}_{AB2}(1 + 0.5 + j0.866)$$

$$= \bar{I}_{AB2}(\sqrt{3}\underline{/30°}) \quad (13\text{-}11)$$

Components of $\bar{I}_B$

From equation (10-8), the positive sequence current is:

$$\bar{I}_{B1} = \bar{I}_{BC1} - \bar{I}_{AB1} \quad (13\text{-}12)$$

From figure A-9:

$$\bar{I}_{BC1} = \bar{I}_{AB1}\underline{/240°} \quad (13\text{-}13)$$

Substitution equation (13-13) into equation (13-12) yields:

$$\bar{I}_{B1} = \bar{I}_{AB1}\underline{/240°} - \bar{I}_{AB1}$$

$$= \bar{I}_{AB1}(\underline{1/240°} - \underline{1/0°})$$

$$= \bar{I}_{AB1}(-1 - 0.5 - j.866)$$

$$= \bar{I}_{AB1}(\sqrt{3}\underline{/-150°}) \quad (13\text{-}14)$$

From equation (13-8), the negative sequence current is:

$$\bar{I}_{B2} = \bar{I}_{BC2} - \bar{I}_{AB2} \quad (13\text{-}15)$$

From figure A-10:

$$\bar{I}_{BC2} = \bar{I}_{AB2}\underline{/120°} \quad (13\text{-}16)$$

Substitution of equation (13-16) into equation (13-15) yields:

$$\bar{I}_{B2} = \bar{I}_{AB2}\underline{/120°} - \bar{I}_{AB2}$$

$$= \bar{I}_{AB2} - (\underline{1/120°} - \underline{1/0°})$$

$$\bar{I}_{B2} = \bar{I}_{AB2}(-1 - 0.5 + j.866)$$

$$= \bar{I}_{AB2}(\sqrt{3}\underline{/150°}) \quad (13\text{-}17)$$

Components of $\bar{I}_C$

From equation (10-9), the positive sequence current is:

$$\bar{I}_{C1} = \bar{I}_{CA1} - \bar{I}_{BC1} \quad (13\text{-}18)$$

From figure A-9:

$$\bar{I}_{CA1} = \bar{I}_{AB1}\underline{/120°} \quad (13\text{-}19)$$

$$\bar{I}_{BC1} = \bar{I}_{AB1}\underline{/240°} \quad (13\text{-}20)$$

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Substitution of equations (13-19) and (13-20) into equation (13-18) gives:

$$\begin{aligned}\bar{I}_{C1} &= \bar{I}_{AB1}\underline{/120°} - \bar{I}_{AB1}\underline{/240°} \\ &= \bar{I}_{AB1}\left(\underline{1/120°} - \underline{1/240°}\right) \\ &= \bar{I}_{AB1}(0 + j1.732) \\ &= \bar{I}_{AB1}\left(\sqrt{3}\underline{/90°}\right)\end{aligned} \quad (13\text{-}21)$$

From equation (10-9), the negative sequence current is:

$$\bar{I}_{C2} = \bar{I}_{CA2} - \bar{I}_{BC2} \quad (13\text{-}22)$$

From figure A-9:

$$\bar{I}_{CA2} = \bar{I}_{AB2}\underline{/240°} \quad (13\text{-}23)$$

$$\bar{I}_{BC2} = \bar{I}_{AB2}\underline{/120°} \quad (13\text{-}24)$$

Substitution of equations (13-23) and (13-24) into equation (13-22) gives:

$$\begin{aligned}\bar{I}_{C2} &= \bar{I}_{AB2}\underline{/240°} - \bar{I}_{AB2}\underline{/120°} \\ &= \bar{I}_{AB2}\left(\underline{1/240°} - \underline{1/120°}\right) \\ &= \bar{I}_{AB2}(0 - j1.732) \\ &= \bar{I}_{AB2}\left(\sqrt{3}\underline{/-90°}\right)\end{aligned} \quad (13\text{-}25)$$

Take phase AB as the reference, the line currents can be written in terms of the sequence currents as follows:

$$\begin{aligned}\bar{I}_A &= \bar{I}_{A0} + \bar{I}_{A1} + \bar{I}_{A2} \\ &= \bar{I}_{AB1}\left(\sqrt{3}\underline{/\text{-}30°}\right) + \bar{I}_{AB2}\left(\sqrt{3}\underline{/30°}\right)\end{aligned} \quad (13\text{-}26)$$

$$\begin{aligned}\bar{I}_B &= \bar{I}_{B0} + \bar{I}_{B1} + \bar{I}_{B2} \\ &= \bar{I}_{AB1}\left(\sqrt{3}\underline{/\text{-}150°}\right) + \bar{I}_{AB2}\left(\sqrt{3}\underline{/150°}\right)\end{aligned} \quad (13\text{-}27)$$

$$\begin{aligned}\bar{I}_C &= \bar{I}_{C0} + \bar{I}_{C1} + \bar{I}_{C2} \\ &= \bar{I}_{AB1}\left(\sqrt{3}\underline{/90°}\right) + \bar{I}_{AB2}\left(\sqrt{3}\underline{/\text{-}90°}\right)\end{aligned} \quad (13\text{-}28)$$

To complete the above equations, the positive and negative sequence currents of phase AB should be calculated. From figure A-9, it can be shown that:

$$\bar{I}_{AB1} = (1/3)(\bar{I}_{AB} + a\bar{I}_{BC} + a^2\bar{I}_{CA}) \quad (13\text{-}29)$$

$$\bar{I}_{AB2} = (1/3)(\bar{I}_{AB} + a^2\bar{I}_{BC} + a\bar{I}_{CA}) \quad (13\text{-}30)$$

Where: $a = \underline{1/120°}$ $\quad a^2 = \underline{1/240°}$

Finally, to determine $\bar{I}_{AB}$, $\bar{I}_{BC}$, and $\bar{I}_{CA}$, the information supplied by the systems should be used. This information should consist of at least one of the following sets:

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Set 1:

Power (P) in kW of each phase load with pf=cos(Θ).

Terminal load voltage (V) of each phase.

Then,

$$\bar{I}_{AB} = [P_{AB}/V_{AB}cos(\theta_{AB})]\underline{/0° - \theta_{AB}} \quad (13\text{-}31)$$

$$\bar{I}_{BC} = [P_{BC}/V_{BC}cos(\theta_{BC})]\underline{/\text{-}120° - \theta_{BC}} \quad (13\text{-}32)$$

$$\bar{I}_{CA} = [P_{CA}/V_{CA}cos(\theta_{CA})]\underline{/120° - \theta_{CA}} \quad (13\text{-}33)$$

Set 2:

Load impedance (Z_{LO}) of each phase.

Load terminal voltage (V) of each phase.

Then,

$$\bar{I}_{AB} = \bar{V}_{AB}/\bar{Z}_{AB} \quad (13\text{-}34)$$

$$\bar{I}_{BC} = \bar{V}_{BC}/\bar{Z}_{BC} \quad (13\text{-}35)$$

$$\bar{I}_{CA} = \bar{V}_{CA}/\bar{Z}_{CA} \quad (13\text{-}36)$$

Once the line currents $\bar{I}$ have been determined, the percent voltage drop for each line can be calculated from equation (10-10) as derived in A.10:

$$D\% = IL\left(\frac{zcos(\alpha)}{E}\right) * 100 \quad (10\text{-}10)$$

Where:

L: Cable length.

z: Conductor impedance per phase per foot.

E: Line-to-neutral source voltage.

$$\alpha = \text{-}\theta + \beta \quad (9\text{-}2)$$

In the above equation, Θ is the angle by which the line current I lags the terminal load voltage, and it is calculated from equations (10-7), (10-8), or (10-9) as derived in A.10.

Check applicable ship specification for any system limits on voltage drop due to cable voltage drop.

A.14. DERIVATION OF CABLE RESISTANCES AND REACTANCES

This section contains the derivation of cable resistances and reactances used in tables XI through XVII. These cables reflect conductor diameters of standard AWG in accordance with MIL-DTL-24643.

A.14.1 Calculations of cable resistance. The ideal method of obtaining accurate cable resistance and reactance is by test measurements. In the absence of test data, these parameters can also be determined mathematically.

The method used in this handbook is based upon the one specified in ASTM B258 and ASTM B8. This method may be used to calculate with reasonable accuracy the resistances of any cables of concentric-lay-stranded conductors. The overall approach is as follows:

Calculate DC resistance at 20 °C for solid conductor using ASTM B258.

Adjust for concentric-lay-stranded conductors using ASTM B8.

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Adjust for temperature for which cable service will be designed.

Adjust for AC resistance by multiplying the DC resistance by the (AC/DC) resistance conversion ratio at any desired frequencies (60/400 Hz).

From the ASTM B258, DC resistance at 20 °C of conductor in ohm per 1,000 feet is given by:

$$R_{dc} = 105.35\left(\frac{\rho_o}{\delta d^2}\right) \quad (14\text{-}1)$$

Where:

ρ_o = resistivity of conductor (copper) at 20 °C

= 875.20 Ω lb/mile^2

δ = conductor (copper) density at 20 °C

= 8.89 g/cm^3

d = conductor diameter in mil

Equation (14-1) becomes:

$$R_{dc} = 10371.46\left(\frac{1}{d^2}\right)\Omega/1000ft \quad (14\text{-}2)$$

From equation (14-2), R_{DC} resistances for copper solid conductors are computed and shown in table XVIII below.

TABLE XVIII. DC conductor resistances at 20 °C.

Designation (AWG)	Conductor diameter (mil)	R_{DC} (Ω/10^{3}ft)
16	50.79	4.021
14	64.11	2.523
10	101.88	0.999
9	114.41	0.792
7	144.29	0.498
3	229.39	0.197
1	289.29	0.124
0	324.96	0.098
00	409.63	0.062
000	460.00	0.049
300 MCM	547.72	0.035
400 MCM	643.12	0.025

The data for these solid conductors should be adjusted for concentric-lay-stranded conductors. Due to the lay stranded conductors, the resistance per unit length of a stranded conductor will be slightly greater than that for an equivalent diameter solid conductor. ASTM B8 provides a mathematical method for deriving the multiplying factor that is used to modify DC resistance of a solid conductor for an equivalent concentric-lay-stranded conductor. A lay factor (m_{ind}) is determined for each wire in a concentric-lay-stranded conductor from:

$$m_{ind} = \sqrt{1 + (9.8696/n^2)} \quad (14\text{-}3)$$

Where:

n = length of lay/diameter of wire helical path.

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The lay factor (m) for the complete stranded conductor is the numerical average of the lay factors (m_{ind}) of the individual wires in the conductor. Finally, the increment factor K is calculated from:

$$K = (m - 1) * 100 \quad (14\text{-}4)$$

K is the percentage increase in resistance of a solid conductor for an equivalent concentric-lay-stranded conductor. In lieu of performing many calculations based upon detailed conductor geometry, which would be required by the above method, an increment factor of 2 percent will be used in accordance with ASTM B8. Therefore, the concentric-lay-stranding of a solid conductor results in a nominal increase in electrical resistance of 2 percent. The previous tables for DC resistance of solid conductors can be used to generate a table for concentric-lay-stranded conductors by applying this increment factor.

TABLE XIX. DC stranded conductor resistances at 20 °C.

Designation (AWG)	Number of Strands	R_{DC} ($\Omega/10^3$ft)
16	7	4.101
14	7	2.573
10	7	1.019
9	7	0.808
7	7	0.508
3	19	0.201
1	37	0.126
0	61	0.100
000	61	0.08
0000	61	0.63
300 MCM	91	0.036
400 MCM	127	0.026

Next, R_{DC} should be adjusted to the designed temperature of cable service. Resistance (R) at a selected temperature (t) can be calculated from the following equation:

$$R_t = R_{to}[1 + 0.00393(t - 20)] \quad (14\text{-}5)$$

Where:

R_{to} = wire resistance at 20 °C

t = temperature at which resistance is required

(0.00393) = temperature coefficient of resistance at 20 °C

At t = 65 °C:

$$R_t = R_{to}[1 + 0.00393(65 - 20)]$$

$$R_t/R_{to} = 1.177$$

Therefore, resistances in the preceding table should be multiplied by the above ratio to yield the DC resistances at 65 °C. The results of table XX below are obtained:

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TABLE XX. DC stranded conductor resistances at 65 °C.

Designation (AWG)	R_{DC} ($\Omega/10^3$ft)
16	4.827
14	3.028
10	1.199
9	0.951
7	0.598
3	0.237
1	0.148
0	0.118
00	0.094
000	0.074
300 MCM	0.042
400 MCM	0.031

Similarly, the (R_t/R_{to}) ratio at 45 °C is as follows:

$$R_t = R_{to}[1 + 0.00393(45 - 20)]$$

$$R_t/R_{to} = 1.098$$

The DC stranded-conductor resistances at 45 °C for various cables are tabulated below:

TABLE XXI. DC stranded conductor resistances at 45 °C.

Designation (AWG)	R_{DC} ($\Omega/10^3$ft)
16	4.503
14	2.825
10	1.119
9	0.887
7	0.558
3	0.221
1	0.138
0	0.110
00	0.087
000	0.069
300 MCM	0.040
400 MCM	0.029

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The DC resistances should be converted to AC resistances. The converting factor known as the skin effect ratio (SER) is determined as follows:

Determine the factor F:

$$F = 0.0635598\sqrt{(f\mu/R)} \tag{14-6}$$

Where:

f = System frequency

μ = Wire permeability

R = DC resistance in Ω/mile

Equation (14-6) can be rewritten as:

$$F = 0.027677\sqrt{(f/R)} \tag{14-7}$$

Where the wire permeability μ is assumed to be equal to 1.0000 and the DC resistance R is in ohm per 1,000 feet. Then, from the table below, the skin effect ratios are determined.

TABLE XXII. Skin effect ratio (SER).

F	SER	F	SER	F	SER
0.0	1.00000	1.3	1.01470	2.6	1.20056
0.1	1.00000	1.4	1.01969	2.7	1.22753
0.2	1.00000	1.5	1.02582	2.8	1.25620
0.3	1.00004	1.6	1.03323	2.9	1.28644
0.4	1.00013	1.7	1.04205	3.0	1.31809
0.5	1.00032	1.8	1.05240	3.1	1.35102
0.6	1.00067	1.9	1.06440	3.2	1.38504
0.7	1.00124	2.0	1.07816	3.3	1.41999
0.8	1.00212	2.1	1.09375	3.4	1.45570
0.9	1.00340	2.2	1.11126	3.5	1.49202
1.0	1.00519	2.3	1.13069	3.6	1.52879
1.1	1.00758	2.4	1.15207	3.7	1.56587
1.2	1.01071	2.5	1.17538	3.8	1.60314

With R_{DC} given in the previous tables, the AC resistance at 60 Hz and 400 Hz can be determined using the AC to DC resistance ratios in the following table:

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TABLE XXIII. DC to AC resistance conversion ratios.

Designation (AWG)	AC/DC Ratio at 60 Hz	AC/DC Ratio at 400 Hz
16	1.000	1.000
14	1.000	1.000
10	1.000	1.000
9	1.000	1.000
7	1.000	1.000
3	1.000	1.000
1	1.000	1.033
0	1.000	1.052
00	1.000	1.111
000	1.000	1.175
300 MCM	1.007	1.286
400 MCM	1.015	1.456

Therefore, AC resistance of conductors at 60/400 Hz and at any temperatures can be calculated by multiplying the appropriate (AC/DC) resistance conversion factors by the R_{DC} of concentric-lay-stranded conductors in the previous tables.

TABLE XXIV. AC resistance for conductors at 65 °C.

Designation (AWG)	R_{AC} at 60 Hz ($\Omega/10^3$ft)	R_{AC} at 400 Hz ($\Omega/10^3$ft)
16	4.827	4.827
14	3.028	3.028
10	1.199	1.199
9	0.951	0.951
7	0.598	0.598
3	0.237	0.237
1	0.148	0.153
0	0.118	0.124
00	0.074	0.082
000	0.059	0.069
300 MCM	0.042	0.054
400 MCM	0.031	0.045

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TABLE XXV. AC resistance for conductors at 45 °C.

Designation (AWG)	R_{AC} at 60 Hz ($\Omega/10^3$ft)	R_{AC} at 400 Hz ($\Omega/10^3$ft)
16	4.503	4.503
14	2.825	2.825
10	1.199	1.199
9	0.887	0.887
7	0.558	0.558
3	0.221	0.221
1	0.138	0.143
0	0.110	0.116
00	0.069	0.077
000	0.055	0.065
300 MCM	0.040	0.051
400 MCM	0.029	0.042

These resistance values are used in tables XI through XIV and XVI through XVII. For type LS6SGU cables in table XV, two conductors are in parallel for each phase. Therefore, the resistance per phase is half of the AC resistance (400 Hz) calculated earlier at the required designed temperature of 65 °C. The resulting resistances are shown in table XXVI.

TABLE XXVI. AC resistance for LS6SGU conductors at 65 °C.

Designation	R_{AC} at 400 Hz ($\Omega/10^3$ft)
LS6SGU-100	0.062
LS6SGU-125	0.049
LS6SGU-150	0.041
LS6SGU-200	0.035

A.14.2 Calculations of cable reactances. The determination of cable reactances is relatively complicated since the reactances are not only dependent upon the geometry of conductors in the cable, but also on the external environment of the cable (e.g., conducting armor material and closeness to surrounding steel). Below are the step-by-step calculations for cable reactances per phase. These calculations give a reasonable estimate of the total cable reactances. The actual reactances will differ slightly when all magnetic effects are considered, and only for larger cables at high frequencies.

Step 1.

Determine the conductor geometric mean radius (GMR) as follows:

$$GMR = 0.779(d/2) \tag{14-8}$$

Where d is the conductor diameter.

Step 2.

Determine the conductor geometric mean distance (GMD) as follows:

$$GMD = \sqrt[3]{D_{AB}D_{BC}D_{CA}} \tag{14-9}$$

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Where,

D_{AB}: Distance between centers of conductors A and B

D_{BC}: Distance between centers of conductors B and C

D_{CA}: Distance between centers of conductors C and A

Step 3.

Determine the reactance per phase as follows:

$$X = 0.05292(f/60)log_{10}(GMD/GMR)\ \Omega/1000ft \quad (14\text{-}10)$$

Where f is the frequency at which the reactance is calculated.

From equation (14-10) with conductor characteristics listed in MIL-DTL-24643 the following tables for reactances can be generated.

TABLE XXVII. Reactances for conductors based on AWG.

Designation (AWG)	GMR	GMD	X at 60 Hz ($\Omega/10^3$ft)	X at 400 Hz ($\Omega/10^3$ft)
16	0.0198	0.1300	0.0432	0.2883
14	0.0250	0.1430	0.0401	0.2672
10	0.0397	0.1870	0.0356	0.2374
9	0.0446	0.2620	0.0407	0.2713
7	0.0562	0.3100	0.0392	0.2616
3	0.0893	0.3340	0.0303	0.2021
1	0.1127	0.4070	0.0295	0.1967
0	0.1266	0.4530	0.0293	0.1953
000	0.1596	0.5570	0.0287	0.1915
0000	0.1792	0.6340	0.0290	0.1936
300 MCM	0.2133	0.7480	0.0288	0.1922
400 MCM	0.2463	0.8620	0.0288	0.1919

Again, the reactances for LS6SGU conductors are equal to half of the LSSGU conductor reactances as shown in table XXVIII.

TABLE XXVIII. Reactances for LS6SGU conductors at 65 °C.

Designation	X at 400 Hz ($\Omega/10^3$ft)
LS6SGU-100	0.0977
LS6SGU-125	0.0974
LS6SGU-150	0.0958
LS6SGU-200	0.0968

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CONCLUDING MATERIAL

Preparing activity:
Navy – SH
(Project 6145-2024-011)

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